

Plural Predication and Scope

by
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ABSTRACT

A central question in formal semantics concerns the division of labor between lexical semantics and the compositional system. Given a particular inference pattern, we can ask whether it is encoded in the lexicon or whether it arises from a particular syntactic interaction between different compositional ingredients. This question is especially important in the domain of plural predication, where there are competing lexical and syntactic accounts for a variety of phenomena. This dissertation argues, via a detailed investigation of cumulative inferences, reciprocal alternations and the interpretation of *together*, for a syntactic approach according to which these core inferences in plural predication result from different scope interactions between a limited set of syntactic operator. The dissertation therefore argues for a more impoverished lexicon and a richer syntax in the domain of plural predication.

The scope-based account faces an immediate explanatory challenge: while lexical analyses appeal to lexical generalizations to capture the distribution of readings in each construction, the scope-based account appears to overgenerate unattested readings. I show that these lexical stipulations can be replaced by independently motivated grammatical constraints that cut across different constructions. More specifically, I propose that two such constraints play a key role in plural predication: (i) a generalized economy constraint, inspired by Fox's (2000) scope economy, which rules out LFs when they can be proven to be equivalent to simpler LFs, and (ii) a constraint that requires implicit arguments to take lowest scope (J. Fodor & Fodor, 1980). The result is a framework in which the different ingredients of plural predication are in principle free to take scope in different positions, as long as they don't violate these independently needed constraints.

Finally, the scope-based account has some broader architectural consequences. The first consequence concerns the nature of economy constraints in the grammar. I show that in order to predict the desired distribution of readings, the generalized economy constraint has to be evaluated relative to an encapsulated system which only has access to some of the information that is needed to compute truth-conditions. My account therefore provides a new domain where we see evidence for a modular deductive system (following Fox, 2000; Gajewski, 2002; Fox & Hackl, 2006; Chierchia, 2013, a.o.). The second consequence concerns the division of labor between the lexicon and the syntax. I suggest that the impoverished role of the lexicon in plural predication might reflect broader constraints on what kinds of meanings can be encoded in a single lexical item.

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Chapter 1

Introduction

The main goal of this dissertation is to investigate the building blocks that give rise to a variety of inferences in plural predication. There are several phenomena where we see systematic inferences that relate pluralities to their parts. Two such patterns, which I will discuss in detail, are cumulative inferences, exemplified in (1), and reciprocal alternations, exemplified in (2). In (1), we see that pluralities appear to inherit properties from their parts, such that whenever a predicate is true of *Jane* and true of *Mary*, it also has to be true of the plurality denoted by *Jane and Mary*.¹ In (2), we see that predicates like *hug* and *date* appear to have 1-place variants that require plural arguments, giving rise to systematic reciprocal inferences that relate them to the two-place counterparts.

- (1) **Cumulative inferences:** (Link, 1983; Krifka, 1992; Kratzer, 2007, a.o.)
- a. Jane danced $\wedge$ Mary danced $\rightarrow$ Jane and Mary danced
 - b. Jane lifted the piano $\wedge$ Mary lifted the piano $\rightarrow$ Jane and Mary lifted the piano
 - c. etc.
- (2) **Reciprocal Alternations** (Levin, 1993; Winter, 2018, a.o.)
- a. Jane and Mary hugged $\rightarrow$ Jane hugged Mary $\wedge$ Mary hugged Jane
 - b. Jane and Mary dated $\rightarrow$ Jane dated Mary $\wedge$ Mary dated Jane
 - c. etc.

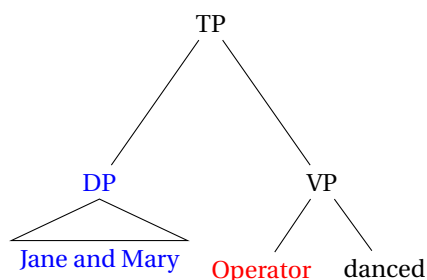
For both patterns, we can ask whether the inference is encoded in the lexical meanings of predicates like *dance* and *hug* (the lexical account) or whether it is due to syntactic operators that modify these lexical predicates (the syntactic account). Under the syntactic account, LFs like (3-b), where no pluralization operator is present between the plural DP and the predicate, lack the cumulative inference. On the other hand, LFs like (3-a) where there is a pluralization operator

¹I assume here that *and*, at least on one reading, takes two individuals of type *e* and returns their sum.

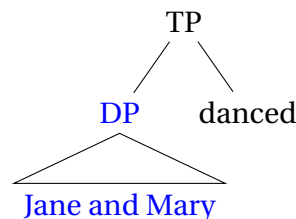
below the plural DP give rise to a cumulative inference. Lexical accounts, on the other hand, predict that cumulative inferences are always present regardless of the syntactic configuration, since the relevant inference is encoded into the lexical meanings of the predicates themselves (Krifka, 1992; Kratzer, 2007).

(3) **Structures under a syntactic account of cumulative inference:**

a. **LF with cumulative inference**



b. **No cumulative inference**

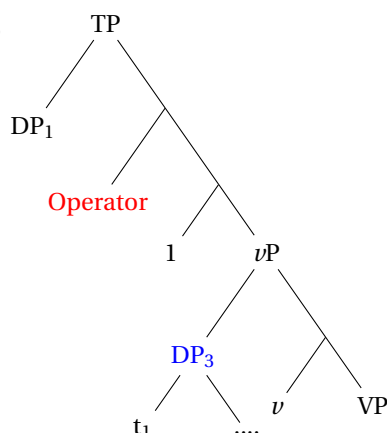


The syntactic account therefore predicts that if we can ensure an LF like (3-b), where no pluralization operator applies below the plural DP, we should see that cumulative inferences are blocked. Given that pluralization operators appear to be silent, the challenge is to identify ways to detect whether or not a pluralization operator is present in a given LF. If the pluralization operator could apply freely in any position, including to lexical predicates as in (3-a), then it would be very difficult to ensure an LF like (3-b), where there is no operator between the predicate and the plural DP. However, if the distribution of this operator is restricted on principled grounds, such that it can apply only in designated syntactic positions, then the syntactic account predicts that we will see a relationship between the availability of a cumulative inference and whether or not the plural DP is interpreted in one of the designated positions.

One reason to expect that the distribution of pluralization operators is restricted to certain syntactic positions comes from the strong theta criterion, which predicts that lexical predicates select for and merge directly with their arguments, therefore blocking any modifiers from applying to the predicate below the position where the external argument is merged (Hirsch, 2017, a.o.). Pluralization operators therefore can't apply directly to lexical predicates and must instead apply only to predicates derived by movement. The syntactic account of cumulative inferences, coupled with this strong theta criterion, therefore predicts that plural DPs which remain in situ, such as DP₃ in (4), have to take scope below the pluralization operator and do not give rise to cumulative inferences.²

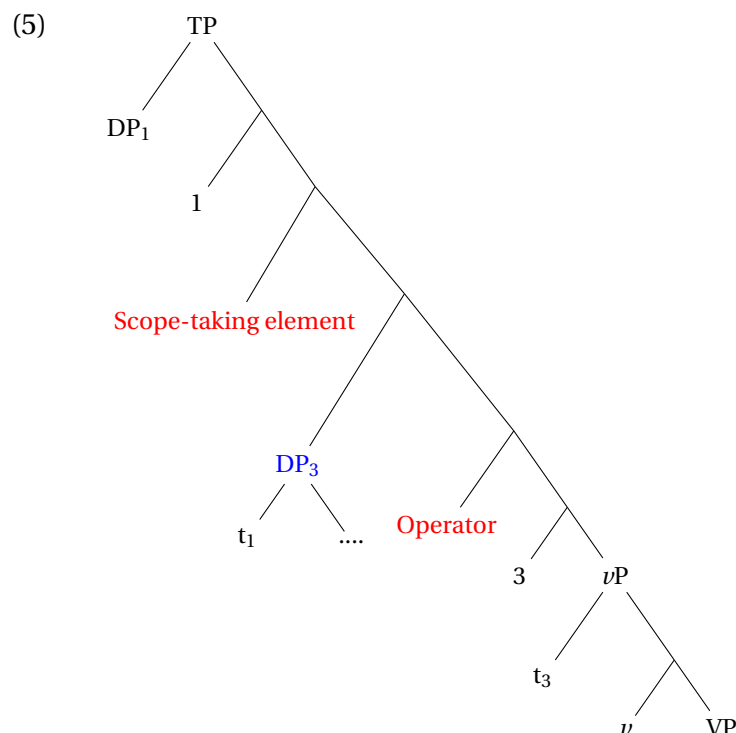
²The restriction that pluralization operators can't apply to lexical predicates might ultimately follow from a different principle than the strong theta criterion. What's important here is that we will find systematic evidence for this relationship between whether or not the relevant plural DP moves and whether we get a cumulative inference over that DP. This relationship provides evidence for a syntactic account, where the availability of cumulative inferences depends on where pluralization operators can apply. The strong theta criterion simply provides a principled reason to expect that the distribution of pluralization operators is restricted in this way.

- (4) **Prediction of the syntactic account, coupled with a strong theta criterion: no cumulative inference over DP₃**



In the first part of the dissertation, I provide two types of evidence that we in fact see this relationship between the availability of a cumulative inference over a plural DP and whether that DP moves from its base position. First, I argue that the Lebanese Arabic double subject construction forces certain plural DPs to remain in situ. I show that these plural DPs don't license cumulative inferences, thus providing evidence that these inferences are not encoded in the lexicon. I then show that LFs that give rise to cumulative inferences show processing signatures of syntactic movement, thus providing corroborating evidence from processing that cumulative inferences are parasitic on movement. Taken together, these two pieces of evidence support a syntactic analysis of cumulative inferences, where we expect to see a relationship between the syntactic position of a plural DP and whether that DP gives rise to a cumulative inference.

Given the conclusion that cumulative inferences are generated solely by syntactic pluralization operators, we can ask what happens when plural DPs are interpreted in different scope positions, relative to these operators and to other scope-taking elements. For example, given a particular scope-taking element, we can ask what reading we get when the plural DP moves from its base position to a position above a pluralization operator and below this scope-taking element, as shown in (5).



less disambiguate against distributive readings? I propose that the answer is provided by an independently motivated grammatical economy constraint that rules out LFs which are truth-conditionally equivalent to simpler LFs. I show that this economy constraint rules out the scope configurations which give rise to unattested distributive readings with *together*, while also making further novel predictions about the distribution of different readings with *together*. The result is a predictive scope-based theory that relies on independently needed grammatical constraints, rather than lexical stipulations, to account for the disambiguating role of *together*.

This account also allows us to probe how economy constraints interact with the grammar more generally. In particular, I show that in order for the economy constraint to predict the full distribution of readings with *together*, it has to be evaluated relative to an encapsulated formal system that does not have access to all of the information that is needed to compute truth-conditions (following Fox, 2000, for scope economy). More specifically, the system that evaluates economy has to be blind to various aspects of the context and to certain information regarding the identity of individual-denoting DPs, like *Jane* and *Mary*. I show that these modular properties are shared with other economy constraints like Fox's (2000) scope economy, suggesting that there is a generalized grammatical economy constraint that subsumes both scope economy and the economy effects governing the distribution of *together*. I show that this generalized economy constraint is also needed to predict the distribution of readings in the Lebanese Arabic double subject construction, thus providing additional support for the role of economy in constructions which disambiguate between different readings in plural predication.

Finally, the dissertation turns to reciprocal alternations, the second inference pattern mentioned above. I show that just like cumulative inferences, the availability of a reciprocal reading in a sentence like *Jane and Mary hugged* requires the plural DP *Jane and Mary* to move from its base position. I therefore argue that these covert reciprocals involve the same pluralization operators that generate cumulative inferences. More specifically, I propose, following Hackl (2002), that sentences like *Jane and Mary hugged* have a covert reciprocal anaphor in object position, which has to be bound by a higher pluralization operator.

One key challenge for this covert reciprocal analysis is that there are systematic differences in meaning between covert and overt reciprocals (Gleitman, Gleitman, Miller, & Ostrin, 1996; Winter, 2018, a.o.). I show that these seemingly lexical differences in fact follow from an independently needed constraint on scope, requiring implicit arguments to take narrowest scope (J. Fodor & Fodor, 1980). This constraint predicts that covert reciprocals require the relevant pluralization operator to take narrowest scope, while the overt counterparts allow for additional scope positions.

We therefore see that covert reciprocals, just like *together*, serve a disambiguating role. Namely, they isolate a low-scope construal and remove the ambiguity we get with overt re-

ciprocals, which can be bound from a variety of scope positions. Moreover, the different readings of *together* and the differences between covert and overt reciprocals rely on the same assumptions regarding clausal architecture. Taken together, these inquiries provide a unified picture of the different scope positions in which plural DPs and their associated pluralization operators can be interpreted.

The scope-based syntactic framework I propose raises some questions about the role of the lexicon in plural predication and beyond. The different phenomena I consider suggest that the lexicon itself does not encode any inferences that relate pluralities to their parts. Why should the lexicon be constrained in this way? I explore some possible answers to this larger question in the conclusion. Within plural predication, one possibility is that lexical predicates only have singular individuals in their domain and that predication of pluralities must therefore always be mediated by syntactic operators. Looking beyond plural predication, I also explore the broader idea that there are universal constraints on lexicalization, which prevent a meaning from being encoded in the lexicon when it can be independently generated in the syntax.

The rest of the introduction summarizes the dissertation in a bit more detail, highlighting how each chapter contributes to the broader picture presented above.

1.1 The sources of cumulative inferences (Ch.2)

The goal of chapter 2 is to provide evidence from the Lebanese Arabic double subject construction (DSC) that the prediction of the syntactic account stated in (4) is borne out. Building on [Ionin and Matushansky's 2002](#) arguments from Russian comitatives, this chapter shows that the DSC reveals a systematic relationship between whether or not a plural DP moves from its base position and whether we get a cumulative inference over that plural DP. To see this with a basic example, consider the contrast between a simple case of the double subject construction in (7-a) and the counterpart with standard conjunction in (7-b).

- (7) a. **Double subject construction** ([Mohammad, 1989](#); [Aoun, Benmamoun, & Sportiche, 1994](#)):

Rasha raʔasit **hiyye w Hadi**.
Rasha danced.3SG.F **her** **and Hadi**
 'Rasha and Hadi danced together.'

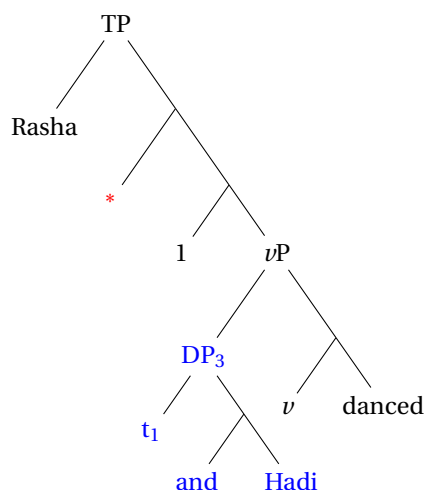
- b. **Standard conjunction:**

Rasha w Hadi raʔaso.
Rasha and Hadi danced.3PL
 'Rasha and Hadi danced.'

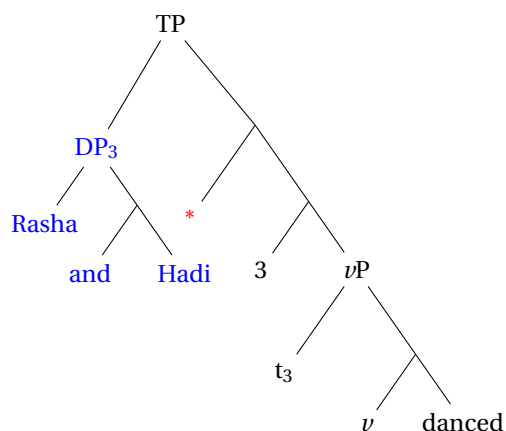
I argue that the only difference between (7-a) and (7-b) is the syntactic position of the conjoined DP *Rasha and Hadi*. In both cases, the conjoined DP is base-generated in spec, ν P. In (7-a), only the first conjunct moves to spec,TP, leaving behind a resumptive pronoun which is pronounced in its base position. On the other hand, in (7-b), the entire conjunction moves to spec,TP. This difference in the syntactic position of the conjoined DP turns out to correlate with a difference in interpretation. (7-b), just like conjunction in English, is true as long as Rasha and Hadi danced, whether they did so *together* or separately, thus licensing a cumulative inference. (7-a), on the other hand, is only true if Rasha and Hadi danced together, thus blocking the cumulative inference.

I propose that this difference in meaning between (7-a) and (7-b) follows if we assume that predicates are lexically collective and that cumulative inferences are due to the pluralization operator *, which can apply only to predicates derived by movement (as predicted by the Strong Theta Criterion). The resulting LFs for the double subject construction and for standard coordination are given in (8-a) and (8-b) respectively. In (8-a), the conjoined DP remains in its base position and therefore has to be interpreted below *, generating a collective reading. In (8-b), the conjoined DP moves to a derived position and * applies below it, generating a reading which is true as long as Rasha and Hadi danced and licensing the cumulative inference.

(8) a. **DSC:**



b. **Standard Conjunction:**



A major question at this point is why the conjoined DP in (7-a) can't move covertly from its base position to a derived position where it can take scope above *. I argue that while this movement is not available in (7-a), it is in fact available in certain environments, where we see cumulative readings over the lower subject in the DSC. I return to this issue in chapter 6, where I show that the licensing of covert movement of the lower subject is governed by the same generalized economy constraint that is needed to account for the disambiguating role of English *together*.

Finally, looking beyond cumulative inferences, we can use the DSC to probe whether or not other inferences depend on the presence of syntactic pluralization operators. For example, reciprocal alternations, like cumulative inferences, are not licensed over the lower subject in the double subject construction. This is illustrated in (9): in (9-a) and (9-b), we see that the verb *kiss* in Lebanese Arabic exhibits a reciprocal alternation with standard coordination; on the other hand, the variant in the DSC in (9-c) is ungrammatical, showing that the DSC does not license a covert reciprocal. This pattern can be explained if covert reciprocals require the relevant plural DP to take scope above a pluralization operator. The lower subject in the DSC cannot do so, since it remains below all pluralization operators. The account therefore predicts that the lower subject in (9-c) should not license a covert reciprocal.

- (9) a. **Two-place predicate:**
Karim be:s Rasha.
Karim kissed.3SG.M Rasha.
Karim kissed Rasha.
- b. **Covert reciprocal with standard coordination:**
Karim w Rasha be:so.
Karim and Rasha kissed.3PL.
Karim and Rasha kissed.
- c. **No covert reciprocal with the DSC:**
*Karim be:s howwe w Rasha.
Karim kissed him and Rasha.

The Lebanese Arabic DSC therefore provides evidence for a syntactic analysis of reciprocal alternations, where they involve the same pluralization operators responsible for cumulative inferences. In chapter 7, I provide a detailed syntactic account of reciprocal alternations that predicts the ungrammaticality of (9-c).

1.2 Distributivity facilitates ACD resolution (Ch.3)

Chapter 3 provides an independent way to corroborate the conclusion from the DSC that cumulative inferences are parasitic on movement.³ Namely, this chapter shows that definite plural DPs, in contexts that only support distributive construals, show processing signatures of syntactic movement. This processing study therefore supports the conclusion from chapter 2 that lexical predicates are inherently collective and that cumulative inferences are due solely to pluralization operators that require movement to apply.

³This chapter is based on joint work with Kate Kinnaird and Martin Hackl.

More specifically, [Hackl, Koster-Hale, and Varvoutis \(2012\)](#) show that sentences with Antecedent-Contained Deletion (ACD) are judged to be more natural when the DP hosting the ACD site is headed by *every* (10-a), compared to a definite singular baseline (10-b). Relying on previous work showing that ACD requires movement, Hackl et al. propose that the difficulty of ACD in (10-b) is due to the fact that the ACD site (*did*) is the only cue for this movement. They then argue that the facilitation effect of *every* is due to the fact that in object position, quantifiers like *every* have to move from their base position for type reasons, independently of ACD. Unlike the definite singular in (10-b), the quantifier *every* itself provides an additional cue for movement in (10-a), thus making it easier to identify the relevant LF with movement that is required to license ACD.

- (10) a. Jane read every book that Mary did.
 b. Jane read the book that Mary did.

In this study, we provide evidence that definite plurals under distributive construals similarly facilitate ACD. In particular, we show that in contexts that support a distributive reading, participants judge sentences with ACD to be more natural when the object is headed by a definite plural, compared to a definite singular baseline. We therefore conclude that just like *every*, definite plurals in object position need to move independently of ACD in order to give rise to distributive readings.

Taken together, chapter 2 and chapter 3 therefore provide evidence for a syntactic analysis of cumulative inferences. Now that we have seen a variety of evidence that these inferences are generated by pluralization operators, we can ask what types of readings these operators are predicted to give rise to when they take scope below plural DPs in a variety of derived positions.

1.3 A scope-based theory of *together* (Ch.4)

Chapter 4 begins to address this question of how the different ingredients of plural predication interact to generate a variety of readings by focusing on the different readings of *together*. In particular, English *together* appears to disambiguate by isolating a variety of non-distributive readings, which are exemplified in (11) ([Lasersohn, 1990](#); [Schwarzschild, 1994](#); [Moltmann, 2004](#), a.o.). I propose a unified scope-based analysis of *together*, where these different readings correspond to different positions where *together* can take scope.

- (11) a. **Collective *together*:** Jane and Mary lifted the piano together.
 b. **Simultaneous *together*:** Jane and Mary stood up together. $\approx$ Jane and Mary stood up at the same time.

- c. **Proximate *together*:** John and Bill's corpses lay together. $\approx$ John and Bill's corpses lay next to each other.
- d. **Cumulative *together*:** Jane and Mary together stole exactly 10 books. $\approx$ The sum of the number of (distinct) books that Jane stole and that Mary stole is 10.

Drawing on evidence from a variety of languages where *together* is overtly decomposed, I propose that English *together* has a similar compositional semantics. In particular, I propose that *together* is always composed of a comitative coordinator *with* and a reciprocal anaphor, yielding a coordinated DP *x with the other*. I assume, following [Ionin and Matushansky \(2002\)](#), that the comitative *with* is semantically equivalent to sum-forming conjunction, such that it takes two individual arguments and returns their sum. Additionally, I adopt a compositional analysis of reciprocals, where a reciprocal like (12-a) is decomposed into a covert pluralization operator, which can be thought of as a covert *each*, and a *the other* DP in object position (12-b) ([Heim, Lasnik, & May, 1991](#)).

- (12) a. Jane and Mary hugged each other.
 b. **Reciprocal Paraphrase:** Jane and Mary **each** hugged **the other**.

We can now see how putting these ingredients together generates the collective reading of *together*. I propose that the coordinated DP denoted by *together* is base-generated in spec, ν P. When the *x with the other* coordination remains in its base position, it has to take scope below all pluralization operators, generating the same collective reading we saw in the DSC. The resulting paraphrase is given in (13-b): collective *together* essentially blocks distributivity due to higher pluralization operators by forcing a plurality to be reintroduced in spec, ν P.

- (13) a. Jane and Mary danced together.
 b. **Collective paraphrase:**
 Jane and Mary are **each** such that **her and the other** danced (collectively).

The other attested readings of *together* given in (11) can now be derived by having the *x with the other* DP move covertly to different scope positions. On this analysis, *together* therefore isolates the different readings that plural DPs give rise to when they are interpreted in different scope positions.

I will illustrate the logic of the scope-based analysis here by showing how it derives the simultaneous reading. In particular, the simultaneous reading follows straightforwardly from allowing the conjoined DP *x with the other* to move above the lowest pluralization but below an existential quantifier over times denoted by Aspect. By forcing distributivity to take scope below this existential quantifier over times, *together* gives rise to the desired *same time* reading we saw

in (11-b). This is shown in the paraphrase in (14).

(14) **Paraphrase of the simultaneous reading:**

Jane and Mary are each such that **there is a time** such that **her and the other each** stood up at that time.

At this point, I have argued that *together* essentially denotes a coordinated DP that takes scope in different positions, giving rise to a variety of readings. This raises the immediate question of how *together* disambiguates by blocking the distributive readings that we can observe with standard conjunction. For example, in a scenario where Jane and Mary each danced separately, *Jane and Mary danced* is true, while *Jane and Mary danced together* clearly isn't. Since *together* simply blocks distributivity due to higher pluralization operators, the analysis predicts that *together* will generate this unattested distributive reading if it takes scope above a lower pluralization operator, as shown in the paraphrase in (15). In order to prevent this unattested LF from being generated, we therefore need a way to constrain which positions *together* is allowed to move to.

(15) **Paraphrase of the unattested distributive reading:**

Jane and Mary are **each** such that **her and the other each** danced.

I argue that LFs like (15) are ruled out by a grammatical economy constraint. In a nutshell, the LF corresponding to the distributive reading in (15) turns out to be equivalent to the result of deleting *together* in (16). Namely, both (15) and (16) are true iff each of Jane and Mary danced. I therefore argue that (15) is ruled out by general economy considerations.

(16) **Paraphrase of the result of deleting *with the other* from (15):**

Jane and Mary are **each** such that **she (each)** danced.

1.4 Economy and the distribution of *together* (Ch.5)

At first sight, it appears that this notion of competition between more complex and simpler LFs can follow from a pragmatic notion of redundancy, motivated by the Gricean Maxim of Manner (Katzir, 2007; Meyer, 2013). As discussed above, this will allow us to rule out the unattested distributive LF with *together*, since it is equivalent to the result of deleting *together*. The main challenge, however, is that this constraint appears to be too strong, ruling out even attested LFs with *together*. For example, the collective LF with *together*, paraphrased in (17-a), turns out to be equivalent to the result of deleting both *together* and the higher pluralization operator, as shown in (17-b). In particular, both LFs are true iff Jane and Mary danced collectively.

(17) **Economy appears to predict that even the collective LF is ruled out:**

a. **Collective paraphrase:**

Jane and Mary are **each** such that **her and the other** danced collectively.

b. **Result of deleting both the pluralization operator and *together* from the collective LF:**

Jane and Mary are such that they danced collectively.

I argue that this issue is resolved if we treat economy as a grammatical constraint that is evaluated relative to an encapsulated formal system (in parallel to what [Fox, 2000](#), argues for scope economy). I propose a generalized economy constraint which rules out LFs which can be proven to be equivalent to simpler LFs. Under this view, it is not enough for an LF to be truth-conditionally equivalent to a simpler LF in order to be ruled out. Rather, the encapsulated deductive system has to be able to prove this equivalence, using only a subset of the information that goes into computing truth-conditions. More specifically, I propose that when proving equivalence for the purposes of economy, the deductive system is blind to the identity of individual-denoting DPs, as illustrated in [\(18\)](#).

(18) a. Mary and Jane danced together.

b. **What the deductive system sees (at most):**

X and Y danced together. (where X and Y can denote plural individuals)

The logic of the argument is as follows. The problem in [\(17\)](#) is that *together* undoes the contribution of the higher pluralization operator by blocking distributivity. Deleting both *together* and the pluralization operator therefore yields the same collective meaning. I argue that this problem is specific to cases where the cardinality of the subject is 2. When the cardinality of the subject is greater than 2, *together* allows for partially distributive readings, where the predicate applies to non-singular parts of the subject plurality [\(12\)](#) ([Schwarzschild, 1994](#), a.o.). The LF corresponding to this reading turns out to satisfy economy: if we delete both *together* and the pluralization operators, we will get a collective reading which is distinct from the partially distributive one.

(19) **Context:** The students are divided into pairs for the competition and they can choose whether to dance as a pair or to dance individually. All of the pairs decided to dance together.

a. The students danced together.

Now, when the subject is *Jane and Mary*, the partially distributive reading is equivalent to the collective one, since *Jane and Mary* can't be divided into non-singular parts. We therefore see that the LF with *together* is only redundant when the cardinality of the subject is 2. I therefore

show that the problem in (17) is resolved if generalized economy is blind to the identity of the subject, such that it considers the possibility that the two conjuncts denote plural individuals, as shown in (18). The distribution of readings with *together* therefore provides novel evidence that certain constraints are evaluated with respect to an encapsulated deductive system (Fox, 2000; Gajewski, 2002; Fox & Hackl, 2006; Chierchia, 2013).

More generally, I argue that generalized economy predicts that *together* is ruled out when the relevant predicate it modifies is cumulative, such that whenever it is true of an individual *j* and an individual *m*, it is also true of the plurality formed from these two individuals. This is because cumulative predicates, by allowing pluralities to inherit the property from their parts, end up collapsing the partially distributive LF with the LF where the predicate applies directly to the entire plurality. For the collective and the simultaneous reading, the relevant predicate is not cumulative, as shown in (20-a) and (20-b). These readings are therefore correctly predicted to be licensed with *together*. On the other hand, the unattested distributive reading corresponds to a distributive predicate which by definition is true of a plurality whenever it is true of its parts (20-c). Generalized economy therefore successfully rules out this unattested distributive reading with *together*.

- (20) a. **Collective reading:** Licensed since the predicate is not cumulative.
 Jane danced $\wedge$ Mary danced $\nrightarrow$ Jane and Mary jointly danced.
- b. **Simultaneous reading:** Licensed since the predicate is not cumulative.
 Jane stood up $\wedge$ Mary stood up $\nrightarrow$ Jane and Mary stood up at the same time.
- c. **Unattested distributive reading:** Ruled out since the predicate is cumulative.
 Jane danced $\wedge$ Mary danced $\rightarrow$ Each of Jane and Mary danced.

Finally, I turn to cumulative readings of *together* with 2-place predicates, where generalized economy predicts that whether or not *together* is licensed depends on certain properties of the object DP. As shown in (21), the predicate modified by *together* is cumulative when the object is an upward-entailing quantifier like *more than 10* (21-a) but not cumulative when the object is a non-monotonic quantifier like *exactly 10* (21-b). Generalized economy therefore predicts that cumulative *together* is licensed with non-monotonic objects but blocked with upward-entailing objects.

- (21) **Cumulative *together*:**
- a. **With more than n:** (Jane and Mary together read more than 10 books.)
 Jane read more than 10 books $\wedge$ Mary read more than 10 books $\rightarrow$ The sum of the number of (distinct) books that Jane read and that Mary read is exactly 10
 (Cumulative)

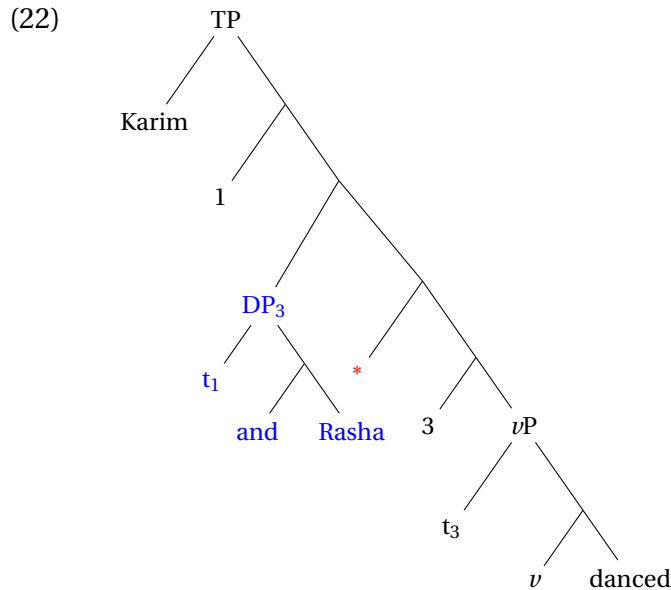
- b. **With exactly n:** (Jane and Mary together read exactly 10 books.)

Jane read exactly 10 books $\wedge$ Mary read exactly 10 books $\nrightarrow$ The sum of the number of (distinct) books that Jane read and that Mary read is exactly 10. (**Not Cumulative**)

I argue that upward-entailing objects that initially appear to be ruled out by generalized economy fall into two classes. The first class, which includes *more than n*, *all* and *at least n*, can give rise to an implicature that strengthens the relevant predicate and makes it non-cumulative. Generalized economy therefore predicts that cumulative *together* should be licensed with these objects when this implicature can be computed. The second class, which includes definite plural DPs and collective nouns, does not allow for the relevant strengthening. Cumulative *together* is therefore predicted to remain blocked with these objects. More generally, I argue that the crucial factor determining whether the required implicature is licensed is whether the object gives rise to homogeneity. I argue that the prediction of generalized economy is borne out. Namely, objects which give rise to homogeneity don't license cumulative *together*.

1.5 Economy and the double subject construction (Ch. 6)

After showing how generalized economy predicts the distribution of readings with *together*, I return to the DSC and explore how this economy constraint might be extended to account for the licensing conditions on covert movement of the lower subject. As discussed above, in order to derive the fact that the DSC blocks distributive readings, we have to assume that the lower subject can't move covertly to an intermediate position between spec, ν P and spec,TP, where it can take scope above the pluralization operator *. The LF that would generate this unattested distributive reading is given in (22).



I suggest that this LF is ruled out by similar economy considerations, since it is equivalent to the simpler LF in (8-b), corresponding to standard conjunction, where the conjoined DP moves directly to spec,TP and gives rise to the same distributive reading. However, the competition between the LF in (22) and the one in (8-b) does not follow straightforwardly from the generalized economy constraint given in chapter 5, since (17) can't be derived from (22) by deletion. I therefore tentatively suggest a possible reformulation of the constraint that would allow it to extend to the DSC, but I leave a full investigation of how economy applies to the DSC to future work.

Looking beyond the simple example in (22), I argue that the double subject construction does allow the lower subject to move covertly under certain conditions to generate cumulative readings with two-place predicates. More specifically, I show that the same class of objects that license a cumulative reading with *together* also license a cumulative reading in the DSC, as expected if the two constructions are sensitive to the same economy considerations. At the same time, there are important differences between the two constructions that pose a challenge for a fully unified analysis. For example, while *together* licenses a simultaneous reading, the DSC doesn't. I therefore leave it to future work to investigate in more detail whether the parallels between the two constructions can ultimately be captured by a unified economy constraint.

1.6 A scope-based theory of reciprocal alternations (Ch.7)

Finally, chapter 7 proposes a novel syntactic analysis of reciprocal alternations. In particular, as with the case of cumulative inferences, both lexical and syntactic analyses of this alternation, exemplified in (23), have been proposed. On the lexical analysis (e.g. Winter, 2018), *hug* in

(23-b) is in fact a 1-place collective predicate that has a distinct lexical entry from the 2-place variant in (23-a). The 1-place and 2-place variants are then related to each other through logical relationships in the lexicon. On the other hand, the syntactic analysis (e.g. Hackl, 2002) assumes that there is only one lexical entry for *hug* as a two-place predicate and (23-b) involves a covert reciprocal in object position.

- (23) a. Jane and Mary hugged each other.
b. Jane and Mary hugged.

As mentioned above, the Lebanese Arabic double subject construction provides evidence for a syntactic analysis where the covert reciprocal in (23-b) involves the same pluralization operators that are needed for overt reciprocals (Sauerland, 1998; Beck, 2001) and for cumulative inferences (Beck & Sauerland, 2000). A major challenge for the syntactic approach is that the overt and covert reciprocal variants, exemplified in (23-a) and (23-b) respectively, appear to have different truth-conditions (Gleitman et al., 1996; Winter, 2018, a.o.). For example, in a context where Jane hugged Mary at one time and Mary hugged Jane at a different time, (23-a) is true, while (23-b) isn't true. The covert reciprocal in (23-b) therefore appears to have a simultaneity requirement that the overt counterpart in (23-a) doesn't have (Winter, 2018).

A simple syntactic analysis where (23-a) and (23-b) have the same LFs therefore fails to predict these differences in truth-conditions. I propose that the differences in meaning between overt reciprocals and their covert counterparts follow from an independently motivated constraint that requires implicit arguments to take narrowest scope. I therefore argue that covert reciprocals like (23-b) are simply reciprocals that have to be bound at the lowest possible scope position.

For example, I argue that the simultaneity requirement described above is the result of the reciprocal being bound below the existential quantifier over times denoted by Aspect. Again, I assume a decompositional analysis of reciprocals into a distributivity operator and a *the other* DP in object position. When the relevant distributivity occurs above the existential quantifier denoted by Aspect, we get a non-simultaneous reading, paraphrased in (24-a). On the other hand, when distributivity occurs below Aspect, we get the simultaneous reading in (24-b). The low-scope restriction on covert reciprocals therefore ensures that they only give rise to the LF in (24-b), where the reciprocal binding occurs below Aspect, thus explaining their simultaneity requirement.

- (24) a. **Reciprocal binding above Aspect:** Jane and Mary **each** is such that **there is a time** such that she hugged the other at that time.
b. **Reciprocal binding below Aspect:** **There is a time** such that **each** of Jane and Mary hugged the other at that time. (Simultaneous reading)

The same simultaneous reading that I derived by having *together* take scope below Aspect can therefore be isolated by the low-scope requirement on covert reciprocals. We therefore begin to see a unified picture of what readings plural DPs give rise to when they are interpreted in different positions, relative to pluralization operators and relative to scope-taking elements like Aspect.

More generally, this scope-based account assumes that the same pluralization operators responsible for cumulative inferences are a crucial ingredient in the LFs for both overt and covert reciprocals. I present novel data arguing that we can diagnose the presence of these pluralization operators with covert reciprocals by showing that covert reciprocals give rise to homogeneity and non-maximality, inferences whose source has been attributed to these operators (Schwarzschild, 1994; Bar-Lev, 2025). I argue that alternative lexical accounts have to stipulate these additional inferences one by one and conclude that these inferences therefore provide evidence for the scope-based analysis.

Chapter 2

The sources of cumulative inferences

This chapter expands on [Wehbe \(2023\)](#), published in the proceedings of Sinn und Bedeutung.

2.1 Introduction

Natural language predicates universally give rise to cumulative inferences when they apply to pluralities. Starting with one-place predicates, the cumulative inference is illustrated by the examples in (1) and (2). Informally, this inference can be characterized as follows: if individual *a* danced and an individual *b* danced then the plurality formed by *a* and *b* danced. In fact, the predicate *dance* can be replaced with any other one-place predicate and the analogous inference holds ([Krifka, 1992](#); [Kratzer, 2007](#), a.o.).

- (1)
 - a. Jane danced.
 - b. Mary danced.
 - c. $(a) \wedge (b) \rightarrow$ Jane and Mary danced.
- (2) **Context:** There are 2 students, Jane and Mary.
 - a. Jane danced.
 - b. Mary danced.
 - c. $(a) \wedge (b) \rightarrow$ The two students danced.

A similar inference pattern can also be observed with 2-place predicates ([Kratzer, 2007](#), a.o.). This is illustrated in (3): here, we can sum up both arguments of the 2-place predicate *saw* to get the cumulative inference in (3-c). I will start by restricting my focus only to cumulative inferences with 1-place predicates, but I return later in the chapter and generalize the discussion to 2-place predicates like *saw*.

- (3)
- a. Jane saw Mary.
 - b. Sue saw Bill.
 - c. $(a) \wedge (b) \rightarrow$ Jane and Sue saw Mary and Bill.

The apparent universality of this inference pattern raises the immediate question of what the source of this generalization is. In particular, what prevents us from having a lexicon where English *dance* is replaced with an alternative predicate *dance'*, which does not license this inference, as illustrated in the hypothetical example in (4). Here, when there is a plural subject X , X *dance'* is only true if the individuals that are part of X danced together, thus blocking the cumulative inference.

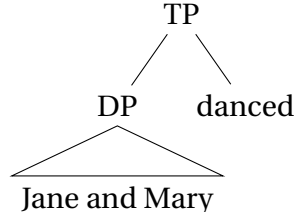
- (4) **Context:** There are two students, Jane and Mary.
- a. **Jane danced'** is true iff Jane danced.
 - b. **Mary danced'** is true iff Mary danced.
 - c. **The two students dance'** is true iff Jane and Mary danced together.
 - d. $(a) \wedge (b) \not\rightarrow (c)$

One view, the lexical cumulativity view, accounts for this generalization by simply stipulating a universal generalization on lexical predicates that rules out predicates like *dance'* from being lexicalized. This view therefore argues that cumulativity is a universal property of lexical predicates (Krifka, 1992; Kratzer, 2007, a.o.). For 1-place predicates of individuals, the lexical cumulativity universal is stated in (5): (5) ensures that whenever a predicate is true of two individuals it is also true of their sum, the plurality formed from the two individuals. It is straightforward to see how the inferences in (1) and (2) follow from (5): as shown in (6) for (1), the cumulativity universal predicts that whenever the predicate is true of Jane and true of Mary, it is also true of *Jane and Mary*, which I assume denotes the sum of Jane and Mary.

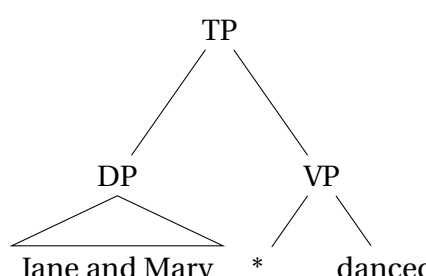
- (5) **Cumulativity universal:** For any lexical predicate $P_{\langle e, t \rangle}$, $\forall x, y[(P(x) \wedge P(y)) \rightarrow P(x \oplus y)]$
- (6)
- a. $\llbracket \text{Jane danced} \rrbracket = 1$ iff $\llbracket \text{dance} \rrbracket(j) = 1$
 - b. $\llbracket \text{Mary danced} \rrbracket = 1$ iff $\llbracket \text{dance} \rrbracket(m) = 1$
 - c. $\llbracket \text{Jane and Mary danced} \rrbracket = 1$ iff $\llbracket \text{dance} \rrbracket(j \oplus m) = 1$
 - d. **By the lexical cumulativity universal:**
 $\llbracket \text{dance} \rrbracket(j) = 1 \wedge \llbracket \text{dance} \rrbracket(m) = 1 \rightarrow \llbracket \text{dance} \rrbracket(j \oplus m) = 1$

Another view, which I'll call the syntactic cumulativity view, assumes that lexical predicates themselves are not inherently cumulative, but there is a cumulative operator $*$ in the syntax that applies to predicates and makes them cumulative (Link, 1983; Beck & Sauerland, 2000, a.o.). On this view, lexical predicates are taken to have collective truth-conditions when they apply

to pluralities. An LF without the * operator, like (7-a), will therefore not license a cumulative inference. In (7-a), the predicate will apply directly to the plurality denoted by *Jane and Mary*, giving rise to the collective truth-conditions in (7-b). It is clear that (7-a) does not support a cumulative inference: if Jane and Mary danced separately, it is true that *Jane danced* and that *Mary danced*, yet (7-a) is false.

- (7) a. 
- b. $\llbracket (7-a) \rrbracket = 1$ iff $\llbracket \text{dance} \rrbracket (j \oplus m) = 1$ iff Jane and Mary danced together.

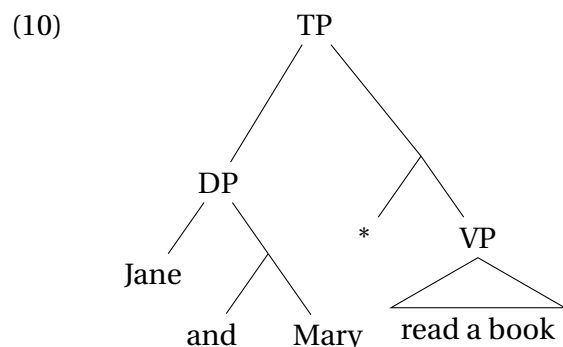
On this view, the desired cumulative inference is furnished by LFs where the pluralization operator * applies to the predicate before it applies to the plurality. In particular, Link's (1983) * operator closes the extension of a predicate under sum formation, ensuring that the starred predicate is true of a plurality whenever it can be divided into parts which the unstarred predicate is true of. At this point it suffices to see that given a predicate P and a sum of two atomic individuals $x \oplus y$, *P will be true iff either P is true collectively of $x \oplus y$ or if P is true of x and y individually (8-a). The LF in (8-b), where * applies to *dance*, is therefore true as long as Jane and Mary danced, whether they did so together or separately. On this view, we therefore see that the cumulative inference is due to the syntactically represented operator *. In particular, while the LF without * in (7-a) did not license a cumulative inference, the LF with * in (8-b) does: if *Jane danced* and *Mary danced* are true then (8-b) is necessarily true.

- (8) a. For any predicate P and any two atomic individuals x and y
 $\llbracket (*)(P) \rrbracket (x \oplus y) = 1$ iff $\llbracket P \rrbracket (x \oplus y) = 1 \vee (\llbracket P \rrbracket (x) = 1 \wedge \llbracket P \rrbracket (y) = 1)$
- b. 
- c. $\llbracket (8-b) \rrbracket = 1$ iff Jane and Mary danced (together or separately).

Regardless of whether we assume lexical cumulativity or syntactic cumulativity as the source of cumulative inferences in simple examples like (1), it has been argued that there are cases of cumulative inferences over complex predicates where lexical cumulativity alone is not enough.

In particular, consider the cumulative inference in (9): here, we get a cumulative inference over the complex predicate *read a book*, which involves a quantifier in object position. Now, as has been argued by Beck and Sauerland (2000) among others, assuming that the lexical predicate *read* is inherently cumulative does not help with deriving this inference, since the cumulative inference is over a derived predicate that has the indefinite *a book* in its scope. On the other hand, applying the syntactically represented * operator to the derived predicate *read a book* predicts the inference (10).

- (9) a. Jane read a book.
 b. Mary read a book.
 c. (9-a) $\wedge$ (9-b) $\rightarrow$ Jane and Mary read a book



Given that the * operator is independently needed to account for cumulative inferences over derived predicates, we are left with two competing views on cumulative inferences. The first view, which is the one argued for by Kratzer (2007), is that lexical cumulativity is responsible for cumulative inferences over lexical predicates and pluralization operators are responsible for cumulative inferences over derived predicates. The alternative syntactic view is one where there is no lexical cumulativity and syntactic operators like * are the only sources of cumulative inferences, as assumed by Beck and Sauerland (2000) for example.

1. **Lexical cumulativity:** All lexical predicates are inherently cumulative and pluralization operators are only needed for cumulative inferences over derived predicates.
2. **Syntactic cumulativity:** Lexical predicates are not inherently cumulative and syntactically represented pluralization operators are the only sources of cumulative inferences.

If the syntactic analysis is correct, we should in principle be able to create syntactic configurations where the * operator is unable to apply to the predicate before it applies to the plurality, thus blocking the cumulative inference. I show in this chapter that this is exactly what happens in the Lebanese Arabic double subject construction, where the availability of a cumulative inference

over a given plural DP depends on that DP moving from its base position. I argue that this pattern provides evidence against lexical cumulativity, which can't predict any relationship between syntax and the availability of a cumulative inference. Instead, I adopt a syntactic account where pluralization operators are the only sources of cumulative inferences and propose that the distribution of these operators is restricted, such that they can only apply to predicates derived by movement.

As discussed in the introduction, the fact that pluralization operators are restricted to applying to predicates derived by movement is crucial in allowing us to distinguish the syntactic account from the lexical one. If these operators were free to apply to the lexical predicate itself, then it would not be possible, with current tools, to diagnose whether or not the operators are necessary for cumulative inferences. This raises the question of whether there is a principled reason that pluralization operators are restricted in this way. I return to this issue in section 2.6, where I suggest that this restriction might follow from a strong theta criterion, which predicts that lexical predicates must apply directly to their arguments, before they merge with any modifiers like *.

In the first part of this chapter, I flesh out the argument for the syntactic theory of cumulative inferences from the Lebanese Arabic double subject construction. I provide a brief overview of this argument here.

The argument builds on [Ionin and Matushansky \(2002\)](#), who argue from the alternation between singular and plural comitatives in Russian that distributivity is associated only with the spec,TP position. Consider the examples in (11-a) and (11-b). The singular comitative in (11-a) gives rise to a collective reading: (11-a) is not true in a scenario where Alexandra and Boris danced alone or with different people. On the other hand, the counterpart in (11-b) is true as long as both Alexandra and Boris danced. Ionin and Matushansky argue that in both (11-a) and (11-b), *Aleksandra s Borisom* starts out as a comitative conjunction, which just like standard conjunction denotes the sum of Aleksandra and Boris. In (11-a), the first conjunct moves to spec,TP, while in (11-b) the entire coordinated DP moves, thus explaining the difference in word order between (11-a) and (11-b). They propose that the restriction to collective readings with singular comitatives is due to distributivity being associated only with the spec,TP position: since the coordinated DP is not in spec,TP in (11-a), it can't get distributed over.

- (11) a. **Aleksandra** tancevala **s Borisom**.
 Alexandra-NOM danced.SG.F with Boris-INST
 'Alexandra danced with Boris.'
- b. **Aleksandra s Borisom** tancevali
 Alexandra-NOM with Boris-INST danced-PL
 'Alexandra and Boris danced.'

Although Ionin and Matushansky do not explicitly argue against lexical cumulativity, their proposal for the contrast between (11-a) and (11-b) is incompatible with lexical cumulativity. In particular, in (11-a), the predicate *dance* applies to the comitative conjunction and does not give rise to a cumulative inference: (11-a) is not true if Jane Aleksandra and Boris danced but separately. On the other hand, the plural comitative in (11-b) does license a cumulative inference.

Ionin and Matushansky's account of the contrast between (11-a) and (11-b) as resulting from a difference in the syntactic position of the comitative conjunction therefore requires abandoning lexical cumulativity. At the same time, one might imagine alternative analyses for (11) that are compatible with lexical cumulativity. For example, it could be that the *with*-phrase in (11-a) is a VP-modifier and that the semantics of *with* ensures the desired collective reading. The goal of this chapter is to provide novel data from the Lebanese Arabic double subject construction that provides robust evidence for Ionin and Matushansky's conclusion that the availability of cumulative inferences is sensitive to syntactic position.

The Lebanese Arabic double subject construction exhibits an alternation with standard conjunction that bears a striking resemblance to (11). The double subject construction (Mohammad, 1989; Aoun et al., 1994) is exemplified in (12-a), where there is thematically one subject (*Rasha and Hadi*), but two realized DPs: the higher subject *Rasha* and the lower subject *her and Hadi*. Crucially, the double subject construction only allows for collective readings: (12-a), like (11-a), is only true if Rasha and Hadi danced together. This differs from the case with standard coordination in (12-b), which like (11-b) is true as long as both Rasha and Hadi danced.

- (12) a. **Rasha** raʔasit **hiyye w Hadi**.
 Rasha danced.3SG.F **her and Hadi**
 'Rasha and Hadi danced with each other.'
- b. **Rasha w Hadi** raʔaso.
 Rasha and Hadi danced.3PL
 'Rasha and Hadi danced.'

I argue that this construction shows that the availability of cumulative inferences depends on the syntactic position of the plural DP. In particular, I argue that the only difference between (12-a) and the counterpart with standard coordination in (12-b) is the position of the conjoined DP: in (12-a), *Rasha* moves to spec,TP, leaving behind a resumptive pronoun in its base position, while in (12-b) the entire conjoined DP moves to spec,TP. This difference in syntactic position correlates with a difference in interpretation: just as with the comitative example in Russian, we see that while (12-b) licenses a cumulative inference, the double subject construction in (12-a) doesn't.

Going beyond the basic pattern that Ionin and Matushansky show in (11), I argue that there is

a systematic relationship in the Lebanese Arabic double subject construction between whether or not the conjoined DP moves and whether we can get a cumulative inference over that DP. I therefore propose that pluralization operators, such as * and its two-place counterpart **, are the only sources of cumulative inferences and that these operators can only apply to predicates derived by movement. I argue that in (12-a), the conjoined subject must remain in its base position, therefore predicting that the cumulative inference is blocked over the lower subject. I show that this predicts a range of additional data beyond the simple case in (12), thus providing systematic evidence against the lexical cumulativity universal.

While I leave the discussion of various additional data for later in the chapter, I want to highlight one particular piece of evidence that illustrates this relationship between movement and the availability of cumulative inferences. In particular, the proposal I outline above predicts that if the conjoined DP *her and Hadi* can move to an intermediate position, it should license a cumulative inference, since * will be able to apply to the resulting derived predicate. I argue that this is exactly what happens when auxiliaries precede the verb, allowing the conjoined DP to move to an intermediate position between the auxiliary and the verb. To see this, consider the example in (13): here the conjoined DP *her and Hadi* precedes the verb and the verb shows plural rather than singular agreement. I argue that this indicates that in (13), *her and Hadi* has moved to an intermediate position. Notice that unlike in the basic case in (12-a), the double subject construction in (13) does license a cumulative inference: (13) is true if Rasha danced yesterday and Hadi danced yesterday.

- (13) Rasha ke:nit hiyye w Hadi ʕam yiriso mbe:riħ.
 Rasha was.3SG.F her and Hadi PROG dance.3PL yesterday
 ‘Rasha and Hadi were dancing yesterday.’

- (14) **Cumulative inference:**

Rasha was dancing yesterday $\wedge$ Hadi was dancing yesterday $\rightarrow$ (13)

The contrast between (12-a) and (13) therefore supports the conclusion that the licensing of cumulative inferences is parasitic on movement. In particular, (13) shows that there is nothing inherent to the double subject construction that blocks the cumulative inference in (12-a). Rather, the licensing of the cumulative inference depends on the position of the conjoined DP in the double subject construction.

In the second part of the chapter, I turn to discussing some further issues that will be particularly important for the rest of the dissertation. First, in order to predict the fact that cumulative inferences are blocked over the lower subject in the double subject construction in (12-a), I have to assume that the lower subject can’t move covertly to a position above a pluralization operator. This raises interesting questions about the licensing conditions on covert movement. I show

that under certain conditions beyond the basic case in (12-a), the lower subject can in fact move covertly and therefore does license the cumulative inference. I argue that whether or not the cumulative inference is possible depends on certain semantic properties of the associated LF, thus providing evidence that the licensing of covert movement is sensitive to interpretation (following Fox, 2000). I will return to this issue in detail in chapter 6, where I argue that the distribution of readings in the double subject construction is governed by an independently motivated economy constraint.

The second major consequence of my analysis is that we can use the double subject construction to probe the division of labor between lexical semantics and pluralization operators beyond the specific case study of cumulative inferences. I discuss two additional case studies: (i) the source of homogeneity and (ii) the nature of reciprocal alternations. For both cases, I show that the double subject construction provides evidence for a syntactic analysis where these phenomena are due to the same pluralization operators that are responsible for cumulative inferences. While this has been argued independently for homogeneity (Schwarzschild, 1994; Bar-Lev, 2025), the syntactic analysis of covert reciprocals faces well-known challenges (see Winter, 2018). I therefore return to reciprocal alternations in chapter 7 and argue for a modified syntactic analysis that addresses these challenges.

2.2 Introducing the double subject construction

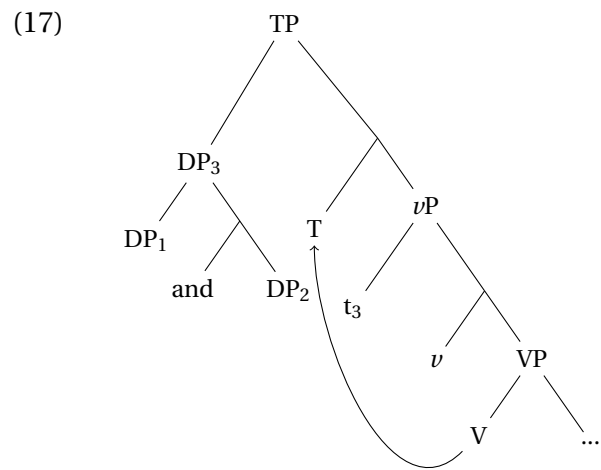
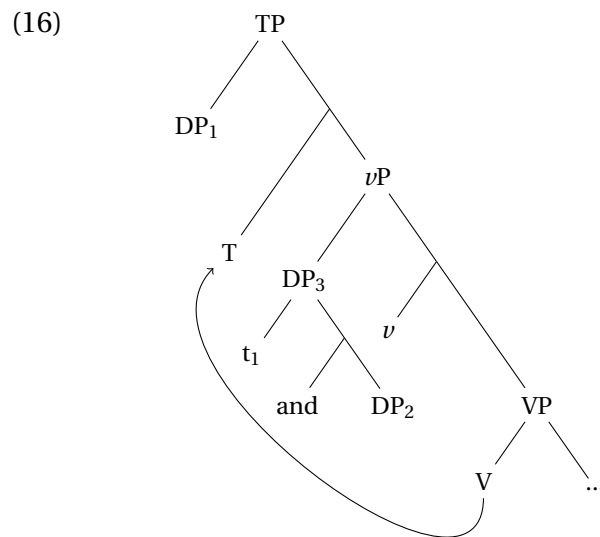
2.2.1 Syntactic properties

Before getting into deriving the meaning differences between the double subject construction and standard conjunction, I want to start by giving some background about the syntactic properties of the double subject construction, which will be relevant for making the argument that there is a relationship between movement and the availability of cumulative inferences. As mentioned in the introduction, the argument against lexical cumulativity from the Lebanese Arabic double subject construction rests on the assumption that the only difference between the double subject construction and standard coordination is the syntactic position of the conjoined DP subject. The goal of this section will therefore be to provide evidence that this is in fact the only difference between (15-a) and (15-b) and to find ways to probe the syntactic position of the conjoined DP in the double subject construction.

- (15) a. **Rasha** raʔasit **hiyye w Hadi**.
 Rasha danced.3SG.F **her** **and Hadi**
 ‘Rasha and Hadi danced with each other.’

- b. **Rasha w Hadi** raʔaso.
Rasha and Hadi danced.3PL
 ‘Rasha and Hadi danced.’

More specifically, I propose that the conjoined DP is always base-generated in spec, ν P. In the basic example of the double subject construction, the first conjunct moves to spec,TP, leaving behind a resumptive pronoun which is pronounced in the base position (16). On the other hand, in the counterpart with standard coordination, the whole conjoined DP moves to spec,TP and nothing is pronounced in the base position (17). Additionally, the verb moves from V to T in both. This derives the word orders we observe in (15-a) and (15-b).^{1 2}



We see that the different syntactic structures proposed in (16) and (17) correlate with different agreement on the verb *dance*. In particular, in the double subject construction, where only the first conjunct *Rasha* moves to spec,TP, the verb agrees with the first conjunct, showing third person singular feminine agreement. On the other hand, in (17), where the entire conjunction moves to spec,TP, the verb agrees with the conjoined DP, showing third person plural agreement. In fact, when auxiliaries are present, we see a tight connection between the position of the lower subject and agreement on the verb, which I argue provides evidence that the conjoined DP starts

¹Note that word order is somewhat flexible in Lebanese Arabic, so other more marked word orders are available in (15-a) and (15-b). I will not discuss the issue of how alternative word orders are derived here.

²While this explains the word order in simple cases like (15-a), when we have a transitive verb, the lower subject follows the entire VP. One possibility is that the specifier of ν P is simply linearized on the right, therefore predicting that it follows the entire VP. Of course, we want to understand how this fits in with a predictive linearization algorithm, but this might be what needs to be said about the word order in the Lebanese Arabic double subject construction. In chapters 4 and 5, I show that a parallel issue arises for my analysis of English *together*, which can also be resolved if we assume that certain lower specifiers are pronounced on the right.

out in spec, ν P and that there is optionality with respect to whether the entire conjunction or only the first conjunct moves to spec,TP.

To see this, I will look at how the double subject construction manifests itself in sentences where we have the auxiliary *ke:n*. First, consider the basic example with this auxiliary in (18): here we see that both the auxiliary and the main verb show agreement with the subject *Karim*. Now, when the subject is a conjoined DP, we see a clear interaction between the position of the conjoined DP and the availability of plural agreement on each of the auxiliary and the main verb. This pattern is shown in the examples in (19)-(21): (i) in (19) the conjoined DP intervenes between the auxiliary and the verb and the auxiliary agrees with the first conjunct, while the main verb agrees with the entire conjunction³; (ii) in (20) the conjoined DP follows the verb and both the auxiliary and the main verb agree with the first conjunct; (iii) in (21), where we have a standard case of a conjoined DP subject, the conjoined DP precedes both the auxiliary and the verb and they both agree with the entire conjunction.

- (18) Karim ke:n ʕam yelʕab.
Karim was.3SG.M PROG play.3SG.M.
Karim was playing.
- (19) Karim ke:n/*ke:no howwe w Hadi ʕam yelʕabo.
Karim was.3SG.M/*was.3SG.PL him and Hadi PROG play.3PL.
Karim and Hadi were playing.
- (20) Karim ke:n/*ke:no ʕam yelʕab/*yelʕabo howwe w Hadi.
Karim was.3SG.M/*was.3SG.PL PROG play.3SG.M/*play.3PL him and Hadi.
Karim was playing with Hadi.
- (21) Karim w Hadi ke:no/*ke:n ʕam yelʕabo/*yelʕab.
Karim and Hadi was.3PL/*was.3SG.M PROG play.3PL/*play.3SG.M.
Karim and Hadi were playing.

This relationship between the position of the conjoined DP and agreement can be accounted for as follows: there are two agreement probes, one corresponding to the verb and one corresponding to the auxiliary. When the closest DP to the probe is a conjoined DP, the probe can either target the first conjunct or the entire conjunction. Whatever the probe agrees with then has to move to the specifier corresponding to that probe. In (19) the lower probe agrees with the conjoined DP, moving it to an intermediate position and the higher probe targets only the first conjunct, resulting in *Karim* in the highest subject position. Since the lower probe targets the conjoined

³Note that (19) also allows for singular agreement on the main verb, but the variant with singular agreement feels more marked and requires focus on the conjoined DP. One possibility is that this marked variant corresponds to a syntactic structure where the verb stays in V and where the subject in spec, ν P therefore precedes it. I will set this marked variant aside for our purposes here.

DP, we get plural agreement on the verb and since the higher probe targets the first conjunct, we get singular agreement on the auxiliary. As for (20), the lower subject stays in its base position and both probes target only the first conjunct. We therefore see singular agreement on both the auxiliary and the main verb. Finally, in (21), both probes target the entire conjunction, therefore resulting in a standard conjunction construction and in plural agreement on both the auxiliary and the verb.

The pattern in (19)-(21) therefore shows that the double subject construction is simply the result of a general optionality with respect to whether agreement probes target the first conjunct of a conjoined DP or the entire conjoined DP, resulting in movement of whichever DP they target. On this view, what I've been calling standard conjunction in (15-b) and (21) simply corresponds to a special case where all of the agreement probes target the entire conjoined DP. We therefore have evidence for a shared base structure for standard conjunction and the double subject construction, where they both start out with the conjoined DP in spec, ν P.

Under this analysis, there is a question of how movement of the first conjunct alone to spec, TP is licensed. In particular, movement out of a conjoined DP violates a well-known constraint on movement, the coordinate structure constraint (CSC) (Ross, 1967). One possibility is that the presence of a resumptive pronoun in the first conjunct is what allows us to obviate the CSC. In fact, it has been argued that resumptive pronouns in Lebanese Arabic in general allow movement out of islands (Aoun, Choueiri, & Hornstein, 2001). Another possibility is that the relationship between the higher subject and the first pronoun is due to binding, where the higher subject does not in fact originate in spec, ν P but simply binds the pronoun in the first conjunct of the lower subject. I will remain agnostic about whether the relationship between the two subjects is movement or binding for now, but I return briefly to this issue in section 2.3.3, where I provide independent evidence that covert movement out of the second conjunct is also licensed, raising additional questions of how this construction obviates the CSC.

There is other evidence that the lower subject in the double subject construction is base-generated in a subject position, rather than being a modifier. For example, the lower subject can control the embedded subject position of obligatory subject control verbs, like *try*, which require that the subject of the matrix clause co-refers with the embedded subject. This is illustrated in (22). In (22-a), the embedded subject has to be controlled by the higher subject *Karim*. In (22-b), we see that when the DP *Hadi* is contained in a verbal modifier, the embedded subject still has to be controlled by the higher subject alone, as shown by the ungrammaticality of plural agreement on *make*. Now, in (22-c), with the DSC, the lower subject *him and Hadi* can control the embedded subject of *make*, as evidenced by the plural agreement on *make*. We therefore have evidence that the conjoined DP is in fact in a subject position in the higher clause in (22-c).

- (22) a. Karim jarrab yħaddir ħasha.
 Karim tried.3SG.M make.3SG.M dinner.
 Karim tried to make dinner.
- b. *Karim jarrab bi mouseħadit Hadi yħaddro ħasha.
 Karim tried.3SG.M with help Hadi make.3PL dinner.
- c. Karim jarrab howwe w Hadi yħaddro ħasha.
 Karim tried.3SG.M him and Hadi make.3PL dinner
 ‘Karim and Hadi tried to make dinner.’

I have therefore argued in this section that both the double subject construction and standard conjunction start out with a base-generated conjoined DP in spec, *v*P. The only difference between the double subject construction and standard coordination is whether the entire conjoined DP moves to spec, TP or whether only the first conjunct does.

2.2.2 Blocking the cumulative inference

In this section, I will present some of the basic meaning differences between the double subject construction and standard conjunction, arguing that cumulative inferences over the conjoined DP are only available when that DP moves overtly from its base position.

- (23) **Movement-Cumulativity Generalization:** Cumulative inferences over a plural DP are only available when that DP moves from its base position.

Starting with the basic unembedded case, repeated in (24), I argued in section 2.2.1 that the conjoined DP stays in its base position. As mentioned in the introduction, we don’t get a cumulative inference over the lower conjoined subject in (24).

- (24) **Rasha** raħasit **hiyye w Hadi.**
Rasha danced.3SG.F **her and Hadi**
 ‘Rasha and Hadi danced with each other.’

To see this in more detail, let’s observe first that the counterpart with standard conjunction in (25) is true both in a scenario where Rasha and Hadi danced together (25-a) and in a scenario where Rasha and Hadi danced separately (25-b). We can therefore see that (25) licenses the desired cumulative inference: if *Rasha danced* is true and *Hadi danced* is true, then (25) is necessarily also true.

- (25) **Rasha w Hadi** raħaso.
Rasha and Hadi danced.3PL
 ‘Rasha and Hadi danced.’
- a. **Scenario 1:** Rasha and Hadi danced together.

- b. **Scenario 2:** Rasha and Hadi danced separately.

(26) **Cumulative inference:** Rasha danced $\wedge$ Hadi danced $\rightarrow$ (25)

Turning to the double subject construction, we see that (24), just like (25), is judged to be true in a scenario where Rasha and Hadi danced together. Unlike (25), (24) is false in a scenario where Rasha and Hadi danced separately.⁴ We therefore see that (24) does not license the relevant cumulative inference: in Scenario 2, it is true that *Rasha danced* and true that *Hadi danced*, yet (24) is false.

(27) Rasha danced her and Hadi.

- | | | |
|----|--|--------------|
| a. | Scenario 1: Rasha and Hadi danced together. | True |
| b. | Scenario 2: Rasha and Hadi danced separately. | False |

(28) **No cumulative inference:** Rasha danced $\wedge$ Hadi danced $\nrightarrow$ (27)

Given the conclusion from section 2.2.1 that the only difference between standard conjunction and the double subject construction is the syntactic position of the conjoined DP, this difference in meaning between the two constructions poses a challenge for lexical cumulativity. In particular, taken at face value, the double subject construction suggests that the availability of a cumulative inference must depend on the syntactic position of the conjoined DP, a conclusion which is incompatible with lexical cumulativity.

Note that this pattern with the double subject construction is not specific to the predicate *dance*. For example, in (29), we see that the double subject construction is only true if Rasha and Karim lifted the piano together, while the counterpart with standard conjunction is also true if they each lifted the piano separately. Similarly, in (30), the double subject construction is only true if Rasha and Karim ate together, while the counterpart with standard conjunction is also true if they each ate separately. We therefore see that the double subject construction always blocks the distributive reading, while standard conjunction allows it.

(29) Rasha himlet l piano hiyye w Karim.
 Rasha lifted.3SG.F the piano her and Karim
 ‘Rasha lifted the piano with Karim.’

- | | | |
|----|--|--------------|
| a. | Scenario 1: Rasha and Karim worked together to lift the piano. | True |
| b. | Scenario 2: Rasha walked in and lifted the piano by herself and then Karim came in later and lifted the piano by himself. | False |

⁴Note that the string in (24) does allow for a distributive reading if there is a pause after the verb, which is generally accompanied by focus on the pronoun in the first conjunct. I discuss this potential problem in Appendix A, where I provide evidence that this is not in fact a confound and is due to a completely distinct construction.

- (30) Karim ʔakal howwe w Rasha.
 Karim ate him and Rasha
 ‘Karim ate with Rasha.’

- a. **Scenario 1:** Karim and Rasha ate together. **True**
 b. **Scenario 2:** Karim and Rasha ate separately. **False**

Looking beyond the simple example of the double subject construction above, we can now ask what happens with respect to the cumulative inference in the more intricate cases where auxiliaries are present and where the lower subject is able to move to an intermediate position between the auxiliary and the main verb. Starting with the case where the lower subject follows both the auxiliary and the main verb and where both agree with the higher subject, we see that the sentence in (31) is only true in Scenario 1, the collective scenario. As for the case of standard conjunction, where the conjoined DP moves all the way to spec,TP in (32), this is true as expected in both the distributive and the collective scenario. Finally, when the lower subject moves to an intermediate position in (33), the sentence is again true in both the distributive and the collective scenario.

- (31) Karim ke:n ʔam yerʔos howwe w Hadi.
 Karim was.3SG.M PROG dance.3SG.M him and Hadi.
 Karim was dancing with Hadi.

- a. **Scenario 1:** Karim and Hadi were dancing together. **True**
 b. **Scenario 2:** Karim and Hadi were dancing separately. **False**

- (32) Karim w Hadi ke:no ʔam yereʔso.
 Karim and Hadi were.3PL PROG dance.3PL.
 Karim and Hadi were dancing.

- a. **Scenario 1:** True **Scenario 2:** True

- (33) Karim ke:n howwe w Hadi ʔam yereʔso.
 Karim was.3SG.M him and Hadi PROG dance.3PL.
 Karim and Hadi were dancing.

- a. **Scenario 1:** True **Scenario 2:** True

We therefore see that both (32) and (33) license a cumulative inference, while (31) doesn't. The conclusion to draw from this pattern is that the cumulative inference requires the conjoined DP to move from its base position overtly: the cumulative inference is only suspended in (31), where the conjoined DP remains in its base position, as evidenced by the word order and by the fact that both the main verb and the auxiliary agree only with the first conjunct.

In the next section, I will give an account of how this relationship between movement and

the availability of a cumulative inference arises. I propose that pluralization operators are the only sources of cumulative inferences, thus abandoning the lexical cumulativity universal. Furthermore, I propose that the distribution of pluralization operators is restricted, such that they can only apply to predicates derived by movement. I show that these two assumptions together predict the relationship between cumulative inferences and movement.

2.3 Proposal

2.3.1 The role and distribution of pluralization operators

I will start by outlining my assumptions regarding the sources of cumulative inferences. I assume that lexical predicates are not inherently cumulative. Lexical predicates can still apply to pluralities, and the lexicon determines when a predicate is inherently true of a plurality. To make things more concrete, consider a predicate like *dance*. This view predicts that the cumulative inference in (34) doesn't hold. Nevertheless, $\llbracket \text{dance} \rrbracket$ can be true of a plurality under certain conditions (those where the plurality collectively danced).

$$(34) \quad \llbracket \text{dance} \rrbracket(a) \wedge \llbracket \text{dance} \rrbracket(b) \not\vdash \llbracket \text{dance} \rrbracket(a \oplus b)$$

Given this view where lexical predicates are not cumulative, pluralization operators must be the sources of cumulative inferences. I assume that pluralization operators, like $*$, are present in the syntax and are solely responsible for cumulative inferences with plural predication. The lexical entry for $*$ is given in (35), where $\bigoplus Q$ denotes the sum of all x s.t. $Q(x)=1$.

$$(35) \quad \llbracket * \rrbracket = \lambda P_{\langle e, t \rangle} . \lambda x . \exists Q_{\langle e, t \rangle} [\forall y [Q(y) = 1 \rightarrow P(y) = 1] \wedge x = \bigoplus Q] \quad \text{Link (1983)}$$

There is a question here regarding the choice between a pluralization operator like $*$ and a simple distributivity operator, shown in (36), in order to derive distributive readings (Higginbotham, 1981; Gillon, 1987; Schwarzschild, 1994, a.o.). For a predicate like *dance*, where the collective reading entails the distributive one (*Jane and Mary danced together* entails that *Jane danced* and *Mary danced*), the $*$ operator and DIST predict the same result, as shown in (37).

$$(36) \quad \llbracket \text{DIST} \rrbracket = \lambda P . \lambda x . \forall y \leq_{AT} x : P(y) = 1$$

- (37) a. $\llbracket * \rrbracket(\llbracket \text{dance} \rrbracket)(a \oplus b) = 1$ iff $\llbracket \text{dance} \rrbracket(a \oplus b) = 1 \vee (\llbracket \text{dance} \rrbracket(a) = 1 \wedge \llbracket \text{dance} \rrbracket(b) = 1)$
iff $\llbracket \text{dance} \rrbracket(a) = 1 \wedge \llbracket \text{dance} \rrbracket(b) = 1$
(Since the first disjunct, *a and b danced together*, entails the second disjunct).
b. $\llbracket \text{DIST} \rrbracket(\llbracket \text{dance} \rrbracket)(a \oplus b) = 1$ iff $\llbracket \text{dance} \rrbracket(a) = 1 \wedge \llbracket \text{dance} \rrbracket(b) = 1$

In order to show that we in fact need the pluralization operator $*$ rather than a distributivity operator, we have to look at predicates like *lift the piano* where the unpluralized predicate does not give rise to distributive entailments. In particular, given that there is no lexical cumulativity, we can assume that before any pluralization operators apply, $\llbracket \text{lift the piano} \rrbracket$ has the truth-conditions in (38-a) when applied to a plurality. Now, it's clear that this does not entail that a and b individually lifted the piano (38-b). Applying the $*$ operator and the distributivity operator to *lift the piano* will therefore result in different truth-conditions: for a plurality $a \oplus b$, applying DIST will make the sentence true only if a and b individually lifted the piano, while applying $*$ will result in weaker truth-conditions where the sentence is true if either a and b individually lifted the piano or if they lifted it together.

- (38) a. $\llbracket \text{lift the piano} \rrbracket(a \oplus b) = 1$ iff a and b lifted the piano together.
 b. $\llbracket \text{lift the piano} \rrbracket(a \oplus b) \not\vdash \llbracket \text{lift the piano} \rrbracket(a) \wedge \llbracket \text{lift the piano} \rrbracket(b)$
- (39) a. $\llbracket * \rrbracket(\llbracket \text{lift the piano} \rrbracket)(a \oplus b) = 1$ iff
 $\llbracket \text{lift the piano} \rrbracket(a \oplus b) = 1 \vee (\llbracket \text{lift the piano} \rrbracket(a) = 1 \wedge \llbracket \text{lift the piano} \rrbracket(b) = 1)$
 b. $\llbracket \text{DIST} \rrbracket(\llbracket \text{lift the piano} \rrbracket)(a \oplus b) = 1$ iff $\llbracket \text{lift the piano} \rrbracket(a) = 1 \wedge \llbracket \text{lift the piano} \rrbracket(b) = 1$

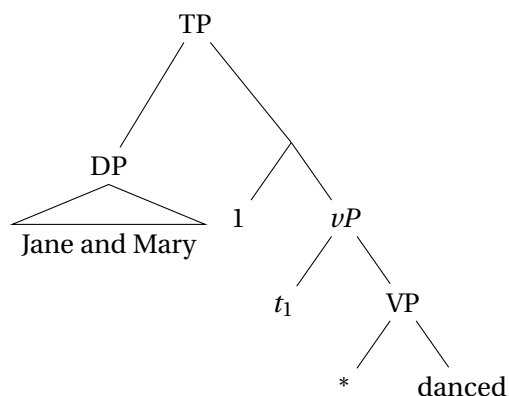
Applying the $*$ operator therefore predicts that a sentence like (40) will have a weak reading where it is true in both a distributive and a collective scenario. It has been argued that sentences like (40) in fact have this weak reading, thus providing evidence for the pluralization operator $*$ (Schwarzschild, 1994; Bar-Lev, 2025, a.o.). Evidence for this underspecification view comes from examples like (41): in this scenario, the sentence is neither true when *lift the piano* is interpreted distributively, as in (41-b) nor when it is interpreted collectively, as in (41-c). Yet, the sentence in (41-a) can be felicitously used to describe a scenario where Sue does not care whether Jane and Mary lift the piano together or separately. This contrast can only be accounted for if we assume that *Jane and Mary lifted the piano* in (41-a) has an underspecified reading which is true in both distributive and collective scenarios.

- (40) Jane and Mary lifted the piano.
 a. **Distributive reading:** Jane and Mary each lifted the piano.
 b. **Collective reading:** Jane and Mary lifted the piano together.
- (41) **Scenario:** Sue wants Jane and Mary to either lift the piano together or to each lift the piano separately.
 a. Sue will be happy only if Jane and Mary lift the piano.
 b. #Sue will be happy only if each of Jane and Mary lifts the piano.
 c. #Sue will be happy only if Jane and Mary lift the piano together.

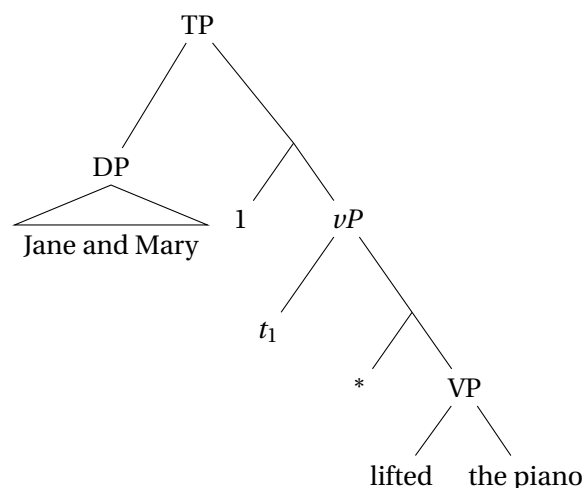
Now that we have syntactically represented pluralization operators as the sources of cumulative inferences, we can ask what the syntactic distribution of these operators is. In order to derive the relationship between cumulative inferences and movement which I outlined in section 2.2, I propose that pluralization operators can only apply to predicates derived by movement. To see the consequences of this restriction, I provide some example LFs in (42) which are predicted to be ruled out by this constraint. In (42-a), the * operator applies directly to the verb *dance*, thus violating the requirement that * apply to a derived predicate. Similarly, in (42-b), * applies to the VP that is base-generated, thus violating the same requirement. Finally, consider the LF in (43): here the * operator applies to a predicate which is derived by moving the subject *Jane and Mary* from its base position in spec, ν P. The * applies between the binder index 1 and the moved DP, thus successfully applying to a predicate which is derived by movement.

(42) **Ungrammatical LFs with *:**

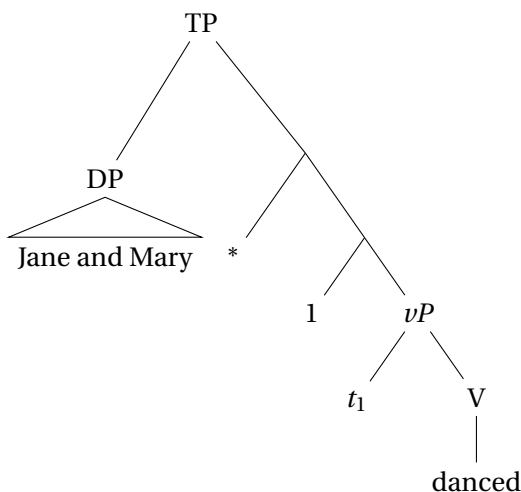
a.



b.



(43) **Grammatical LF with *:**



Stated more formally, the constraint on the distribution of pluralization operators is given in (44), requiring that the node that * merges with has a binder index as one of its daughters. This constraint is satisfied in (43) where the binder index 1 is a daughter of the node that * merges with, but not in the LFs in (42), where there is no binder index below *. In the rest of the chapter, I will continue to use ‘predicates derived by movement’ as shorthand to describe the distribution of pluralization operators predicted by (44).⁵

- (44) **Syntactic constraint on pluralization operators:** Pluralization operators can only be merged with a node that has a binder index as one of its two daughters.

There is of course a question here of what the source of this constraint is. In particular, why should the distribution of * be sensitive to whether the predicate it combines with is derived by movement or not? I will return to this issue in section 2.6, where I discuss a couple of possible

⁵Of course, (44) predicts that if a binder index could be inserted without movement, then pluralization operators could apply to the resulting predicate. For my purposes here, this distinction between predicates derived by movement and nodes which have a lambda index as one of their daughters will not be relevant.

ways to derive this constraint from more general considerations.

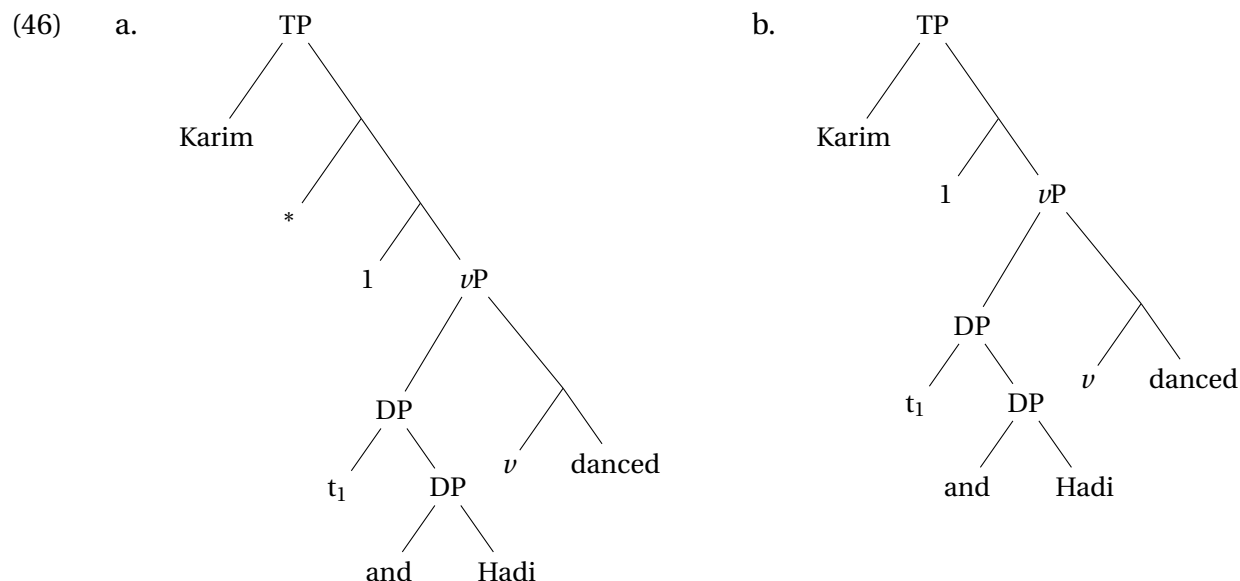
At this point, we have all of the ingredients necessary to derive the different readings that arise in the double subject construction. The rest of this section will show that the assumptions proposed here derive the patterns we have seen so far, in addition to further cases where one or both of the individual conjuncts in the double subject construction are plural.

2.3.2 Deriving the relationship between movement and cumulative inferences

I will start by showing what the assumptions in section 2.3.1 predict for the basic case of the double subject construction in (45). As discussed in section 2.2.1, I assume that the lower subject remains in its base position in (45), as evidenced by the fact that it follows the verb and that the main verb agrees only with the higher subject. An immediate question arises here: Why can't the lower subject undergo covert movement from its base position in (45)? For now, I will simply assume that covert movement of the lower subject is not available but I return to this issue in section 2.3.5 and again in section 2.4 when I discuss two-place predicates.

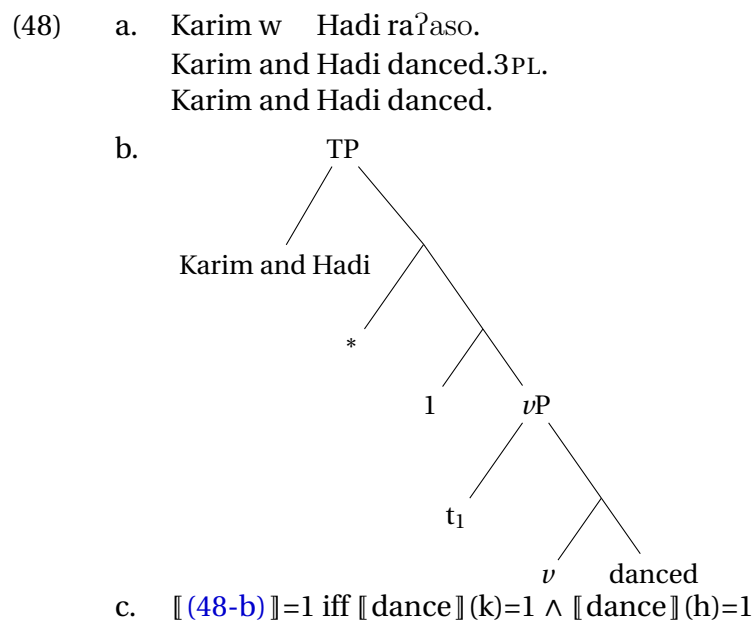
- (45) Karim raʔas howwe w Hadi.
 Karim danced.3SG.M him and Hadi.
 Karim and Hadi danced together.

Given that the lower subject has to stay in its base position and that * can only apply to predicates derived by movement, the possible LFs for (45) are given in (46): in (46-a), the * operator applies below the higher subject and in (46-b), there is no pluralization operator. LFs where there is a * operator below the lower subject that applies directly to the lexical predicate *dance* are ruled out by the constraint on *. As shown in (47), both (46-a) and (46-b) are predicted to be true iff Karim and Hadi danced together. In particular, when the higher subject denotes an atomic individual as in (46-a), the contribution of * is trivial and (46-a) is simply true if the lexical predicate is true of the plurality $k \oplus h$, giving rise to the desired collective truth-conditions.



(47) $\llbracket (46-a) \rrbracket = \llbracket (46-b) \rrbracket = 1$ iff $\llbracket \text{dance} \rrbracket (k \oplus h) = 1$ iff Karim and Hadi danced together.

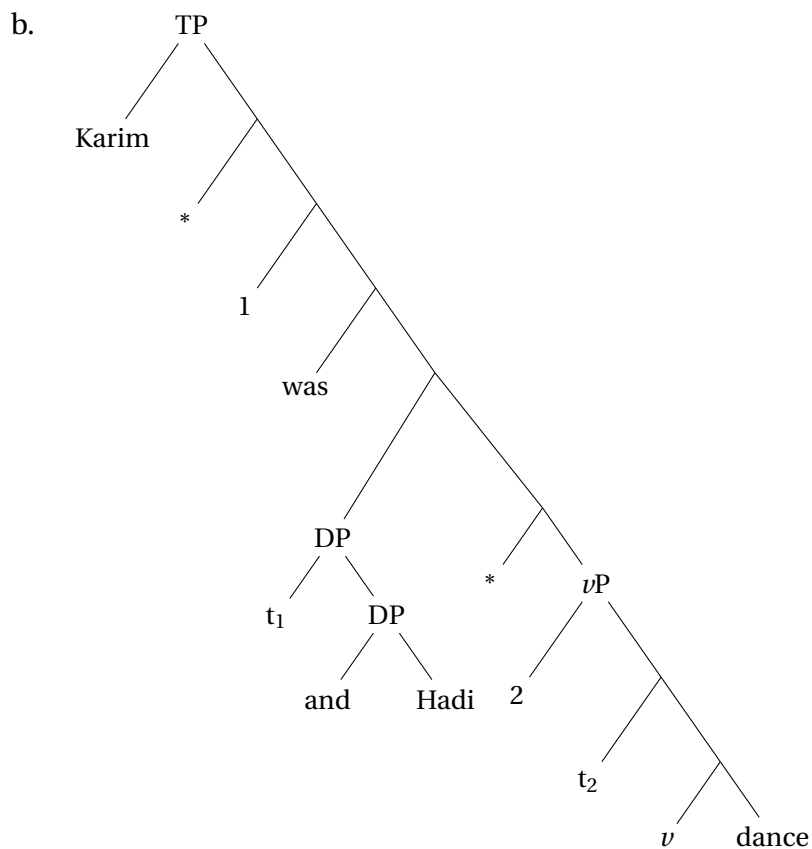
As for standard conjunction, since the entire conjunction moves to spec,TP, this will allow for the LF in (48-b), where * takes scope below the conjoined DP. The resulting truth-conditions with the * operator are given again in (48-c): (48-b) is predicted to be true iff Karim danced and Hadi danced. We therefore correctly predict that (48-a), unlike its double subject counterpart in (45), is true in a scenario where Karim and Hadi danced separately.



Finally, the double subject construction is predicted to behave just like standard conjunction with respect to cumulative inferences when the lower subject can move to an intermediate

position, licensing the * operator to apply to the derived predicate below it. In section 2.2, I argued that this is exactly what happens when the auxiliary *ke:n* is present, as in (49-a): here the main verb agrees with the lower subject but the auxiliary agrees with the higher subject, indicating that the lower subject has moved to an intermediate position, corresponding to the main verb agreement probe, but not all the way to spec,TP. The corresponding LF for (49-a), where * applies below the conjoined DP is given in (49-b) and the resulting truth-conditions are paraphrased in (49-c).

- (49) a. Karim *ke:n* howwe w Hadi *ʕam yereʔso*.
 Karim was him and Hadi PROG dance.3PL.
 Karim and Hadi were dancing.



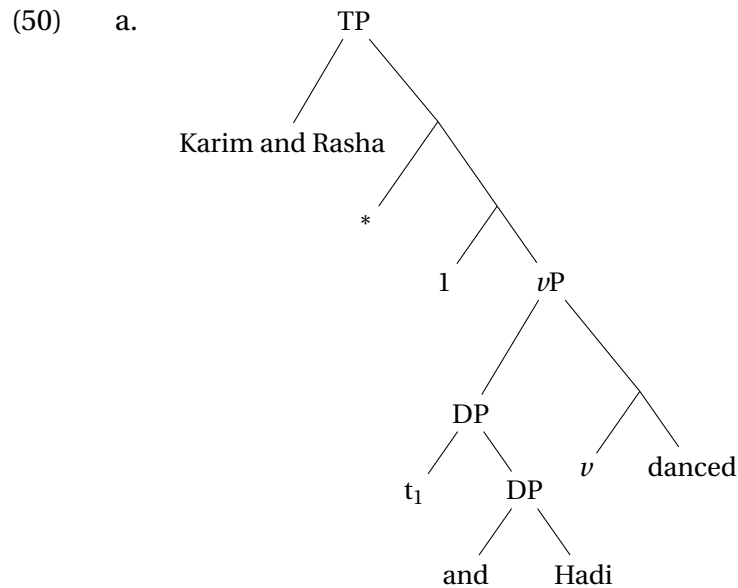
- c. $\llbracket (49-b) \rrbracket = 1$ iff Karim was dancing and Hadi was dancing.

We therefore see that both standard conjunction and the subcases of the double subject construction where the lower subject moves license cumulative inferences: (49-b) is true if Karim was dancing and Hadi was dancing and (48-b) is true if Karim danced and Hadi danced. On the other hand, when the lower subject remains in its base position, * can't apply below it and we correctly predict that the cumulative inference is blocked.

2.3.3 When one conjunct is plural

In the previous subsection, I focused only on cases where the two conjuncts in the double subject construction denote atomic individuals and showed that the account proposed here predicts the relationship between movement and the availability of cumulative inferences with respect to the lower subject. The goal of this section will be to consider the predictions when one of the conjuncts denotes a plural individual. I show that while the double subject construction blocks cumulative inferences over the lower conjunction as a whole, it does license cumulative inferences over the individual conjuncts when they're plural. I argue that this is predicted by the current analysis if we assume that the individual conjuncts move from their base position, while the conjunction as a whole has to stay in spec, ν P.

Starting with the case of the first conjunct being plural, when * takes scope below the higher subject, we predict that a distributive reading should be available over the higher subject. For example, consider the LF in (50-a), where the higher subject is the conjoined DP *Karim and Rasha*. The predicted truth-conditions for (50-a) are given in (50-b): (50-a) is predicted to be true if either all 3 of Rasha, Karim and Hadi danced together or if Rasha and Hadi danced together and Karim and Hadi independently danced together. We therefore see that we can get distributivity over the higher plurality but not the lower subject: (50-a) is predicted to be false in a scenario where each of Rasha, Karim and Hadi danced separately.



- b. $\llbracket (50-a) \rrbracket = 1$ iff $(*(\lambda x. \llbracket \text{dance} \rrbracket (x \oplus h)))(r \oplus k) = 1$ iff
 $\llbracket \text{dance} \rrbracket (r \oplus h \oplus k) = 1 \vee (\llbracket \text{dance} \rrbracket (r \oplus h) = 1 \wedge \llbracket \text{dance} \rrbracket (k \oplus h) = 1)$
 iff Either Rasha and Karim and Hadi all danced together or Rasha and Hadi danced together and Karim and Hadi danced together.

The LF in (50-a) predicts the correct truth-conditions for the double subject construction when the higher subject is plural. In particular, (51) is true in the purely collective scenario 1 and in scenario 2, where there is distributivity only over the higher conjunct, but not in the purely distributive scenario 3.

- (51) Karim w Rasha raʔaso henne w Hadi.
Karim and Rasha danced.PL them and Hadi.
Karim and Rasha danced with Hadi.
- a. **Scenario 1:** Rasha, Hadi and Karim all danced together. **True**
 - b. **Scenario 2:** Karim and Rasha each danced with Hadi separately. **True**
 - c. **Scenario 3:** Rasha, Hadi and Karim each danced separately. **False**

With respect to cumulative inferences, we therefore see that we get a cumulative inference with respect to the higher subject, as shown in (52-a), but not over the entire conjunction *Karim and Rasha and Hadi* (52-b). This is exactly what the current account predicts: while the higher subject moves and can take scope above *, the lower conjunction stays in its base position, therefore blocking a cumulative inference over the entire conjoined DP.

- (52) a. **Cumulative inference with respect to the higher subject:**
Karim danced him and Hadi $\wedge$ **Rasha** danced her and Hadi $\rightarrow$ **Karim and Rasha** danced them and Hadi
- b. **No cumulative inference with respect to the entire conjoined DP:**
Karim danced $\wedge$ Rasha danced $\wedge$ Hadi danced $\nrightarrow$ Karim and Rasha danced them and Hadi

Turning to the case where only the second conjunct is plural, we see that just like the first conjunct, we can get a distributive reading over the second conjunct but not over the entire conjunction. For example, (53) is true in the purely collective Scenario 1, and in Scenario 2, where we have distributivity only over the second conjunct, but not in the purely distributive Scenario 3.

- (53) **Context:** There are two students, Student 1 and Student 2.

Rasha raʔasit hiyye w l tle:miz.
Rasha danced.3SG her and the students.

Rasha danced with the students.

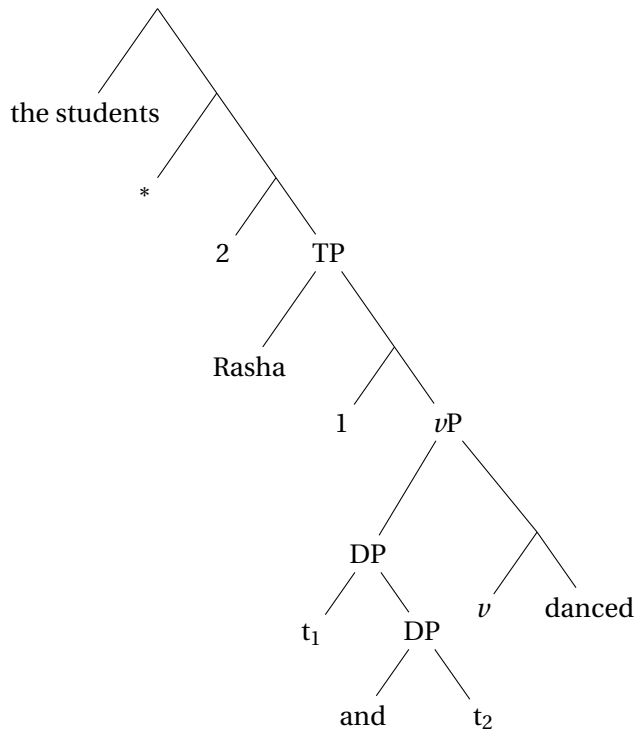
- a. **Scenario 1:** Rasha and Student 1 and Student 2 danced together. **True**
- b. **Scenario 2:** Rasha danced with Student 1 and Rasha separately danced with Student 2. **True**

c. **Scenario 3:** Rasha, Student 1 and Student 2 each danced separately.

False

The desired reading for (53) is predicted if we assume that the second conjunct can move covertly from its base position, allowing * to take scope below it. The corresponding LF is given in (54): (54) is predicted to be true if either each of the two students are such that Rasha danced with them or if the two students and Rasha all danced together. We therefore again predict a cumulative inference over the second subject alone when it is plural but not over the entire conjunction.

(54)



The availability of covert movement of the second conjunct in (54) again raises the issue of how this CSC-violating movement is licensed. While I won't be able to offer a satisfactory answer to why the CSC can be violated here even in the absence of a resumptive pronoun, I will provide independent evidence that covert movement of the second conjunct in the double subject construction is possible. In particular, when the two conjuncts are quantifiers, we can get both surface scope readings where the quantifier in the first conjunct scopes over the one in the second conjunct and an inverse scope reading where the quantifier from the second conjunct takes the higher scope. This is illustrated in (55). In order to get the inverse scope reading in (55), the second conjunct has to QR to a position above the first conjunct, thus providing independent evidence that covert movement of the second conjunct in the double subject construction is licensed.

- (55) 3 tle:miz jtamafo henne w kel ?isterz.
 3 students met them and every teacher
Surface scope reading: 3 students are such that they met with every teacher.
Inverse scope reading: Every teacher is such that 3 students met with them.

2.3.4 When both conjuncts are plural

In addition to licensing distributive readings over the first conjunct and over the second conjunct individually, when both conjuncts are plural, we can get a co-distributive reading with respect to the two conjuncts. To see this, consider the example in (56): here, (56-a) is true in the scenario in (56), where each actor danced with a singer and each singer danced with an actor.

- (56) **Scenario:** Everyone had to pick partners to dance with. There were 10 actors and 10 singers and the actors all decided to pick singers as their partners.
- a. l mmasli:n ra?aso henne w l mitirbi:n.
 the actors danced.3PL them and the singers
 ‘The actors danced with the singers.’

Crucially, (56-a) still differs in meaning from the counterpart with standard coordination, which doesn’t impose any requirement on who danced with who. For example, in the scenario in (57), where all of the actors and singers danced but only some actors danced with singers, the double subject construction in (57-a) is not true, while the counterpart standard coordination (57-b) is. The double subject construction therefore seems to be imposing a restriction here which can’t be captured by simply allowing distributivity over the conjunction as a whole. Instead, we seem to be maintaining the collective reading with respect to the pairs of actor-singers while allowing us to distribute to individual events of actors and singers dancing together.

- (57) **Scenario:** Everyone had to pick partners to dance with. There were 10 actors and 10 singers and 4 of the actors decided to pick singers as their partners, while the other 6 danced with other actors, leaving the other 6 singers to pick each other as partners.
- a. #l mmasli:n ra?aso henne w l mitirbi:n.
 The actors danced.3PL them and the singers
 ‘The actors danced with the singers.’
- b. l mmasli:n w l mitirbi:n ra?aso.
 The actors and the singers danced.3PL
 ‘The actors and the singers danced.’

I will show in this section that this co-distributive reading in (56-a) is predicted if we allow both conjuncts to move, creating a derived 2-place predicate that the 2-place pluralization operator **

can apply to. The resulting reading will therefore be parallel to the co-distributive readings we see with two-place lexical predicates like *saw* when they take two plural arguments (Scha, 1981). For example, (58) is true if every student killed a mosquito and every mosquito was killed by a student. In order to capture these cumulative readings, Krifka (1986) proposes that a 2-place version of the * operator is needed, which he calls **. The operator ** closes the extension of a 2-place predicate under sum-formation. The lexical entry for ** is given in (59).

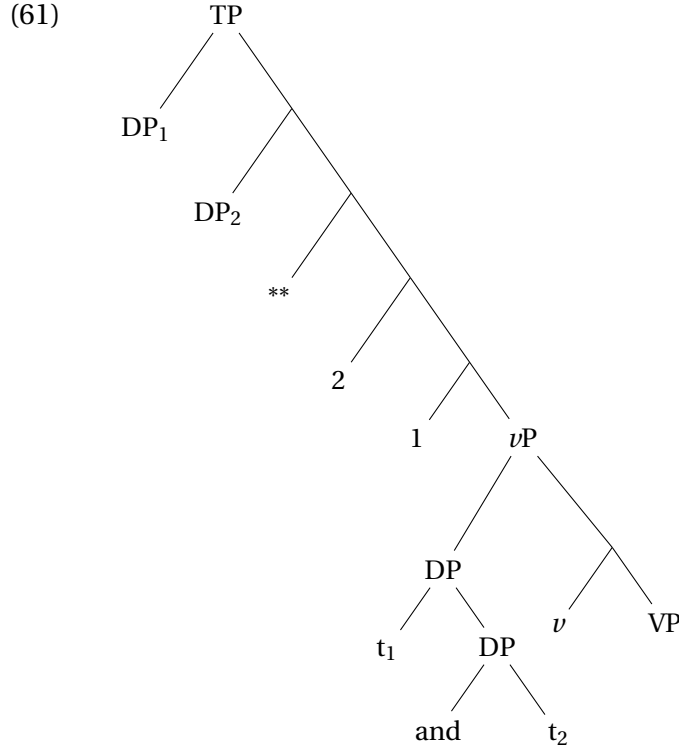
(58) The students killed the mosquitos.

$$(59) \quad \llbracket ** \rrbracket = \lambda f_{\langle e, \langle e, t \rangle \rangle} . \lambda x . \lambda y . \exists g [\forall x' , y' [g(x')(y') = 1 \rightarrow f(x')(y') = 1] \wedge \\ x = \oplus \lambda x . \exists y [g(x)(y)] \wedge y = \oplus \lambda y . \exists x [g(x)(y)]]$$

Applying $\llbracket ** \rrbracket$ to the predicate $\llbracket \text{killed} \rrbracket$ predicts the co-distributive reading of (58). The truth-conditions are given in (60): there is a set of mosquitos which were killed by students in $\llbracket \text{the students} \rrbracket$ and the sum of individuals in that set adds up to $\llbracket \text{the mosquitos} \rrbracket$, and there is a set of students which killed mosquitos in $\llbracket \text{the mosquitos} \rrbracket$ and the sum of individuals in that set adds up to $\llbracket \text{the students} \rrbracket$. In other words, (60) is correctly predicted to be true if and only if every student killed a mosquito and every mosquito was killed by a student.

$$(60) \quad \llbracket ** \rrbracket (\llbracket \text{killed} \rrbracket) (\llbracket \text{the mosquitos} \rrbracket) (\llbracket \text{the students} \rrbracket) = 1 \text{ iff} \\ \exists g [\forall x' , y' [g(x')(y') = 1 \rightarrow \llbracket \text{killed} \rrbracket (x')(y') = 1] \wedge \llbracket \text{the mosquitos} \rrbracket = \oplus \lambda x . \exists y [g(x)(y)] \\ \wedge \llbracket \text{the students} \rrbracket = \oplus \lambda y . \exists x [g(x)(y)]]$$

The co-distributive reading therefore suggests that ** is at play in the cases where both conjuncts in the double subject construction are plural. I argue that the data in this section can be accounted for if we assume that the second conjunct covertly moves to a position higher than the pluralization operator, tucking in below the higher subject and creating a two-place predicate that the cumulativity operator ** can apply to. The resulting LF is given in (61), where ** applies to a two-place predicate created by predicate abstraction over the indices 1 and 2 and both conjuncts scope above **. This LF is analogous to structures proposed by Sauerland (1998) and Beck and Sauerland (2000) to account for certain co-distributive readings which arise over derived predicates.



Consider now the example in (56-a) (*the actors danced them and the singers*). The truth-conditions in (62) are predicted: (56-a) is true iff every actor danced or is part of a plurality of actors that danced with at least one singer and every singer danced or is part of a plurality of singers that danced with at least one actor. Therefore, in a situation where some of the actors danced with each other and not with singers, (56-a) is correctly predicted not to be true. The co-distributive reading over the two conjuncts is therefore correctly predicted.

(62) $\llbracket \text{the actors danced them and the singers} \rrbracket = 1$ iff
 $(** \lambda y. \lambda x. \llbracket \text{dance} \rrbracket (x \oplus y))(\llbracket \text{actors} \rrbracket)(\llbracket \text{the singers} \rrbracket) = 1$

To summarize, we saw in this section and in section 2.3.3 that while distributive readings are blocked over the lower conjunction in the double subject construction, the construction allows for distributive readings over the individual conjuncts, in addition to co-distributive readings when both conjuncts are plural. I argued that these different readings are all predicted by the account proposed here if we allow the second conjunct to move covertly from its base position, while ensuring that the entire conjoined DP stays in spec, ν P. The assumption that the entire conjoined DP can't move covertly from its base position is therefore crucial in predicting the desired reading. I turn to this issue in section 2.3.5, where I discuss how to ensure the correct conditions on when covert movement is licensed.

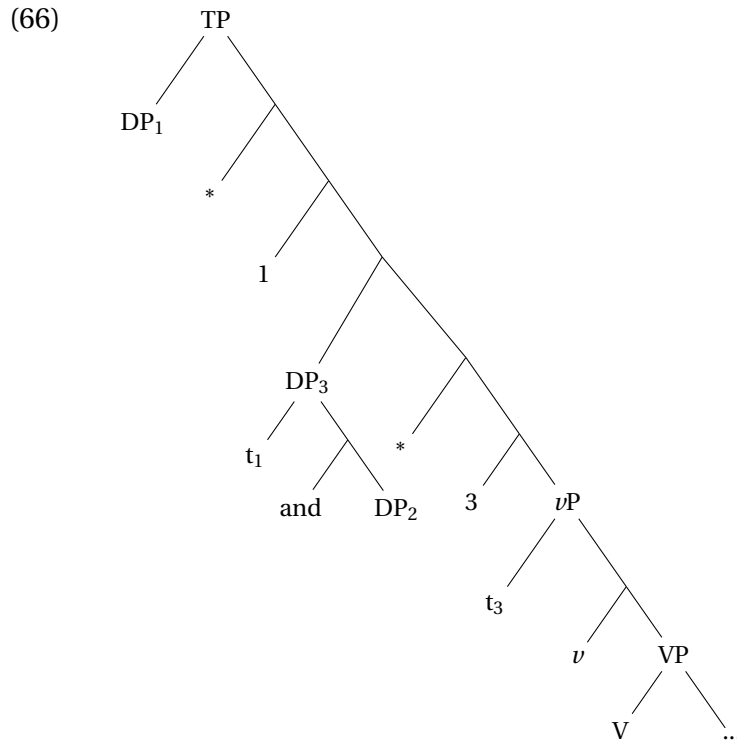
2.3.5 Why can't the entire lower subject move covertly?

In section 2.3.2, I argued that the lower subject can move overtly to an intermediate position and license a distributive reading when auxiliaries like *ke:n* are present, as in (63). Then, in section 2.3.3, I argued that the second conjunct can move covertly from the lower subject, licensing a distributive reading over the second conjunct alone but not the entire conjoined DP. Given the availability of covert movement in (64), there is a question of why covert movement of the entire lower subject is blocked in (65). In particular, why can't the lower conjoined DP in (65) move covertly to an intermediate position and license a distributive reading in the same way that overt movement of the lower subject does in (63)?

- (63) Karim *ke:n* howwe w Hadi *ʕam yereʔso*.
Karim was him and Hadi PROG dance.3PL.
Karim and Hadi were dancing.
- (64) Karim *raʔas* howwe w l *tle:miz*.
Karim danced.3SG.M him and the students.
Karim danced with the students.
- (65) Karim *raʔas* howwe w Hadi.
Karim danced.3SG.M him and Hadi.
Karim and Hadi danced together.

In this subsection, I will present a possible solution to this issue, which I will ultimately reject in section 2.4. Namely, I show in this section that if the Ban on Improper Movement is generalized to subextraction (following Sakai, 1994; Abels, 2007), the correct licensing conditions on covert movement will be predicted for all of the cases we have seen so far. I will then show in section 2.4 that this purely syntactic constraint is not able to predict the correct licensing conditions on covert movement when we consider the behavior of two-place predicates in the double subject construction. I therefore argue in section 2.4 that this syntactic constraint has to be replaced with a constraint on covert movement that is sensitive to semantic interpretation.

In what follows, I show that the Ban on Improper Sub-extraction blocks the lower subject from moving covertly in (65), while allowing for the required movements in (63) and (64). The LF for (65) that we want to block is given in (66). As argued in section 2.2.1, the singular agreement we see on the verb in the double subject construction in (65) corresponds to the agreement probe agreeing with the first conjunct and moving it to spec,TP. Therefore, in order for the entire conjoined DP to move from its base position, it has to QR to an intermediate position before the first conjunct moves to spec,TP, as shown in the LF in (66).



To derive the LF in (66), the lower subject needs to undergo A-bar movement to an intermediate position, after which the first conjunct undergoes A-movement to spec,TP. This kind of derivation has been independently argued to be unlicensed, violating a generalized version of the ban on improper movement which targets sub-extraction. In particular, chains of movement where A movement targets a sub-constituent of a phrase which has undergone A-bar movement have been argued not to be licensed (67) (Sakai, 1994; Abels, 2007; Keine, 2020).

- (67) **Improper sub-extraction:** A-extraction out of an A-bar moved constituent is not possible.

Note that this ban on the LF in (66) relies on an assumption that the first step of movement where the conjoined DP moves to an intermediate position is an instance of A-bar movement. If the conjoined DP can move to an intermediate A-position between spec, ν P and spec,TP, there would be no violation of (67) and we would be able to get a distributive reading in the double subject construction. This therefore commits us to assuming that there are no intermediate A-positions available in the simple example of the double subject construction in (65). This account therefore predicts that if intermediate A-positions are made available, intermediate movement of the lower subject will become available and the distributive reading will be licensed.

I want to suggest that this is exactly what happens in (63), repeated below. Here, the presence of the auxiliary makes available an intermediate A position that the subject can move to, corresponding to the agreement probe that controls agreement on the main verb. In (68), the

conjoined DP moves to this intermediate A-position, licensing a distributive reading, and the first conjunct then moves to a higher A-position, spec,TP, without violating the Ban on Improper Movement.

- (68) Karim ke:n howwe w Hadi ʕam yereʔso.
 Karim was him and Hadi PROG dance.3PL.
 Karim and Hadi were dancing.

Finally, nothing rules out covert movement of the second conjunct in (64). Since only the first conjunct undergoes A-movement to spec,TP, the second conjunct is free to undergo A-bar movement without violating (67). I have therefore shown that the ban on improper sub-extraction in (67) rules out the unattested distributive LF for the simple case of the double subject construction, while allowing covert movement in the desired cases where distributive readings are in fact available.

2.4 Cumulative inferences with two-place predicates

So far, I have only looked at the behavior of 1-place predicates like *dance* in the double subject construction. Based on a variety of patterns with these 1-place predicates, I argued that pluralization operators must apply to predicates derived by movement and that covert movement of the lower subject in the double subject construction is blocked by the ban on improper movement. This allowed us to predict that cumulative inferences are in general blocked over the lower subject, since it stays in its base position and takes scope below *.

The goal of this section is to try to extend the account to two-place predicates like *eat* and *lift*. I show that the behavior of these predicates in the double subject construction poses a puzzle for the present account. In particular, it appears that two-place cumulative inferences are sometimes licensed over the lower subject in the double subject construction but that this depends crucially on certain semantic properties of the object DP.

I show that this puzzle forces us to replace any purely syntactic restriction on covert movement with a constraint that is sensitive to semantic properties of the sentence. More specifically, I propose that the licensing of covert movement of the lower subject must depend on whether the resulting LF generates truth-conditions that are distinct from simpler alternative LFs. This type of constraint is familiar in the literature. For example, Fox's (2000) scope economy argues that the availability of QR is sensitive to whether or not covert movement generates different truth-conditions than an alternative LF where there is no movement. I suggest that the movement of the lower subject is constrained by a similar trans-derivational economy constraint that compares the truth-conditions of the LF where the lower subject moves covertly to those

of alternative LFs with standard conjunction. I only offer a very informal sketch at the end of this section of how this type of constraint can predict the desired distribution of cumulative inferences with 2-place predicates, as I will return to discussing this economy constraint in much more detail in the following chapters.

2.4.1 Introducing 2-place cumulativity

The cumulative inference with 2-place predicates is exemplified in (69). Here we see a cumulative inference with respect to a pair of two argument positions: if *a saw b* is true and *c saw d* is true, then *a and c saw b and d* also has to be true. As in the case of 1-place cumulative inferences, the predicate *saw* in (69) can be replaced with any other 2-place predicate, giving rise to the analogous inference.

- (69) a. Jane saw Mary.
 b. Bill saw Sue.
 c. $(a) \wedge (b) \rightarrow \text{Jane and Bill saw Mary and Sue.}$

The lexical account argues that cumulative inferences over two-place predicates, exemplified in (69), are due to lexical generalizations over predicate meanings, just as we saw with 1-place predicates in the introduction. For 2-place predicates, the statement of the lexical cumulativity universal is given in (70) (Scha, 1981; Kratzer, 2007, a.o.). It is straightforward to see how (70) predicts the inference in (69): as shown in (71-d), the lexical cumulativity universal predicts that when the predicate $\llbracket \text{saw} \rrbracket$ is true of when applied first to *m* and then to *j* and is true when applied first to *s* and then to *b*, then it also has to be true of when applied first to $m \oplus s$ and then to $j \oplus b$. Therefore, the inference in (69) is predicted if we assume that the predicate $\llbracket \text{saw} \rrbracket$ applies directly to its two arguments, as shown in (71).

(70) **2-place lexical cumulativity:**

For any 2-place predicate $P_{\langle e, \langle e, t \rangle \rangle}$,

$$\forall x, x', y, y' [(P(x)(y) = 1 \wedge P(x')(y') = 1) \rightarrow P(x \oplus x')(y \oplus y') = 1]$$

- (71) a. $\llbracket \text{Jane saw Mary} \rrbracket = 1$ iff $\llbracket \text{saw} \rrbracket (m)(j) = 1$
 b. $\llbracket \text{Bill saw Sue} \rrbracket = 1$ iff $\llbracket \text{saw} \rrbracket (s)(b) = 1$
 c. $\llbracket \text{Jane and Bill saw Mary and Sue} \rrbracket = 1$ iff $\llbracket \text{saw} \rrbracket (m \oplus s)(j \oplus b) = 1$
 d. **Predicted Inference:** $\llbracket \text{saw} \rrbracket (m)(j) = 1 \wedge \llbracket \text{saw} \rrbracket (s)(b) = 1 \rightarrow \llbracket \text{saw} \rrbracket (m \oplus s)(j \oplus b) = 1$

The syntactic account, on the other hand, again maintains that the lexical meanings of 2-place predicates are collective and therefore do not inherently give rise to the cumulative inference in (69). Rather, the 2-place cumulativity operator **, which we saw in section 2.3.4, is taken to

be the only source of 2-place cumulative inferences (Sternefeld, 1998; Beck & Sauerland, 2000). The $**$ operator closes the extension of a 2-place predicate under sum formation, ensuring that $\llbracket ** \rrbracket (\llbracket \text{saw} \rrbracket)$ for example is true of $m \oplus s$ and $j \oplus b$ if the lexical predicate $\llbracket \text{saw} \rrbracket$ is true of m and j and of s and b . In order to derive the cumulative inference in (69), we therefore have to assume that *Jane and Bill saw Mary and Sue* has an LF where $**$ applies to the predicate, as shown in (72-a). Now, (72-b) follows from the definition of $**$, therefore predicting the inference from *Jane saw Mary* and *Bill saw Sue* to (72-a).

- (72) a. $\llbracket \text{Jane and Bill } [** \text{ saw}] \text{ Mary and Sue} \rrbracket = 1$ iff $\llbracket ** \rrbracket (\llbracket \text{saw} \rrbracket) (m \oplus s) (j \oplus b) = 1$
 b. $(\llbracket \text{saw} \rrbracket) (m) (j) = 1 \wedge \llbracket \text{saw} \rrbracket (s) (b) = 1 \rightarrow \llbracket ** \rrbracket (\llbracket \text{saw} \rrbracket) (m \oplus s) (j \oplus b) = 1$

In addition to the cumulative readings we see with conjunctions, we see parallel cumulative readings when the object is a definite plural (73-a) or a numeral quantifier (73-b). Again, since the syntactic account predicts that the lexical meanings of the predicates themselves are collective, in order for (73-a) and (73-b) to be true in the given cumulative scenarios, $**$ has to apply to the predicate before it applies to its arguments.

- (73) a. Jane and Mary lifted the pianos.
Cumulative Scenario: Jane lifted half of the pianos and Mary independently lifted the other half of the pianos.
 b. Jane and Mary lifted 10 pianos.
Cumulative Scenario: Jane lifted 5 pianos and Mary independently lifted another 5 pianos.

I assume that the restriction that $*$ applies only to predicates derived by movement is a more general one that applies to all pluralization operators, including the 2-place pluralization operator $**$, an assumption which I will argue for in section 2.4.2.⁶ Given this assumption, the present account will predict that the cumulative readings in (73-a) and (73-b) require movement of both the subject and the object from their base position.

- (74) **Restriction on pluralization operators:** Pluralization operators, like $*$ and $**$, can only apply to predicates derived by movement.

Starting with the definite plural object, the predicted LF corresponding to the cumulative reading is given in (75-a). In (75-a), both the subject and the object move from their base position, creating a 2-place derived predicate that $**$ can apply to. The truth-conditions in (75-b) are

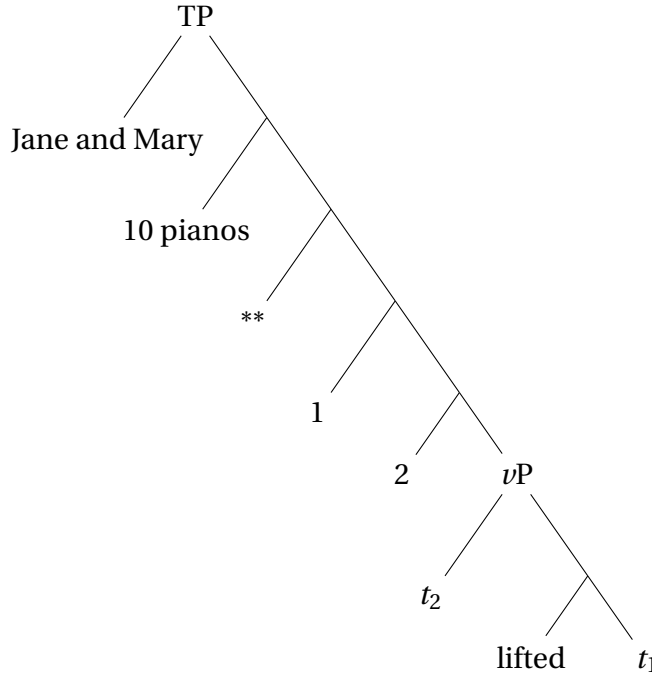
⁶I will argue in section 2.4.2 that this assumption is necessary in order to rule out the unattested distributive readings with predicates like *lift the piano* in the double subject construction. In particular, I show that if we allow $**$ to apply to the 2-place predicate *lift*, this will incorrectly predict that a sentence like *Jane lifted the piano her and Mary* in the double subject construction will be true if Jane and Mary lifted the piano separately.

predicted: (75-a) is predicted to be true iff Jane and Mary cumulatively lifted the pianos.

- (75) a.
-
- b. $\llbracket (75-a) \rrbracket = 1$ iff $(\llbracket ** \rrbracket (\llbracket \text{lift} \rrbracket)) (\llbracket \text{the pianos} \rrbracket) (j \oplus m) = 1$
iff Jane and Mary cumulatively lifted the pianos.

Finally, the LF and corresponding truth-conditions for the cumulative reading when there is a numeral quantifier in object position are given in (76). Here, again the object and subject move to create a 2-place predicate which $**$ applies to. The resulting truth-conditions are given in (76-c): (76-a) is predicted to be true if there is a plurality of x consisting of 5 pianos such that Jane and Mary cumulatively lifted x . This derives the desired cumulative reading in (73-b), while maintaining again that $**$ applies only to predicates derived by movement.

(76) a.



- b. $\llbracket 10 \text{ pianos} \rrbracket = \lambda P. \exists x : |x| = 10 \wedge * \llbracket \text{piano} \rrbracket (x) = 1 \wedge P(x) = 1$
c. $\llbracket (76\text{-a}) \rrbracket = 1$ iff $\exists x : \llbracket \text{pianos} \rrbracket (x) \wedge |x| = 10 \wedge (\llbracket ** \rrbracket (\llbracket \text{lift} \rrbracket)) (\llbracket x \rrbracket) (j \oplus m) = 1$
iff there are 10 pianos that Jane and Mary cumulatively lifted.

2.4.2 2-place predicates in the double subject construction

If we assume that the restriction on pluralization operators proposed in section 2.3 applies to **, we predict that just like *, ** can't apply to the lower subject in the double subject construction. To see this with an example, consider (77).⁷ The constraint on improper movement predicts that the lower conjunct *him and Hadi* has to stay in its base position. Given that * and ** can apply only to predicates derived by movement, neither of these pluralization operators can apply to a predicate below the conjoined DP. We therefore predict that (77) should neither have a cumulative nor a distributive reading with respect to *Karim and Hadi*. More specifically, I will consider the three types of scenarios in (77), where the counterpart with standard conjunction can be true, and look at what our account predicts for the double subject construction.

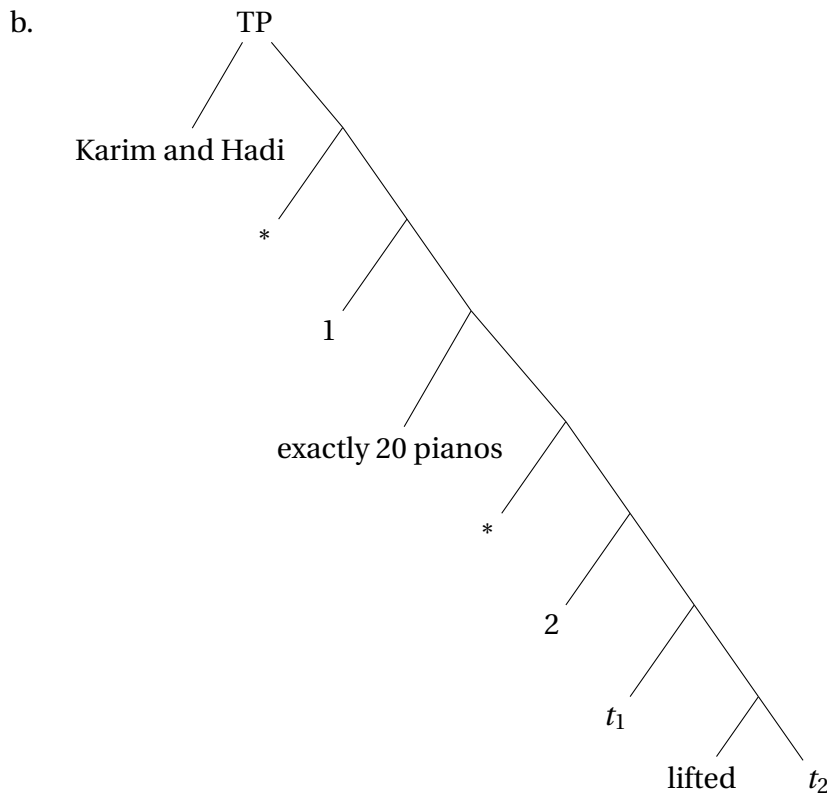
- (77) Karim himil howwe w Hadi ʕishreen piano ʕalʔad.
Karim lifted him and Hadi 20 piano exactly.
a. **Distributive Scenario:** Karim and Hadi each independently lift 20 pianos.

⁷I look at cumulative readings with numeral quantifier objects here, rather than definite plural objects, since definite plural objects appear to behave differently from other objects in the DSC, not licensing the cumulative reading. I return to this difference in chapter 6, where I show that it is predicted by the economy constraint I propose.

- b. **Cumulative Scenario:** Karim lifted 10 pianos and Hadi independently lifted 10 other pianos.
- c. **Collective Scenario:** Karim and Hadi are working together to lift pianos. They go to each piano and lift it together. They do this for exactly 20 pianos.

Starting with the distributive scenario in (77-a), in order to generate this reading with a standard conjoined subject like (78-a), the subject has to move to a position above *exactly 20 pianos* and * has to apply between *Karim and Hadi* and *exactly 10 pianos*, giving rise to a reading which is true if each of Karim and Hadi lifted exactly 20 pianos. The corresponding LF that derives the desired distributive reading is given in (78-b). We therefore predict that the double subject construction should not have this distributive reading, as it also requires movement of the lower subject *Karim and Hadi*.

(78) a. Karim and Hadi lifted exactly 20 pianos.

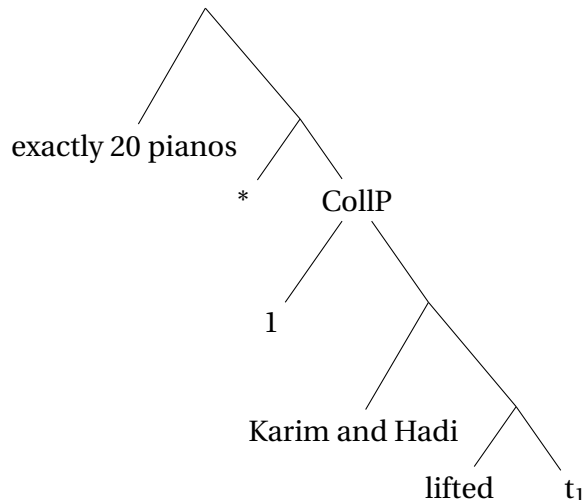


Turning to the cumulative reading in (77-b), we saw in section 2.4.1 that this reading requires both the subject and the object to move, creating a 2-place derived predicate that ** can apply to. We therefore again predict that the double subject construction does not allow for this reading, as it requires the lower subject to move.

Finally, the collective reading in (77-c) can be generated without moving the subject. To see this, consider the LF in (79-a): here we see that * takes scope below *exactly 20 pianos* but

that there is no pluralization operator below the subject. Now, given that predicates like *lift* are assumed to inherently have collective truth-conditions, the predicate below * will have the collective meaning in (79-b). The higher * then allows for distributivity over *exactly 20 pianos*, predicting that (79-a) is true in the scenario in (77-c) where Karim and Hadi jointly lifted each of 20 pianos. Given that the collective reading in (77-c) can be generated without moving the subject, we predict that the double subject construction should allow for this reading.

(79) a.



b. $\llbracket \text{CollP} \rrbracket = \lambda x. \text{Karim and Hadi jointly lifted } x.$

To summarize, I considered the example in (80) of the double subject construction with a 2-place predicate and looked at the three types of scenarios in (80). I showed that in order for (80) to be true in (80-a) and (80-b), the subject has to move from its base position, while the collective reading in (80-c) can be generated without movement of the lower subject. Given that I argued in section 2.3 that the ban on improper movement forces the lower subject in its base position, we therefore predict that (80) is only true in the collective scenario in (80-c).

(80) Karim himil howwe w Hadi ʕishreen piano ʕalʔad.
 Karim lifted him and Hadi 20 piano exactly.
 Karim and Hadi lifted exactly 20 pianos together.

- a. **Distributive Scenario:** Karim and Hadi have never met each other. They each independently lift 20 pianos. **Prediction:** False.
- b. **Cumulative Scenario:** Karim and Hadi have never met each other. Karim lifted 10 pianos and Hadi independently lifted 10 other pianos. **Prediction:** False.
- c. **Collective Scenario:** Karim and Hadi are working together to lift pianos. They go to each piano and lift it together. They do this for exactly 20 pianos. **Prediction:** True.

While the predictions that (80) is false in the distributive scenario and true in the collective

scenario are both borne out, the prediction that the sentence is false in the cumulative scenario isn't. In particular, (80) is in fact true in a scenario where each of Karim and Hadi lifted 10 pianos, even when Karim and Hadi did so completely independently. This poses a serious challenge to our account of the double subject construction, which at this point does not make a distinction between the cumulative scenario in (80-b) and the distributive one in (80-a), as they both require movement of the lower subject.

One might think that this problem can be resolved by assuming that the 1-place * operator and the 2-place ** operator in fact have different distributions. In particular, we saw that the LF which makes the sentence true in the distributive scenario in (80-a) requires * to take scope below the lower subject and above *20 pianos*, while the LF which makes the sentence true in the cumulative scenario requires ** to take scope below the lower subject. Now, suppose that the restriction that pluralization operators apply only to predicates derived by movement applies only to the 1-place * but not to **. Then, we can maintain the results in section 2.3 and predict that the double subject construction is false in (80-a), while allowing ** to apply directly to the 2-place predicate *lift* in (80), predicting the desired cumulative reading.

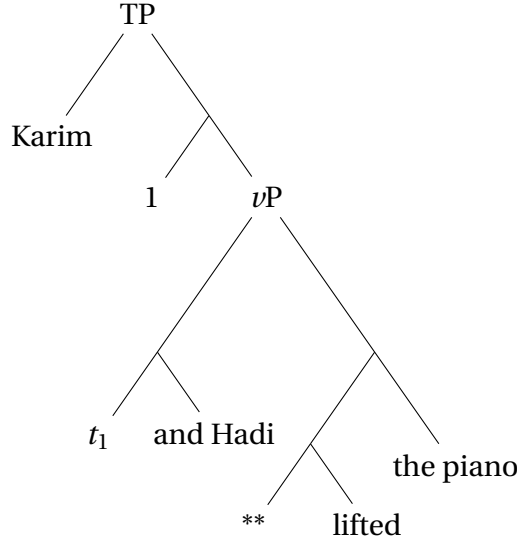
This relatively straightforward solution incorrectly predicts that the differences in meaning between standard conjunction and the double subject construction are restricted to 1-place predicates like *dance*. More specifically, if we allow ** to apply directly to two-place predicates like *lift*, we would predict that the double subject construction will have a distributive reading when *lift* takes a singular object. This is clearly an incorrect prediction: as shown in (81), *lift the piano* just like *dance* only allows for collective readings in the DSC.

(81) Karim himil l piano howwe w Hadi.
Karim lifted the piano him and Hadi.
Karim and Hadi lifted the piano together.

- | | |
|---|--------------|
| a. Scenario 1: Each of Karim and Hadi lifted the piano separately. | False |
| b. Scenario 2: Karim and Hadi lifted the piano together. | True |

To see how the unattested distributive reading of (81) can be derived with **, consider the LF in (82-a), where ** merges with the lexical predicate *lift*. The corresponding truth-conditions in (82-b): (82-a) is predicted to be true in both the collective Scenario 2 and the distributive Scenario 1. This is because ** closes the extension of $\llbracket \text{lift} \rrbracket$ under sum formation, ensuring that if $\llbracket \text{lift} \rrbracket$ is true of $\llbracket \text{the piano} \rrbracket$ and k and true of $\llbracket \text{the piano} \rrbracket$ and h then $** (\llbracket \text{lift} \rrbracket)$ has to be true of $\llbracket \text{the piano} \rrbracket$ and $k \oplus h$, as shown in (82-c).

(82) a.



- b. $\llbracket (82-a) \rrbracket = 1$ iff $\llbracket ** \rrbracket(\llbracket \text{lift} \rrbracket)(\llbracket \text{the piano} \rrbracket)(k \oplus h) = 1$
 iff $\llbracket \text{lift} \rrbracket(\llbracket \text{the piano} \rrbracket)(h) = 1 \wedge \llbracket \text{lift} \rrbracket(\llbracket \text{the piano} \rrbracket)(k) = 1 \vee \llbracket \text{lift} \rrbracket(\llbracket \text{the piano} \rrbracket)(h \oplus k) = 1$
 iff Either Hadi lifted the piano alone and Karim lifted the piano alone or Karim and Hadi lifted the piano together.
- c. $\llbracket \text{lift} \rrbracket(\llbracket \text{the piano} \rrbracket)(k) \wedge \llbracket \text{lift} \rrbracket(\llbracket \text{the piano} \rrbracket)(h) \rightarrow \llbracket ** \rrbracket(\llbracket \text{lift} \rrbracket)(\llbracket \text{the piano} \rrbracket)(k \oplus h)$

To predict the cumulative reading of (83-a), while maintaining that (83-b) only has a collective reading, we therefore have to assume that ****** can apply below the lower subject in (83-a), generating the desired cumulative reading, but can't apply below the lower subject in (83-b), where it would generate an unattested reading that is true in a distributive scenario. The licensing of ****** below the lower subject therefore has to depend on the semantic properties of the object DP, which can't be achieved by placing purely syntactic constraints on the distribution of ****** and on the ability of the lower subject to move from its base position.

- (83) a. Karim ħimil howwe w Hadi ħishreen piano ħalʔad.
 Karim lifted him and Hadi 20 piano exactly.
 Karim and Hadi lifted 20 pianos together.
- b. Karim ħimil l piano howwe w Hadi.
 Karim lifted the piano him and Hadi.
 Karim and Hadi lifted the piano together.

To summarize, this section makes the crucial point that the current account is unable to distinguish the attested cumulative reading in (83-a) from the unattested distributive reading of (83-b): either ****** can apply below the lower subject, in which case both readings are predicted to be available or ****** can't apply below the lower subject, making both readings unavailable.⁸

⁸One possibility, which I did not discuss here, is that when both arguments of a 2-place predicate are plural, the cumulative reading is encoded into the lexical semantics of the predicate. This would allow us to maintain that

2.4.3 A semantic constraint on QR

The puzzle outlined in section 2.4.2 will be explored in detail in chapter 6 after discussing a similar issue that arises for my analysis of English *together* in chapters 4 and 5. At this point, I will simply give a very brief sketch of what the required solution should look like.

I argue that the fact that distributive readings continue to be blocked in (83-b) shows that we need to maintain the syntactic restriction proposed on the distribution of * and ** in section 2.3, repeated in (84) below. Now, if we assume that the lower subject has to remain in its base position in both (83-a) and (83-b) due to the ban on improper movement, we would correctly account for the blocking of the distributive reading in (83-b) but incorrectly predict that the cumulative reading in (83-a) is also blocked. Instead, I want to suggest that whether or not the lower subject can QR from its base position is mediated by a different constraint that is not purely syntactic and that makes reference to semantic properties of the sentence, allowing us to predict the required connection between whether or not the subject can move and the type of object (*20 pianos* vs *the piano*).

- (84) **Restriction on pluralization operators:** Pluralization operators, like * and **, can only apply to predicates derived by movement.

More specifically, I will argue that the movement of the lower subject is mediated by a trans-derivational economy constraint, similar to Fox's scope economy, that compares the truth-conditions generated by the double subject construction and by standard coordination. Put informally for now, the LF where the lower subject QRs from its base position in (83-b) would make (83-b) true both in scenarios where Karim and Hadi lifted the piano together and scenarios where they lifted it separately. The resulting truth-conditions are therefore exactly equivalent to those generated by the entire conjunction simply moving to spec,TP, as in the case of standard coordination in (85). The economy constraint will therefore rule out QR of the lower subject in (83-b), since the resulting reading is the same as the one which can be generated with standard conjunction without having to QR the subject.

- (85) Karim w Hadi hemlo l piano.
Karim and Hadi lifted.3PL the piano.

two-place predicates are lexically collective when only one of their arguments is plural, as in (83-b), while deriving the cumulative reading in (83-a) without requiring a pluralization operator to apply below the lower subject. This solution will not end up working either, when we consider the full distribution of readings with 2-place predicates in the DSC. In particular, I show in chapter 6 that definite plural objects, unlike numeral quantifiers like (83-a), don't license cumulative readings in the DSC. This shows that the licensing of a cumulative inference can't depend on whether the object is plural or singular. Rather, I show that the differences between definite plurals and numeral quantifiers is the result of an economy constraint that determines when the object can move covertly to a position above **.

Karim and Hadi lifted the piano.

On the other hand, I will argue that the LF where the lower subject moves out of its base position and takes scope above ** in (83-a) is not ruled out by this economy, since the resulting truth-conditions are still different from the counterpart where the entire conjunction moves to spec,TP, without the additional step of covert movement.

While I won't be able to flesh out the formal details of the economy comparison here, we can see that when the object is a numeral quantifier like *exactly 20 pianos*, we can allow a cumulative reading with respect to the conjunctive DP and the object, while maintaining a difference in truth-conditions between the double subject construction and standard coordination. In particular, the standard conjunction in (86), in addition to being true in the collective and cumulative scenarios in (87-b) and (87-c), is also true in the distributive scenario in (87-a), where each of Karim and Hadi lifted exactly 20 pianos. On the other hand, the double subject construction counterpart is not true in the distributive scenario. Therefore, despite allowing the lower subject to QR and generate the cumulative reading, we still maintain a difference with standard conjunction, where the entire conjunction moves all the way to spec,TP, allowing for an additional distributive reading.

- (86) Karim w Hadi hemlo fashreen piano falʔad.
Karim and Hadi lifted 20 pianos exactly.
Karim and Hadi lifted exactly 20 pianos.

- (87) a. **Distributive Scenario:** Karim and Hadi have never met each other. They each independently lift 20 pianos.
Standard Conjunction: True **Double Subject:** False
- b. **Cumulative Scenario:** Karim and Hadi have never met each other. Karim lifted 10 pianos and Hadi independently lifted 10 other pianos.
Standard Conjunction: True **Double Subject:** True.
- c. **Collective Scenario:** Karim and Hadi are working together to lift pianos. They go to each piano and lift it together. They do this for exactly 20 pianos.
Standard Conjunction: True **Double Subject:** True

The resulting picture, which I will outline in much more detail in the following chapters, will be one where covert movement of the lower subject is constrained not by purely syntactic constraints but by a constraint which is sensitive to both the truth-conditions generated by a given LF and by comparisons with truth-conditions of alternative LFs. This will allow us to capture the intricate licensing conditions on the different readings available in the double subject construction, which I showed to depend crucially on semantic properties of the object DP (e.g.

20 pianos vs the piano), rather than only on syntactic properties of the sentence.

2.5 Beyond cumulative inferences

So far, I have shown that the only difference between the DSC and standard coordination is that pluralization operators can apply below the conjoined DP in the latter but not the former. We can therefore use the double subject construction to tease apart the contribution of pluralization operators and lexical semantics beyond cumulative inferences. In particular, if some property of plural predication is due to the lexical semantics of the verb, then the account proposed here predicts that it should behave the same in the DSC and standard coordination, since in both cases the verb applies to the plural conjoined DP. On the other hand, if it is due to the contribution of pluralization operators, then we should not observe it over the lower subject in the DSC which has to scope below pluralization operators. The goal of this section will be to take homogeneity inferences and reciprocal alternations as two such case studies and argue that just like with cumulative inferences, the DSC provides evidence that pluralization operators play a crucial role in both phenomena.

2.5.1 The sources of homogeneity

I'll start by introducing homogeneity and then turn to the interaction between homogeneity and the DSC.⁹ When predicates that allow for distributive readings apply to pluralities, they give rise to an apparent truth-value gap (J. D. Fodor, 1970). I'll start by illustrating this with the distributive predicate *smile* in (88): here the positive counterpart is true iff both Karim and Rasha smiled (88-a), while its negation is only true if neither of them smiled (88-b), leaving a homogeneity gap where neither (88-a) nor (88-b) is true in a scenario where only one of Karim and Rasha smiled.

- (88) a. *Karim and Rasha smiled* is **true iff** both Karim and Rasha smiled.
 b. *It's not the case that Karim and Rasha smiled* is **true iff** neither of them smiled.
 c. **Neither is true iff** only one of Karim and Rasha smiled.

For predicates like *dance* and *lift the piano*, which allow for collective readings, the interaction with homogeneity turns out to be a bit more subtle (Schwarzschild, 1994; Križ, 2015; Bar-Lev, 2025, a.o.). In particular, the presence of a homogeneity gap with these predicates depends on context. For example, in the context in (89), where what's relevant is whether Jane danced and

⁹I will focus in this section only on homogeneity with 1-place predicates. Of course, analogous homogeneity gaps are observed with 2-place predicates (Križ, 2015; Chatain, 2021), but given the questions raised in section 2.4 about the distribution of cumulative readings with 2-place predicates in the DSC, I will set the interaction of homogeneity with 2-place predicates aside.

whether Mary danced, neither (89-a) nor its negated counterpart in (89-b) is true in the scenario in (89) where only one of Jane and Mary danced. We therefore see the same truth-value gap as the one characterized in (88) for *smile* here. Unlike in the case of the distributive predicate *smile*, the homogeneity gap with *dance* can be suspended in the right context. In particular, given a context where only the collective reading of *dance* is relevant (i.e. whether or not Jane and Mary danced together), it appears that we can get a purely collective reading with *dance* that when negated gives rise to a negated collective reading. This is illustrated by the scenario in (90): *Jane and Mary didn't dance* is true in this scenario, despite this being a scenario where one of Jane and Mary danced, violating homogeneity.

- (89) **Scenario:** Bill is wondering whether Jane danced and whether Mary danced at the party yesterday. It turns out that Jane danced and Mary sat on the couch and didn't feel like dancing.
- a. #Jane and Mary danced.
 - b. #Jane and Mary didn't dance.
- (90) **Scenario:** Bill is wondering if Jane and Mary danced together at the party yesterday. It turns out that Jane danced and Mary sat on the couch and didn't feel like dancing.
- a. #Jane and Mary danced.
 - b. Jane and Mary didn't dance.

I assume, following Schwarzschild (1994) and Bar-Lev (2025), that we need to derive a default homogeneity gap for *dance*, just as with *smile* above, and that the suspension of homogeneity is due to a mechanism akin to domain restriction that allows for homogeneity removal in the right context.

We therefore see that just like cumulative inferences, homogeneity appears to be a universal property of lexical predicates.^{10,11} Now, in parallel with the debate regarding the sources of cumulative inferences, there are two types of accounts regarding the source of homogeneity. The lexical account, proposed for example by Križ (2015), attributes the homogeneity gap with respect to lexical predicates like *dance* to a lexical property of predicates. In particular, Križ proposes that lexical predicates give rise to a truth-value gap when applied to pluralities. This

¹⁰Certain predicates like *less than 5 in number* don't give rise to homogeneity. For example, *the students are less than 5 in number* clearly does not entail that no part of *the students* is less than 5 in number. This same class of predicates also doesn't license cumulative inferences. I assume that these are complex predicates and not lexical predicates, as evidenced by their morphological realization and therefore don't consider them to be an exception to either the generalization regarding cumulativity in the introduction or the homogeneity generalization being discussed here.

¹¹Not all plural DPs give rise to homogeneity. For example, numeral quantifiers like *6 students* and quantifiers headed by *all* seem to remove homogeneity. For the purposes of this chapter I will simply be focusing on homogeneity with definite plural DPs and set aside homogeneity removal.

truth-value gap corresponds to assigning the falsity conditions in (91) to lexical predicates.

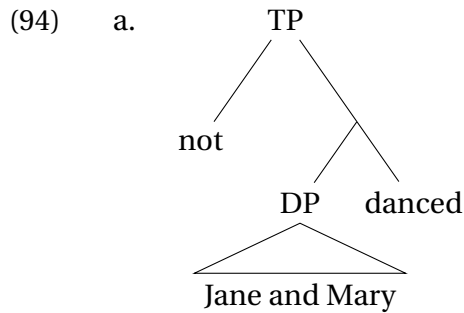
(91) **Lexical homogeneity:** Given a predicate P and a plurality x : $P(x)=0$ iff $\forall y \leq x : P(y) = 0$

Applying this to the example in (88), the truth-conditions in (92) are predicted for *Rasha and Hadi didn't dance*. In particular, (92) is correctly predicted to be true only if neither Rasha nor Hadi danced. Given that the positive counterpart is true if both Rasha and Hadi danced, we therefore have derived the truth-value gap: neither (92) nor its positive counterpart is true in a scenario where only one of Rasha and Hadi danced.

(92) $\llbracket \text{Rasha and Hadi didn't dance} \rrbracket = 1$ iff $\llbracket \text{Rasha and Hadi danced} \rrbracket = 0$ iff $\llbracket \text{dance} \rrbracket (r \oplus h) = 0$ iff $\llbracket \text{dance} \rrbracket (r) = 0 \wedge \llbracket \text{dance} \rrbracket (h) = 0$

Alternatively, we can attribute the truth-value gap in (88) to the same pluralization operators that we argued are responsible for cumulative inferences (Schwarzschild, 1994; Bar-Lev, 2025). For 1-place predicates, this can be done by adding a truth-value gap into the denotation of the pluralization operator, following Bar-Lev (2025)(93). On this view the presence of a cumulative inference and the presence of a homogeneity gap go hand in hand. In particular, given an LF like (94-a), where there is no pluralization operator, the truth-conditions in (95-b) are predicted under negation: (94-a) is simply predicted to be true if Jane and Mary didn't dance together and no truth-value gap is derived.

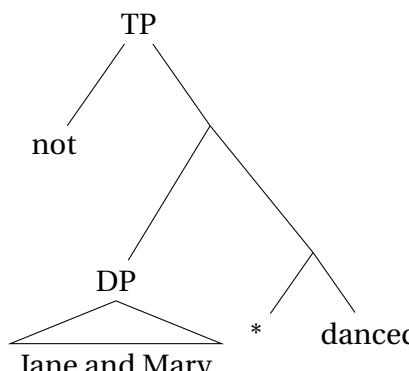
$$(93) \quad \llbracket * \rrbracket (P)(x) = \begin{cases} 1 & \text{iff } \exists Q_{\langle e, t \rangle} [\forall y [Q(y) = 1 \rightarrow P(y) = 1] \wedge x = \oplus Q] \\ 0 & \text{iff } \neg \exists y \leq x : P(y) = 1 \\ \# & \text{otherwise} \end{cases}$$



b. $\llbracket (94-a) \rrbracket = 1$ iff $\llbracket \text{dance} \rrbracket (j \oplus m) = 0$ iff Jane and Mary didn't dance together.

On the other hand, when the pluralization operator $*$ applies to the predicate before it applies to the plurality, we get the desired truth-value gap. This is illustrated in (95): the LF with $*$ in (95-a) is predicted to be true iff neither Jane nor Mary danced. Under the syntactic account, in order to derive the desired homogeneity gap with respect to a given plurality, it has to be the case that $*$

applies below that plurality.

- (95) a. 
- b. $\llbracket (95-a) \rrbracket = 1$ iff $\llbracket * \rrbracket (\llbracket \text{dance} \rrbracket) (j \oplus m) = 0$
iff $\llbracket * \rrbracket (\llbracket \text{dance} \rrbracket) (j) = 0 \wedge \llbracket * \rrbracket (\llbracket \text{dance} \rrbracket) (m) = 0 \wedge \llbracket * \rrbracket (\llbracket \text{dance} \rrbracket) (j \oplus m) = 0$
iff neither Jane nor Mary danced.

Now, the syntactic account and the lexical account make different predictions with respect to how the double subject construction interacts with homogeneity. On the syntactic account, since $*$ is required to get the homogeneity gap, we expect that there should be no homogeneity gap with respect to the lower subject in spec, νP , which has to take scope below $*$. On the other hand, the lexical account predicts that homogeneity should not depend in any way on the syntactic position of the plural DP, thus predicting a homogeneity gap in the double subject construction.

Consider the example in (96), where the double subject construction is negated: (96) is true as long as Karim and Rasha didn't dance together, thus simply giving rise to complementary truth-conditions with its positive counterpart. We therefore see that there is no homogeneity gap here. For example, unlike what we saw with the English counterpart *Jane and Mary didn't dance* in (89), (96) is true in a scenario like (89) where Karim danced but Rasha didn't. Crucially, (96) does not require a context that makes salient the collective reading in order to cancel homogeneity, as shown by the fact that (96), unlike (89-b), is true even when the salient question is whether Karim and Rasha danced individually (as in (89)).

- (96) Karim ma raʔas howwe w Rasha.
Karim NEG dance.PST.PFV.3SG.M him and Rasha.
Karim and Rasha didn't dance together.
- | | |
|--|-------------|
| a. Scenario 1: Rasha and Karim danced but separately. | True |
| b. Scenario 2: Rasha danced but Karim didn't. | True |
| c. Scenario 3: Karim danced but Rasha didn't. | True |

The truth-conditions in (96) are exactly what we expect if homogeneity is due to pluralization operators and not the lexical semantics of predicates. Since $*$ can't take scope below the conjoined

DP in (96), we don't get homogeneity over that DP. On the other hand, building homogeneity into the lexical semantics of *dance* incorrectly predicts that (96) is only true if neither Karim nor Rasha danced.¹² We therefore see that the double subject construction offers a window into the division of labor between lexical semantics and pluralization operators in plural predication beyond cumulative inferences.

2.5.2 The sources of reciprocal alternations

Another interesting case study where the double subject construction can be used to distinguish between lexical and syntactic accounts concerns the sources of reciprocal alternations, exemplified in (97) and (98). In (98) for example, we see that *kiss* can be a 2-place predicate in (98-b) but appears to be a 1-place predicate in (98-a). The 1-place variant has a reciprocal-like meaning but differs from overt reciprocals in certain ways (see Winter (2018) for a detailed discussion).

(97) a. Karim and Rasha dated.

b. Karim dated Rasha.

(98) a. Karim and Rasha kissed.

b. Karim kissed Rasha.

Just as in the case of cumulative inferences and homogeneity discussed above, two types of analysis have been proposed for reciprocal alternations: (i) a lexical analysis, which proposes that predicates like *date* and *kiss* are ambiguous between a 2-place predicate in (97-b) and (98-b) and a 1-place collective predicate in (97-a) and (98-a) (Winter, 2018, a.o.) and (ii) a syntactic analysis, which posits that *date* and *kiss* are always 2-place predicates and that (97-a) and (98-a) have an elided reciprocal anaphor in object position (Hackl, 2002).

If the lexical analysis is correct, we expect that the 1-place variants in (97-a) and (98-a) should be licensed in the double subject construction. This is because the lexical analysis simply assumes that the 1-place variant of predicates like *date* has the collective truth-conditions given in (99). In the double subject construction, this predicate can therefore apply to the lower subject without any intervening pluralization operators, giving rise to the truth-conditions in (99).

(99) $\llbracket \text{date}_{1\text{-place}} \rrbracket (k \oplus r) = 1$ iff Karim and Rasha dated.

¹²This conclusion that pluralization operators are the only sources of homogeneity raises the question of why the homogeneity gap with standard plural predication can't be removed by simply not inserting any pluralization operator. Schwarzschild (1994) and Bar-Lev (2025) assume that pluralization operators apply obligatorily at every position of the right type, but this can't be maintained here, given that I've argued that pluralization operators can only apply to predicates derived by movement. I return to this issue in Appendix B, where I discuss some possible ways to reconcile my analysis of the double subject construction with the fact that suspending homogeneity with underspecified predicates like *dance* requires a specialized context.

The predictions of the syntactic analysis for the double subject construction depend on the analysis we assume for overt reciprocals. One prominent analysis of reciprocals is that they involve the same pluralization operators needed for cumulative inferences (Sauerland, 1998; Beck, 2001). More specifically, this type of analysis assumes that the LF for a reciprocal sentence like (100-a) in fact involves a *the other* DP in object position and a higher pluralization operator that allows for a distributive reading, as shown in the paraphrase in (100-b). Assuming such an analysis, the syntactic analysis of reciprocal alternations will therefore predict that the double subject construction, by blocking pluralization operators from applying below the lower subject, should not license the 1-place variants in (97-a) and (98-a). This is because the LFs for (97-a) and (98-a) will have an elided *each other* in object position and therefore require a pluralization operator to take scope below the conjoined DP subject.

- (100) a. Jane and Mary kissed each other.
 b. Jane and Mary each kissed the other.

In what follows, I will show that the predictions of the syntactic analysis are borne out, at least for certain predicates like *kiss* and *date*. Consider first the behavior of overt reciprocals in the double subject construction in (101): (101) shows that when the higher subject denotes an atomic individual, overt reciprocals are ungrammatical in the double subject construction. This follows from the analysis of overt reciprocals mentioned above, where they require a pluralization operator to apply below the conjoined DP. Since * can't apply below the lower subject in (101), the overt reciprocal is predicted to be ungrammatical.

- (101) a. *Karim be:s howwe w Rasha baʕdon.
 Karim kissed him and Rasha each-other.
 Intended: Karim and Rasha kissed each other.
 b. *Karim saḥab howwe w Rasha baʕdon.
 Karim dated him and Rasha each-other.
 Intended: Karim and Rasha dated each other.

Turning to the 1-place variants in reciprocal alternations, I will again consider two predicates *kiss* and *date*. First, the examples in (102-a) and (102-b) for *kiss* and (103-a) and (103-b) show that the Lebanese Arabic counterparts of *kiss* and *date* participate in the same kind of reciprocal alternation as we saw above in English. Crucially, with both predicates, the 1-place variant is ungrammatical in the double subject construction, as shown in (102-c) for *kiss* and in (103-c) for *date*.

- (102) **Kiss:**

- a. Karim be:s Rasha.
Karim kissed.3SG.M Rasha.
Karim kissed Rasha.
- b. Karim w Rasha be:so.
Karim and Rasha kissed.3PL.
Karim and Rasha kissed.
- c. *Karim be:s howwe w Rasha.
Karim kissed him and Rasha.

(103) **Date:**

- a. Karim sa:ħab Rasha.
Karim dated.3SG.M Rasha.
Karim dated Rasha.
- b. Karim w Rasha sa:ħabo.
Karim and Rasha dated.3PL.
Karim and Rasha dated.
- c. *Karim sa:ħab howwe w Rasha.
Karim dated him and Rasha.

The syntactic analysis of (102-c) and (103-c) as covert reciprocals can straightforwardly explain their ungrammaticality. In particular, just as the overt reciprocals in (101) are ungrammatical in the double subject construction, the covert counterparts are also predicted to be ungrammatical, due to the requirement that a pluralization operator applies below the conjoined DP in order to license the reciprocal object. On the other hand, the ungrammaticality of (102-c) and (103-c) poses a serious challenge for the lexical account which treats the 1-place variants simply as collective predicates. In particular, there is no reason that this collective predicate shouldn't be able to apply directly to the lower subject and generate the desired reading, just as we saw with the collective readings of other predicates like *dance* earlier in the chapter.

The interaction of reciprocal alternations with the double subject construction therefore provides evidence that the LFs for sentences like (104-a) and (104-b) contain a covert reciprocal object.¹³

¹³There are well-known challenges to the covert reciprocal analysis of reciprocal alternations. In particular, the overt reciprocal and its covert counterpart sometimes differ in truth-conditions. Consider the example in (i) from Winter (2018): in the given scenario, the overt reciprocal in (i-b) is true, while the covert counterpart in (i-a) isn't. This is a serious problem for the syntactic analysis which simply assumes that (i-a) has an elided reciprocal object. The goal of chapter 7 will be to propose a modified syntactic analysis of reciprocal alternations which addresses this challenge.

- (i) Context: Jane hugged Mary while Mary was sleeping and then Jane fell asleep and Mary woke up and hugged her.
 - a. #Jane and Mary hugged.
 - b. Jane and Mary hugged each other.

- (104) a. Jane and Mary kissed.
b. Jane and Mary hugged.

2.6 Deriving the constraint on pluralization operators

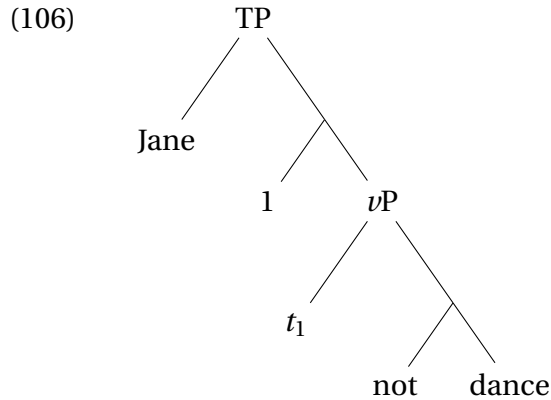
I have argued that the distribution of pluralization operators is restricted, such that they can apply only to predicates derived by movement (105). At this point, this constraint is simply stipulated as a syntactic constraint on the distribution of *. This raises questions about whether there is a systematic reason that the distribution of * is restricted in this way.

- (105) **Syntactic constraint on pluralization operators:** Pluralization operators can only be merged with a node that has a binder index as one of its two daughters.

I show in this section that this constraint follows from a general requirement, proposed by Hirsch (2017), that lexical predicates must apply immediately to their arguments. Hirsch's constraint, which can be thought of as a semantic implementation of the theta criterion, predicts that modifiers in general, including pluralization operators, must apply only to predicates derived by movement, thus predicting my constraint in (105) as a special case.¹⁴

The goal of Hirsch's constraint is to rule out operations like predicate conjunction and predicate negation from applying to lexical predicates, as in the LF in (106). One version of this constraint, proposed by Amir Anvari in lecture notes, is given in (107). The strict linking condition rules out the LF in (106): since *dance* is of type $\langle e, t \rangle$ and does not have a lambda index as one of its daughters, the sister of *dance* must have type e . Instead, *not*, the sister of *dance* has type $\langle \langle e, t \rangle, \langle e, t \rangle \rangle$, thus violating the condition in (107).

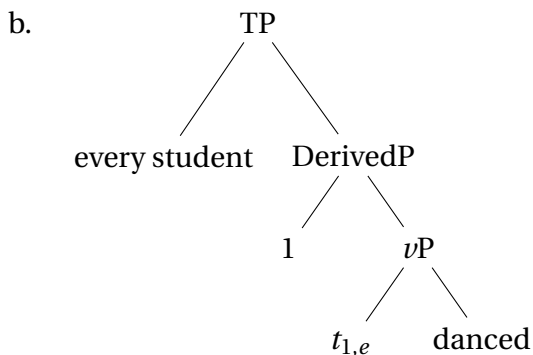
¹⁴So far, I have only considered two options regarding the sources of cumulative inferences: (i) they are due to the lexical semantics of predicates and (ii) they are due to syntactic operators. This leaves open a third possibility that I haven't considered: cumulative inferences are the result of a rule of composition. In Appendix C, I explore an alternative idea for the relationship between movement and cumulative inferences, which attributes cumulative inferences to the composition rule of predicate abstraction, rather than to syntactically present pluralization operators. This will predict that cumulative inferences only arise over predicates that are derived by predicate abstraction, i.e. derived predicates which have a binder index as one of their daughters. While this idea will also capture the connection between movement and cumulative inferences, it remains unclear why predicate abstraction should encode cumulativity in this way. I will set aside this third option for the purposes of this dissertation, but this remains a possible analysis that should be explored in more detail in future work.



- (107) **Strict Linking Condition (SLC):** If an expression α has a meaning of type $\langle\sigma, \tau\rangle$ and is merged with its sister β , β must have a meaning of type σ unless α has a lambda index as one of its daughters.

The exception for expressions that have a lambda index as one of their daughters is needed to allow for appropriate LFs for sentences like (108-a), where the subject denotes a quantifier. Here, the condition in (107) will prevent the predicate *dance* from merging directly with the quantifier: again, the sister of *dance* has to have type e and *every student* has type $\langle\langle e, t \rangle, t\rangle$. On the other hand, the strict linking condition does allow the LF in (108-b) for (108-a), where the quantifier moves from its base position and leaves behind a trace of type e . In (108-b), *dance* merges with an expression of type e , satisfying the SLC. The derived predicate DerivedP also satisfies the SLC: since DerivedP has a lambda index as one of its daughters, it is allowed to merge with a higher type expression like *every student* without violating the SLC.

- (108) a. Every student danced.

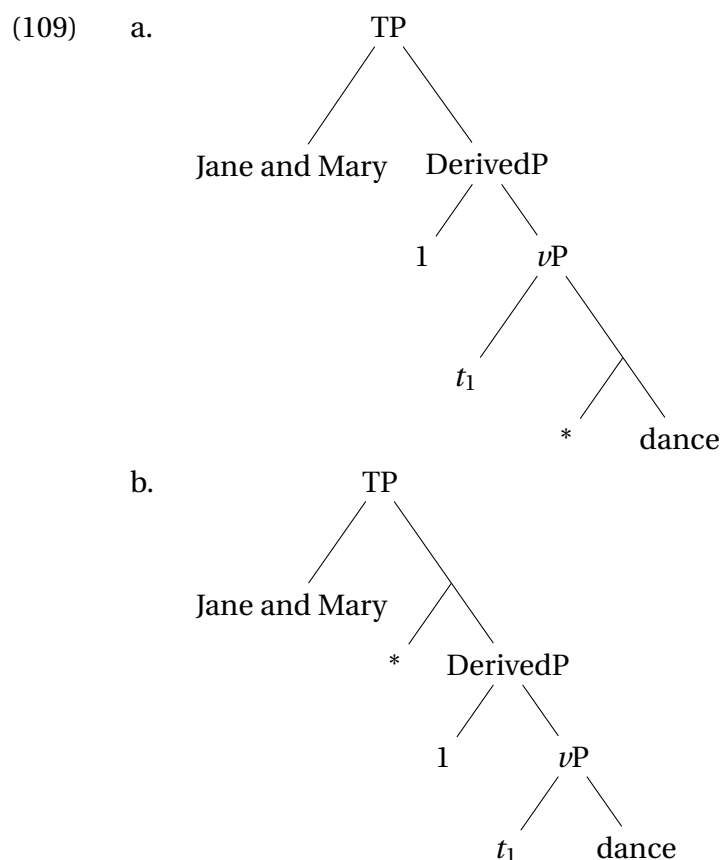


There are many questions regarding how generally applicable the SLC is and whether the formulation given in (107) is the right one, which I won't be able to get into in this dissertation.¹⁵ I will

¹⁵Hirsch's (2017) own formulation states that the requirement applies unless β is internally merged. This would predict the pluralization operators can never be externally merged, incorrectly predicting that they can't even apply to predicates derived by movement. The formulation given in (107), on the other hand, allows for modifiers like * and ** to externally merge with a node that has a binder index as one of its daughters, thus allowing pluralization

simply show in what follows that if we have the condition in (107), which has been independently suggested, then our constraint on the distribution of pluralization operators immediately follows.

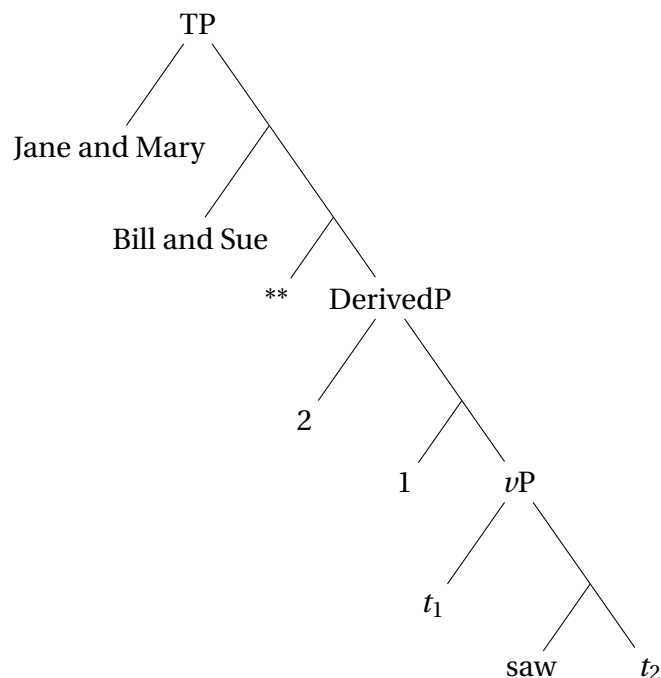
To see this consider the LF in (109-a), where * applies directly to the predicate *dance*. The LF in (109-a) clearly violates the SLC: the sister of *dance* has type $\langle\langle e, t \rangle, \langle e, t \rangle\rangle$ and not the required type *e*. Now, consider the LF in (109-b), where * instead applies to the predicate derived by moving the subject from spec, ν P to spec, TP. Unlike (109-a), (109-b) satisfies the SLC: as we saw above, since DerivedP has a binder index as one of its daughters, it is able to merge with *, which has a higher type $\langle\langle e, t \rangle, \langle e, t \rangle\rangle$, without violating the SLC.



The Strict Linking Condition given in (107) therefore predicts that * can only apply to nodes which have a binder index as one of their daughters. The same result generalizes to two-place predicates and **. In particular, given a predicate of type $\langle e, \langle e, t \rangle \rangle$ like *saw*, the SLC predicts that this predicate must merge first with an individual of type *e*. Therefore, **, which has type $\langle\langle e, \langle e, t \rangle \rangle, \langle e, \langle e, t \rangle \rangle\rangle$, won't be able to merge directly with the predicate. As with 1-place *, the SLC allows for ** to apply to a derived predicate, as shown in (110). Here, the predicate that ** applies to, DerivedP, has a lambda index as one of its daughters and therefore merging ** with DerivedP satisfies the SLC.

operators to apply to predicates derived by movement.

(110)



We therefore see that the SLC predicts the constraint that pluralization operators in general can apply only to predicates derived by movement. As mentioned at the beginning of this section, the SLC makes the radical prediction that modifiers in general can't apply directly to lexical predicates and must apply only to predicates derived by movement. It remains to be seen whether the SLC can be maintained in general and whether the predictions beyond pluralization operators are borne out.

2.7 Conclusion

This chapter has argued that the only sources of cumulative inferences are pluralization operators whose syntactic distribution is restricted such that they can apply only to predicates derived by movement. This conclusion provides evidence against an influential approach to cumulative inferences, lexical cumulativity, which takes cumulative inferences to be built into the lexical meanings of all natural language predicates. To argue for this conclusion, I showed that there is an intricate relationship between movement and the availability of cumulative inferences in the Lebanese Arabic double subject construction. In particular, I showed that we can only get cumulative inferences over plural DPs that have moved from their base position. I argued that this interaction between movement and the availability of cumulative inferences can only be explained if we assume that the only sources of cumulative inferences are syntactically present pluralization operators.

In addition to the main conclusion regarding cumulative inferences, this chapter raised

several open issues which will be addressed in future chapters in the dissertation. I want to reiterate some of the issues that will be relevant as I try to develop a unified framework for the role and distribution of pluralization operators in different phenomena in plural predication.

First, section 2.4 raised a very interesting puzzle regarding what licenses movement of the lower subject in the double subject construction. In particular, I argued in section 2.4 that in the case of 2-place predicates, whether or not the lower subject can move does not depend solely on syntactic properties but also on semantic properties of the object DP. This suggests that whatever constrains movement of the lower subject has to be sensitive to the resulting truth-conditions corresponding to the LF with movement. After presenting a similar puzzle with English *together* in chapters 4 and 5, I will argue in chapter 6 that the movement of the lower subject is mediated by a trans-derivational economy constraint, akin to Fox's (2000) scope economy. The distribution of cumulative inferences in the DSC will therefore give us a new case study to investigate the nature of economy constraints in natural language more generally.

Another important open issue, discussed in section 2.5.2, relates to the sources of reciprocal alternations. In particular, I showed that the double subject construction provides evidence for a syntactic analysis of reciprocal alternations involving a covert reciprocal object in sentences like *Jane and Mary hugged*. At the same time, there are several challenges for this syntactic approach that have motivated prominent lexical analyses of these alternations. Chapter 7 will focus on the problem of reciprocal alternations, where I will propose a modified version of the syntactic analysis that addresses these challenges.

Finally, the analysis proposed in this chapter requires us to have a ban on predicates that lexically give rise to cumulative inferences. In particular, the pattern we saw in the double subject construction holds across all predicates, in the sense that no predicate gives rise to cumulative inferences in the double subject construction. To account for the generality of this pattern, it is therefore not enough to remove lexical cumulativity as a universal generalization over lexical meanings. We also have to ban mixed lexicons like the toy example in (111), where the predicate *eat* is not lexically cumulative but the predicate *dance* is. This alternative lexicon would incorrectly predict that the double subject construction gives rise to a collective reading with *eat* but allows for a distributive reading with *dance*. We therefore need to have a general ban on lexically cumulative predicates (112).

(111) **Alternative mixed lexicon:**

- a. $\llbracket \text{eat} \rrbracket (j \oplus m) = 1$ iff Jane and Mary ate together.
- b. $\llbracket \text{dance} \rrbracket (j \oplus m) = 1$ iff Jane and Mary danced (together or separately).

(112) **Ban on lexical cumulativity (1-place):**

There is no lexical predicate P such that $\forall x, y : P(x) = 1 \wedge P(y) = 1 \rightarrow P(x \oplus y) = 1$

I return to this ban in chapter 6, where I suggest that this might follow from a general economy constraint on the lexicon that mediates the division of labor between lexical semantics and the syntax. In short, I will argue that a lexicon which both has pluralization operators and lexically cumulative predicates is ruled out by an economy constraint that optimizes the expressive power of the lexicon, relative to its size. While I won't discuss this constraint in detail at this point, the idea is that a lexicon which has * can generate cumulative predicates by applying * to non-cumulative predicates. The economy constraint I will propose will generally block lexical meanings that can be independently generated in the syntax, thus predicting that lexically cumulative predicates violate economy.

Appendix A: The pause construction

In this appendix, I discuss the issue I mentioned in section 2.2.2 regarding a possible confound with the double subject construction. In particular, when there is a pause before the second subject and focus on the pronoun, the same string as the double subject construction does allow for the distributive reading that I've been assuming is unattested. While I won't have an analysis of what's happening when we have this pause, I will try to argue in this section that this is a distinct construction that does not in fact undermine the claims that I've made so far about the double subject construction. The different readings we get with and without the pause are given in (113): without the pause, we get the collective reading which is false in the scenario in (113), but with the pause, we get a distributive reading which is true in the given distributive scenario.

(113) **Scenario:** Rasha and Hadi danced separately.

a. **Without pause:**

Rasha raʔasit hiyye w Hadi.
 Rasha danced.3SG.F her and Hadi.
 Rasha danced with Hadi.

False

b. **With pause:**

Rasha raʔasit, hiyye w Hadi.
 Rasha danced.3SG.F, her and Hadi.
 Rasha danced, along with Hadi.

True

The first piece of evidence that these are two distinct constructions comes from the interaction of the two constructions with negation. In particular, as we saw in section 2.5, the double subject construction can be negated, giving rise to a negated collective reading in (114-a). On the other hand, the counterpart with a pause before the conjunction in (113-b) results in oddness when we try to negate it (114-b). Therefore under negation, the string in (114-a) only gives rise to the

negated collective reading that is predicted in the double subject construction. Interestingly, there is a different way to form the construction with a pause under negation. This is shown in (115-b), where the pause is followed by the *not A and not B* conjunction illustrated in (115-a).

(114) a. **Negation without pause:**

Rasha ma raʔasit hiyye w Hadi.
 Rasha not danced.3SG.F her and Hadi.
 Rasha didn't dance with Hadi.

b. **Negation with pause:**

??Rasha ma raʔasit, hiyye w Hadi.
 Rasha not danced.3SG.F, her and Hadi.
 Intended: Neither Rasha nor Hadi danced.

(115) a. la Rasha w la Hadi raʔaso.
 not Rasha and not Hadi danced.3SG.PL.
 Neither Rasha nor Hadi danced.

b. Rasha ma raʔasit, la hiyye w la Hadi.
 Rasha not danced.3SG.F, not her and not Hadi.
 Rasha didn't dance and neither did Hadi.

The pattern in (114) and (115) therefore provides evidence that the construction involving the pause is distinct from the double subject construction. One possibility is that this alternative construction is due to ellipsis, where it in fact involves two separate clauses as shown in (116), with the VP in the second clause being elided. Now, assuming that this is VP-ellipsis and that negation can't be elided, this allows us to predict the patterns in (114) and (115) with negation. In particular, (114-b) is predicted to correspond to (117-b), where only *dance* is elided in the second sentence. The second sentence in (117-b) contradicts the first one, thus explaining the oddness of (114-b). On the other hand, in (117-c), assuming that only *dance* is elided predicts the desired reading, as shown in the paraphrase in (117-c): (117-c) is true if neither Rasha nor Hadi danced.

(116) Rasha raʔasit. hiyye w Hadi raʔaso.
 Rasha danced.3SG.F. her and Hadi danced.
 Rasha danced, along with Hadi.

(117) a. Rasha danced. Her and Hadi danced.
 b. #Rasha didn't dance. Her and Hadi danced.
 c. Rasha didn't dance. Not her and not Hadi danced.

(118) a. #Rasha didn't dance. Both her and Hadi did.
 b. Rasha didn't dance. Neither her nor Hadi did.

Interestingly, we see a similar pattern in English with *along with*. The standard comitative *with*, just like the double subject construction, allows only for a collective reading. On the other hand, *along with*, just like the construction with the pause in Lebanese Arabic, allows for a distributive reading when there is a pause after the verb. In parallel to the pattern we saw with negation, while the standard *with* comitative can be negated and gives rise to a negated collective reading, the *along with* construction gives rise to oddness under negation (120-b). While I don't have an analysis of the difference between *along with* and *with* in English, this parallel points towards a more general analysis of what we find with the pause in Lebanese Arabic.

- (119) a. Jane danced with Mary.
b. Jane danced, along with Mary.
- (120) a. Jane didn't dance with Mary.
b. ??Jane didn't dance, along with Mary.

Aside from the pattern under negation, we see similarly that other embedded environments, such as the scope of negative quantifiers (121) and in *wh*-questions (122), don't license the construction with the pause. This again provides additional evidence that we are dealing with two completely different constructions with different distributions. The ellipsis analysis suggested above can account for (121) and (122) straightforwardly: if the construction with the pause involves two different clauses with ellipsis in the second clause, it shouldn't be possible to embed the construction at all.

(121) **Under negative quantifiers:**

a. **Without Pause:**

maḥada raʔas howwe w Hadi.
Nobody danced him and Hadi.
Nobody danced with Hadi.

b. **With Pause:**

??wala telmi:z raʔas, howwe w Hadi.
no student.3SG.M danced.3SG.M, him and Hadi.
Intended: No student danced and neither did Hadi.

(122) **In a *wh*-question:**

a. **Without Pause:**

Min raʔas howwe w Hadi?
Who danced him and Hadi.
Who danced with Hadi.

b. **With Pause:**

??Min raʔas, howwe w Hadi?

Who danced, him and Hadi?

Intended: Who danced other than Hadi?

While a complete lack of embedding would provide strong evidence for the ellipsis analysis, the pause construction is licensed in certain embedding conditions, such as the antecedents of conditionals. This is exemplified in (123): we have a pause after the verb in (123) and the antecedent is interpreted distributively, just as we saw in the unembedded case. The availability of embedding in (123) therefore raises problems for a straightforward analysis in terms of ellipsis.

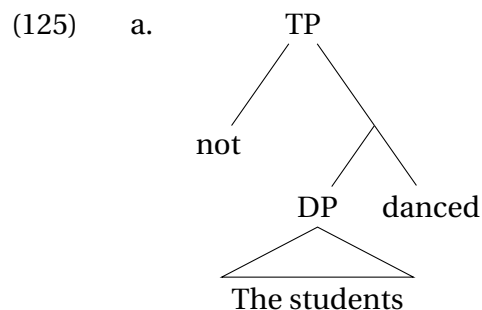
- (123) Iza Rasha raʔasit, hiyye w Hadi, raḥ yenbesit Karim.
If Rasha danced, her and Hadi, will happy.3SG.M Karim.
If Rasha danced, along with Hadi, Karim will be happy.

I will leave a full analysis of the pause construction to future work. What's important for the purposes of this chapter is that I have provided some evidence here that this construction is distinct from the double subject construction, both in its distribution and with respect to how the sentence is uttered (i.e. the pause). I will therefore take the data we saw in this section as providing justification for setting aside this potential confound for the purposes of analyzing the double subject construction, as I've done in this chapter.

Appendix B: Homogeneity and the obligatoriness of *

In section 2.5.1, I used the double subject construction to provide additional evidence for Bar-Lev's (2025) conclusion that pluralization operators are the only sources of homogeneity. This conclusion raises the question of why the homogeneity gap with standard plural predication can't be removed by simply not inserting any pluralization operator. For example, consider the example in (124): out of the blue, (124) seems to be only true if none of the students danced. Now, if there was an LF for (124) where * simply did not apply below the subject, as in (125-a), we would predict that (124) has a reading where it is true as long as the students didn't all dance together.

- (124) It's not the case that the students danced.
→ None of the students danced.



- b. $\llbracket (125\text{-a}) \rrbracket = 1$ iff $\llbracket \text{danced} \rrbracket (\llbracket \text{the students} \rrbracket) = 0$ iff the students didn't dance together.

In order to avoid generating this type of reading, Bar-Lev (2025), following Schwarzschild (1994), assumes that pluralization operators obligatorily apply at every position of the appropriate type. This predicts that a predicate can't apply to a plurality without * first applying to the predicate, thus ruling out the LF corresponding to the unattested reading in (125-a).

- (126) **Obligatoriness of *:** Pluralization operators must be inserted as sisters to every predicate.

In this chapter, I have argued that we need to restrict the distribution of pluralization operators such that they have to apply only to predicates derived by movement. If this is correct, then we can no longer maintain that * obligatorily applies to every predicate, so the strong constraint in (126) has to be abandoned.

One solution to this issue is to maintain that * applies obligatorily below the subject in *Jane and Mary danced* by assuming that *Jane and Mary* has to move and that * obligatorily applies to every predicate derived by movement. This updated constraint is given in (127).

- (127) **Obligatoriness of * (updated):** Pluralization operators have to be inserted as sisters to every predicate derived by movement.

The updated constraint in (127) predicts that we get a homogeneity gap over any DP that moves. We can therefore predict the desired homogeneity gap for (124), only if we assume that the subject has to obligatorily move from its base position to spec,TP. In particular, given a simple sentence with a 1-place predicate like (128-a), we can assume that the subject has to move from spec,νP to spec,TP, therefore requiring * to apply to the predicate derived by movement below the higher subject. This allows us to predict the desired falsity conditions in (128-c), predicting that the negated counterpart of (128-a) is only true if none of the students danced.

- (128) a. The students danced.

- b.
-
- ```

graph TD
 TP --> the_students[the students]
 TP --> vP
 vP --> 1
 vP --> danced
 1 --> t1
 1 --> danced
 style TP fill:none,stroke:none
 style the_students fill:none,stroke:none
 style vP fill:none,stroke:none
 style 1 fill:none,stroke:none
 style danced fill:none,stroke:none
 style t1 fill:none,stroke:none

```
- c.  $\llbracket (128-b) \rrbracket = 0$  iff  $\neg \exists y : y \leq \llbracket \text{the students} \rrbracket \wedge \llbracket \text{danced} \rrbracket (y) = 1$   
iff None of the students danced.

While this seems to be a reasonable solution to (128-a), where we can assume that movement of the subject from spec,  $\nu P$  to spec, TP is obligatory, there is a question of how to extend this to other cases of homogeneity, for example with respect to objects. Again, if the object is able to stay in its base position in (129), we predict that there should be an LF where \* doesn't apply below the object and where (129) simply has collective truth-conditions, where it is true as long as Jane didn't lift the books together. In order to predict the desired homogeneity gap for (129), we therefore have to assume that the object obligatorily moves from its base position in (129). This raises some difficult questions regarding what triggers this obligatory movement and why the conjoined DP in the double subject construction is exempt from this obligatory movement that we seem to have to assume for all other plural DPs.

- (129) Jane didn't lift the books.  
 $\rightarrow$  Jane lifted none of the books.

The second possible solution is to maintain that there are in fact LFs for sentences like (130-a) and (130-b) where no pluralization operator applies below the plural DP. This predicts that there is a distinct collective reading for (130-a) and (130-b) and that when the LF without \* is negated, there will be no homogeneity.

- (130) a. Jane and Mary didn't dance.  
b. Jane and Mary didn't lift the piano.

On the face of it, this analysis goes against the observation that these sentences, when uttered out of the blue, give rise to a homogeneity gap. Nevertheless, we saw in section 2.5.1 that in certain contexts, these sentences can have a distinct collective reading that does not give rise to homogeneity (Schwarzschild, 1994; Bar-Lev, 2025). To see this with another example, consider the context in (131): here, (131-a) can be true and conveys that Jane and Mary didn't



dance together. In particular, the homogeneity gap here would entail that (131-a) is only true if neither Jane nor Mary danced. The context in (131) therefore contradicts homogeneity, since it is common ground that Jane danced with Bill, yet the sentence in (131-a) is felicitous.

(131) **Context:** Everyone knows that Jane danced with Bill, but I'm wondering if she also danced with Mary. Someone says:

- a. Jane and Mary didn't dance.
- b. # Jane didn't dance and Mary didn't dance.

Similarly, the example (132) shows that in the right context, *Jane and Mary danced* can have a purely collective reading when embedded in the antecedent of conditionals.<sup>16</sup> In particular, in the context in (132), the conditional in (132-a) can only be true if the antecedent is interpreted collectively. The underspecified reading, where the antecedent is true regardless of whether Jane and Mary danced together or separately (132-b), makes (132-a) false, since John will only be happy if they dance together.

(132) **Scenario:** John will be very sad if Jane dances alone and Mary dances alone, but he will be very happy if they dance together.

- a. If Jane and Mary dance, John will be very happy. **True**
- b. If Jane dances and Mary dances, John will be very happy. **Not True**

We therefore saw evidence that in certain contexts, we can get a distinct collective reading that is predicted if \* does not apply below the plural DP. Schwarzschild and Bar-Lev attribute this reading to a collective cover that can be thought of as restricting what the pluralization operator can sum up. Schwarzschild argues that this collective cover requires a specialized context which makes salient that what is relevant is whether or not the predicate is true collectively of the plurality (i.e. whether or not *Jane and Mary danced together*). We can therefore simply assume here that the collective reading we get in (132-a) and in (131-a) corresponds to the LF where the plural DP doesn't move from its base position and where \* therefore can't be inserted below it. Of course, this leaves the question of why the collective reading in (132-a) seems to require a supporting context, something which I don't have an answer to at this point.

We therefore see that the availability of an LF without movement where the predicate applies directly to the plurality without an intervening \* correctly predicts that there is a distinct collective reading for sentences like *Jane and Mary danced*. Before concluding this discussion, I want to point out that the pattern we saw with *dance* does not generalize to all predicates that seem

<sup>16</sup>The antecedent of conditionals is an environment from which homogeneity does not appear to project (Spector, 2013b; Križ, 2015), thus allowing us to examine the different readings of (128-a) without the possible confound of homogeneity.

to allow for a collective reading in the double subject construction. For example, consider the predicate *eat* in the double subject construction in (133): (133) is true only in a collective scenario where Karim and Rasha ate together. Assuming that the predicate applies directly to the plurality in (133), as I have argued in this chapter, we can conclude from (133) that the unpluralized predicate *eat* has the truth-conditions in (134) when applied to a plurality like  $j \oplus m$ .

- (133) Karim ?akal      howwe w    Rasha.  
 Karim ate.3SG.M him      and Rasha.  
 Karim and Rasha ate together.

- (134)  $\llbracket \text{ate} \rrbracket(j \oplus m) = 1$  iff Jane and Mary ate together

We therefore predict that just like with *dance*, *Jane and Mary ate* should have a distinct LF that gives rise to the collective reading in (134). Unlike with *dance*, this prediction is not borne out here. Starting with homogeneity, we see that in a parallel context to (131), where we saw that homogeneity can be suspended with *dance*, *eat* nevertheless gives rise to a homogeneity inference. In particular, (135-a) is not true in a scenario where Jane ate but not with Mary. Similarly, the example in (136) shows that *Jane and Mary ate* lacks a distinct collective reading when embedded in the antecedent of a conditional, unlike *Jane and Mary danced* in (132). In particular, (136-a) is not true in the scenario in (136), while the counterpart where the collective reading is picked out with *together* is true (136-b).

- (135) **Context:** Everyone knows that Jane ate with Bill, but I'm wondering if she also ate with Mary. Someone says:
- a. # Jane and Mary didn't eat.
  - b. # Jane didn't eat and Mary didn't eat.
- (136) **Scenario:** John will be very sad if Jane eats alone and Mary eats alone, but he will be very happy if they eat together.
- a. If Jane and Mary ate, John will be very happy. **Not True**
  - b. If Jane and Mary ate together, John will be very happy. **True**

This very interesting difference between *eat* and *dance* is puzzling for both the analysis of the collective reading of *dance* as being due to a distinct LF and for Schwarzschild's and Bar-Lev's analyses of the collective reading as being due to the collective cover. In particular, there is no reason to expect either the availability of a collective cover or whether or not \* has to apply below the plural DP to depend on the choice of predicate. I leave it for future work to figure out what this difference might be between predicates like *dance* and predicates like *eat*.

## Appendix C: Cumulative inferences from rules of composition

This appendix explores an alternative theory of the relationship between cumulative inferences and movement, which attributes cumulative inferences to a rule of composition, predicate abstraction, rather than to syntactic operators. This idea that cumulative inferences are due to composition rules has been argued for by [Schmitt \(2019\)](#), who builds cumulative inferences into the rule of function application, thus predicting that whenever a predicate applies to a plurality it generates a cumulative inference. While I won't be able to do justice to Schmitt's arguments for her system of cumulative composition, I want to suggest that the relationship between movement and cumulative inferences I argued for here can be accounted for if we assume that cumulative inferences arise only from the rule of predicate abstraction. This will allow us to predict that only predicates derived by movement give rise to cumulative inferences.

To see this, consider first the predicate abstraction rule from [Heim and Kratzer \(1998\)](#) in (137). Now, restricting ourselves first only to abstraction that creates a predicate of type  $\langle e, t \rangle$ , we can replace the standard predicate abstraction rule with the one in (138), which applies the same modification that the syntactic operator  $*$  does to the predicate derived by the standard abstraction rule in (137). The abstraction rule in (138) now necessarily produces a predicate that will generate the desired cumulative inference.

(137) **Predicate Abstraction Rule:**

Let  $\alpha$  be a branching node with daughters  $\beta$  and  $\gamma$ , where  $\beta$  is an index  $i$ . Then, for any variable assignment  $g$ ,  $\llbracket \alpha \rrbracket^g = \lambda x. \llbracket \gamma \rrbracket^{g^{x/i}}$

(138) **Cumulative Predicate Abstraction (type  $\langle e, t \rangle$ ):** Let  $\alpha$  be a branching node with daughters  $\beta$  and  $\gamma$ , where  $\gamma$  is of type  $t$  and  $\beta$  is an index  $i$  that binds a trace of type  $e$ . Then, for any variable assignment  $g$ ,  $\llbracket \alpha \rrbracket^g = *(\lambda x. \llbracket \gamma \rrbracket^{g^{x/i}})$ ,  
where  $*(P) = \lambda x. \exists Q_{\langle e, t \rangle} [\forall y [Q(y) = 1 \rightarrow P(y) = 1] \wedge x = \bigoplus Q]$

The cumulative abstraction rule therefore allows us to capture the relationship between movement and cumulative inferences with respect to 1-place predicates as follows. When a plural DP moves, it triggers predicate abstraction, creating a cumulative predicate. On the other hand, since there is no lexical cumulativity and no syntactic pluralization operators, when lexical predicates apply to plural DPs by standard function application, no cumulative inference is predicted.

There is a question of how to generalize this rule to cumulative inferences over predicates that take multiple arguments. For example, we saw that cumulative inferences over 2-place predicates, exemplified in (139), require the two-place pluralization operator  $**$  to apply to

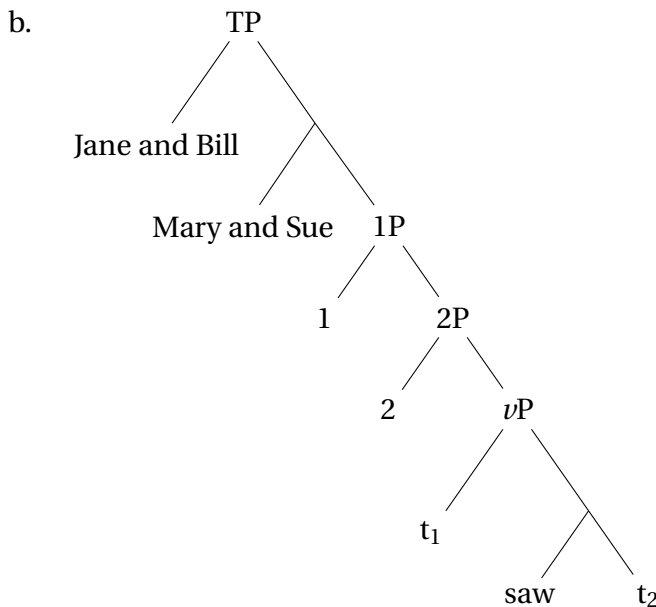
the two-place predicate, closing its extension under sum formation with respect to both of its arguments. The predicate abstraction rule can be generalized such that regardless of the type of the index's sister and the type of the trace that the index binds, the resulting predicate is closed under sum formation. While I won't offer a general statement of this rule here, I will show what this will look like for predicate abstraction that creates a two-place predicate of type  $\langle e, \langle e, t \rangle \rangle$ . The predicate abstraction rule corresponding to two-place cumulativity is given in (140).

(139) Jane saw Mary  $\wedge$  Bill saw Sue  $\rightarrow$  Jane and Bill saw Mary and Sue.

(140) **Cumulative Predicate Abstraction (type  $\langle e, \langle e, t \rangle \rangle$ ):** Let  $\alpha$  be a branching node with daughters  $\beta$  and  $\gamma$ , where  $\gamma$  is of type  $\langle e, t \rangle$  and  $\beta$  is an index  $i$  that binds a trace of type  $e$ . Then, for any variable assignment  $g$ ,  $\llbracket \alpha \rrbracket^g = **(\lambda x. \llbracket \gamma \rrbracket^{g^{x/i}})$ , where  $**(\mathbf{P}) = \lambda x. \lambda y. \exists Q [[\forall x', y' [Q(x')(y') = 1 \rightarrow \mathbf{P}(x')(y') = 1] \wedge x = \oplus \lambda x. \exists y [Q(x)(y)] \wedge y = \oplus \lambda y. \exists x [Q(x)(y)]]$

Now, we can generate the desired cumulative inference in (139) as follows. Consider the LF in (141-b) where both the subject and the object move, creating a derived 2-place predicate. Now the first step of predicate abstraction will give rise to a 1-place predicate whose argument is closed under sum formation (141-c) and the second step of predicate abstraction, corresponding to the index 1, will result in a 2-place cumulative predicate (141-c). This 2-place cumulative predicate will allow us to derive the desired cumulative inference in (139): as shown in the derivation in (141-f), (141-b) is predicted to be true whenever *Mary saw Jane* and *Sue saw Bill* are both true.

(141) a. Jane and Bill saw Mary and Sue.



- c.  $\llbracket 2P \rrbracket^g = *(\lambda x. g(1) \text{ saw } x)$
- d.  $\llbracket 1P \rrbracket = **(\lambda y. *(\lambda x. y \text{ saw } x))$
- e.  $\llbracket TP \rrbracket = 1$  iff  $**(\lambda y. *(\lambda x. y \text{ saw } x))(m \oplus s)(j \oplus b) = 1$
- f. Since  $**$  closes the extension of the two-place predicate under sum formation:  
 $(\lambda y. *(\lambda x. y \text{ saw } x))(m)(j) = 1 \wedge (\lambda y. *(\lambda x. y \text{ saw } x))(s)(b) \rightarrow \llbracket TP \rrbracket = 1$   
 Applying the predicates in each conjunct to their first argument:  
 $*(\lambda x. m \text{ saw } x)(j) = 1 \wedge *(\lambda x. s \text{ saw } x)(b) = 1 \rightarrow \llbracket TP \rrbracket = 1$   
 Since  $*$  is vacuous when the resulting predicate applies to an atomic individual:  
 $m \text{ saw } j \wedge s \text{ saw } b \rightarrow \llbracket TP \rrbracket = 1$

We therefore see that building cumulativity into the predicate abstraction rules allows us to derive the desired cumulative inferences, while straightforwardly predicting that cumulative inferences arise only over predicates derived by movement. Assuming that there is no optionality when applying rules of composition (i.e. that cumulative predicate abstraction replaces the standard predicate abstraction rule), this analysis would commit us to cumulative inferences being obligatory whenever movement occurs. As we saw in section 2.5 when discussing homogeneity, this seems to be a desirable result, but further work is needed to assess whether there is any evidence to distinguish the view presented in this subsection from one where pluralization operators are syntactically represented and apply only to predicates derived by movement.



# Chapter 3

## Distributivity facilitates ACD resolution

This chapter is based on joint work with Kate Kinnaird and Martin Hackl, published in the proceedings of the Amsterdam Colloquium ([Wehbe, Kinnaird, & Hackl, 2024](#)).

### 3.1 Introduction

In chapter 2, I argued that the availability of cumulative inferences is parasitic on movement in the Lebanese Arabic double subject construction. I showed that this relationship can be made sense of only if (i) predicates themselves are not inherently cumulative and (ii) the pluralization operators responsible for cumulative inferences are restricted to applying only to predicates derived by movement. This raises the question of how general the relationship between movement and cumulative inferences is. In particular, are the ban on lexical cumulativity and the restriction on the distribution of pluralization operators specific to Lebanese Arabic or do they constitute universal properties of natural language? The goal of this chapter is to begin to answer this question by finding an independent way to probe the relationship between movement and cumulative inferences beyond the DSC.

More specifically, the restriction on pluralization operators proposed in chapter 2 predicts that a definite plural DP like *the two students* has to move from its base position in order for a sentence like (1-a) to be true in the given distributive scenario. In this chapter, I will present joint work with Martin Hackl and Kate Kinnaird which shows that definite plurals under distributive construals show processing signatures of syntactic movement, thus providing additional evidence for the relationship between movement and the availability of cumulative inferences.

- (1) **Distributive Scenario:** Jane talked to Student 1 on Monday and to Student 2 on Friday.
- a. Jane talked to the two students.

To probe the relationship between movement and distributivity, we rely on a diagnostic from [Hackl et al. \(2012\)](#), who show that antecedent-contained deletion (ACD) provides a way to diagnose whether a given object DP has to move. To see the logic of this diagnostic, let's observe first that sentences involving ACD, such as (2-a), are harder to parse than counterparts with no ellipsis (2-b), as shown experimentally by [Hackl et al. \(2012\)](#). Now, prominent analyses of ACD require the object DP, *the politician that Mary did* in (2-a), to move from its base position in order to license the ellipsis ([Sag, 1976](#), a.o.). Hackl et al. argue that it is this requirement for movement that makes ACD in (2-a) difficult to parse: simplifying a bit for the moment, the only cue for movement in (2-a) is the ellipsis site at the end of the sentence, thus making it difficult to identify the desired movement structure that licenses ACD.

- (2) a. Jane talked to the politician that Mary did.
- b. Jane talked to the politician that Mary invited.

Now, Hackl et al. hypothesize that if there is a second cue, independent of the presence of ACD, that the object needs to move, it should become easier to license ACD. It has been argued that when quantifiers like *every* are in object position, they have to move covertly in order to avoid a type mismatch. If this is true, Hackl et al. reason, then we should expect that when the object hosting the ACD site is headed by *every*, it will be easier to license ACD, since the object will need to move regardless of whether we have ellipsis. Hackl et al. show that ACD resolution is in fact facilitated when the object is headed by *every*. In particular, they show that ACD in (3), where *the* is replaced by *every* is easier to parse than the counterpart with a definite singular object in (2-a). They argue that this is because *every* provides a cue in its own right, independent of ACD, that the object in (3) needs to move.

- (3) Jane talked to every politician that Mary did.

The paradigm described above provides a diagnostic for whether a given object DP has to move. We therefore use this paradigm to test whether distributive readings over plural DPs require movement. In particular, if the conclusion above that distributivity requires movement is correct, then we expect that definite plural DPs under distributive construals facilitate ACD in the same way that *every* does in (3). For example, we expect that in a distributive context like (4), where Jane could only talk to the politicians one-at-a-time, the definite plural object in (4-b) will require syntactic movement in its own right, independent of the presence of ACD. Therefore, just like *every* in (3), the definite plural object in (4-b) should provide an independent cue for movement, thus resulting in a facilitation effect on ACD resolution, relative to the definite singular baseline in (4-a).



- (4) **Scenario:** A different politician was visiting the school which Jane and Mary attend every day this week.
- a. Jane talked to the politician that Mary did.
  - b. Jane talked to the politicians that Mary did.

In short, we find that definite plurals in distributive contexts in fact facilitate ACD resolution, providing an additional argument for the conclusion from the Lebanese Arabic double subject construction that there is a relationship between movement and the availability of cumulative inferences.

The rest of the chapter is structured as follows. Section 3.2 describes the logic of using ACD resolution to probe for syntactic movement in more detail. Section 3.3 presents our experimental design and results. Section 3.4 discusses the findings and the implications for our theory of the sources of cumulative inferences. Finally, section 3.5 concludes.

## 3.2 Probing for the correct LF via ACD resolution

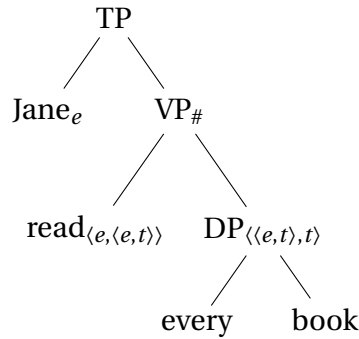
The present study follows the approach of Hackl et al. (2012) to look for a processing signature of covert DP-movement. The basic idea in Hackl et al. is to use the relative ease with which comprehenders are able to resolve antecedent contained deletion (ACD) as a proxy for how and when they can determine the correct LF-position of the DP hosting the ACD site. To appreciate their logic, consider the factorial template in (5) in which the presence or absence of ACD is crossed with two types of determiners on the hosting DP - the quantificational determiner *every* or the non-quantificational determiner *the*.

- (5) Jane read {every<sub>+quant.</sub>, the<sub>-quant.</sub>} book that Mary { did<sub>+ACD</sub>, wrote<sub>-ACD</sub>}.

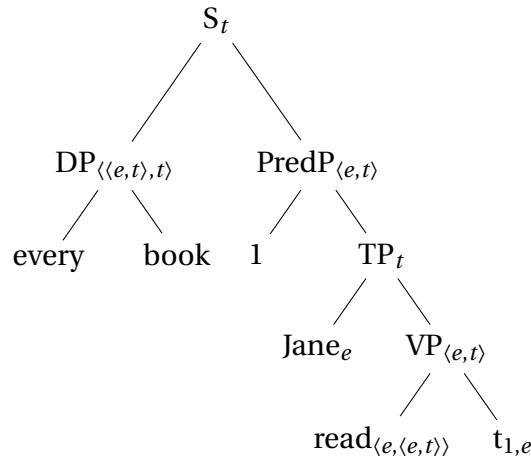
Canonical analyses of ACD predict that for a sentence to exhibit ACD, it has to have a structural description (LF) where the elided VP is not included inside the matrix VP, which serves as its antecedent (Sag, 1976). This can be achieved if the object DP hosting the ACD site is moved to a position higher than the matrix VP, thereby "undoing" antecedent containment (Sag, 1976; Williams, 1977; see Jacobson, 1992 for an alternative). Importantly, while both quantificational and non-quantificational DPs need to be moved when they host an ACD site, Hackl et al. assume that only the former require it on their own, irrespective of whether or not they host an ACD site. In particular, if we adopt the standard assumption that quantificational DPs denote predicates of type  $\langle\langle e, t \rangle, t\rangle$  (Barwise & Cooper, 1981), then an LF for a sentence like (6-a) where the object stays in its base position is predicted to give rise to a type mismatch (6-c). In order to have an interpretable LF for (6-a) under these assumptions, the object therefore needs to move from its

base position, leaving behind a trace of type  $e$  (6-d) (Heim & Kratzer, 1998).

- (6) a. Jane read every book.  
 b.  $\llbracket \text{every book} \rrbracket = \lambda P_{\langle e, t \rangle}. \forall x : \llbracket \text{book} \rrbracket (x) = 1 \rightarrow P(x) = 1$   
 c. **Uninterpretable LF:**



- d. **Interpretable LF:**



We can therefore conclude that *every* in object position has to move in order to give rise to an interpretable LF, independent of whether the DP headed by *every* hosts an ACD site. From a procedural perspective, this means that the [+quant, +ACD] variant of (5) contains two triggers for the parser to identify the correct LF involving movement of the object, namely that the object is headed by *every* and that the object hosts an ACD site. On the other hand, the [-quant, +ACD] variant contains only one, namely the ACD site itself, since the definite singular object does not by itself require movement.

Adopting a linking hypothesis that connects the number of cues for a particular LF with the ease with which that LF can be identified leads Hackl et al. to expect a facilitation effect for *every* on ACD resolution. More specifically, resolving ACD is a complex process, which involves (i) postulating an LF without antecedent containment (i.e. involving covert movement of the object DP), (ii) identifying the antecedent VP and (iii) retrieving it from memory when computing the meaning of the relative clause. The [+ACD] variants of (5) are therefore expected to be more costly

than their [-ACD] counterparts. It is also expected, however, that when the host DP is projected from *every* this cost is reduced and ACD resolution is facilitated since a critical component of resolving ACD - identifying an LF without containment - is triggered independently from and earlier than the ACD site. By contrast, projecting the host DP from *the* does not provide a separate trigger for moving the object DP above the matrix VP. Thus, in this condition, the cue for such an LF is provided by the ACD site alone, at a stage when a parse without movement is likely entrenched and so relatively costly to abandon.

Combining these two types of considerations leads to the prediction of a particular type of interaction such that the [+quant,+ACD] condition is relatively less costly than its [-quant,+ACD] counterpart. This facilitation effect has been documented, among other things and most relevantly for our purpose here, in naturalness ratings in Hackl et al. and it has been replicated in a series of subsequent work including Gibson et al. (2015), who offer an alternative account of the facilitation effect.<sup>1</sup>

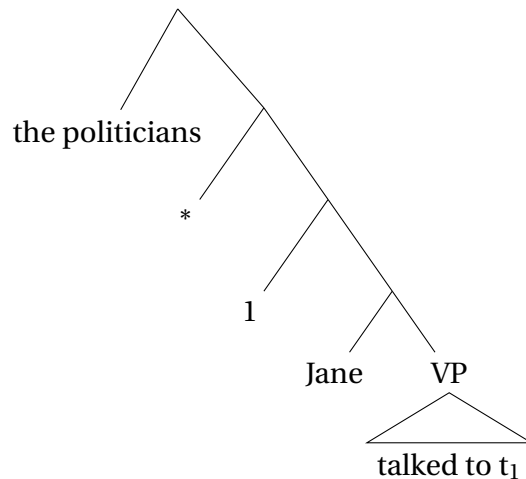
The present project extends this logic to distributive readings of plural definites. Concretely, if distributive readings require a syntactically represented pluralization operator that can only apply to a derived predicate, then a definite plural object DP, just like *every* in (5), triggers movement in order to get the desired reading. Thus, we expect a similar facilitation effect to Hackl et al.'s observation for *every* for distributively interpreted definite plurals.

More concretely, if the constraint on the distribution of pluralization operators proposed in chapter 2 is a universal constraint, we expect that for a sentence like (7-b) to be true in a distributive context like (7-a), the object *the politicians* has to move from its base position in order to allow for \* to apply below it. In particular, if the object stays in its base position, *the politicians* in (7-b) has to be interpreted collectively, just like the lower subject in the double subject construction. Under this collective reading, (7-b) would only be true if Jane talked to the politicians together, making (7-b) incompatible with the given context. Therefore, in the context in (7-a), a cooperative hearer will have to assign to (7-b) an LF where the object moves from its base position, allowing for the intended distributive reading.

- (7) a. **Context:** Each day this week, one politician visited the campus.  
 b. **Sentence:** Jane talked to the politicians.  
 c. **Predicted LF:**

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<sup>1</sup>The facilitation effect has also been documented in self-paced reading data in Hackl et al. and both types of observations have been replicated and extended to a variety of other expressions that are hypothesized to trigger upstream movement in their own right (Hackl, Koster-Moeller, & Gottstein, 2009; Breakstone, Cremers, Fox, & Hackl, 2011; Kotek, 2019).



The analysis proposed in chapter 2 therefore predicts that given a distributive context like (7-a), definite plurals, just like *every* in object position, provide an independent cue for an LF with movement. This predicts that, just like *every*, definite plural objects hosting an ACD site should lead to a facilitation effect on ACD resolution in distributive contexts.

### 3.3 Distributivity and ACD resolution: an experimental study

#### 3.3.1 Design and materials

We adopt the logic of Hackl et al. to probe whether definite plural DPs in object position undergo movement when they are interpreted distributively. We focus on a particular instance of a distributive reading where the different atomic parts of the relevant plural DP participated in the event in question at different times. We will refer to this as a *one-at-a-time* reading. Our target sentences, which parallel the format in (5), are paired with contexts that strongly encourage a *one-at-a-time* interpretation. An example is given in (8), where we crossed +/-ACD and the type of DP (definite plural vs. singular).

- (8) **Context:** 25 international students are invited for a campus visit at an Ivy league university, and they get the chance to speak with professors one-on-one.

**Sentence:** The English literature professor met with...

- the humanities **students**<sub>PL</sub> that the philosophy professor **did**<sub>+ACD</sub>
- the humanities **student**<sub>Sg</sub> that the philosophy professor **did**<sub>+ACD</sub>
- the humanities **students**<sub>PL</sub> that the philosophy professor **interviewed**<sub>.ACD</sub>
- the humanities **student**<sub>Sg</sub> that the philosophy professor **interviewed**<sub>.ACD</sub>  
... due to the dean's insistence

To establish a baseline for our design (seeking to replicate Hackl et al.), we also ran a variant of the study where the definite plural in (8) is replaced with *every* (e.g. *every humanities student*), keeping everything else constant. In this variant, +/-ACD is therefore crossed with whether the object DP is headed by *every* or *the*, as shown in (9).

(9) **Context:** 25 international students are invited for a campus visit at an Ivy league university, and they get the chance to speak with professors one-on-one.

**Sentence:** The English literature professor met with...

- a. **every** humanities student that the philosophy professor **did**<sub>+ACD</sub>
  - b. **the** humanities student that the philosophy professor **did**<sub>+ACD</sub>
  - c. **every** humanities student that the philosophy professor **interviewed**<sub>-ACD</sub>
  - d. **the** humanities student that the philosophy professor **interviewed**<sub>-ACD</sub>
- ... due to the dean's insistence

Items were presented in PCIBex (Zehr & Schwarz, 2023) using a counterbalanced Latin Square design, with each participant being assigned to one of four lists. Each of the baseline and target experiments consisted of 32 fillers and 32 target items. Item order was randomized for every participant. Filler items were of similar complexity and length to target items. As exemplified in (8) and (9), participants first saw a context sentence followed by a target sentence, and then were prompted to rate the target sentence's naturalness relative to the context on a 7-point Likert scale. Following their selection, the context sentence and target sentence were removed from the screen. Participants were then given a Yes/No comprehension question asking about information conveyed by the target sentence. To probe for whether participants are paying attention to the crucial distinction between plural and singular definite DPs, some of our comprehension questions were sensitive to this distinction (e.g. for (8): *Did the English literature professor meet with only one student?*).

Our hypothesis is that if the analysis in chapter 2 is correct and generalizes cross-linguistically, the cost of ACD with both *every* and definite plurals should be lower than with definite singulars. This predicts that in both the baseline and the target experiment, we should observe an interaction effect between DP type and +/-ACD, corresponding to a lower difference between the acceptability ratings of +ACD and -ACD in the definite plural and *every* conditions than in the definite singular condition.

### 3.3.2 Results

For each of the baseline and target experiments, we recruited 80 monolingual English speakers, after exclusions, using the crowd-sourcing platform Prolific. Participants were excluded if they

failed any of our three practice items or if they answered less than 8 out of our 12 attention check items correctly (36 for the baseline and 19 for the target experiment). Participants were paid approximately 5 USD.

Mean ratings for the baseline and target experiments, with 95 percent confidence interval-based error bars, are displayed in Figure 3.1. We report the maximally-specified linear mixed-effects model (conditioned on convergence), predicting naturalness ratings as a function of [+/-ACD], DP type (every vs. definite singular for the baseline experiment and definite plural vs. definite singular for the target experiment), and their interaction. We include random slopes of DP type and [+/-ACD] for participants and a random slope of [+/-ACD] for items. The model reveals a significant interaction between [+/-ACD] and DP type for both the baseline experiment and the target experiment ( $p < 0.0001$  and  $p < 0.05$ , respectively). Analyses were performed in R Statistical Software (v 4.2.2; R Core Team 2022) using the lme4 package (Bates, Mächler, Bolker, & Walker, 2015).

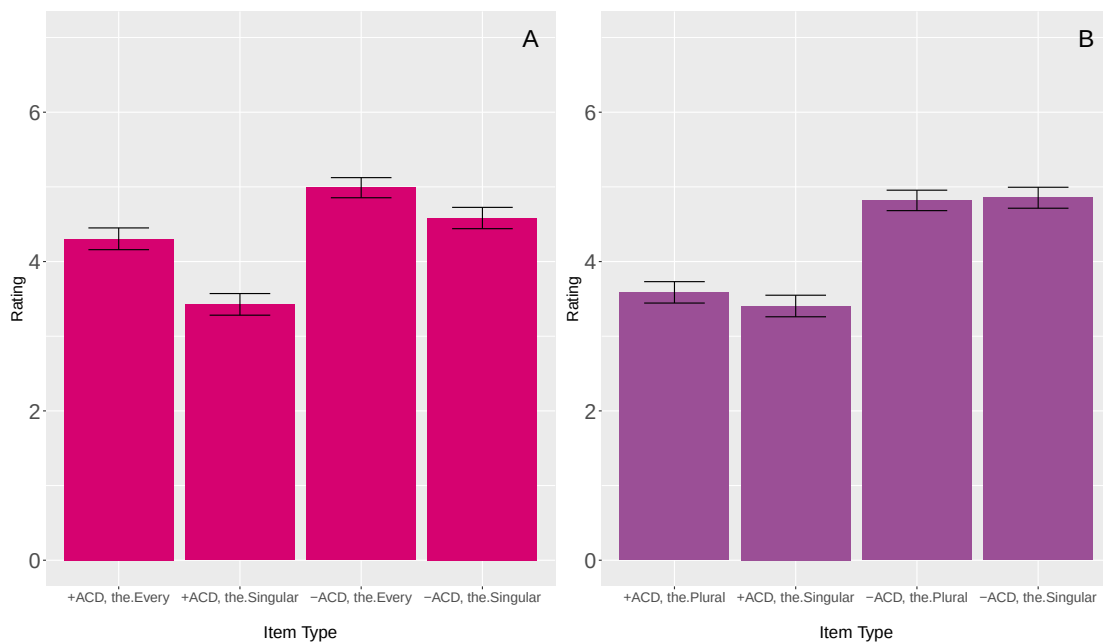


Figure 3.1: Mean naturalness ratings for Baseline Experiment in A (N=80) and Target Experiment in B (N=80)

### 3.4 Discussion

Our baseline experiment with *every* replicates the results in Hackl et al. In particular, we see that the presence of ACD results in a marked degradation of naturalness, but critically, we observe an interaction indicating that ACD resolution is facilitated with *every*, when compared to the

definite singular. Following Hackl et al., we take this to support the conclusion that *every* in object position has to move for reasons independent of ACD.

Turning to the target experiment, we see a parallel facilitation effect of the definite plural on ACD resolution, manifesting in a significant interaction between +/-ACD and DP type. Adopting, again, the reasoning in Hackl et al., we take this to show that the distributive reading over a plural object DP in our environment requires movement of that DP. Interestingly, as is evident in Figure 3.1, the size of the interaction effect with the definite plural is smaller than that with *every* (Model estimate= 0.2188 for the target experiment and 0.47188 for the baseline experiment). This difference in effect size could be due to several factors. One possibility is that the distinction between *the students* and *the student* is easier to miss for our participants than the distinction between *every student* and *the student*. This would result in more noise in the target experiment, thereby lowering the size of the effect. To explore this possibility, we excluded participants who failed to correctly answer more than one-third of comprehension questions which were sensitive to the singular/plural distinction. This resulted in the effect size going up to 0.26906 and the p-value going down to 0.0194 from 0.0457.<sup>2</sup> This is as expected on the view that part of the difference in effect size is due to participants missing the singular/plural distinction in the target experiment.

Another possible contributing factor to the difference in effect size is that the information needed to recognize the trigger for movement in the two experiments is different. In the case of *every*, it is simply recognizing that *every* occurs in object position. On the other hand, the relevant information in the case of the definite plural crucially includes the distributive context (assuming that definite plurals have to move only under the distributive construal). Facilitation is therefore only predicted for those participants who keep the context in mind and attempt to generate a matching distributive reading for the target sentence.

Setting aside the difference in effect size, we can conclude that definite plurals under distributive construals, just like *every* in object position, require movement independent of the presence of an ACD site. Specifically, definite plurals under our *one-at-a-time* contexts exhibit a facilitation effect on ACD resolution. We take this to show that the definite plural is perceived by comprehenders as a trigger for movement in our environment, allowing the parse where antecedent containment has been undone to be identified before the ACD site is reached. This follows from theories where pluralization operators are always syntactically represented and where movement of the object DP to a position above the relevant operator is required to generate the desired distributive reading, as argued in the analysis of the Lebanese Arabic double subject construction in chapter 2.

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<sup>2</sup>Excluding participants at 25% and 50% thresholds resulted in similar changes in effect size and p-value, although simpler models were required for convergence.

We therefore have evidence that both the absence of lexical cumulativity and the constraint that pluralization operators apply only to derived predicates are universal and apply beyond the Lebanese Arabic double subject construction.

### 3.5 Conclusion

In this chapter, we provided evidence for the generalization argued for in chapter 2 that cumulative inferences over a plural DP are only available if that DP moves from its base position. In particular, we showed that definite plurals in object position, when interpreted distributively, show processing signatures of syntactic movement, therefore providing evidence for the connection between movement and the availability of distributive readings. These results raise several interesting questions, which I will mention below.

First, our results raise interesting questions in the context of previous processing studies of distributivity. For example, [Dotlačil and Brasoveanu \(2021\)](#) argue for a distinction between lexical and phrasal distributivity based on differences in how easy the corresponding readings are to access. In particular, they show that the distributive reading over an indefinite exemplified in (10-a) is dis-preferred relative to the collective reading, while there is no parallel difficulty for the distributive reading in (10-b). [Dotlačil and Brasoveanu](#) take this as evidence that these two kinds of distributive readings are due to distinct sources: the reading in (10-a) requires a distributivity operator to apply to a derived predicate that includes the indefinite, while the distributive reading in (10-b) follows from lexical cumulativity.

- (10)    a.    The students read a book./ Dist. reading: The students each read a book.  
         b.    The students lifted the piano./Dist. reading: The students each lifted the piano.

The arguments from both this chapter and chapter 2 against lexical cumulativity raise a problem for explaining the difference between (10-a) and (10-b) in terms of a distinction between phrasal and lexical cumulativity. In particular, the account I've been pursuing commits us to analyzing both (10-b) and (10-a) in terms of a syntactically present pluralization operator, requiring an alternative analysis of the contrast in (10). Interestingly [Bar-Lev \(2025\)](#) recently proposed an alternative explanation for the difficulty of accessing the distributive reading in (10-a). He argues that (10-a) gives rise to an implicature that the students between them didn't read more than one book, which therefore does not allow for (10-a) to be true in a scenario where each student read a different book. To make (10-a) true in such a scenario, the implicature therefore has to be canceled, explaining the difficulty of accessing the desired distributive reading in (10-a).

Another question that the present study leaves open is how crucial the distributive context is in generating the observed facilitation effect for definite plurals in object position. In particular,



one could in principle imagine two explanations for our results: (i) definite plurals in object position always have to move, regardless of what readings they give rise to and (ii) definite plurals only have to move to be true in distributive scenarios. Given that all of our items were paired with distributive scenarios, it is not possible to tell these two explanations apart at this point. We therefore plan to investigate in future work whether definite plurals in collective scenarios give rise to the same facilitation effect as definite plurals in distributive scenarios or whether they in fact behave like definite singulars, as predicted if movement is only necessary for distributivity.

Looking beyond the specific case-study investigated in this chapter, our study lays the foundation for a new tool to compare frameworks where plural predication is mediated by syntactic operators (Beck & Sauerland, 2000, a.o.), in which movement plays a crucial role, to frameworks where we don't expect to see systematic syntactic correlates of plural predication (Winter, 2000; Schmitt, 2019, a.o.). Looking ahead, we hope to use this new tool to investigate the role of movement in generating different readings in plural predication. For example, non-local cumulative readings have been argued to require movement of the relevant DPs to a derived scope position (Beck & Sauerland, 2000), but certain accounts, including the more recent plural projection framework (Schmitt, 2019) are able to generate these non-local readings without movement. Our tool can help to distinguish the predictions of these two theories.



# Chapter 4

## A scope-based theory of *together*

### 4.1 Introduction

At this point, I have provided two independent arguments that cumulative inferences have to be generated by pluralization operators and that lexical predicates give rise to collective truth-conditions when they apply to pluralities. This raises the question of how a variety of other readings, beyond the collective and the distributive ones discussed in previous chapters, are generated. To address this question, I focus in this chapter on how the different readings of sentences with *together* are generated. Namely, *together* appears to disambiguate by blocking distributive construals: *Jane and Mary lifted the piano* is true both in the collective scenario in (1-a) and the distributive one in (1-b), while the counterpart with *together* is only true in (1-a).

- (1) a. **Scenario 1:** Jane and Mary jointly lifted the piano.
- b. **Scenario 2:** Each of Jane and Mary lifted the piano separately.

In addition to this collective reading in (2-a), *together* has been argued to have distinct simultaneous (2-b), proximate (2-c) and cumulative (2-d) readings (Lasersohn, 1990; Schwarzschild, 1994; Moltmann, 2004, a.o.). I argue in this chapter that *together* does not lexically encode these different readings (contra Lasersohn, 1990). Instead, I propose that *together* essentially introduces a coordinated DP that forces pluralities to be interpreted in different scope positions, resulting in the different readings we see in (2). By blocking distributive readings, *together* therefore reveals how plural DPs interact scopally with pluralization operators and other scope-taking elements to generate different non-distributive readings.

- (2) a. **Collective *together*:** Jane and Mary lifted the piano together.
- b. **Simultaneous *together*:** Jane and Mary stood up together.  $\approx$  Jane and Mary stood up at the same time.

- c. **Proximate *together*:** John and Bill's corpses lay together.  $\approx$  John and Bill's corpses lay next to each other.
- d. **Cumulative *together*:** Jane and Mary together ate exactly 10 cookies.  $\approx$  The sum of the number of cookies that Jane ate and that Mary ate is 10.

More specifically, I propose an analysis of *together* that takes seriously the fact that in several languages, such as Lebanese Arabic and Farsi, *together* is overtly decomposed into the comitative *with* and a reciprocal anaphor. For example, in Lebanese Arabic, the preposition *maʕ* is the counterpart of English comitative *with* (3-a) and *baʕed* is a reciprocal anaphor (3-b). The Lebanese Arabic counterpart of *together*, *maʕ baʕed* is therefore overtly composed from the comitative *maʕ* and the reciprocal anaphor *baʕed*. Crucially, *maʕ baʕed* gives rise to the full range of readings we saw in (2) for English *together*, as shown in (4).

(3) a. **Comitative in LA:**

Karim ʔakal maʕ Hadi.  
 Karim ate.3SG with Hadi.  
 Karim ate with Hadi.

b. **Reciprocals in LA:**

Karim w Hadi she:fo baʕed.  
 Karim and Hadi saw each-other.  
 Karim and Hadi saw each other.

(4) **Together in LA:**

a. **Collective together:**

Karim w Hadi ʔakalo maʕ baʕed.  
 Karim and Hadi ate.3PL with each-other.  
 Karim and Hadi ate together.

b. **Simultaneous together:**

Karim w Hadi weʔfo maʕ baʕed.  
 Karim and Hadi stood-up.3PL with each-other.  
 Karim and Hadi stood up at the same time.

c. **Proximate together:**

l jitte:n milʔiyyin maʕ baʕed.  
 the corpses.DU placed.PRT.3PL with each-other.  
 The two corpses lay together.

d. **Cumulative together:**

Karim w Hadi maʕ baʕed ʔakalo ʕasher tiffihat.  
 Karim and Hadi with each-other ate.3PL 10 apples.

Karim and Hadi together ate 10 apples.

Existing analyses of *together* either focus only on a subset of these readings (Schwarzschild, 1994), resort to lexical ambiguity in order to derive the different readings (Lasnik, 1990), or argue against a genuine four-way ambiguity (Moltmann, 2004). Furthermore, no existing analysis of *together* is able to offer an appropriate compositional semantics that transparently explains the decomposition of *together* into *with* and a reciprocal anaphor. This chapter therefore sets out to propose such an analysis.

I show that if we assume Ionin and Matushansky's (2002) analysis of comitatives as conjunctions and a compositional analysis of reciprocals (Heim et al., 1991), the analysis of *together* as *with each other* predicts the different readings in (2) as a result of *together* taking scope in different positions. On this view, *together* essentially introduces an *x with the other* DP in spec, *vP*. When this DP remains in its base position, we get a collective reading, and when it moves covertly to different positions, it generates the various other readings in (2). This proposal therefore crucially relies on the conclusion from chapters 2 and 3 that lexical predicates are not inherently cumulative, allowing for collective readings to arise by simply having the predicate apply to a plurality below all pluralization operators.

The scope-based analysis by itself does not explain why *together* disambiguates against distributive readings. Since *together* simply introduces a comitative conjunction, the proposal initially predicts that sentences with *together* should give rise to the same range of readings as sentences with standard conjunction, including unattested distributive readings. I propose that this disambiguation, rather than coming from the lexical meaning of *together*, is the result of an independently motivated economy constraint that rules out LFs where the contribution of *together* is trivial. I show that this constraint correctly predicts a range of restrictions on what readings *together* can give rise to. The resulting account therefore preserves a unified compositional semantics for *together*, while still deriving the fact that *together* blocks certain readings that are available when it is not inserted.

In this chapter, I provide a preliminary version of the analysis, where I assume that distributive readings are due to a syntactically present distributivity operator, DIST. This simplified version makes it possible to present the core results of the proposal but faces several issues related to the economy constraint, which I will focus on in more detail in chapter 5. More specifically, chapter 5 proposes a generalized economy constraint that rules out LFs which are truth-conditionally equivalent to simpler LFs. I argue that in order to predict the desired distribution of readings with *together*, this economy constraint has to be modular, such that it is evaluated relative to an encapsulated deductive system (following Fox, 2000). Moreover, I show that once we replace DIST with \*, this constraint makes more fine-grained predictions regarding the distribution of

readings with *together*. Before turning to the details of the proposal, I will start by giving an informal overview in section 4.2.

## 4.2 Informal overview

### 4.2.1 *together* as *with each other*

The goal of this section is to offer a preliminary simplified version of the proposal and to highlight its main components. As mentioned in the introduction, I propose that *together* is the spell-out of *with each other*.

I assume, following Ionin and Matushansky (2002) that *with* is semantically equivalent to *and*: it takes two individuals as arguments and returns their sum. Furthermore, I adopt two assumptions argued for in chapters 2 and 3: (i) lexical predicates give rise to collective readings when they apply to pluralities and (ii) covert operators that generate distributive readings can apply only to predicates derived by movement.

Given these assumptions, the difference in meaning between comitatives and standard conjunction can simply be attributed to the scope of the coordinated DP. With standard conjunction, exemplified in (5), the conjoined DP moves from spec, $\nu$ P to spec,TP. This allows for the insertion of a covert operator, which we can think of for now as a covert *each*, below the conjoined DP, giving rise to a distributive reading. On the other hand, with English comitatives, only the first conjunct moves to spec,TP. Therefore, since the covert *each* has to apply to a predicate derived by movement, it can't apply below the conjoined DP, predicting that the comitative only has the collective reading illustrated in (6).

- (5) Jane and Mary danced.
- a. **Distributive LF paraphrase:** Jane and Mary are each such that she danced.
  - b. **Predicted truth-conditions:** True iff Jane and Mary each danced.
- (6) Jane danced with Mary.
- a. **LF paraphrase:** Jane is such that her and Mary danced.
  - b. **Predicted truth-conditions:** True iff Jane and Mary danced together.

As for reciprocals, I follow Heim et al. (1991) and assume a compositional analysis where *each other* is decomposed into a distributivity operator *each* that modifies the predicate and a *the other* DP in object position. The resulting LF is informally paraphrased in (7-a) and the predicted truth-conditions are given in (7-b).

- (7) Jane and Mary saw each other.

- a. **LF paraphrase:** Jane and Mary each saw the other.
- b. **Predicted truth-conditions:** True iff Jane saw Mary and Mary saw Jane.

Putting these two ingredients together, I propose that *together* results in a comitative conjunction *x and the other*, where the first conjunct moves to the higher subject position. The paraphrase of the resulting LF is given in (8-a), where the covert *each* takes scope below the higher subject but the lower conjunction again remains in its base position. Again, this forces the predicate to apply directly to the conjoined DP below *each*, yielding the desired collective truth-conditions in (8-b).<sup>1</sup>

(8) Jane and Mary danced together.

- a. **LF paraphrase:** Jane and Mary are each such that her and the other danced.
- b. **Predicted truth-conditions:** True iff Jane and Mary danced together.  
(since *dance* applies to the conjunction of Jane and Mary below *each*, this gives rise to the lexically specified collective truth-conditions)

Turning to the other readings of *together*, I will show that they reveal how plural DPs give rise to a variety of interpretations when interpreted in different scope positions. For example, the simultaneous reading follows straightforwardly if we allow the comitative conjunction *x with the other* to take scope above the lowest distributivity operator and below perfective Aspect, which I assume denotes an existential quantifier over times.

To see how this simultaneous reading is generated, consider first the sentence without *together* in (9). I assume that in order to get a reading for (9) where Jane and Mary stood up at different times, there has to be a distributivity operator that takes scope above an existential quantifier over times. This is shown in the LF paraphrase in (9-a), where the covert *each* takes scope below the subject but above the existential quantifier. When *together* takes scope below this higher distributivity operator, the simultaneous reading is predicted. To see this consider the LF in (10-a): here the conjunction *x and the other* in bold takes scope below the existential quantifier over times while still allowing for a covert *each* below it. The resulting simultaneous truth-conditions are paraphrased in (10-b). We therefore see that when *together* is allowed to

<sup>1</sup>At this point, one might already wonder how this analysis can explain certain differences between *together* and *with each other* in English with respect to how they interact with collective predicates like *meet* and *agree*. This contrast is illustrated in (i). I will focus only on underspecified predicates like *lift the piano* when discussing collective *together* and set these predicates aside for now. I return to this challenge again in chapter 6 when I discuss the differences between *together* and the DSC, but I will have to largely leave this issue with collective predicates as an open question for future work.

- (i) a. Jane and Mary agreed with each other.
- b. \*Jane and Mary agreed together.

take scope in higher positions, it can give rise to readings beyond the lexically specified collective one.

(9) Jane and Mary stood up.

- a. **LF paraphrase:** Jane and Mary are each such that there is a time such that she stood up at that time.
- b. **Predicted truth-conditions:** True iff Jane stood up at some time and Mary stood up at some time.

(10) Jane and Mary stood up together.

- a. **LF paraphrase:** Jane and Mary are each such that there is a time such that **her and the other** each stood up at that time.
- b. **Predicted truth-conditions:** True iff there is a time such that Jane and Mary each stood up at that time.

Turning to the proximate reading, I propose that it similarly reveals how pluralization operators interact with scope-taking elements in the clause. More specifically, I propose that the proximate reading reveals that there is an existential quantifier over locations in the clausal structure, parallel to the existential quantifier over times denoted by *Aspect*. While this existential quantifier over locations, unlike *Aspect*, has not been independently argued for, I argue that it is crucially needed to account both for the proximate reading of *together* and for the differences between covert and overt reciprocals in chapter 7. This raises the question of whether we can find independent evidence for this quantifier beyond plural predication, which I will leave as an open issue for the purpose of this thesis.

I show that the proximate reading is generated when *together* takes scope below both *Aspect* and the existential quantifier over locations. The LF paraphrase for the proximate reading is given in (11-a), and the resulting truth-conditions are given in (11-b): (11) is correctly predicted to be true if there is a time and location such that the two corpses each lay at that time and location.

(11) The two corpses lay together.

- a. **LF paraphrase:** The two corpses are each such that there is a time and location such that them and the other corpse each lay at that time and at that location.
- b. **Predicted truth-conditions:** There is a time and location such that the two corpses each lay at that time and location.

Finally, I leave the more complicated details of the cumulative reading for section 4.4, but I provide a brief sketch here. I argue that *together* takes scope below a higher covert *each* but



above a lower cumulativity operator that generates the desired cumulative reading. In particular, I assume the cumulative reading for a sentence like (12) corresponds to an LF like (12-a), where the cumulativity operator applies to the predicate before it applies to its arguments. On the other hand, the distributive reading is again derived by a covert *each* applying below the higher subject. Turning to the contribution of *together*, I assume that the cumulative reading corresponds to an LF where the comitative conjunction takes scope below a higher covert *each* but above the cumulativity operator, as shown in the LF paraphrase in (13-a). This predicts the desired cumulative truth-conditions in (13-b), since the lower conjunction ensures that the cumulative predicate applies to the conjunction denoted by *Jane and Mary*.

(12) Jane and Mary ate 10 cookies.

- a. **Cumulative LF:** Jane and Mary are such that they Cumulative(ate) 10 cookies.
- b. **Distributive LF:** Jane and Mary are each such that she ate 10 cookies.

(13) Jane and Mary together ate 10 cookies.

- a. **LF paraphrase:** Jane and Mary are each such that her and the other cumulative(ate) 10 cookies.
- b. **Predicted truth-conditions:** True iff Jane and Mary cumulatively ate 10 cookies.

The sketch provided in this section shows how the different readings of *together* can be derived by allowing *together* to take scope in different positions, relative to the covert *each*, to the cumulativity operator and to the existential quantifiers over times and locations. The central idea is that, just like what we saw in the double subject construction in chapter 2, *together* corresponds to a comitative conjunction *with the other* that forces the subject to be interpreted below certain pluralization operators, thereby blocking a higher distributive reading.

## 4.2.2 The role of economy

So far, I have assumed that *together* can freely take scope in different positions, generating different readings. It is straightforward to see that this scope-based analysis alone does not account for the disambiguating role of *together*. Namely, *together* on my account simply blocks distributivity due to higher pluralization operators by introducing a plurality *x with the other* below them. At this point, nothing prevents us from nevertheless deriving a distributive reading with *together* by inserting a distributivity operator below *together*, as shown in (14). In order to account for the crucial fact that *together* disambiguates against the distributive reading, we therefore need to supplement this account with a constraint that governs the distribution of *together*.

(14) **Unattested LF:**

- a. **Distributive LF:** Jane and Mary are each such that her and the other each danced.
- b. **Predicted truth-conditions:** Each of Jane and Mary danced.

I argue that this issue can be resolved if we adopt an economy constraint that governs the distribution of *together*. In this chapter, I outline a narrow version of this constraint, which rules out LFs where deleting *together* does not change the truth-conditions. To see how this blocks the unattested LF, consider again the paraphrase for the distributive LF given in (14-a). Now, notice that deleting *and the other* from (14-a) results in the LF paraphrased in (15-a) which gives rise to the same distributive truth-conditions in (15-b).

- (15) a. **(14-a) without *and the other*:** Jane and Mary are each such that she (each) danced.  
b. **Predicted truth-conditions:** Each of Jane and Mary danced.

We have therefore seen a sketch of how the unattested LF in (14-a) can be ruled out by our economy constraint. Notice that the other readings of *together* derived in section 4.2.1 are all licensed by this narrow version of the economy constraint. For example, consider the LF for the collective reading of *together* given in (16-a). Here, deleting *with the other* results in the LF paraphrased in (16-b), which is simply true iff Jane and Mary each danced. Thus, we see that (16-a) and (16-b) have different truth-conditions, predicting that (16-a) is licensed by the economy constraint. The same reasoning applies to the other readings of *together* where deleting *with the other* results in a different higher distributive reading. For example, in (17) we see that deleting *with the other* from the LF for the simultaneous reading results in an LF where distributivity over the conjoined DP takes place above the existential quantifier over times, predicting a possibly non-simultaneous reading.

- (16) a. **Collective LF:** Jane and Mary are each such that her and the other danced.  
b. **(16-a) without *and the other*:** Jane and Mary are each such that she danced.
- (17) a. **Simultaneous LF:** Jane and Mary are each such that there is a time such that her and the other each stood up.  
b. **Without *and the other*:** Jane and Mary are each such that there is a time such that she (each) stood up.

I therefore argue that the economy constraint on *together* allows for all of the desired readings of *together*, while ruling out LFs that predict an unattested distributive reading. After working out the final version of this account in chapter 5, I show that this economy constraint predicts additional restrictions on when cumulative *together* is licensed. In particular, I show that the economy constraint correctly predicts that the licensing of cumulative *together* depends on

certain semantic properties of the object DP. On the other hand, an alternative account where the distribution of *together* is mediated by purely syntactic constraints is unable to explain how the licensing of cumulative *together* depends on semantic properties of the object.

Finally, while I won't get into this in detail until the end of chapter 5, I want to briefly note that this economy constraint can be seen as part of a larger class of economy constraints in the grammar that play a crucial role in ruling out unattested LFs. In particular, I show that the economy constraint on *together* is modular in the same way as other economy constraints, such as scope economy (Fox, 2000). More specifically, I argue, following Fox (2000) and Fox and Hackl (2006) for other phenomena, that this constraint is evaluated at a level of representation which only has access to certain 'logical' information.

To see an example of the modular properties of economy, consider the sentence with *together* in (18-a). Here, it is common ground that neither Jane nor Mary alone lifted the piano, thus making (18-a) contextually equivalent to the counterpart without *together* in (18-b). Despite the contribution of *together* being trivial in the context in (18), (18-a) is completely acceptable. I argue, based on several cases including examples like (18), that this is because the computation of redundancy is blind to contextual information. Going beyond context-dependency, I show in chapter 5 that there are other parameters that go into evaluating the actual truth-conditions of a given LF but that are not considered for the purpose of proving truth-conditional equivalence in the economy constraint. Again, this discussion will be parallel to arguments in Fox (2000) and Fox and Hackl (2006) that certain constraints are evaluated relative to a formal system that can only see some of the information that goes into evaluating truth-conditions.

- (18)    **Context:** It is common knowledge that neither Jane nor Mary alone can lift the piano.
- a.    Jane and Mary lifted the piano together.
  - b.    Jane and Mary lifted the piano.

This economy constraint on the distribution of *together* therefore offers a novel case-study to investigate the properties of economy constraints in the grammar more generally. One question that arises immediately is whether the economy constraint on *together* might follow from a more general economy constraint in the grammar that subsumes other constraints such as scope economy. For example, we might propose a general economy constraint that rules out LFs which are logically equivalent to simpler LFs that result from deleting one or more nodes. At first sight, this generalized constraint seems to be too strong and in fact rules out the collective LF with *together*. In particular, the LF for the collective reading in (19-a) is equivalent to the LF where both *each* and *and the other* are deleted, paraphrased informally in (19-b). Both (19-a) and (19-b) are true iff Jane and Mary danced together.

- (19) a. **LF for the collective reading:**  
Jane and Mary are each such that them and the other danced.
- b. **LF without *each* and without *and the other*:**  
Jane and Mary are such that they danced.

It is therefore not straightforward to derive our economy constraint from more general grammatical economy considerations. I will return to this issue in chapter 5, where I argue that the desired distribution of *together* can in fact be made to follow from a generalized economy constraint, once appropriate assumptions are made regarding what information this constraint has access to when evaluating equivalence between LFs.

## 4.3 A compositional analysis of collective *together*

### 4.3.1 A syntactic theory of distributivity

I argued in chapter 2 and chapter 3 that in order to get a distributive reading over a plural DP, that DP has to move from its base position. In chapter 2, I proposed that this connection between movement and distributivity can be captured if we assume (i) that lexical predicates are inherently collective and that we therefore need syntactic operators to generate distributive readings and (ii) that the syntactically present pluralization operators responsible for these readings, like \* and \*\*, can apply only to predicates derived by movement (20).

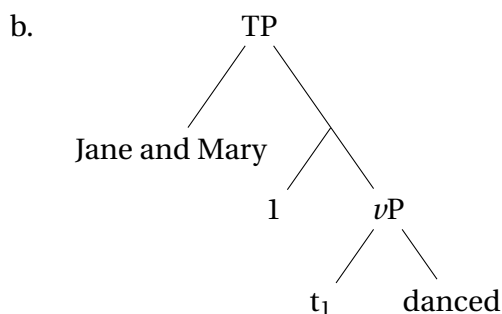
- (20) **Syntactic restriction on pluralization operators:** Pluralization operators like \* and \*\* can only apply to predicates derived by movement.

As mentioned in section 4.2, I will maintain these assumptions in this chapter but assume for simplicity that readings which are true in distributive scenarios and cumulative scenarios are generated by the distributivity operator DIST and the co-distributivity operator Co-DIST, rather than \* and \*\*.

Focusing only on distributive readings with 1-place predicates like *dance* for now, these assumptions predict that an LF like (21-b) for *Jane and Mary danced*, where DIST is not inserted, will have the collective truth-conditions in (21-c). In order to generate a reading for (21-a) which is true in a scenario where each of Jane and Mary danced separately, we therefore have to have an LF where DIST applies below the conjoined DP. Since, following (20), DIST can only apply to predicates derived by movement, DIST will have to be inserted above the binder 1 rather than directly above the predicate. The resulting LF is given in (22-a) and the resulting truth-conditions are given in (22-c): (22-a) is predicted to be true iff each of Jane and Mary danced (whether they

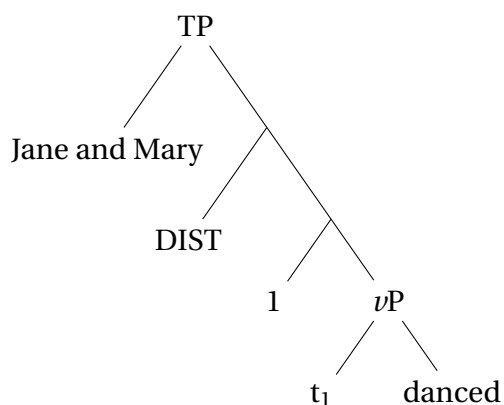
did so together or separately).

(21) a. Jane and Mary danced.



c.  $\llbracket (21-b) \rrbracket = 1$  iff  $\llbracket \text{dance} \rrbracket (j \oplus m) = 1$  iff Jane and Mary danced together.

(22) a.



b.  $\llbracket \text{DIST} \rrbracket = \lambda P. \lambda x. \forall y \leq_{AT} x : P(y) = 1$

c.  $\llbracket (22-a) \rrbracket = 1$  iff  $\llbracket \text{dance} \rrbracket (j) = 1 \wedge \llbracket \text{dance} \rrbracket (m) = 1$  iff Jane danced and Mary danced

### 4.3.2 *together as with the other*

We can now turn to the compositional analysis of *together*. First, as mentioned in section 4.2, I assume that *with* has the same lexical entry as individual conjunction, as shown in (23) (following Ionin and Matushansky, 2002). Two immediate questions arise for this analysis of English *with*, which are similar to questions raised in chapter 2 about the Lebanese Arabic double subject construction. First, it is clear that sentences with comitatives and with standard conjunctions have different truth-conditions. For example, while (24-b) is true if Jane and Mary danced together or separately, (24-a) is only true if they danced together. Additionally, the overt material in (24-a) does not suggest that *with* takes two individuals as arguments as seems plausible to assume for *and* in (24-b).

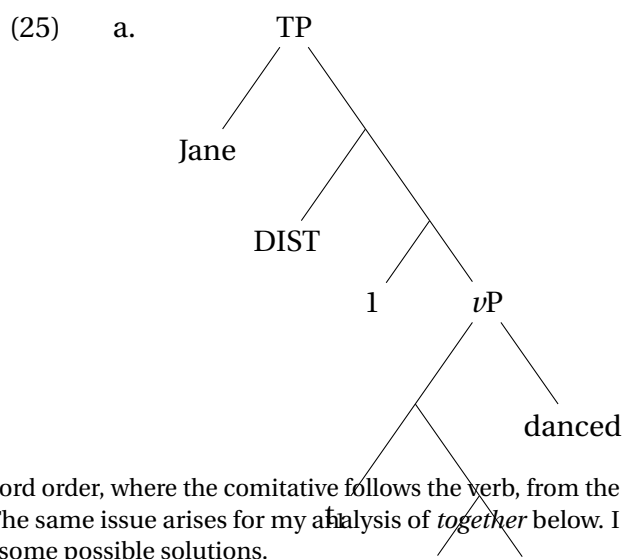
(23)  $\llbracket \text{with} \rrbracket = \lambda x. \lambda y. x \oplus y$

(24) a. Jane danced with Mary.

- b. Jane and Mary danced.

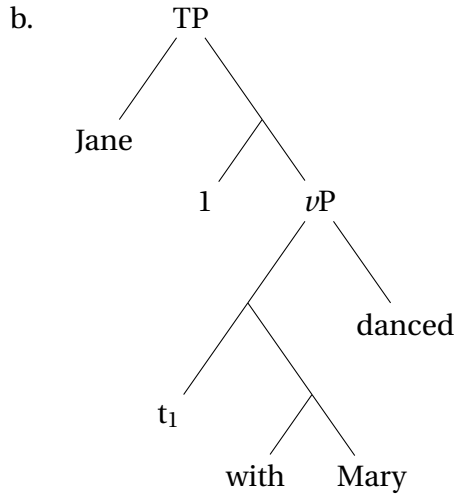
Ionin and Matushansky (2002) propose that in (24-a), *Jane with Mary* starts out as a comitative conjunction, after which the first conjunct, *Jane*, moves to spec,TP. Furthermore, they propose that the distributivity operator only applies below DPs in spec,TP, thus predicting that (24-a) lacks a distributive reading over the comitative conjunction. On the other hand, in (24-b), the entire conjunction moves to spec,TP therefore licensing a distributive operator below it and predicting that (24-b) is true in a scenario where each of Jane and Mary danced separately. As reiterated in section 4.3.1, I will assume here that rather than having distributivity be associated specifically with spec,TP, there is a more general requirement that DIST and co-DIST apply only to predicates derived by movement. Unlike Ionin and Matushansky's proposal, this looser requirement on pluralization operators will allow for distributive readings in intermediate positions, which will be crucial for deriving the simultaneous, proximate, and cumulative readings of *together* in section 4.4.

The possible LFs for (24-a) are therefore given in (25-a) and (25-b): either there is no distributivity operator or a distributivity operator applies below the higher moved subject in spec,TP. Both (25-a) and (25-b) are simply predicted to be true iff the predicate  $\llbracket \text{dance} \rrbracket$  is inherently true of the plurality  $j \oplus m$ .<sup>2</sup> Since DIST can't take scope below the lower subject in spec, $\nu$ P, there can be no distributive reading over the comitative conjunction.<sup>3</sup> Given my assumption that predicates are lexically collective, this analysis therefore predicts that (25-a) and (25-b) are true iff Jane and Mary danced together.



<sup>2</sup>There is a question here of how to derive the correct word order, where the comitative follows the verb, from the LFs in (25-a) and (25-b), where *with Mary* is in spec, $\nu$ P. The same issue arises for my analysis of *together* below. I return to this in Appendix A of chapter 5, where I discuss some possible solutions.

<sup>3</sup>One might wonder at this point whether we can get the same results from treating *with Mary* as a modifier that takes a predicate P and returns a predicate of the form  $\lambda x.P(x \oplus m)$ , thus ensuring that the original predicate applies to the sum of  $j$  and  $m$ . I discuss this possibility in Appendix B in chapter 5, where I show that the main insights of the analysis of *together* proposed here can be recaptured with this alternative analysis of comitatives. Nevertheless, I present some conceptual considerations that argue for favoring Ionin and Matushansky's coordination analysis.



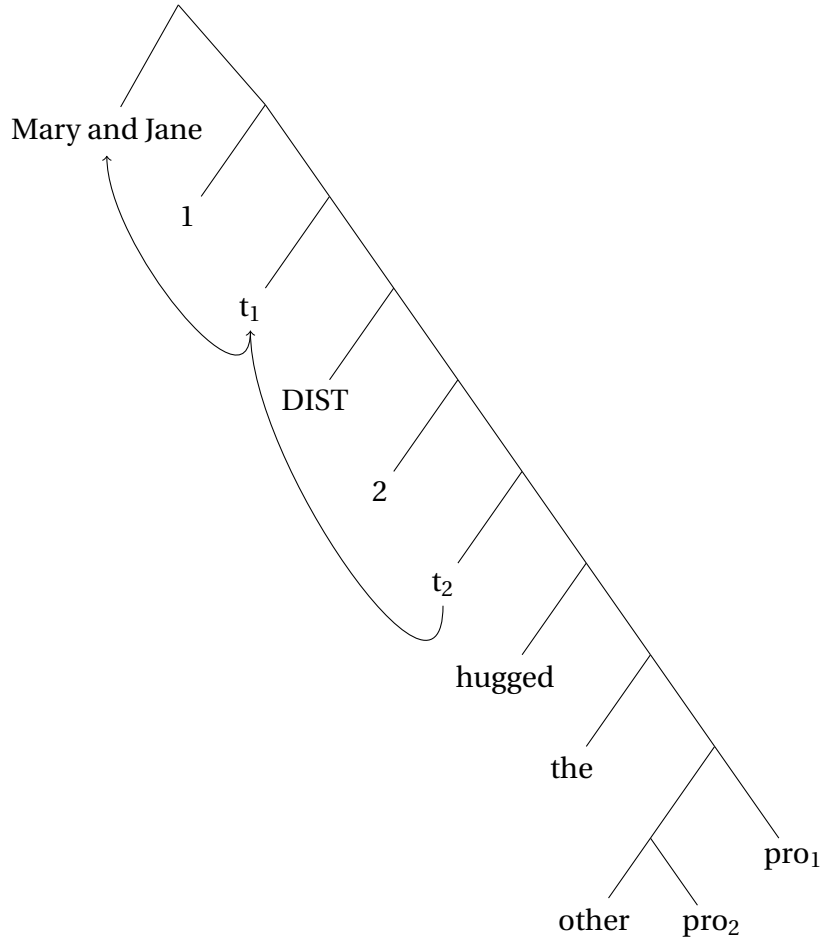
(26)  $\llbracket (25-a) \rrbracket = \llbracket (25-b) \rrbracket = 1$  iff  $\llbracket \text{dance} \rrbracket(j \oplus m) = 1$  iff Jane and Mary danced together.

There is a question here of why the comitative conjunction in (25-a) can't move from spec,  $\nu P$ , licensing a distributivity operator below it. In fact, Ionin and Matushansky argue that this is exactly what happens with Russian comitatives, which do allow for distributive readings in certain configurations. At this point, I will simply assume that there is a syntactic requirement on comitative conjunction in English that requires only the first conjunct to move to spec, TP, while the conjoined DP remains in its base position.

As for the reciprocal component of the analysis of *together*, I assume a compositional analysis of reciprocals, following Heim et al. (1991), where *each other* is decomposed into a distributivity operator and a *the other* DP in object position. More specifically, I adopt the implementation of this analysis from Sauerland (1998). The resulting LF is given in (10), where the subject moves twice, first binding the first argument of *other* distributively and then binding the second argument collectively. The denotation for *the other* is given in (12), where *other* takes two individuals as arguments, a contrast and a range, and returns that set of individuals which are part of the range but don't overlap with the contrast. As a result, when *other* takes an individual  $j$  as its first argument (contrast) and a plurality  $j \oplus m$  as the second (range),  $\llbracket \text{the other} \rrbracket$  denotes the individual  $m$ , which results from removing  $j$  from  $j \oplus m$  (12-c). Now, the resulting truth-conditions for (10) are given in (13): (10) is predicted to be true iff each of Jane and Mary hugged the other, i.e. iff Jane hugged Mary and Mary hugged Jane.

(27) a. Jane and Mary hugged each other.

b.



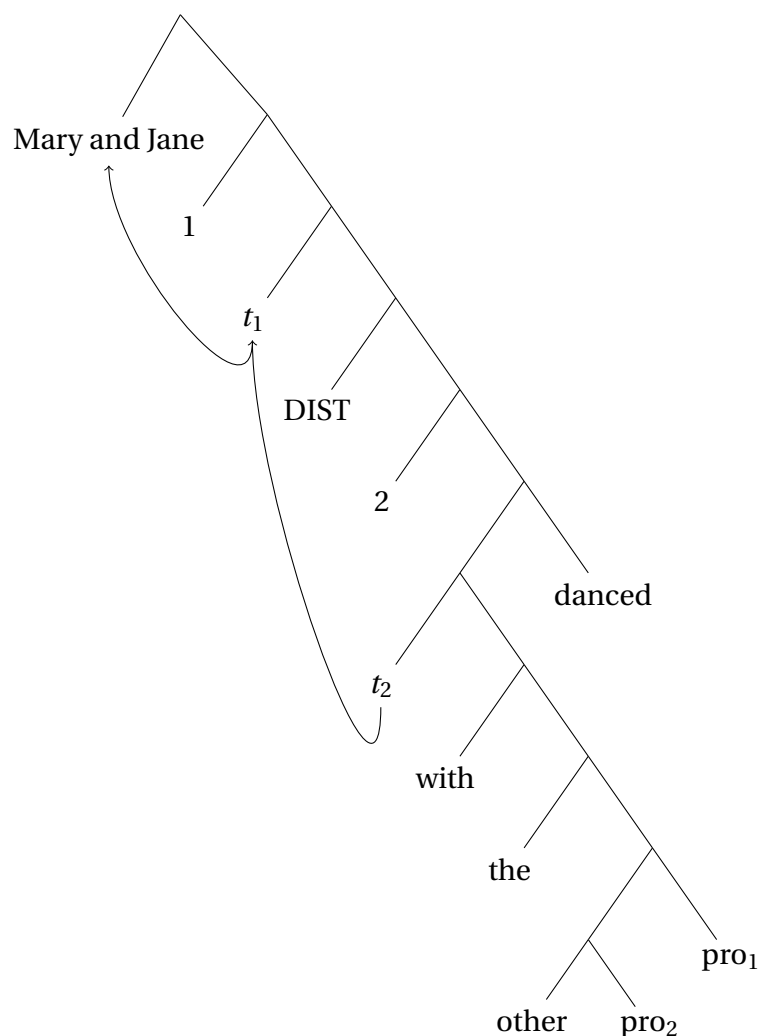
- (28) a.  $\llbracket \text{other} \rrbracket = \lambda x. \lambda y. \lambda z. z \leq y \wedge \neg \exists x' : x' \leq x \wedge x' \leq z$   
 b.  $\llbracket \text{the} \rrbracket (\llbracket \text{other} \rrbracket (x)(y)) = y \ominus x$ , defined iff  $x < y$ ,  
 (where  $y \ominus x$  is the maximal individual  $z$  s.t.  $z \leq y \wedge \neg \exists x' : x' \leq x \wedge x' \leq z$ )  
 c.  $\llbracket \text{the} \rrbracket (\llbracket \text{other} \rrbracket (j)(j \oplus m)) = m$
- (29)  $\llbracket \text{Mary and Jane hugged each other} \rrbracket = 1$  iff  
 $\lambda y. ((\text{DIST}(\lambda x. \llbracket \text{hugged} \rrbracket (x)(y \ominus x))) (y)) (m \oplus j) = 1$  iff  
 $(\llbracket \text{hugged} \rrbracket (m)(j) \wedge \llbracket \text{hugged} \rrbracket (j)(m))$

Putting together the contribution of *with* and of *the other*, we can derive the collective reading of *together* by forcing the comitative conjunction *x with the other* to remain in its base position, where it has to take scope below all pluralization operators. The corresponding LF is given in (30-b) and the truth-conditions are given in (30-c): (30-b) is predicted to be true iff  $\llbracket \text{dance} \rrbracket$  is inherently true of the plurality  $j \oplus m$ , since the comitative conjunction takes scope below DIST. In other words (30-b) is true iff Jane and Mary danced together, thus deriving the desired collective reading.



(30) a. Jane and Mary danced together.

b.

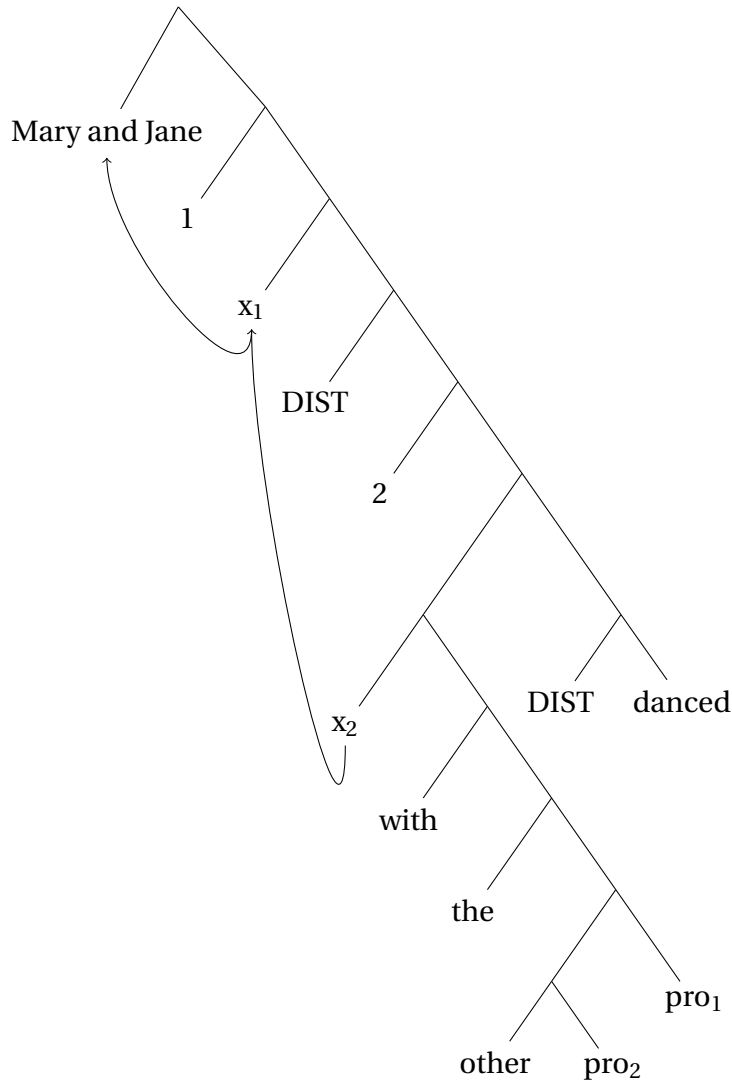


c.  $\llbracket (30-b) \rrbracket = 1$  iff  $\llbracket \text{dance} \rrbracket (m \oplus j) = 1 \wedge \llbracket \text{dance} \rrbracket (j \oplus m) = 1$   
iff  $\llbracket \text{dance} \rrbracket (j \oplus m) = 1$  iff Jane and Mary danced together.

Again, to derive the fact that (30-a) is not true if Jane and Mary danced separately, we have to make sure that there is no LF for (30-a) where *together* takes scope above a lower distributivity operator. For example, the LF in (31-a), where the distributivity operator applies to the predicate before it applies to *together*, is true iff each of Jane and Mary danced (31-b).

(31) **Unattested LF:**

a.



b.  $\llbracket (31-a) \rrbracket = 1$  iff  $\llbracket \text{dance} \rrbracket(j) = 1 \wedge \llbracket \text{dance} \rrbracket(m) = 1$

The LF in (31-a) is again ruled out by my constraint that DIST can apply only to predicates derived by movement: DIST merges directly with the lexical predicate *dance* in (31-a) and is therefore predicted to be ruled out. There is again a question here of why the *with the other* conjunction can't move from its base position, where DIST would be allowed to take scope below it and generate the unattested reading in (31-b).

We will see in section 4.4 that in order to predict the other readings of *together*, the *with the other* conjunction has to move and take scope in higher positions. We therefore can't maintain that *together* lacks a distributive reading because it has to remain in its base position. In section 4.5, I argue that no purely syntactic constraint on the distribution of *together* will be able to derive its desired distribution. Instead, I propose that the distribution of *together* is mediated by an economy constraint that is sensitive to the truth-conditions corresponding to the LF with

*together*.

To summarize, I proposed in this section that *together* is decomposed into a *with*-coordinator and a *the other*-DP, taking seriously the fact that in several languages *together* is decomposed into a comitative and a reciprocal anaphor. I showed that the collective reading of *together* is predicted by this analysis if we force the *with*-conjunction to take scope below all pluralization/distributivity operators. In the next section, I will turn to showing how the different readings of *together* can be predicted by stranding the *with*-conjunction in different scope positions.

## 4.4 Deriving the higher scope readings

In this section, I show how the simultaneous, proximate and cumulative readings of *together* are derived by allowing the comitative coordination *x with the other* to move from its base position. As outlined in section 4.2, I propose that the three readings correspond to three different scope positions for *together*: (i) the proximate reading in (32) is derived when *together* moves above the lowest distributivity operator but remains below the existential quantifiers over times and locations corresponding to Aspect and to a lower Location head respectively; (ii) the simultaneous reading in (33) is derived when *together* takes scope below Aspect but above the existential quantifier over locations; (iii) the cumulative reading in (34) is derived when *together* takes scope above a 2-place co-distributivity operator.

(32) **Proximate reading:**

- a. **Scenario:** At the morgue, one employee put a corpse on the table in the corner and another employee independently put a different random corpse on the same table.
- b. The two corpses lay together on the table.

(33) **Simultaneous reading:**

- a. **Scenario:** There is a crowd of people watching a play. Jane was sitting in the front seat and Mary was sitting at the very back, and Jane and Mary don't know each other. When one of the actors started cursing at the other actors, Jane and Mary both stood up and walked out independently, Jane from the door at the front, and Mary from the door at the back.
- b. Jane and Mary stood up together.

(34) **Cumulative reading:**

- a. **Scenario:** Jane walked in and ate 5 of the 20 cookies in a bowl on the table. Later, Mary, who doesn't know Jane, walked in and ate 5 other cookies.
- b. Jane and Mary together ate 10 of the cookies.

Despite the appearance of the readings in (32)-(34) being distinct readings from the collective one discussed in section 4.3, it is possible that it is simply an illusion that *together* gives rise to genuine ambiguity. In particular, focusing on the simultaneous and the proximate readings for now, it is possible that different predicates have different lexically specified truth-conditions for when they are true primitively of a plurality. For example, it is possible that while  $\llbracket \text{dance} \rrbracket$  gives rise to the collective truth-conditions in (35-a) when it applies to a plurality,  $\llbracket \text{stood up} \rrbracket$  simply requires that the members of the plurality stood up at the same time (35-b) and  $\llbracket \text{lay} \rrbracket$  requires that the members of the plurality lay at the same location (35-c). Under this view, the appearance of different readings of *together* simply follows from different lexical specifications of when a predicate can be true of a plurality.

- (35)    a.  $\llbracket \text{dance} \rrbracket(j \oplus m) = 1$  iff Jane and Mary collectively danced.  
           b.  $\llbracket \text{stood up} \rrbracket(j \oplus m) = 1$  iff Jane and Mary stood up at the same time.  
           c.  $\llbracket \text{lay} \rrbracket(c_1 \oplus c_2) = 1$  iff corpse 1 and corpse 2 lay at the same location.

In fact, the alternative view sketched above is very close to what Moltmann (2004) proposes for the simultaneous and proximate readings of *together*. In particular, Moltmann proposes that *together* simply demands that there is a single event that the members of the plurality are agents of. Moltmann then suggests that what counts as a single event depends on the choice of predicate: a single event of dancing whose agents are Jane and Mary is one where Jane and Mary danced together, while a single event of standing up is one where Jane and Mary stood up at the same time. Lasersohn (1990), on the other hand, assumes that the meaning of *together* is ambiguous between a collective, a simultaneous and a proximate reading. In what follows, I will argue, in line with Lasersohn's analysis, that there is genuine ambiguity with *together*. Unlike Lasersohn, who posits lexical ambiguity in the meaning of *together*, I will argue that this ambiguity is best treated as a scope ambiguity, allowing us to maintain a uniform lexical meaning for *together*.

#### 4.4.1 Evidence for genuine ambiguity

Starting with the simultaneous reading, I will show that the same sentence can have both a collective reading and a simultaneous reading, thus providing evidence that there is a distinct simultaneous reading. In particular, consider the example in (36). I will argue that (36) has a simultaneous reading which is true as long as Jane and Mary called Sue at the same time, even if they called her completely independently (36-a). In addition, I show that (36) has a collective reading where it is only true if Jane and Mary jointly called Sue from the same phone.

- (36)    Jane and Mary called Sue together.

- a. **Simultaneous Reading:** Jane and Mary called Sue at the same time.
- b. **Collective reading:** Jane and Mary called Sue from the same phone, at the same time, and with joint goals.

First, we can observe that (36) is true in both simultaneous and collective scenarios. In particular, (36) is true in the scenario in (37), where there was no joint planning or action on the part of Jane and Mary and where they simply called Sue at the same time. (36) is also true in the collective scenario in (38), where Jane and Mary were a team and jointly called Sue from the same phone. At this point, it is not clear whether there is genuine ambiguity in (36). In particular, it is possible that (36) simply has a single underspecified meaning which is true in both of the scenarios in (37) and (38).

- (37) **Simultaneous Scenario:** There was a contest where Jane and Mary each had to get to a phone and call Sue, whose number they were supposed to have memorized. Jane and Mary each reach their phone at the same time and they call Sue simultaneously, so they tie in the contest. Someone, describing what happened says:
  - a. Jane and Mary called Sue together.
- (38) **Collective scenario:** There was a contest, where participants were paired up and they had to find the phone and be in the same room and call Sue from the same phone and both talk to her. Jane and Mary successfully find each other and the phone and call Sue and both talk to her.
  - a. Jane and Mary called Sue together.

To rule out the possibility of a single weaker meaning for (36), we therefore have to embed the sentence in an environment which is not upward-entailing and see whether the two readings survive there. For example, consider the example in (39), where the sentence with *together* is embedded in the restrictor of a universal quantifier. If the restrictor had a single weak meaning which is true both if the couple called Sue at the same time from different phones or if they called Sue together from the same phone, then the universal statement would be incorrectly predicted to be false in the scenario in (39). In particular, the weaker meaning would make the restrictor in (39) true as long as a couple called Sue at the same time, thus predicting that (39-a), just like (39-b), is false in the collective scenario. The fact that (39-a) is true in (39) therefore provides evidence that (36) has a collective reading that is distinct from the simultaneous one, thus providing evidence for genuine ambiguity in the meanings of sentences with *together*.

- (39) **Collective Scenario:** There are different couples participating in the competition. The couple gets a prize if they can find one phone together and call Sue from the same phone

and talk to her. Several couples achieved this and got a prize. One couple each found a different phone and called Sue at the same time but they did not get a prize.

- a. Every couple who called Sue together got a prize.
- b. #Every couple who called Sue at the same time got a prize.

Following a parallel logic, we can argue for the presence of a proximate reading that is distinct from the collective one. For example, (40) is true in a scenario where each of John and Bill was separately tied up, while being in proximity of each other (40-a). (40) is also true in what might be called a collective reading, where there is one rope tying John and Bill together (40-b). Again, we can show that these are distinct readings by embedding *together* in the restrictor of *every*: the fact that (41-a) is true in the given scenario shows that *tied up together* has a collective reading which is distinct from the proximate one.

(40) John and Bill were tied up together.

- a. **Proximate Scenario:** John was tied up and Bill was separately tied up next to him.
- b. **Collective scenario:** There was one rope tying John and Bill together.

(41) **Collective Scenario:** There was a competition with pairs of teammates. They were tied up in a room and had to escape. Some pairs were separately tied up next to each other and some were tied up with one rope together. Those who were separately tied up next to each other managed to escape but no pair who were tied with one rope managed to escape.

- a. Every pair who was tied up together failed to escape.
- b. # Every pair who were tied up next to each other failed to escape.

As for the cumulative reading of *together*, it is more straightforward to show that this reading is distinct from the other readings. In fact, Moltmann (2004), who denies the ambiguity in the other readings, maintains that cumulative *together* is distinct from collective *together*. For example, we can show that (42) has a collective reading, paraphrased in (42-a) and a distinct cumulative reading, paraphrased in (42-b). To see this, consider the scenario in (43): here, the conditional in (43-a) is only predicted to be true if there is a distinct collective reading for (42) and (43-b) is only predicted to be true if (42) has a distinct cumulative reading. We can therefore conclude that (42) is ambiguous between the reading in (42-a) and the one in (42-b).

(42) Jane and Mary lifted exactly 10 pianos together.

- a. **Collective reading:** There are exactly 10 pianos such that Jane and Mary lifted each of them together.
- b. **Cumulative reading:** The sum of the number of (distinct) pianos that Jane lifted and

that Mary lifted is exactly 10.

- (43) **Scenario:** There is a competition where students are split into teams of 2 and have to lift pianos of different sizes. For each piano, students can choose to lift it alone or together with their teammate. There is a teamwork prize that is given to teams that lift each of exactly 10 pianos with their teammate and a silver prize that is given to teams that lift a total of exactly 10 pianos with their teammates, whether they lifted each of them together or separately.
- a. **Collective reading:** If Jane and Mary lifted exactly 10 pianos together, they will win the teamwork prize.
  - b. **Cumulative reading:** If Jane and Mary lifted exactly 10 pianos together, they will win the silver prize.

#### 4.4.2 Simultaneous *together*

As mentioned above, I propose that the simultaneous reading of *together* is predicted when *together* takes scope above the lowest distributivity operator but below a higher distributivity operator above Aspect. Assuming that perfective aspect denotes an existential quantifier over times, the simultaneous reading is derived when the *x with the other* is interpreted in the scope of this existential quantifier, as shown in the paraphrase in (44).

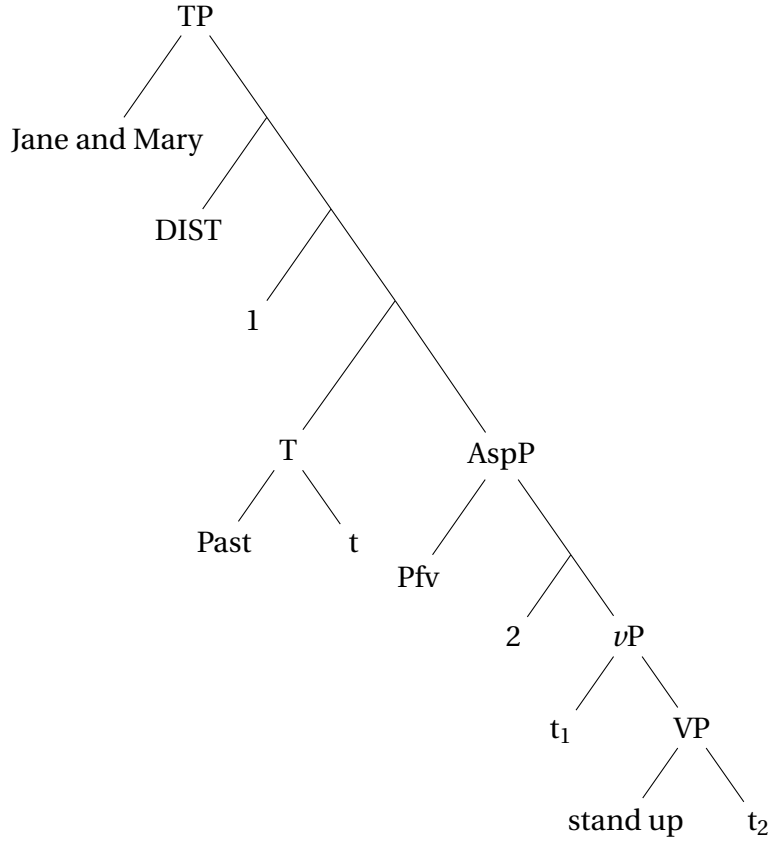
- (44) **Paraphrase of the simultaneous reading:**  
Jane and Mary each is such that **there is a time** such that **her and the other** each stood up at that time.

More concretely, I assume that predicates take, in addition to their individual arguments, a time interval as an argument and that predicates like *stood up*, which don't seem to have a collective reading, can only be true of atoms (45-a). As for the contribution of tense and aspect, I take tense to be referential (37-b) (Partee, 1973) and I assume that perfective aspect existentially quantifies over the time interval that tense picks out (37-c) (Bhatt & Pancheva, 2005, a.o.). The resulting LF for the simple sentence in (46-a) without *together* is given in (46-b), where the subject moves to spec,TP and where there is a distributivity operator above Aspect. This allows for (46-b) to be true if Jane and Mary stood up at different times, as shown by the predicted truth-conditions in (46-c).

- (45) a.  $\llbracket \text{stood up} \rrbracket = \lambda t. \lambda x : x \in AT. x \text{ stood up at } t.$   
 b.  $\llbracket \text{Past} \rrbracket^c = \lambda t : t < t_c. t$   
 c.  $\llbracket \text{Pfv} \rrbracket = \lambda P. \lambda t. \exists t' \subseteq t : P(t') = 1$

(46) a. Jane and Mary stood up.

b.



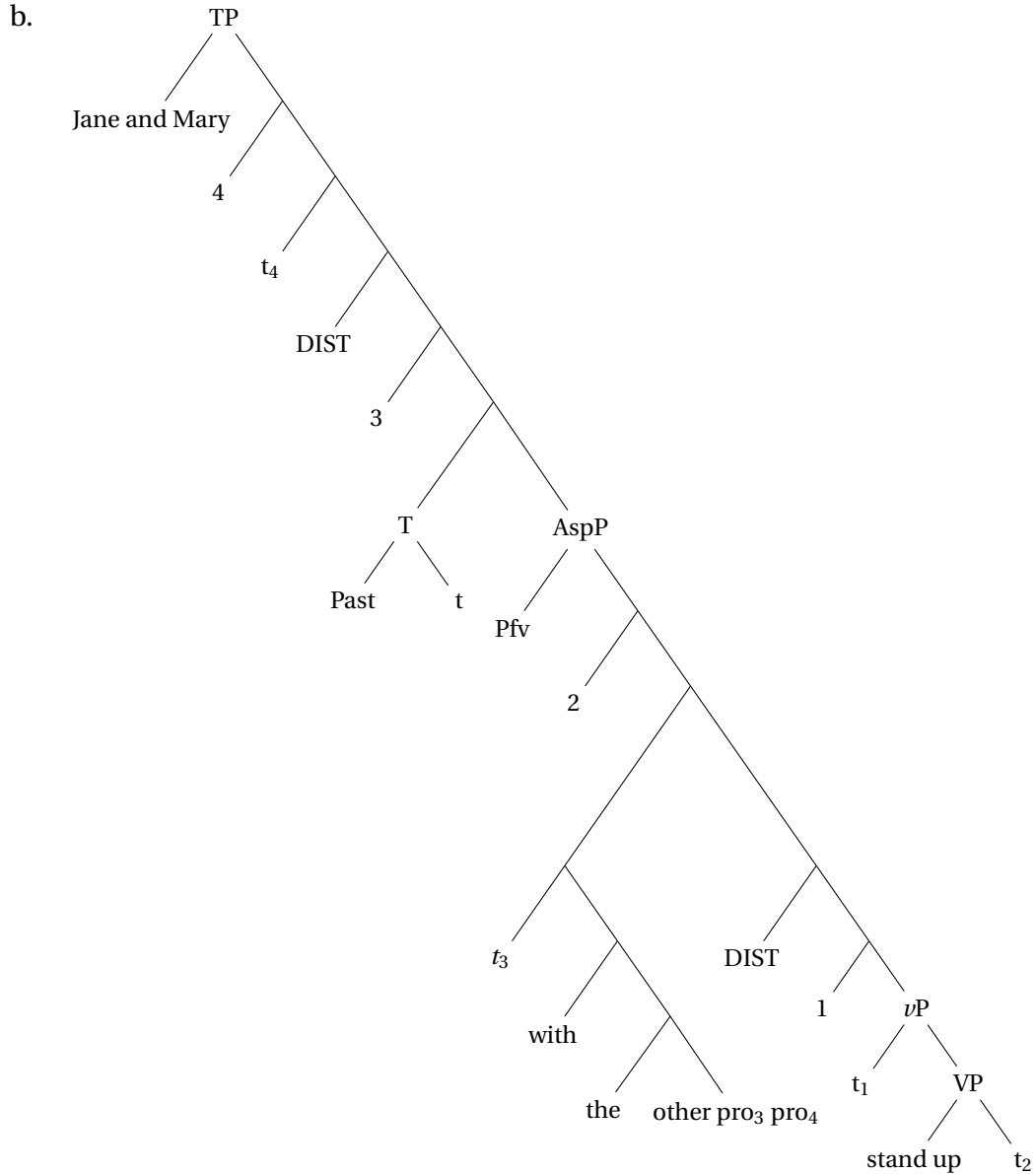
c.  $\llbracket \text{Jane and Mary stood up} \rrbracket = 1$  iff

$$\exists t' \subseteq t : \llbracket \text{stood up} \rrbracket(t')(j) = 1 \wedge \exists t'' \subseteq t : \llbracket \text{stood up} \rrbracket(t'')(m) = 1$$

Now, consider the LF in (47-b) for the counterpart with *together*. Here, the *x with the other* conjunction moves from spec, νP to a position below Aspect but above a lower distributivity operator. The conjoined DP is therefore interpreted in the scope of the existential quantifier over times denoted by Aspect, thus ensuring the desired simultaneous reading in (47-c). In particular, (47-b) is predicted to be true iff there is a time such that each of Jane and Mary stood up at that time. We have therefore derived the simultaneous reading while keeping the same lexical entry for *together*.

(47) a. Jane and Mary stood up together.





c.  $\llbracket (47-b) \rrbracket = 1$  iff  $\exists t' \subseteq t: \llbracket \text{stood up} \rrbracket (t')(j) = 1 \wedge \llbracket \text{stood up} \rrbracket (t')(m) = 1$

One immediate question that arises for this analysis is why with certain predicates, the simultaneous reading seems to be more difficult to get than the collective one. For example, (48-a) and (48-b) don't seem to allow a simultaneous reading very easily. In particular, the easily accessible reading for (48-a) is one where Jane and Mary sat together and ate, rather than simply eating at the same time. Similarly, when uttered out of the blue, (48-b) is interpreted as asserting that Jane and Mary jointly did a dance. Nevertheless, I think that given the right context, sentences like (48-a) and (48-b) do allow for a simultaneous reading. For example, consider the scenario in (49): here it is felicitous to utter (49-a) or (49-b) to convey that Jane and Mary were eating at the same time but sitting at different tables.

- (48) a. Jane and Mary ate together.  
b. Jane and Mary danced together.
- (49) **Scenario:** Jane and Mary hate each other and they never sit together when eating dinner. I go to dinner at the dining hall every night, and I try to sit with Jane or Mary if one of them is there for dinner. Usually, Jane and Mary eat dinner at different times so I just eat with whichever one is there when I go.
- a. Jane and Mary were eating dinner together today, so I had to choose which one to sit with.
- b. Jane and Mary ate dinner together today, so I had to choose which one to sit with.

While Moltmann's (2004) lexical analysis can hardwire constraints on which predicates allow for a simultaneous reading into the lexical meanings of different predicates, the scope-based account predicts that the simultaneous reading should in principle be available with any predicate that takes a time interval as one of its arguments. Therefore, the syntactic account has to resort to independent mechanisms to explain why certain readings are dispreferred with certain predicates. While I don't have a complete explanation for these restrictions, I showed above that the simultaneous reading is available even with predicates where it is not the most salient reading. One possibility is to assume that there is a preference for *together* to remain in its base position. With predicates like *stood up*, which only have atoms in their extension, the LF where *together* remains in its base position gives rise to a contradiction, since the predicate *stood up* can't apply directly to the *x with the other* conjunction. It is therefore reasonable to assume that the simultaneous reading is more easily accessible with these predicates because the corresponding LF where *x with the other* moves is the only one which avoids a contradiction. On the other hand, with *ate* and *danced*, this reading is obscured by a more easily accessible LF where the coordinated DP does not move.

#### 4.4.3 Proximate *together*

We can now turn to the proximate reading of *together*, repeated in (50). I show in this section that we can use a strategy for the proximate reading which is parallel to the one used for the simultaneous reading in section 4.4.2. In particular, I propose that in addition to the existential quantifier over times contributed by Aspect, there is a lower existential quantifier over location and that when *together* blocks distributivity over both Aspect and Location, the spatial proximity reading can be derived. While the contribution of perfective Aspect has been discussed extensively in the literature, the existential quantifier over location that I rely on in this section has not been independently argued for. On the scope-based analysis proposed here, the proximity

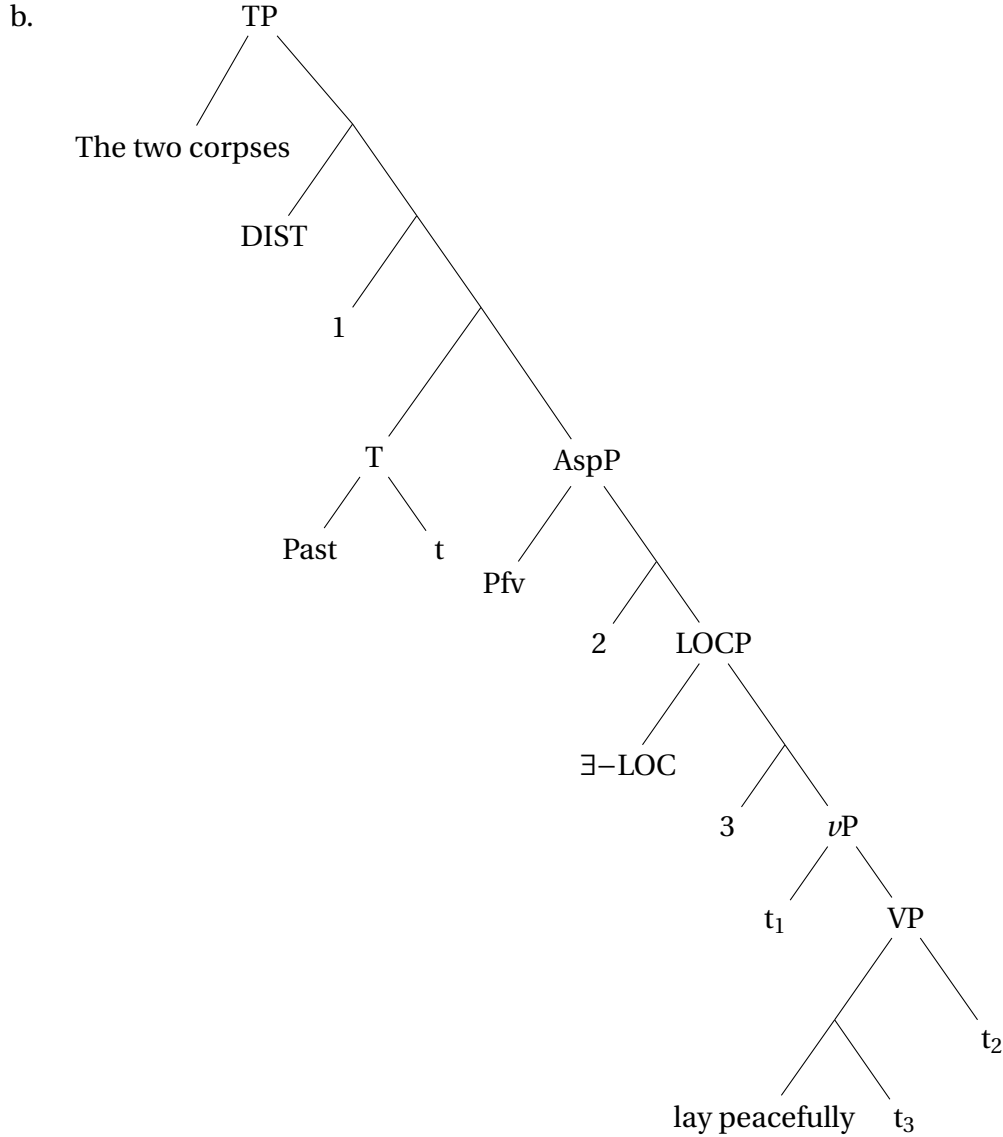
reading with *together* therefore reveals that there has to be such a quantifier in the clause so that we can get a *same location* reading when *together* is interpreted below it. This raises the question of whether we can find independent evidence for this quantifier beyond the interpretation of *together*. While I mostly leave this issue for future work, I return to this in chapter 7 and argue that the same existential quantifier over location is needed to account for certain truth-conditional differences between overt and covert reciprocals.

(50) The two corpses lay together.

More concretely, I propose that predicates take an additional location argument, as shown in (51-a) for *lay*. The lexical entry for the existential quantifier over location is given in (51-b). I'll return to this issue in more detail in section 4.4.4, but I assume for our purposes here that this existential quantifier over location takes scope below Aspect. Let's start by looking at what this predicts for a simple example like (52-a) without *together*. The LF for (52-a) is given in (52-b) and the resulting truth-conditions are given in (52-c): (52-b) is predicted to be true iff for each of the two corpses there is a time and a location such that they lay peacefully at that time and at that location.

- (51) a.  $\llbracket \text{lay} \rrbracket = \lambda l. \lambda t. \lambda x : x \in AT. x \text{ lay at time } t \text{ in location } l.$   
 b.  $\llbracket \exists\text{-LOC} \rrbracket = \lambda P. \exists l : P(l) = 1$

(52) a. The two corpses lay peacefully.



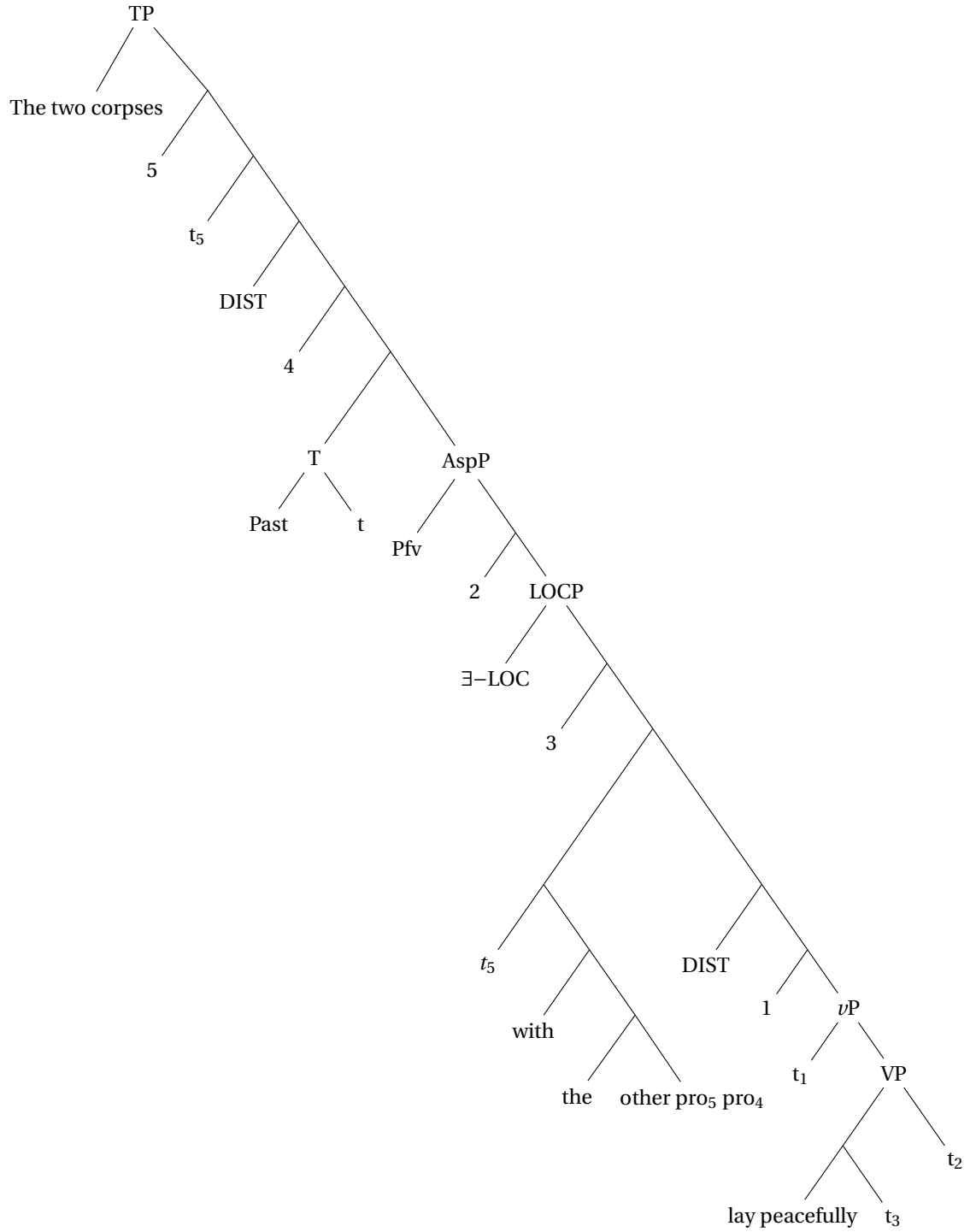
c.  $\llbracket (52-b) \rrbracket = 1$  iff

$$\exists t', l' : \llbracket \text{lay peacefully} \rrbracket (l')(t')(c_1) = 1 \wedge \exists t'', l'' : \llbracket \text{lay peacefully} \rrbracket (l'')(t'')(c_2) = 1$$

Turning to the counterpart with *together* in (53-a), when *x with the other* moves below both Asp and  $\exists$ -LOC but above a lower DIST, we get the desired proximate reading. This LF for (53-a) is given in (53-b) and the corresponding truth-conditions are given in (53-c): (53-b) is predicted to be true if there is a time and a location such that each of the two corpses lay at that time and at that location. We have therefore derived the desired proximate reading of *together*.

(53) a. The two corpses lay together.

b.



c.  $\llbracket (53-b) \rrbracket = 1$  iff  $\exists l' \exists t' \subseteq t : \llbracket \text{lay} \rrbracket(l')(t')(c_1) = 1 \wedge \llbracket \text{lay} \rrbracket(l')(t')(c_2) = 1$

Note that given the additional existential quantifier over location, in order to get the pure simultaneous reading derived in the last section, *together* has to take scope above a DIST operator between Pfv and  $\exists$ -LOC but below a higher DIST operator, therefore blocking distributivity over the existential quantifier over times contributed by Aspect but allowing distributivity above

∃-LOC.

Before moving on, I want to note that there is a question here as to why with the time argument, we get a *same-time* reading, while with the location argument, we get a proximate reading rather than a *same place* reading. In particular, a full parallel with the *stood up together* simultaneous reading would lead us to predict that the two corpses in (53-a) literally occupy the same space, which is not the right reading here.

I do not have a full theory of how the relevant granularity is determined here, but it is reasonable to assume that how coarse-grained the location and temporal intervals are depends on the lexical semantics of the predicate, together with contextual factors. In particular, even with what we've been calling the simultaneous reading, there can be contexts which do not require strict temporal overlap. For example, (54-b) is true in the scenario in (54-a) even though here Jane and Mary did not stand up at the same moment. I therefore don't believe that there is a systematic asymmetry between the temporal and spatial domain. Rather, contextual factors determine what counts as *same time* and *same location* for different predicates.

- (54) a. **Scenario:** There is a crowd of people watching a play. Jane was sitting in the front seat and Mary was sitting at the very back, and Jane and Mary don't know each other. When one of the actors started cursing at the other actors, Jane stood up and walked out and then a few seconds later Mary stood up and walked out. The rest of the audience stayed till much later when the situation escalated even further.
- b. Jane and Mary stood up together.

#### 4.4.4 No proximity without simultaneity

Assuming that Aspect always takes scope above ∃-LOC, the syntactic account predicts an asymmetry between the simultaneous and proximate readings. Namely, when *together* takes scope below LOC in order to derive the proximate reading, it also necessarily takes scope below Aspect. As a result, the proximate reading is predicted to block distributivity over both times and locations, thus predicting that proximity is not possible without simultaneity. In other words, the scope-based analysis proposed here predicts that we can get a spatial-temporal proximity reading for *together* but never a pure spatial proximity reading. I briefly show in this section that this prediction is borne out and outline how this provides evidence for the scope-based account proposed here and against lexical accounts which have to stipulate this pattern as part of the lexical meaning of *together*.

Moltmann (2004) observed that *together* lacks a pure spatial proximity reading which doesn't also require temporal simultaneity. This is illustrated in the examples in (55) and (56). In (55), we see that *together* can have a purely simultaneous reading which does not require any

spatial proximity: (55-a) is true in a scenario where Jane and Mary stood up at the same time in different rooms. (56) shows that the converse, a pure proximity reading which does not require simultaneity, is not attested. In particular, the scenario in (56) is one where the corpses lay in the same location as each other but at different times. The sentence with *together* in (56-a) is clearly infelicitous in this scenario. We can therefore conclude that there is no proximate reading of *together* that doesn't also require simultaneity.

(55) **Context:** Sue is watching Mary and Jane on security cameras. She sees that Jane and Mary stood up at the same time from their respective couches.

a. Jane and Mary stood up together.

(56) **Context:** John's corpse lay on the table yesterday, and was later removed. Then Bill's corpse was placed on the table next to where John's corpse was.

a. # John's corpse and Bill's corpse lay together.

On both Lasersohn's and Moltmann's lexical accounts of *together*, there is no principled explanation for why we see this asymmetry between the spatial and the temporal domain. On the other hand, this type of asymmetry is necessary on the syntactic account, assuming that Aspect and LOC have fixed syntactic positions. In particular, either Aspect takes scope below location, in which case we would incorrectly predict that the simultaneous reading requires proximity but not vice versa. Alternatively, as we have assumed here, Aspect takes scope above location and we correctly predict that the proximate reading requires simultaneity. Therefore, the only thing that has to be stipulated on the syntactic account is the direction of the asymmetry between locations and times: assuming that there is a fixed order of Aspect and LOC, there will necessarily be an asymmetry whose manifestation will depend on which quantifier takes higher scope. I therefore take the presence of such an asymmetry as evidence for the scope-based account I propose above.

#### 4.4.5 Cumulative *together*

The final reading that we saw with *together* in the introduction is the cumulative reading, exemplified in (57). The cumulative reading seems most natural when *together* precedes the verb, as in (57-a), but it is also available when *together* follows the VP (57-b). Setting the issue of word order aside for now, we see in (57) that (57-a) and (57-b) have a reading where they simply indicate that the total number of politicians that Jane and Mary shook hands with is exactly 10.

(57) **Context:** There are 10 politicians at this conference. Jane walks in and shakes hands with 5 of the 10 politicians. Then Jane walks out, and Mary, who doesn't know Jane, walks in

and shakes hands with the other 5 politicians.

- a. Jane and Mary together shook hands with exactly 10 politicians.
- b. Jane and Mary shook hands with exactly 10 politicians together.

Before I look at the contribution of *together* in generating cumulative readings, I will first provide some background assumptions on how these cumulative readings are generated in general without *together*. In particular, consider the variant of (57-a) without *together* in (58). Unlike (57-a), (58) has both a cumulative reading, paraphrased in (58-a) and a distributive reading paraphrased in (58-b). The contribution of *together* therefore seems to be to block the distributive reading and allow only for the cumulative one.

(58) Jane and Mary shook hands with exactly 10 politicians.

- a. **Cumulative reading:** The sum of the number of (distinct) politicians that each of Jane and Mary shook hands with is 10.
- b. **Distributive reading:** Each of Jane and Mary shook hands with exactly 10 politicians.

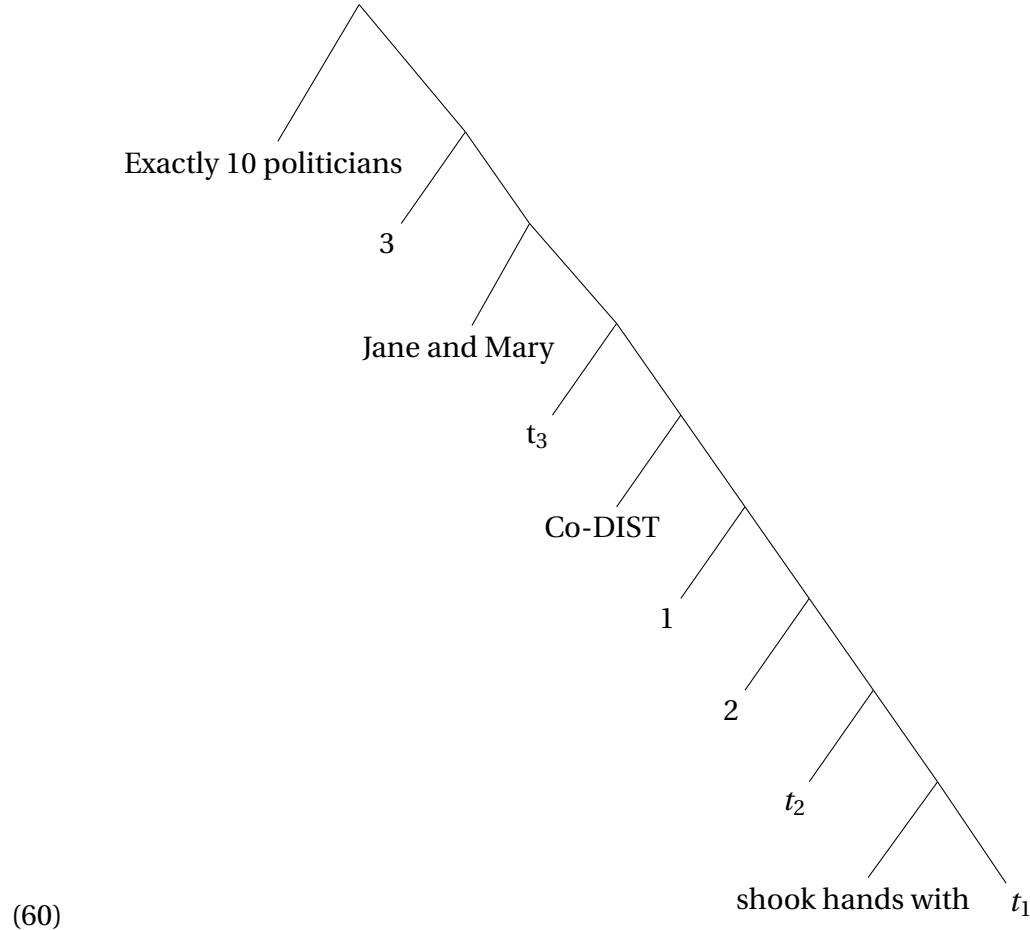
To see how the cumulative reading is generated, recall that I'm assuming crucially that lexical predicates give rise to collective truth-conditions when they apply directly to pluralities. The cumulative reading in (58-a), which does not require any joint action on the part of Jane and Mary, therefore has to be generated via a syntactic operator. For cumulative readings, this can be achieved by applying the two-place pluralization operator \*\* to the relevant 2-place predicate (Scha, 1981). Again, for simplicity, I will start by assuming that cumulative readings are due to a 2-place co-distributivity operator rather than \*\*. The lexical entry for this co-distributivity operator is given in (59-a). To see this with an example, consider the predicted reading when Co-DIST applies to the predicate *talked* in (59-b) before it applies to its arguments. The resulting truth-conditions are given in (59-c): (59-c) is true iff each of Jane and Mary talked to at least one of Bill and Sue and each of Bill and Sue talked to at least one of Jane and Mary. This predicts the desired cumulative reading: for example, (59-c) is true in a context where Jane talked to Bill and Mary talked to Sue.

- (59) a.  $\llbracket \text{Co-DIST} \rrbracket =$   
 $\lambda P. \lambda y. \lambda x. \forall y' \leq_{AT} y [\exists x' \leq_{AT} x : P(y')(x') = 1] \wedge \forall x' \leq_{AT} x [\exists y' \leq_{AT} y : P(y')(x') = 1]$
- b. Jane and Mary talked to Bill and Sue
- c.  $(\llbracket \text{Co-DIST} \rrbracket (\llbracket \text{talked} \rrbracket)) (b \oplus s) (j \oplus m) = 1$  iff  
 $\forall y' \leq_{AT} b \oplus s [\exists x' \leq_{AT} j \oplus m : \llbracket \text{talked} \rrbracket (y')(x') = 1] \wedge \forall x' \leq_{AT} j \oplus m [\exists y' \leq_{AT} b \oplus s : \llbracket \text{talked} \rrbracket (y')(x') = 1]$

As argued in chapter 2, I assume that all pluralization/distributivity operators apply only to



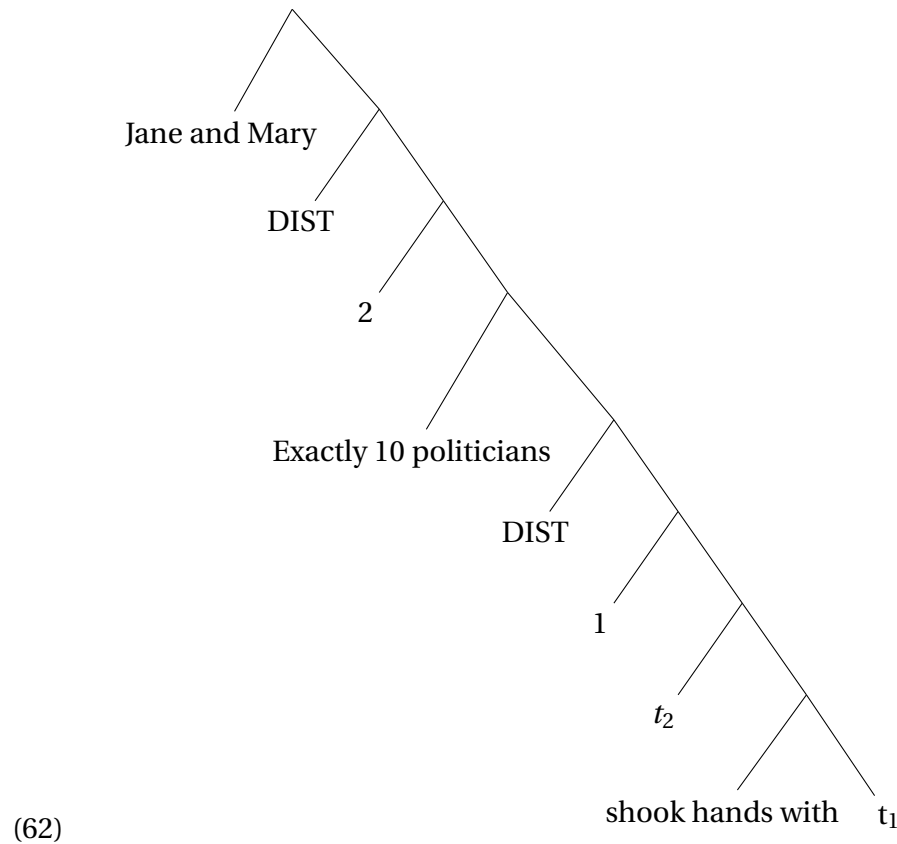
predicates derived by movement. Therefore, in order to generate the cumulative reading for (58), both the subject and the object have to move in order to create a 2-place derived predicate that Co-DIST can apply to. This LF is given in (60). Assuming the denotation in (61-a) for *exactly 10*, the truth-conditions in (61-b) are predicted: (60) is predicted to be true iff there are exactly 10 politicians such that at least one of Jane and Mary shook hands with each of them and each of Jane and Mary shook hands with at least one of them. This therefore predicts the desired cumulative reading.



- (61)
- a.  $\llbracket \text{exactly } 10 \rrbracket =$   
 $\lambda P_{\langle e, t \rangle}. \lambda Q_{\langle e, t \rangle}. \exists x [|x| = 10 \wedge P(x) = 1 \wedge Q(x) = 1] \wedge \neg \exists x [|x| = 11 \wedge P(x) = 1 \wedge Q(x) = 1]$
- b.  $\llbracket (60) \rrbracket =$   
 1 iff  $\exists z : |z| = 10 \wedge \llbracket \text{politicians} \rrbracket(z) \wedge \forall y' \leq_{AT} z [\exists x' \leq_{AT} j \oplus m : \llbracket \text{shook hands with} \rrbracket(y')(x')] \wedge$   
 $\forall x' \leq_{AT} j \oplus m [\exists y' \leq_{AT} z : \llbracket \text{shook hands with} \rrbracket(y')(x')] \wedge$   
 $\neg \exists z' : |z'| = 11 \wedge \llbracket \text{politicians} \rrbracket(z') \wedge \forall y' \leq_{AT} z' [\exists x' \leq_{AT} j \oplus m : \llbracket \text{shook hands with} \rrbracket(y')(x')] \wedge$   
 $\forall x' \leq_{AT} j \oplus m [\exists y' \leq_{AT} z' : \llbracket \text{shook hands with} \rrbracket(y')(x')]$

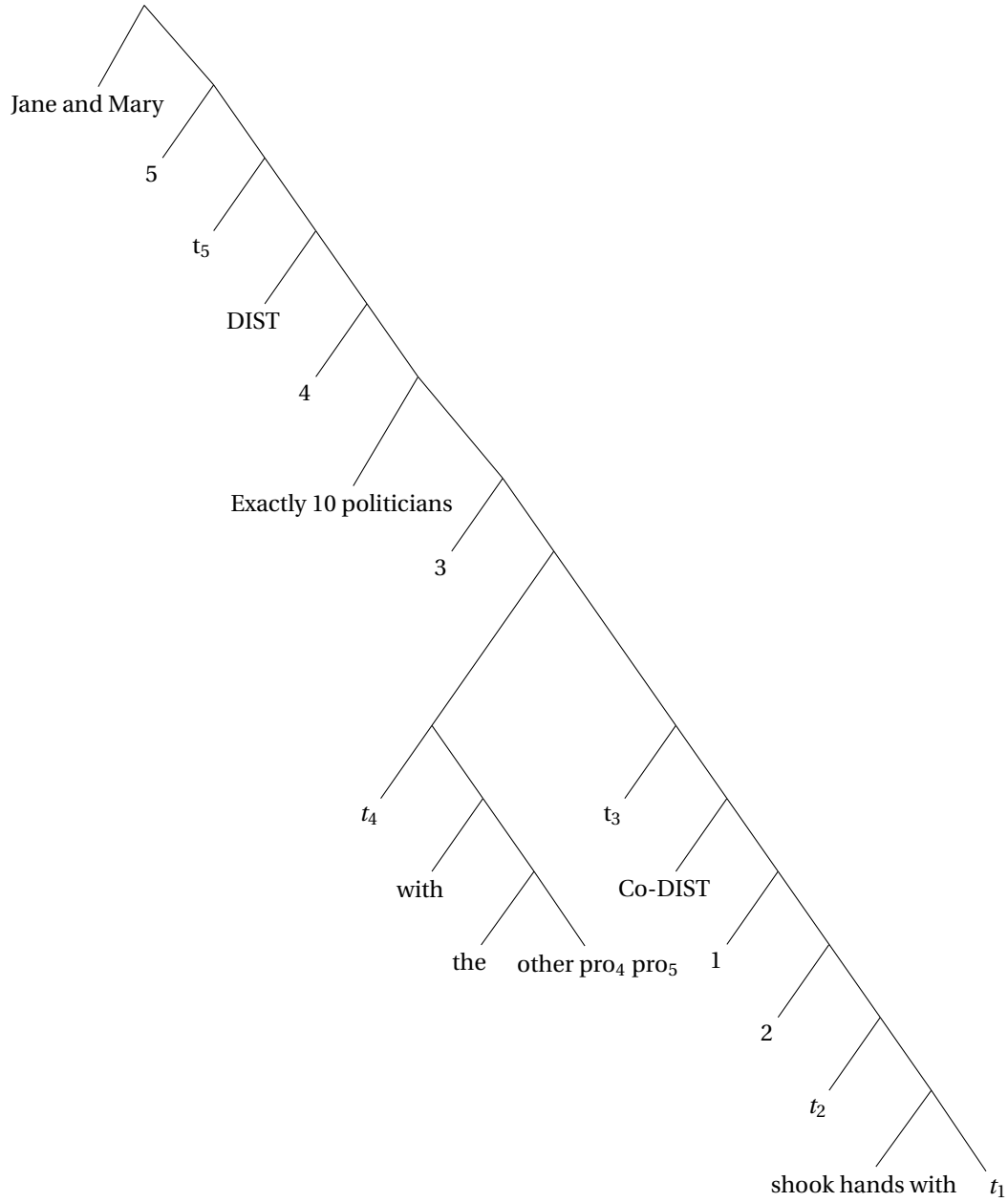
As for the distributive reading in (58-b), this can be derived by having a distributivity operator

take scope above the quantifier *exactly 10*, as shown in the LF in (62). (62) is predicted to be true iff for each of Jane and Mary there are exactly 10 politicians such that she shook hands with each of them.



As for the counterpart with *together* in (63-a), we can generate the desired cumulative reading by having the comitative conjunction *x with the other* take scope below a higher Distributivity operator but above the Co-DIST operator in (60). The relevant LF is given in (63-b) and the corresponding truth-conditions in (63-c): (63-b) is predicted to be true iff there are exactly 10 politicians such that Jane and Mary cumulatively shook hands with them.

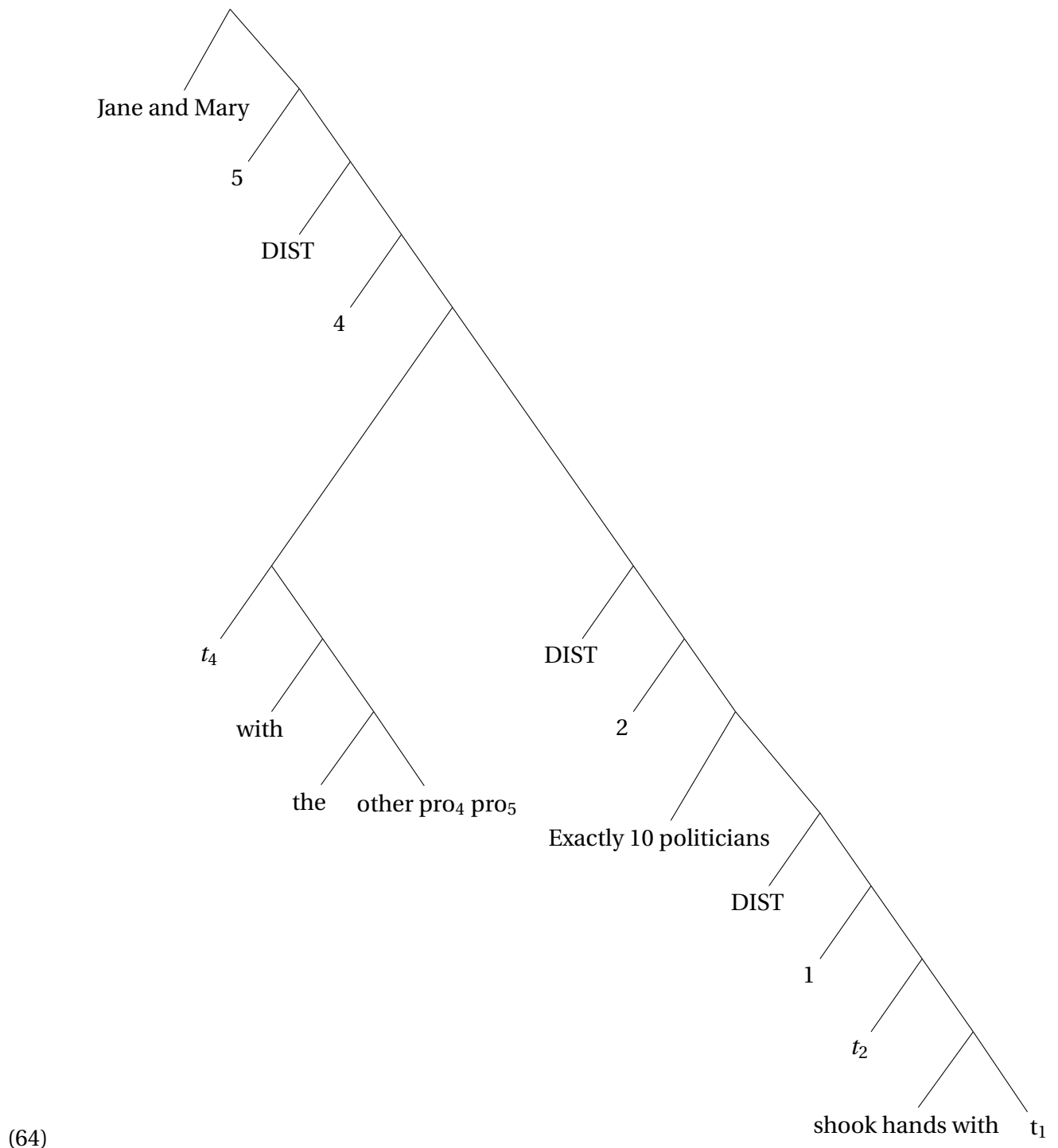
- (63) a. Jane and Mary together shook hands with exactly 10 politicians.



b.

c.  $\llbracket (63-a) \rrbracket = 1$  iff  $\llbracket \text{Exactly 10 politicians} \rrbracket (\lambda y. \text{Co-DIST} \llbracket \text{shook hands with} \rrbracket (y)(j \oplus m))$

Again there is an immediate question here of why *together* can't similarly take scope above the higher DIST operator in (62), where it would give rise to an unattested distributive reading. This unattested LF is given in (64). (64), like (62), is predicted to be true iff each of Jane and Mary shook hands with exactly 10 politicians. If we allow this LF for (63-a), we therefore incorrectly predict that, just like the counterpart without *together*, (63-a) has both a distributive and a cumulative reading. Again, I will propose in section 4.5 that an economy constraint on the insertion of *together* rules out the LF in (64), while allowing for the one in (63-b), therefore predicting that (63-a) only has a cumulative reading.



#### 4.4.6 Motivation for a semantic constraint

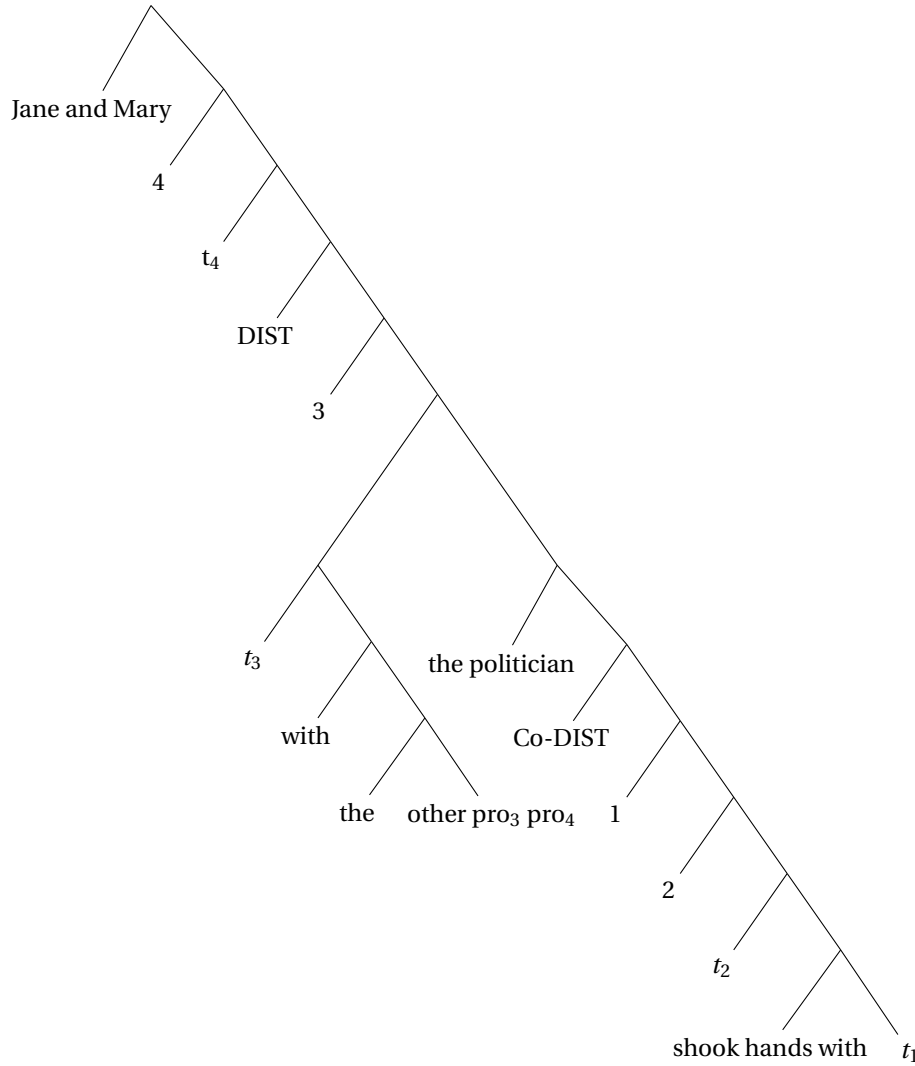
Before moving on to laying out the economy constraint on *together*, I want to show in this section that the distribution of *together* has to be constrained in a way that is sensitive to semantic properties of the object DP. In particular, given what we have seen so far, it is possible to

formulate, at least by stipulation, a purely syntactic constraint on the distribution of *together*. For example, if we assume that Co-DIST, Aspect, Location and the various distributivity operators all have fixed positions in the clause, then we can stipulate that *together* can't take scope above the highest distributivity operator that takes scope above all of them. This will ensure that *together* lacks the higher distributive reading in all of the examples we have seen so far, allowing us to rule out unattested readings of *together*. Of course, this would raise questions about what the source of this constraint might be. Setting aside possible conceptual issues, the goal of this section is to show that a purely syntactic constraint will incorrectly predict that two-place predicates with definite singular objects like (65) allow for distributive readings with *together*.

(65) Jane and Mary shook hands with the politician together.

- a. **Distributive Scenario:** Jane shook hands with the politician and then Mary independently shook hands with the politician. **False**
- b. **Collective Scenario:** Jane and Mary approached the politician together and jointly shook hands with him. **True**

In particular, we saw in section 4.4.5 that in order to derive the cumulative reading of *together*, we have to allow for *together* to take scope above a co-distributivity operator. I will show here that if we allow *together* to similarly take scope above Co-DIST in (65), we would incorrectly predict that (65) has a reading where it is true in the distributive scenario in (65-a). To see this, consider the potential LF for (65) in (66-a). Here, just like in the cumulative reading derived in section 4.4.5, the *with the other* conjunction takes scope above co-DIST. Now, given that the object [the politician] denotes an atomic individual, the truth-conditions predicted by co-DIST are the same as those that we would get by simply distributing over the subject. This is shown in the predicted truth-conditions in (66-b): (66-a) is predicted to be true iff each of Jane and Mary shook hands with the politician. The LF in (66-a) therefore derives an unattested distributive reading.



- (66) a.
- b.  $\llbracket (66-a) \rrbracket = 1$  iff  $\forall y \leq_{AT} \llbracket \text{the politician} \rrbracket [\exists x \leq_{AT} j \oplus m : \llbracket \text{shook hands with} \rrbracket (y)(x) = 1] \wedge \forall x \leq_{AT} j \oplus m [\exists y \leq \llbracket \text{the politician} \rrbracket : \llbracket \text{shook hands with} \rrbracket (y)(x) = 1]$   
iff  $\llbracket \text{shook hands with} \rrbracket (\llbracket \text{the politician} \rrbracket)(j) = 1 \wedge \llbracket \text{shook hands with} \rrbracket (\llbracket \text{the politician} \rrbracket)(m) = 1$   
iff each of Jane and Mary shook hands with the politician.

We can therefore conclude that whether or not *together* can move above Co-DIST has to depend on semantic properties of the object DP. Namely, when the object is a numeral quantifier as in (58), movement above Co-DIST is needed to derive the desired cumulative reading. On the other hand, when the object is a definite singular as in (66-a), movement above Co-DIST has to be blocked in order to avoid generating an unattested distributive reading. I therefore take this contrast to provide compelling evidence that the distribution of *together* is not mediated by a purely syntactic constraint that determines which syntactic positions it can move to. Rather, the distribution has to also be sensitive to the semantic properties associated with different LFs. In the next section I will argue that an economy condition on *together* provides exactly the right

type of constraint and predicts the attested distribution of *together*.

## 4.5 The role of economy

In this section, I will propose that the distribution of *together* is governed by an economy constraint that rules out LFs where deleting *with the other* does not change the truth-conditions. At this point, I assume that this constraint applies only to the deletion of *with the other* and illustrate how this straightforwardly rules out unattested distributive LFs, while allowing for all of the attested readings of *together* discussed in sections 4.3 and 4.4. I return to the role of economy in more detail in chapter 5, where I propose a generalized economy constraint, inspired by Fox's (2000) scope economy, that rules out any LFs which are equivalent to simpler LFs.<sup>4</sup>

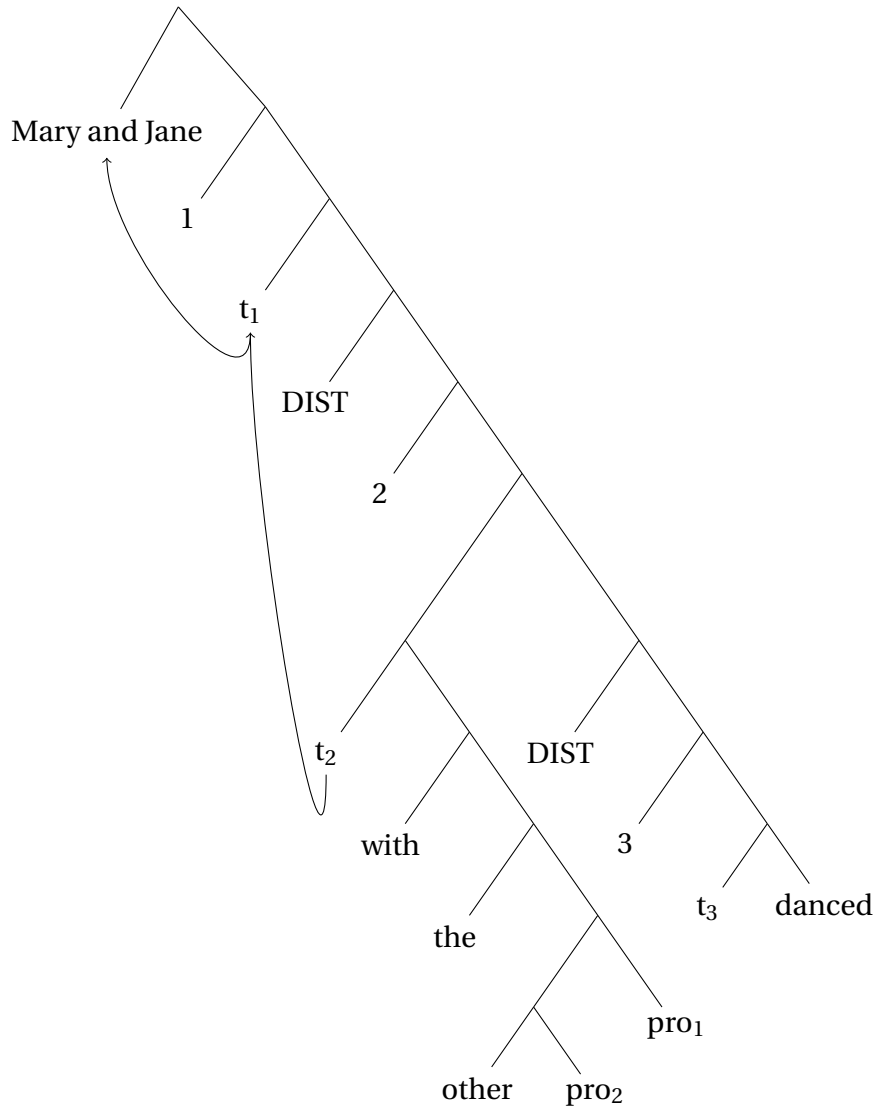
Consider again the LF in (67), repeated from (31), which derives an unattested distributive reading with *together*. In particular, the *with the other* conjunction in (67-a) takes scope above a DIST operator, without taking scope below Aspect or LOC. The resulting truth-conditions are given in (67-b): (67-a) is predicted to be true iff each of Jane and Mary danced. As mentioned in section 4.3, it is crucial to rule out this reading with *together*. Otherwise, we end up predicting that the *x with the other* conjunction has the same range of readings that standard conjunction has.

### (67) Unattested LF:

---

<sup>4</sup>At this point, it might seem that this economy constraint is very similar to pragmatic redundancy principles, motivated by the Gricean Maxim of Manner, which have been proposed to rule out LFs which have structurally simpler alternatives that are contextually equivalent to them (Katzir, 2007; Meyer, 2013; Haslinger, 2024a, a.o.). I argue in chapter 5 that the economy constraint with *together* has to crucially be modular in nature, such that it is blind to the context and to a variety of other information. The resulting constraint therefore needs to be a grammatical economy constraint, like scope economy, rather than a pragmatic one.

a.



b.  $\llbracket (67-a) \rrbracket = 1$  iff  $\llbracket \text{dance} \rrbracket(j) = 1 \wedge \llbracket \text{dance} \rrbracket(m) = 1$

I propose that there is an economy constraint, given in (68), that rules out LFs in which *with the other* can be deleted without changing the truth-conditions.

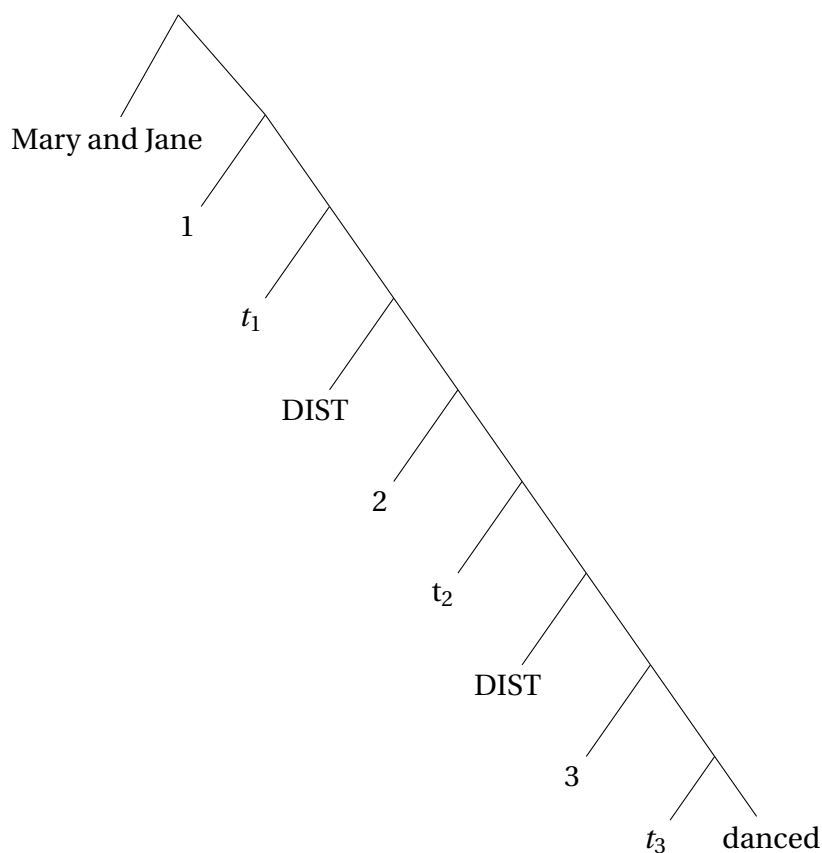
(68) **Economy constraint on *together*:** An LF  $\phi$  with [ with[the [other  $\text{pro}_i$   $\text{pro}_j$ ]]] is ungrammatical if it is truth-conditionally equivalent to the LF that results from deleting [ with[the [other  $\text{pro}_i$   $\text{pro}_j$ ]]] from  $\phi$ .

To see how (68) rules out the unattested LF in (67-a), consider the LF in (69-a), which results from deleting the node dominating *with the other* from (67-a). Just like (67-a), (69-a) is predicted to be true iff each of Jane and Mary danced (69-b). We therefore see that deleting *together* from (67-a) results in equivalent truth-conditions. Thus, the economy constraint in (68) successfully rules



out the LF in (67-a), correctly predicting that *together* does not have the resulting distributive reading.

(69) a.

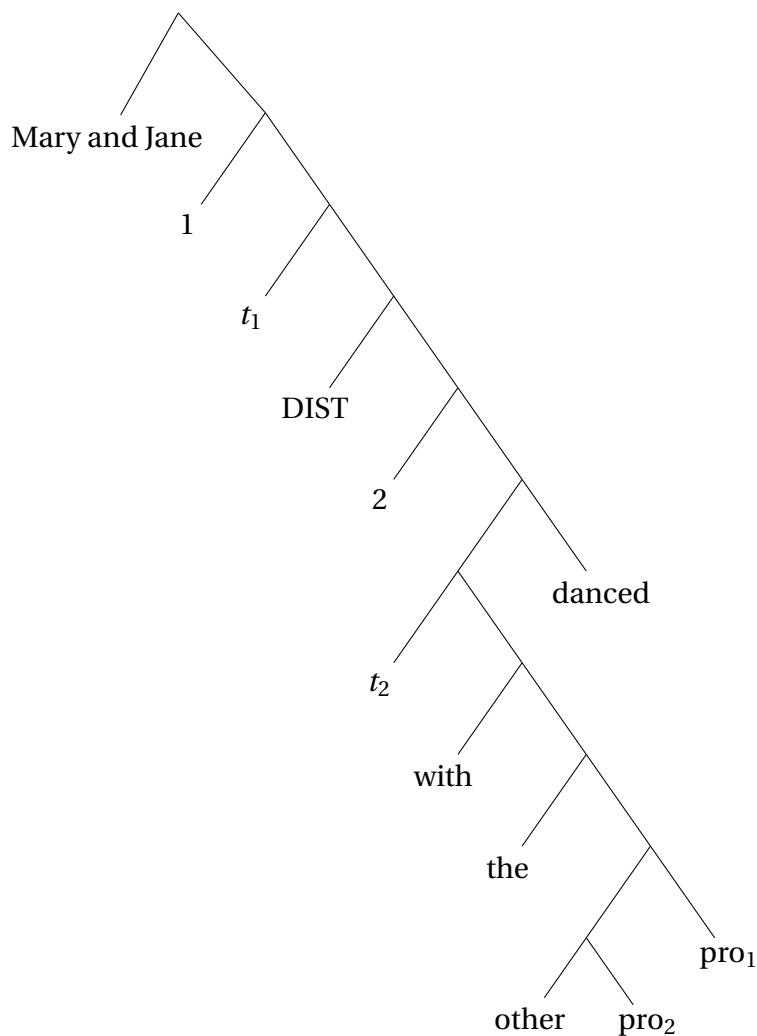


b.  $\llbracket (69-a) \rrbracket = 1$  iff  $(\llbracket \text{dance} \rrbracket(j)=1 \wedge \llbracket \text{dance} \rrbracket(m)=1)$   
iff Jane danced and Mary danced

At this point, there is a question of whether (68) can be seen as a special case of a more general economy constraint that rules out LFs which are truth-conditionally equivalent to simpler LFs. In particular, I show in chapter 5 that the economy constraint on *together* shares key properties with other economy constraints that have been proposed in the literature, such as scope economy (Fox, 2000), thus providing motivation for a more general principle. Unfortunately, as things stand now, this generalized economy principle seems to rule out all of the readings of *together*, including the attested collective reading. For example, the same collective truth-conditions obtained in (70-a) can arise with an LF where both *with the other* and DIST are deleted, thus violating this more general economy constraint. I return to this issue in chapter 5 and argue that the final version of the analysis with \* allows us to formulate a modular generalized economy constraint that successfully predicts the desired distribution of *together*.

(70) **Collective LF with *together*:**

a.

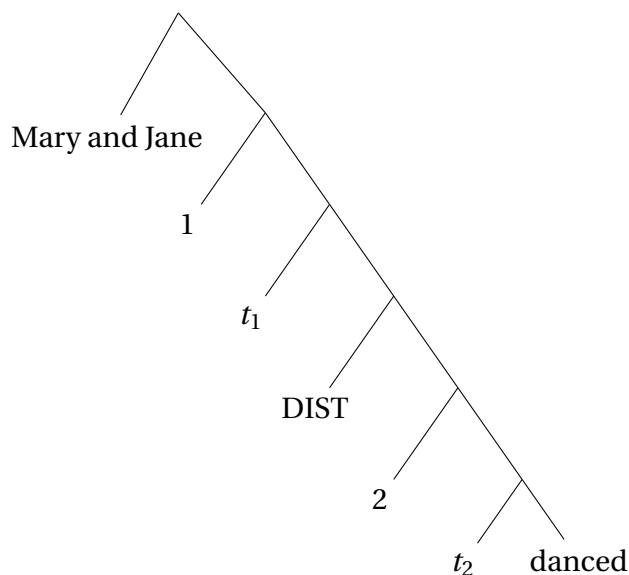


b.  $\llbracket (70-a) \rrbracket = 1$  iff  $\llbracket \text{dance} \rrbracket (j \oplus m) = 1$  iff Jane and Mary danced together.

As for the other attested readings of *together*, I will show below that they are all licensed by the economy constraint, as desired. Starting with the collective reading, the LF with *together* is repeated in (70-a), which derives the collective truth-conditions in (70-b). Deleting *with the other* results in the LF in (71-a), which is predicted to be true iff each of Jane and Mary danced (71-b). We therefore see that deleting *with the other* changes the truth-conditions here, thus predicting that (70-a) is licensed by the economy constraint.

(71) **Result of deleting *together*:**

a.

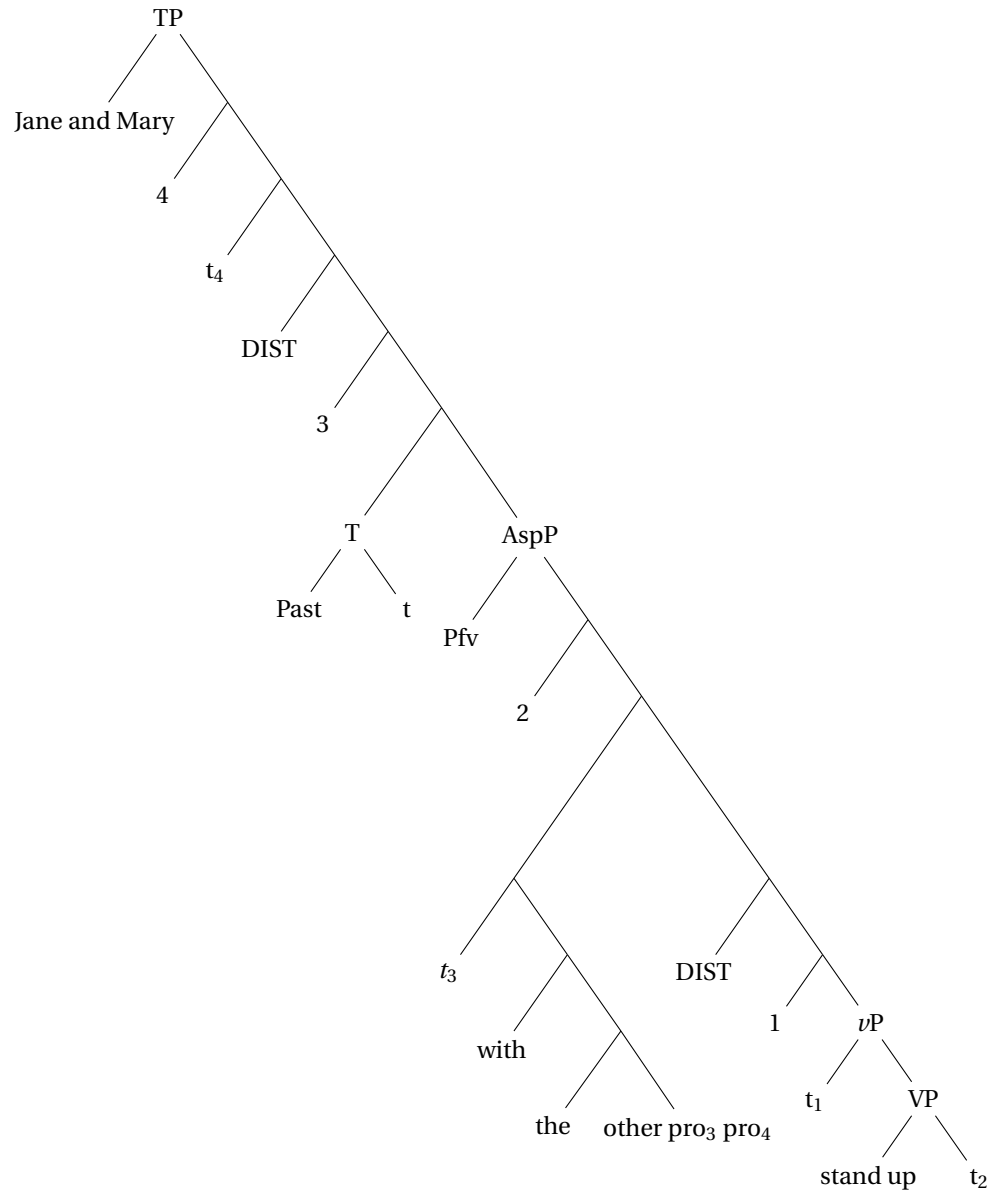


b.  $\llbracket (71-a) \rrbracket = 1$  iff  $(\llbracket \text{dance} \rrbracket(j)=1 \wedge \llbracket \text{dance} \rrbracket(m)=1)$  iff Jane danced and Mary danced.

The same reasoning extends to the simultaneous and the proximate readings: in both cases, deleting *together* results in distinct truth-conditions. Since both of these readings involve the same type of existential quantification but in different domains, I will simply show this for the simultaneous reading here. Consider the LF for the simultaneous reading in (72-a) and the result of deleting *together* in (73-a). While (72-a) is true only if Jane and Mary stood up at the same time (72-b), the counterpart without *together* in (73-a) is true as long as they each stood up at some time. The economy constraint therefore successfully licenses the simultaneous LF in (72-a).

(72)

a.

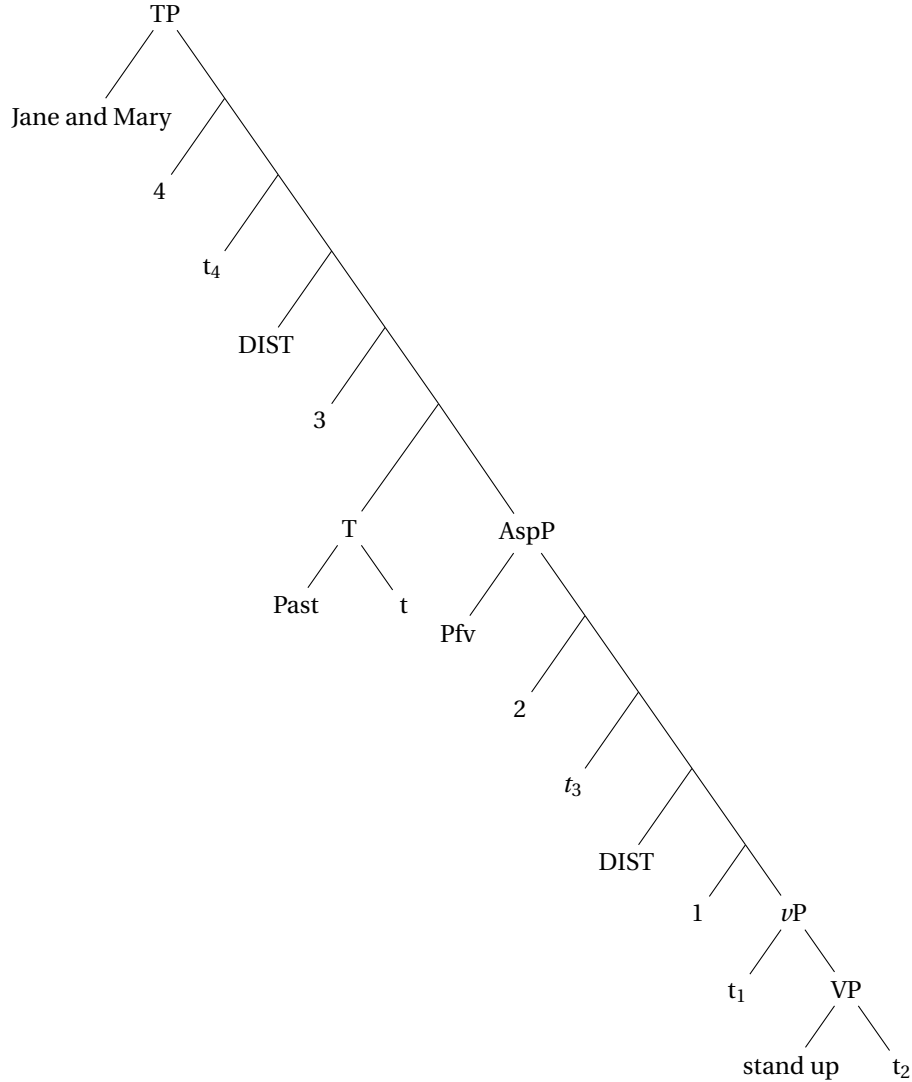


b.  $\llbracket (72-a) \rrbracket = 1$  iff  $\exists t' \subseteq t$ :  $\llbracket \text{stood up} \rrbracket (t')(j) = 1 \wedge \llbracket \text{stood up} \rrbracket (t')(m) = 1$

(73)

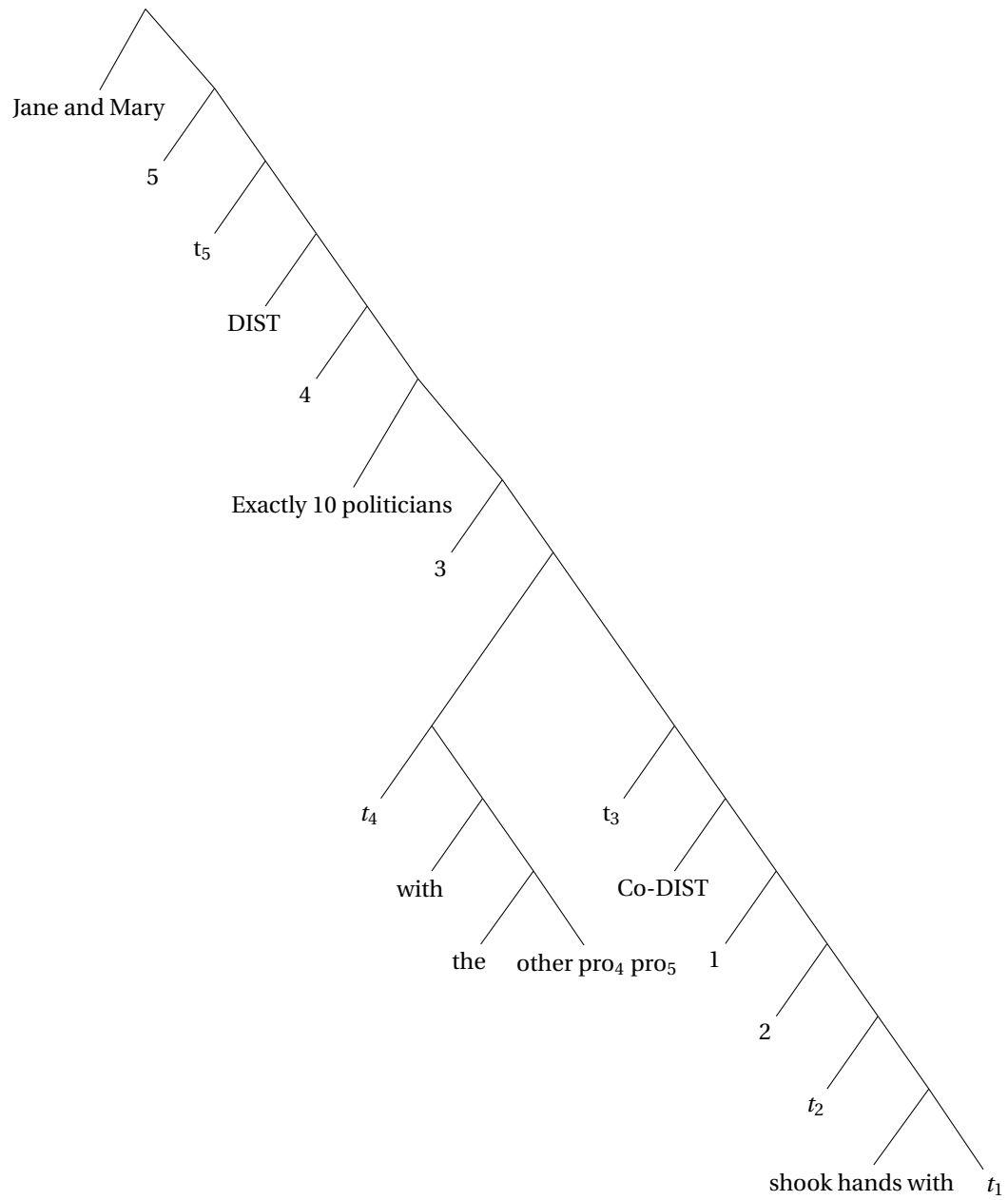
**Result of deleting *together*:**

a.

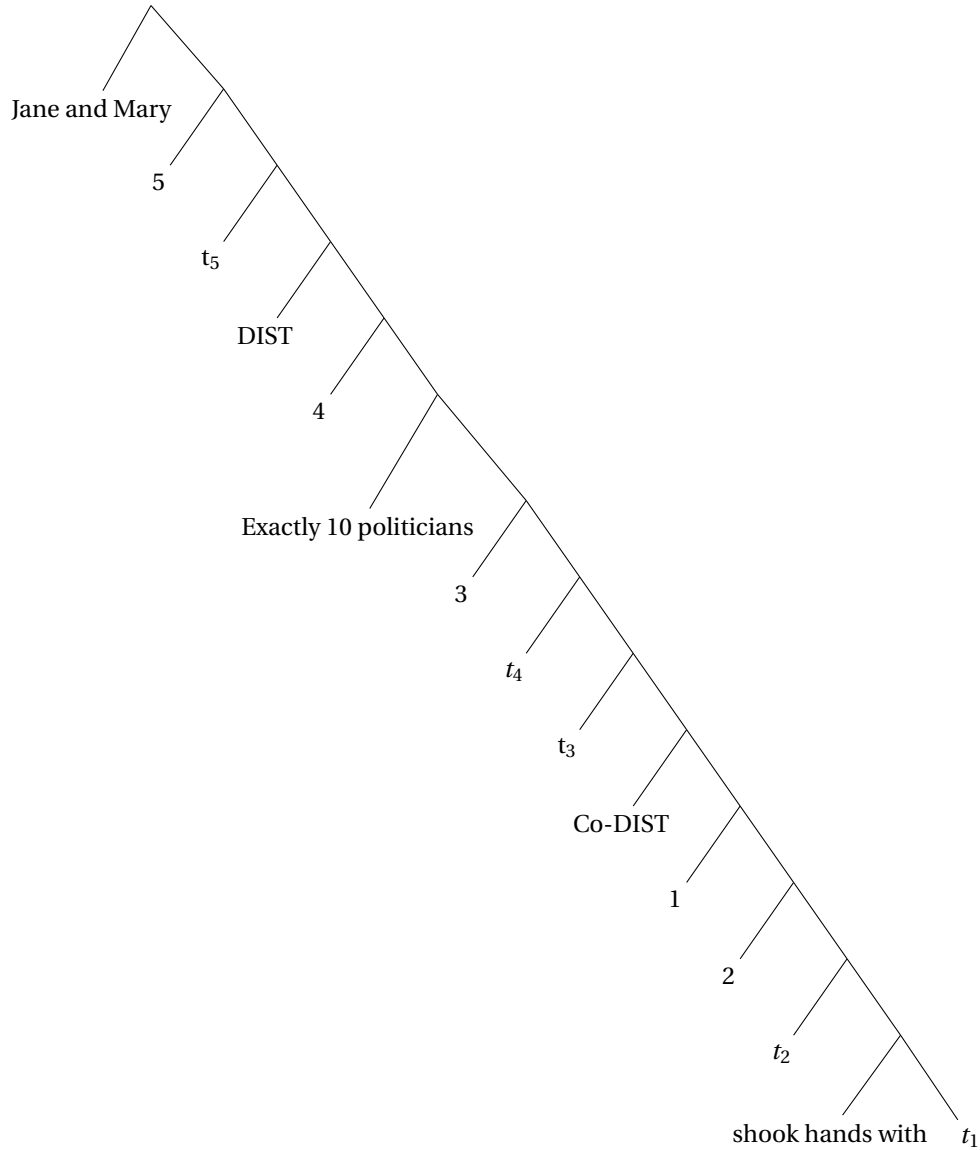


b.  $\llbracket (73\text{-a}) \rrbracket = 1$  iff  $\exists t' \subseteq t : \llbracket \text{stood up} \rrbracket(t')(j) = 1 \wedge \exists t'' \subseteq t : \llbracket \text{stood up} \rrbracket(t'')(m) = 1$

Finally, we can similarly see that the economy constraint licenses the cumulative LF for *together*, when there is a plural DP in object position. For example, when the object is *exactly 10 politicians*, as in (74-a), the LF with *together* is true iff Jane and Mary cumulatively shook hands with exactly 10 politicians (74-b), while the counterpart without *together* in (75-a) is true iff each of Jane and Mary shook hands with exactly 10 politicians. We therefore again see that the LF where *with the other* is deleted gives rise to different truth-conditions, thus predicting that (74-a) satisfies the economy constraint.

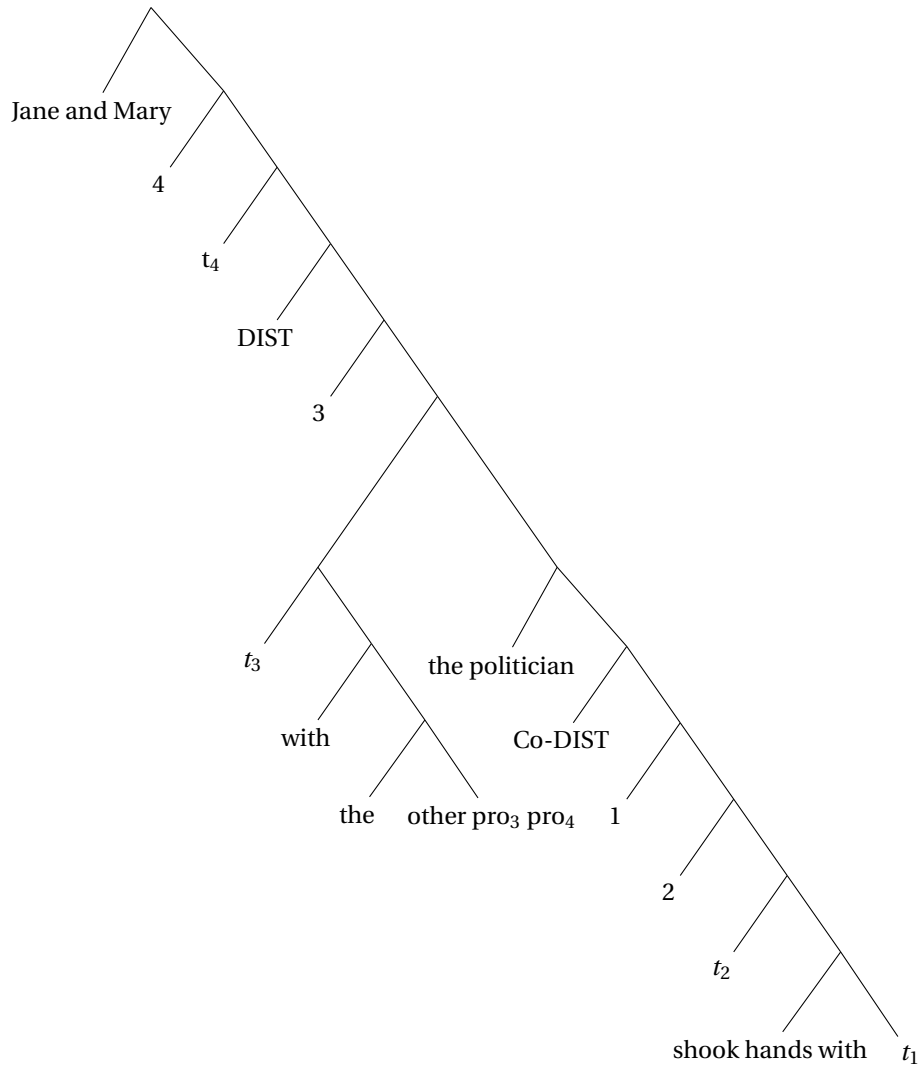


- (74) a.
- b.  $\llbracket (74\text{-a}) \rrbracket = 1$  iff  $\llbracket \text{Exactly 10 politicians} \rrbracket (\lambda y. (\text{Co-DIST}(\llbracket \text{shook hands with} \rrbracket)))(y)(j \oplus m))$



- (75) a.
- b.  $\llbracket (75-a) \rrbracket = 1$  iff  $\llbracket \text{Exactly 10 politicians} \rrbracket (\lambda y. (\text{Co-DIST}(\llbracket \text{shook hands with} \rrbracket))(y)(j)) = 1$   
 $\wedge \llbracket \text{Exactly 10 politicians} \rrbracket (\lambda y. (\text{Co-DIST}(\llbracket \text{shook hands with} \rrbracket))(y)(m)) = 1$   
 iff Jane shook hands with exactly 10 politicians and Mary shook hands with exactly 10 politicians.

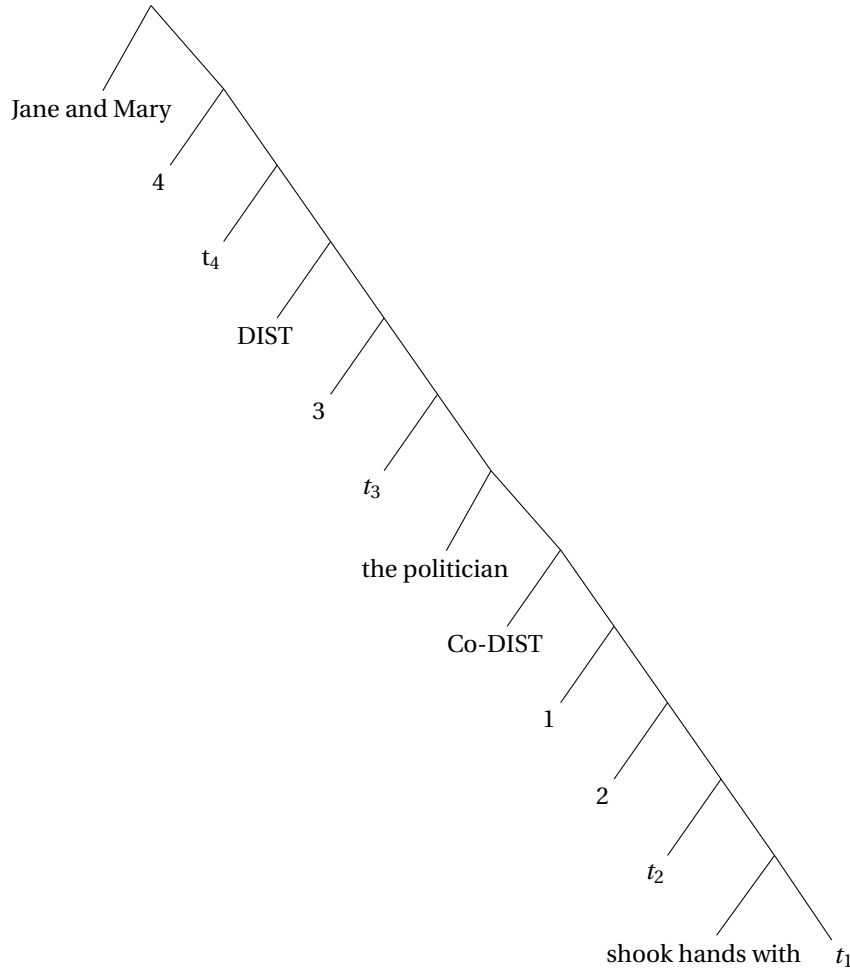
Crucially, we saw at the end of section 4.4 that whether or not the LF with *together* taking scope above Co-DIST is licensed has to depend on semantic properties of the object DP. In particular, when the object is a definite singular, we saw that allowing *with the other* to move above Co-DIST, as in (76-a), predicts an unattested distributive reading in (76-b), where *Jane and Mary shook hands with the politician together* is incorrectly predicted to be true in a scenario where each of Jane and Mary shook hands with the politician separately.



- (76) a.  
 b.  $\llbracket (76-a) \rrbracket = 1$  iff  
 $\llbracket \text{shook hands with} \rrbracket (\llbracket \text{the politician} \rrbracket)(j)=1 \wedge \llbracket \text{shook hands with} \rrbracket (\llbracket \text{the politician} \rrbracket)(m)=1$   
 iff each of Jane and Mary shook hands with the politician.

Consider the counterpart of (76-a) where *with the other* is deleted in (77-a). Just like (76-a), (77-a) is predicted to be true iff each of Jane and Mary shook hands with the politician (77-b). We can therefore conclude that the economy constraint successfully rules out the unattested LF in (76-a). We therefore see that the economy constraint, unlike a purely syntactic constraint, predicts that whether or not *together* is licensed in a certain position can depend on semantic properties of the sentence, for example whether or not the object is a definite singular or a numeral quantifier.





- (77) a.  
 b.  $\llbracket (77-a) \rrbracket = 1$   
 iff  $\llbracket \text{shook hands with} \rrbracket (\llbracket \text{the politician} \rrbracket)(j) = 1 \wedge \llbracket \text{shook hands with} \rrbracket (\llbracket \text{the politician} \rrbracket)(m) = 1$

At this point, the economy constraint predicts that cumulative *together* is licensed whenever the object does not denote an atomic individual. In particular, we saw in (76-a) and (77-a) that when the object was a definite singular, like *the politician*, the co-distributive reading over the object and the subject and the higher distributive reading end up being equivalent. This is because Co-DIST simply reduces to distributing over the plural subject. The account so far therefore predicts that if we replace the definite singular object with a definite plural, we should again be able to get a cumulative reading with *together*, just as in the case of *exactly 10* above. This prediction is not borne out. In particular, consider the cumulative context in (78) again: while the cumulative *together* is available with a numeral quantifier object (78-a) and with a universal quantifier object (78-c), the definite plural counterpart in (78-b) does not seem to have a cumulative reading.

- (78) **Context:** There are 10 politicians at this conference. Jane walks in and shakes hands with

5 of the 10 politicians. Then Jane walks out, and Mary, who doesn't know Jane, walks in and shakes hands with the other 5 politicians.

- a. Jane and Mary together shook hands with exactly 10 politicians.
- b. #Jane and Mary together shook hands with the politicians at the conference.
- c. Jane and Mary together shook hands with all of the politicians at the conference.

I will return to the pattern in (78) in chapter 5, after I lay out the final version of the proposal with pluralization operators rather than distributivity operators. I provide additional data showing that this pattern is part of a systematic generalization that cumulative *together* is not licensed with objects that give rise to homogeneity. I then argue that the economy constraint is able to predict this generalization, thus providing additional evidence that the licensing of *together* is sensitive to semantic properties of the object.

To summarize, I proposed in this section an economy constraint on the distribution of *together* and showed that this constraint correctly rules out unattested distributive readings of *together* with both 1-place and 2-place predicates. At the same time, this constraint allows for all of the attested readings of *together*, including the collective, simultaneous, proximate and cumulative readings. Finally, I showed that the distribution of cumulative *together* is more subtle than what is predicted by the preliminary version of the proposal given in this section.

## 4.6 Conclusion

The goal of this chapter was to provide a compositional analysis of *together* as the spell-out of a comitative coordinator *with* and a reciprocal anaphor *each other*. This compositional analysis was motivated by the fact that in several languages *together* seems to be overtly spelled out as *with each other*. I showed that this analysis, when paired with an economy constraint on the insertion of *together*, provides an explanatory account of the different readings of *together* while ruling out unattested readings. In particular, I showed that the different readings of *together* correspond to different positions where *together* can take scope, thus allowing us to predict the desired ambiguity without positing lexical ambiguity in the meaning of *together*. At the same time, I proposed that the economy constraint offers a predictive theory of which positions *together* is allowed to take scope in, thus allowing us to rule out unattested distributive readings with *together*. The resulting account is one where there is nothing inherently collective or inherently cumulative in the lexical meaning of *together*. Rather, *together* simply denotes a conjunction *x with the other* and the fact that *together* forces a collective reading in sentences like *Jane and Mary danced together* is due to the interaction between the economy constraint and the different positions that the conjunction can take scope in, relative to pluralization operators and other

scope-taking elements.

The preliminary account provided in this chapter therefore illustrates the core insights of the decompositional analysis of *together*. At the same time, the proposal presented here faces two key problems, mentioned in section 4.5. First, this chapter relied on a specific economy constraint that only looks at LFs where *together* is deleted, raising the question of how general this constraint can be and whether it applies beyond the distribution of *together*. Second, this preliminary account proposed here predicts that cumulative *together* is always licensed and is therefore unable to make the fine-grained distinctions required to allow the licensing of cumulative *together* to depend on the properties of the object DP. The goal of chapter 5 is to address these two issues. I argue that once we replace DIST with \*, we can formulate a generalized economy constraint that is crucially modular and has access to only a subset of the information that is needed to compute truth-conditions. I then show that this modular economy constraint is able to predict the more fine-grained licensing conditions on cumulative *together*, while maintaining the key results from the preliminary analysis presented in this chapter.



# Chapter 5

## Economy and the distribution of *together*

### 5.1 Introduction

In chapter 4, I gave a preliminary version of my proposal for the meaning and distribution of *together* which assumed that distributive readings are due to the distributivity operator DIST. First, I argued for a compositional analysis of *together* and showed that the different readings of *together* result from *together* taking scope in different positions. I then proposed that there is an economy constraint that rules out LFs where the contribution of *together* is trivial and showed that this constraint plays a crucial role in ruling out unattested distributive readings with *together*.

The proposal in chapter 4, however, left two key issues unresolved. The first issue concerns the scope of the economy constraint. Namely, I proposed a specific economy constraint that applies to the deletion of *together*. This raises the question of whether we can derive this constraint from a more general notion of economy in the grammar, one that subsumes other independently motivated economy constraints, such as Fox's (2000) scope economy. The second issue concerns the licensing of cumulative *together*. As discussed at the end of chapter 4, the distribution of cumulative *together* is more fine-grained than what the preliminary economy constraint predicted, raising the question of whether economy is sufficient to predict the full distribution of readings with *together*.

The goal of this chapter is to show that both issues are resolved once we replace the DIST operator with the pluralization operator \*, which I provided independent arguments for in chapter 2. To address the first issue, I propose that the economy constraint on *together* is a special case of a more general economy constraint that rules out LFs which are equivalent to simpler LFs. As discussed in chapter 4, this general economy constraint makes incorrect predictions when paired with the preliminary proposal with DIST, ruling out even attested non-distributive readings with *together*. I show that this issue is resolved if (i) the proposal is recast

with \* and (ii) this general economy constraint is modular, such that it only has access to a subset of the information that goes into computing truth-conditions.

Turning to cumulative *together*, I show that without additional mechanisms, generalized economy incorrectly predicts that cumulative *together* is only licensed when the object is a non-monotonic quantifier. I propose that with upward-monotonic quantifiers, this economy violation can be obviated by computing a local implicature below the pluralization operator \*. This proposal makes an interesting prediction: when this local implicature is blocked, cumulative *together* will no longer be licensed. I argue that this prediction is borne out with certain objects that block the local implicature, such as definite plural objects, thus providing additional evidence for the crucial role of the economy constraint.

The rest of the chapter is structured as follows. Section 5.2 proposes the generalized economy constraint and shows that this constraint needs to be modular in order to predict the desired distribution of readings with *together*. I then show that generalized economy seems to predict that cumulative *together* is only licensed with non-monotonic objects, posing a challenge for the account. Section 5.3 argues that this challenge can be resolved by computing a local implicature below \*. Section 5.4 shows that objects which give rise to homogeneity appear to disallow a cumulative reading with *together*. I argue that this is because homogeneity blocks the local implicature which is necessary to avoid violating the economy constraint, thus predicting the relationship between homogeneity and cumulative *together*. Finally, section 5.5 discusses in more detail the variety of evidence that the economy constraint has to be modular in the same way as scope economy. I therefore suggest in this section that both scope economy and economy with *together* should be seen as special cases of the same generalized economy constraint.

## 5.2 Generalized economy and the distribution of *together*

The goal of this section is to argue that the distribution of readings with *together* can be predicted by a generalized economy constraint, given in (1), which rules out LFs which can be proven to be equivalent to simpler LFs.

- (1) a. **Generalized Economy Constraint:**  
An LF  $\phi$  is ruled out if the deductive system can prove that there is a competitor  $\psi$  such that (i)  $\psi < \phi$  and (ii)  $\forall w, g : \llbracket \psi \rrbracket^{w,g} = \llbracket \phi \rrbracket^{w,g}$
- b. An LF  $\psi < \phi$  if  $\psi$  can be derived from  $\phi$  by deleting a number of nodes.

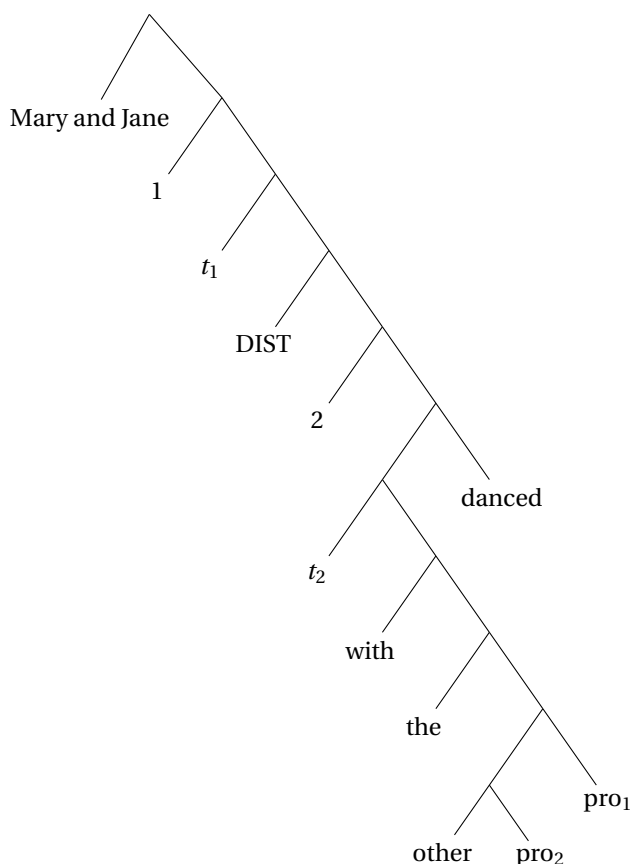
(Adapted from Meyer, 2013, which is in turn based on Katzir, 2007)

At this point, this constraint looks very similar to pragmatic constraints motivated by the Maxim of Quantity and the Maxim of Manner, which rely on the same notion of structural complexity

to rule out LFs which are contextually equivalent to simpler LFs (Katzir, 2007; Meyer, 2013; Haslinger, 2024a, a.o.). I will argue in this section and in section 5.5 that the constraint in (1), unlike these pragmatic constraints, must be thought of as a grammatical constraint that is blind to contextual parameters and to a variety of other information that goes into computing truth-conditions.

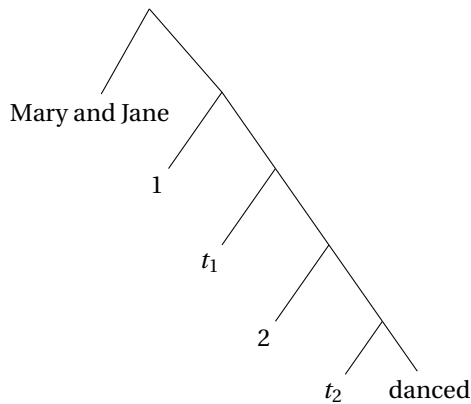
As discussed briefly in chapter 4, the main challenge for applying a constraint like (1) to the distribution of *together* is that (1) seems to be too strong, incorrectly ruling out even the attested readings with *together*. To see this again, consider the LF for the collective reading of *together*, repeated from chapter 4 in (2).

(2) **Collective LF with *together*:**



In order to ensure that (2) satisfies the generalized economy constraint in (1), we have to make sure that there is no deletion alternative for (2) that gives rise to the same collective truth-conditions. Consider the LF in (3), where both *with the other* and DIST are deleted from (2).

(3) **Result of deleting *together* and DIST:**



We can straightforwardly see that (2) and (3) are truth-conditionally equivalent, regardless of the choice of predicate and the choice of subject. In particular, *together* essentially undoes the contribution of the higher DIST in (2). Deleting both DIST and *together* therefore results in the same collective truth-conditions in (3). The collective LF in (2) is therefore ruled out by the generalized economy constraint. In order to predict that the desired collective reading of *together* is licensed, we therefore have to find an alternative LF which generates this reading without violating generalized economy.

In what follows, I will show that replacing DIST with \* allows for the generalized economy constraint to license the collective reading, once we make appropriate assumptions regarding what information the economy constraint has access to when proving that two LFs are equivalent.<sup>1</sup>

### 5.2.1 Recasting the proposal with \*

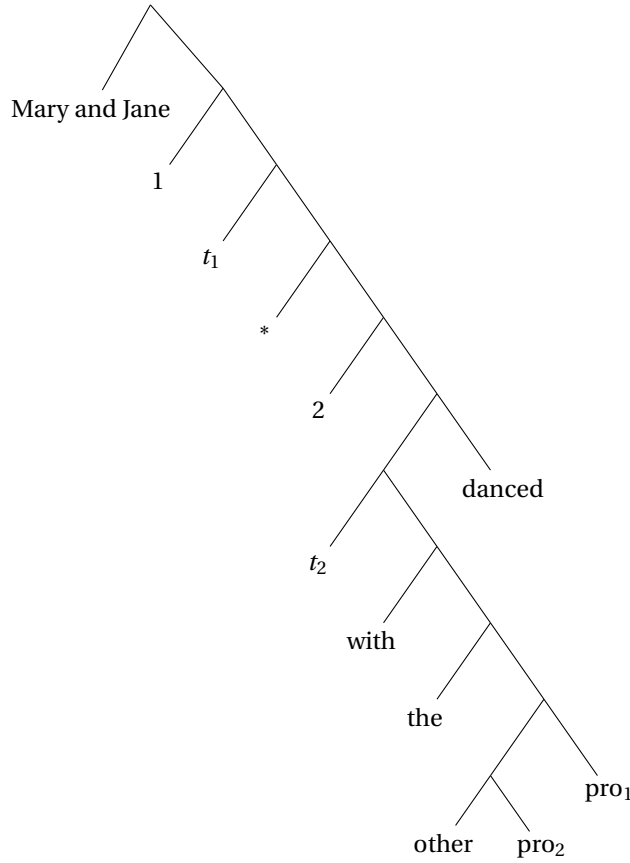
In order to show how replacing DIST with \* resolves the issue with the generalized economy constraint, we have to first show that the different readings generated in chapter 4 can be similarly derived with \*. I show in this section that the shift from DIST to \* results in a complication with respect to how the presupposition of *the other* projects from the scope of \*. I propose that this issue can be resolved by locally accommodating the presupposition below \*, thus allowing us to predict the same truth-conditions that were derived with DIST in chapter 4. I then show that \* predicts additional partially distributive readings with *together*, which will turn out to be crucial when resolving the issue with generalized economy in section 5.2.2.

Consider first the collective LF with \* in (4-a), where *together* takes scope below all pluralization operators.

<sup>1</sup>I set aside the possibility that the lexicon has both DIST and \*. I discuss this issue in more detail in Appendix C, where I show that generalized economy predicts the correct results, even if we allow for a DIST operator in addition to \*.



(4) a.



b.  $\llbracket (4-a) \rrbracket = 1$  iff

$$(\llbracket \text{dance} \rrbracket (m \oplus \llbracket \text{the} \rrbracket (\llbracket \text{other} \rrbracket (\mathbf{m})(j \oplus \mathbf{m}))) = 1 \wedge \llbracket \text{dance} \rrbracket (j \oplus \llbracket \text{the} \rrbracket (\llbracket \text{other} \rrbracket (\mathbf{j})(j \oplus \mathbf{m}))) = 1) \\ \vee \llbracket \text{dance} \rrbracket (m \oplus j \oplus \llbracket \text{the} \rrbracket (\llbracket \text{other} \rrbracket (\mathbf{j} \oplus \mathbf{m})(j \oplus \mathbf{m}))) = 1$$

Now, recall that the *the other* DP is only defined if the first argument of *other* is a proper part of the second one (5-a). While this presupposition is satisfied in the first disjunct in (4-b), where the first argument in bold is a subpart of the second, it is not satisfied in the second disjunct where both arguments are  $j \oplus m$ . In order to determine the resulting truth-conditions corresponding to (4-a), we therefore have to ask how this presupposition projects out of the scope of  $*$ .

(5) a.  $\llbracket \text{the} \rrbracket (\llbracket \text{other} \rrbracket (x)(y)) = y \ominus x$ , defined iff  $x < y$ .

b.  $\llbracket \text{the} \rrbracket (\llbracket \text{other} \rrbracket (j \oplus m)(j \oplus m))$  is undefined because  $j \oplus m$  is not a proper part of itself.

I assume that presuppositions project with a Strong Kleene logic, such that we get the partial truth-table in (6) for disjunction. Since the second disjunct is always undefined, we therefore predict that (4-b) is true when the first disjunct is true and undefined when the first disjunct is false. Strong Kleene therefore predicts the trivalent truth-conditions in (7-a) and the corresponding presupposition in (7-b) for (4-a).

(6) **Strong Kleene disjunction:**

| p | q | $p \vee q$ |
|---|---|------------|
| 1 | # | 1          |
| 0 | # | #          |
| # | # | #          |

- (7) a.  $\llbracket (4-a) \rrbracket = \begin{cases} 1 \text{ iff } \llbracket \text{danced} \rrbracket(j \oplus m) \\ \# \text{ otherwise} \end{cases}$   
b. **Presupposition:** Jane and Mary danced together.

Now, it is clear that this is not a desirable result. In particular, the meaning in (7-a) is a pathological one, which is tautological in any context where the presupposition is met. If this presupposition projects from the scope of negation, we would predict the truth-conditions in (8-b) for (8-a). In particular, (8-a) is incorrectly predicted to be a contradiction, since (8-b) is never true.

- (8) a. Jane and Mary didn't dance together.  
b.  $\llbracket (8-a) \rrbracket = \begin{cases} 0 \text{ iff } \llbracket \text{dance} \rrbracket(j \oplus m) = 1 \\ \# \text{ otherwise} \end{cases}$

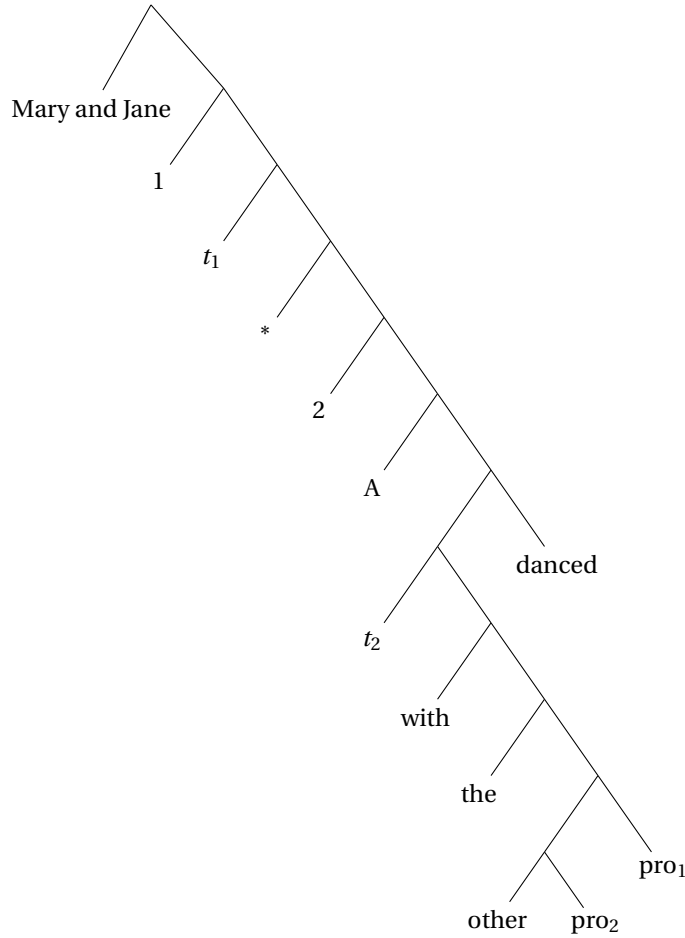
I will therefore assume that the presupposition of *the other* is locally accommodated with an A operator that is inserted in the scope of \*.<sup>2</sup> The lexical entry for the A-operator is given in (9), where A simply collapses falsity and undefinedness. The resulting LF with the A operator is given in (10-b) and the corresponding truth-conditions in (10-c). The A operator makes the second disjunct in (10-c) always false rather than always undefined, thus predicting the desired truth-conditions: (10-b) is true iff Jane and Mary danced together.

- (9)  $\llbracket A \rrbracket(p) = \begin{cases} 1 \text{ iff } p=1 \\ 0 \text{ iff } p=0 \vee p=\# \end{cases}$

- (10) a. Jane and Mary danced together.

<sup>2</sup>I assume that this local accommodation selectively targets the presupposition of *the other*, thus avoiding the undesirable result that other presuppositions in the scope of \* also have to be locally accommodated when *the other* is present. While I won't say more here about how the A operator can be indexed to specific presupposition triggers, see Romoli (2012) for independent evidence that this is needed.

b.



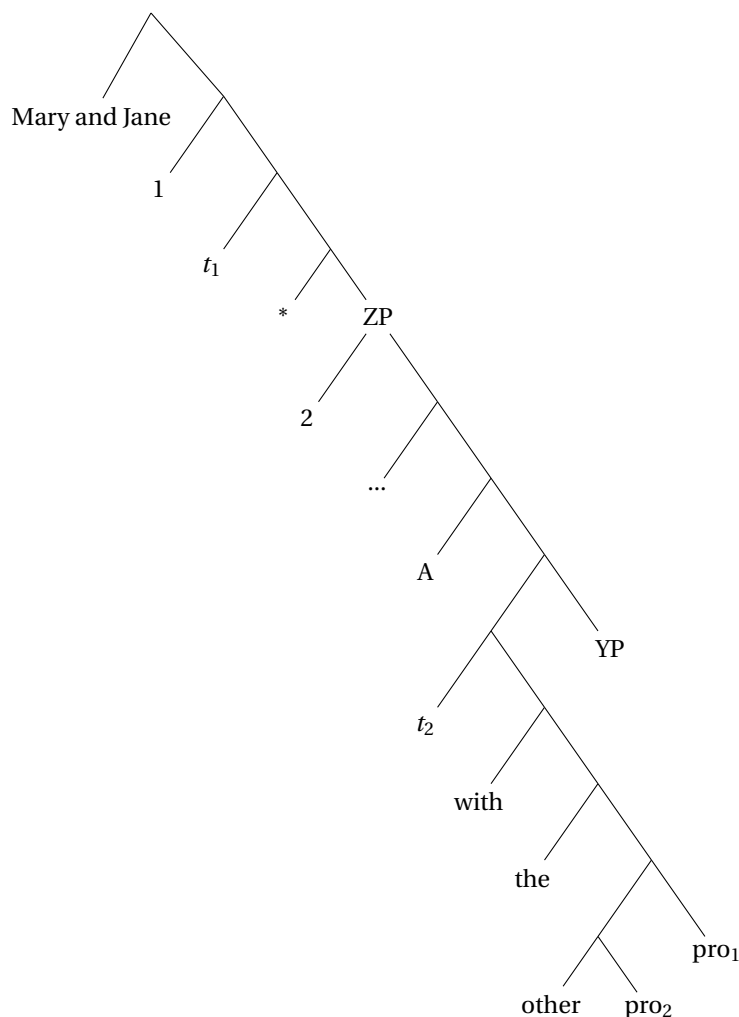
- c.  $\llbracket (10-b) \rrbracket = 1$  iff  $(\llbracket \text{dance} \rrbracket (m \oplus j) = 1 \wedge \llbracket \text{dance} \rrbracket (j \oplus m) = 1) \vee \llbracket A \rrbracket (\llbracket \text{dance} \rrbracket (m \oplus j \oplus (\llbracket \text{the} \rrbracket (\llbracket \text{other} \rrbracket (j \oplus m) (j \oplus m)))) = 1$   
iff  $\llbracket \text{dance} \rrbracket (j \oplus m) = 1$  iff Jane and Mary danced together.
- d.  $\llbracket A \rrbracket (\llbracket \text{dance} \rrbracket (m \oplus j \oplus (\llbracket \text{the} \rrbracket (\llbracket \text{other} \rrbracket (j \oplus m) (j \oplus m)))) = 0$ , since  $\llbracket \text{the} \rrbracket (\llbracket \text{other} \rrbracket (j \oplus m) (j \oplus m))$  is undefined because  $j \oplus m$  is not a proper part of itself.

I have therefore shown that the same collective truth-conditions that we got with DIST can be derived with \*, once we locally accommodate the presupposition of *the other* below \*. This result extends to all of the other readings of *together* discussed in chapter 4. The reason is that regardless of where the *x with the other* DP takes scope, the \* operator will give rise to a disjunction like (10-c), where the disjunct in which the predicate applies directly to the plurality will always be false, making \* essentially equivalent to DIST.

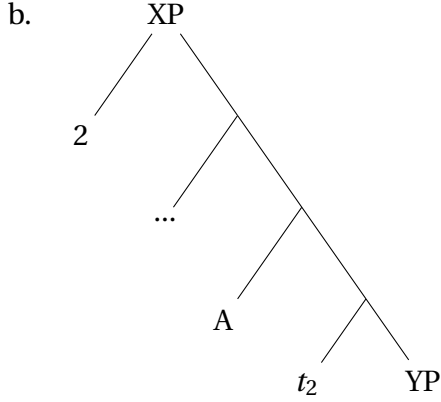
At this point, in order to see how the different results will generalize from the collective LF to the other LFs where *with the other* moves to derived positions, it will be useful to define the different readings in terms of a predicate XP, which will be different depending on where *with the other* takes scope. Consider the schematic LF in (11-a): here *with the other* moves to some

derived position above a phrase YP and potentially below some other scope-taking operators indicated by the “...” in the tree. We can define the predicate ZP in this abstract LF as the resulting predicate when we abstract over the first argument of *other*, as shown in (11-a). Now, it will be useful to define XP as the structure we get by deleting *with the other* from ZP, as shown in (11-b). What *together* does is essentially block distributivity over XP: the LF in (11-a) is true iff  $\llbracket \text{XP} \rrbracket$  is true of the plurality denoted by the subject. For the collective reading,  $\llbracket \text{XP} \rrbracket$  is simply equivalent to the lexical predicate  $\llbracket \text{dance} \rrbracket$  and *together* forces a collective reading as shown in (10-c). On the other hand, with the simultaneous reading for example,  $\llbracket \text{XP} \rrbracket$  is given in (11-d) and *together* ensures that there is a time such that each atomic part of the subject plurality danced at that time.<sup>3</sup> Similarly, when *together* takes scope above a lower two-place pluralization operator \*\*, XP denotes the cumulative predicate in (11-e) and *together* therefore picks out a cumulative reading.

(11) a.



<sup>3</sup>See Appendix D for a more detailed derivation for the simultaneous reading.



c.  $\llbracket (11-a) \rrbracket = 1$  iff  $\llbracket XP \rrbracket (j \oplus m) = 1$

d. For the simultaneous reading:

$$\llbracket XP \rrbracket = \lambda x. \exists t [\forall y \leq_{AT} x : \llbracket V \rrbracket (t)(y) = 1]$$

e. For the cumulative reading:

$$\llbracket XP \rrbracket = \lambda x. (\llbracket ** \rrbracket (\llbracket V \rrbracket)) (\llbracket \text{Object} \rrbracket) (x) = 1$$

So far, all of the examples we've considered had a subject of cardinality 2, like *Jane and Mary*. As observed by [Schwarzschild \(1994\)](#), when the subject denotes an individual of greater cardinality, *together* allows for partially distributive readings. To see this, consider the example in (12): here, *together* allows for a reading which is collective relative to pairs of students but which is partially distributive down to these pairs. In particular, (12-a) is true in the scenario in (12) where each pair of students danced together.

(12) Context: The students are divided into pairs for the competition and they can choose whether to dance as a pair or to dance individually. All of the pairs decided to dance together as a pair.

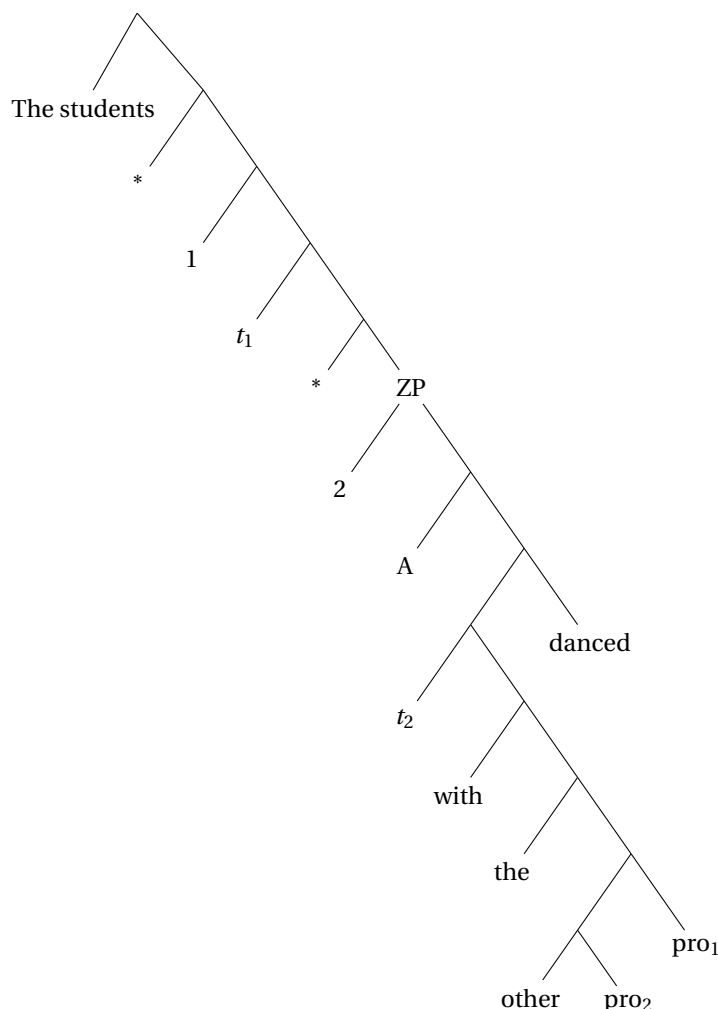
a. The students danced together.

While this reading can't be generated with DIST, we can generate this reading with *\** by inserting another *\** operator above the higher binder index 1 in (11-a).<sup>4</sup> This LF is given in (13-a). Here the higher *\** can allow for partial distributivity to pairs of students and then the *with the other* reciprocal will make sure that for every pair, each member danced with the other, thus giving rise to the desired reading in (12). More specifically, (13-a) is predicted to be true as long as the students can be divided into (non-atomic) parts such that each part danced collectively, as

<sup>4</sup>Note that this reading can be generated by a higher partial distributivity operator, paired with a cover that demands distributivity to specific pairs of students. While this can generate a reading which is true in (12), it will not help with the problem with generalized economy discussed at the beginning of this section. Namely, deleting both *with the other* and the lower pluralization operator will result in the same partially distributive reading that we get with *together* in this case. I will therefore continue to assume that these readings are derived with the pluralization operator *\**. As I show in section 5.2.2, this will allow for LFs like (13-a) to satisfy generalized economy.

shown in (13-b).

(13) a.



b.  $\llbracket (13-a) \rrbracket = 1$  iff  $\llbracket * \rrbracket (\lambda x. \llbracket \text{dance} \rrbracket (x) = 1 \wedge |x| > 1) (\llbracket \text{the students} \rrbracket) = 1$

Again, we can formulate a general statement of this partially distributive reading with respect to different readings of *together*, including the simultaneous, proximate and cumulative readings. In particular, depending on where the *x with the other* DP takes scope, XP, the result of deleting *with the other* from ZP, will denote a different predicate, as defined in (11). Regardless of what XP denotes, the LF where there is another \* operator above 1 will be true iff  $\llbracket \text{the students} \rrbracket$  can be divided into non-atomic subparts, each of which  $\llbracket \text{XP} \rrbracket$  is true of (14-a). For example, with the simultaneous reading, XP denotes the predicate in (11-d) and the partially distributive truth-conditions are given in (14-b). (14-b) correctly predicts that a sentence like *the students stood up together* is true in a scenario where the students are divided into pairs where the members of each pair stood up simultaneously.

(14) a. **General Statement:**

$$1 \text{ iff } \llbracket * \rrbracket (\lambda x. \llbracket XP \rrbracket (x) = 1 \wedge |x| > 1) (\llbracket \text{the students} \rrbracket) = 1$$

b. **Partially distributive simultaneous reading:**

$$1 \text{ iff } \llbracket * \rrbracket (\lambda x. \exists t [\forall y \leq_{AT} x : \llbracket V \rrbracket (t)(y) = 1] \wedge |x| > 1) (\llbracket \text{the students} \rrbracket) = 1$$

As I show in section 5.2.2, these partially distributive readings will turn out to be crucial for deriving the distribution of readings with *together* from the generalized economy constraint.

## 5.2.2 Towards a modular generalized economy constraint

Now that we have established how the different readings of *together* are derived with *\**, we can return to the generalized economy constraint. I will argue that generalized economy is able to license the desired readings with *together* only if the economy calculation is modular. In other words, (15) does not always rule out LFs which are equivalent to a simpler competitor. Rather, it only rules out LFs where this equivalence can be proven by the deductive system.

(15) a. **Generalized Economy Constraint:**

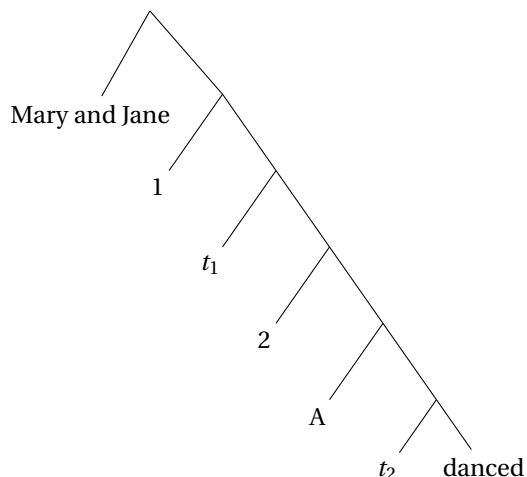
An LF  $\phi$  is ruled out if the deductive system can prove that there is a competitor  $\psi$  such that (i)  $\psi < \phi$  and (ii)  $\llbracket \psi \rrbracket = \llbracket \phi \rrbracket$

b. An LF  $\psi < \phi$  if  $\psi$  can be derived from  $\phi$  by deleting a number of nodes.

This subsection will proceed as follows. First, I show that the collective LF in (10-b) is ruled out by the economy constraint, since deleting *with the other* and *\** yields a truth-conditionally equivalent LF. I then show that the partially distributive LF in (13-a) is licensed by economy when the subject denotes an individual of cardinality greater than 2. When the subject denotes an individual of cardinality 2, the same LF yields a purely collective reading which is equivalent to the result of deleting *with the other* and all pluralization operators. I argue that generalized economy nevertheless does not rule out this LF, because the deductive system does not have access to the cardinality of the subject when proving equivalence for the purposes of the economy constraint. The collective reading can therefore be generated without violating generalized economy. Finally, I show that this result generalizes to the simultaneous and proximate readings of *together* and argue that the unattested distributive reading continues to be blocked by generalized economy.

Starting with the collective LF in (10-b), the same collective reading can be derived by the simpler LF in (16), where both *with the other* and *\** are deleted from (10-b). In order to predict that the collective reading is licensed, we therefore have to find a different LF that generates the same reading without violating generalized economy.

(16)



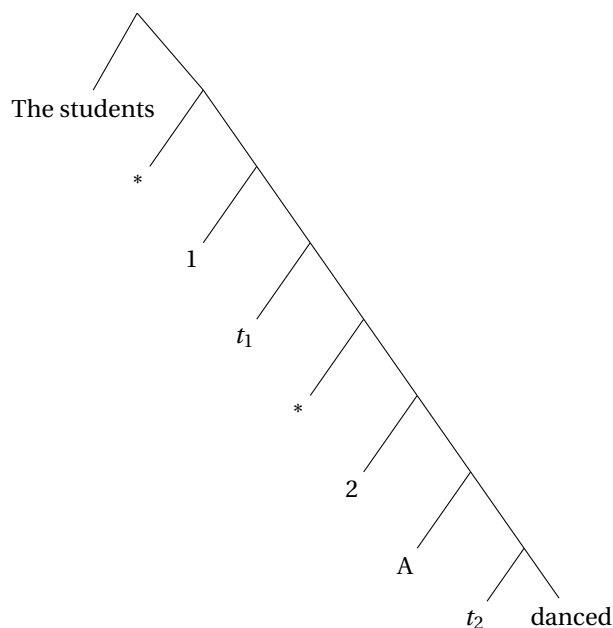
The partially distributive LF in (13-a) turns out to satisfy the generalized economy constraint. In particular, (13-a) is not equivalent to any of its deletion alternatives. To see this, consider first the result of deleting *with the other* from (13-a) in (18-a).

(17) **Partially distributive truth-conditions:**

$\llbracket (13-a) \rrbracket = 1$  iff  $\llbracket * \rrbracket (\lambda x. \llbracket \text{dance} \rrbracket (x) = 1 \wedge |x| > 1) (\llbracket \text{the students} \rrbracket) = 1$

(18) **Result of deleting *with the other*:**

a.



b.  $\llbracket (18-a) \rrbracket = 1$  iff  $\llbracket * \rrbracket (\llbracket \text{dance} \rrbracket) (\llbracket \text{the students} \rrbracket) = 1$

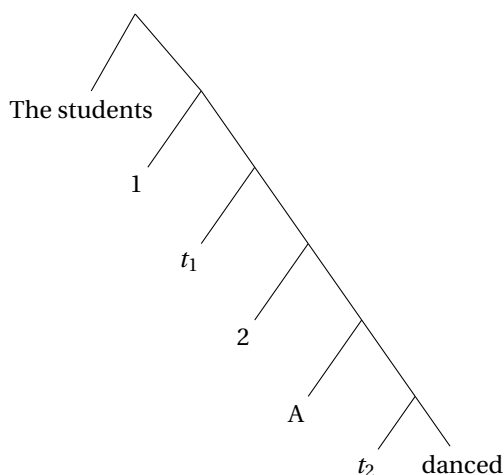
Unlike (17), (18-b) is true as long as each of the students danced, whether they did so together or separately. All of the other deletion alternatives where there is a \* operator below *the students* and where *with the other* is deleted similarly give rise to the underspecified truth-conditions



in (18-b). Finally, if both \* operators are deleted, as in the LF in (19-a), we will get the purely collective truth-conditions in (19-b), which require that all of the students danced together. We can therefore see that there is no deletion alternative of (13-a) that gives rise to the same partially distributive truth-conditions.<sup>5</sup>

(19) **Result of deleting *with the other* and both \* operators:**

a.



b.  $\llbracket (19-a) \rrbracket = 1$  iff  $\llbracket \text{dance} \rrbracket (\llbracket \text{the students} \rrbracket) = 1$

The partially distributive LF in (13-a) therefore satisfies the generalized economy constraint. Crucially, in order for the partially distributive reading in (13-b) to be distinct from the collective one in (19-a) where *with the other* and both \* operators are deleted,  $\llbracket \text{the students} \rrbracket$  has to denote an individual of cardinality greater than 2. Otherwise, when the subject denotes an individual of cardinality 2, like *Jane and Mary*, both the partially distributive and the collective LFs are true iff Jane and Mary danced together. We therefore see that as it stands, the generalized economy constraint predicts that collective *together* is not licensed when the subject denotes an individual of cardinality 2. It is clear that this is not a desired prediction, as evidenced by the fact that *Jane and Mary danced together* does have a collective reading.

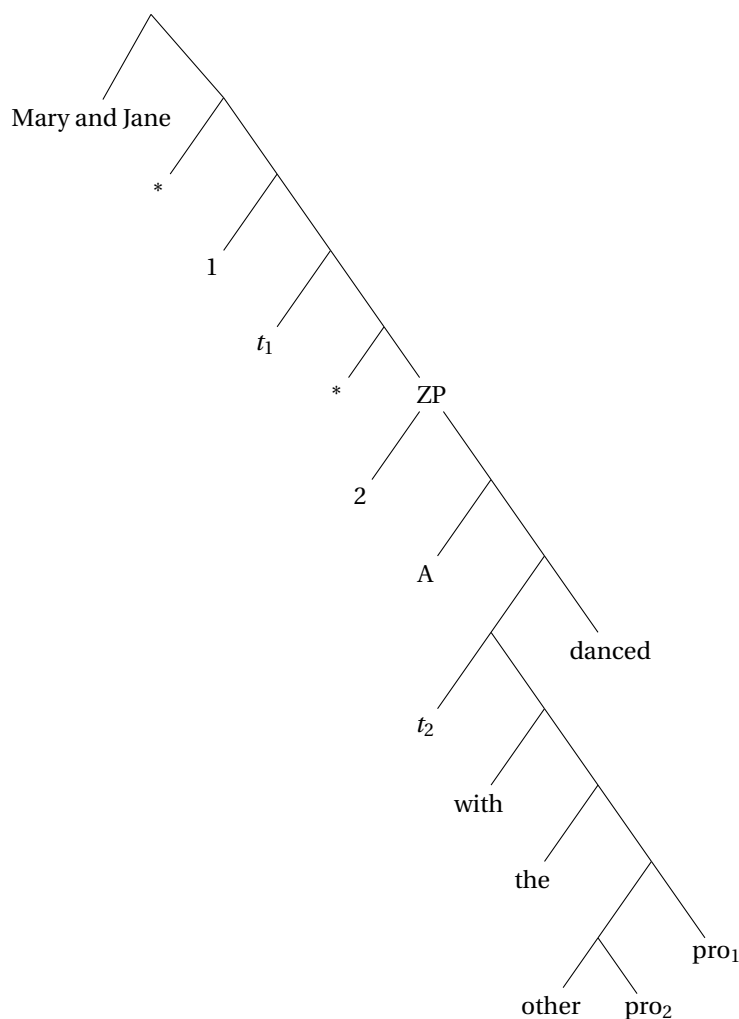
In order to resolve this issue, I propose that the cardinality of the subject DP is not accessible to the deductive system when evaluating whether an LF is ruled out by the economy constraint. More specifically, I will assume that the deductive system is unable to see the identity of DPs that denote individuals of type e. Given a sentence like (20-a), the deductive system will therefore at most have access to the information in (20-b), where X and Y can denote plural individuals of type e.

<sup>5</sup>Note that considering alternatives where the A operator is deleted does not affect this result in any way. In particular, if *with the other* is deleted, then deleting A will not change the truth-conditions. If *with the other* is not deleted, then the resulting truth-conditions will be clearly distinct from (17), since the LF without A will have a presupposition that (17) doesn't have.

- (20) a. Jane and Mary danced together.  
 b. **What the deductive system sees** (at most):  
 X and Y danced together. (where X and Y are possibly plural individuals)

Given this modularity assumption, we can generate an LF that gives rise to the collective reading for (20-a), while satisfying generalized economy. In particular, consider the LF in (21-a), where there is another \* operator above the binder index 1, just as in the partially distributive reading discussed above.

- (21) a.



- b.  $\llbracket (21-a) \rrbracket = 1$  iff  $\llbracket \text{dance} \rrbracket (j \oplus m) = 1$

Now, (21-a) gives rise to the desired collective reading, as shown in (21-b). While the LF in (21-a) is equivalent to the result of deleting *with the other* and both \* operators, this equivalence depends on the fact that *Jane and Mary* denotes an individual of cardinality 2. In particular, as shown above, if we replace *Jane and Mary* with a subject that denotes an individual of cardinality greater than 2, the LF in (21-a) gives rise to a partially distributive reading that is not equivalent

to any of the deletion alternatives. Since the deductive system is blind to the identity of the two conjuncts, it is unable to prove equivalence between the LF in (21-a) and any deletion alternative.

To reiterate, I have argued that the higher \* in (21-a) is independently needed in order to generate partially distributive readings. This partially distributive LF gives rise to a collective reading when the subject denotes an individual of cardinality 2 like *Jane and Mary*. At the same time, since the deductive system is blind to the cardinality of the subject, this collective reading is licensed by the generalized economy constraint, as the resulting partially distributive LF satisfies the economy constraint when the cardinality of the subject is greater than 2. We can therefore conclude that the modular generalized economy constraint correctly predicts that (20-a) has a collective reading.

Looking beyond the example with conjunction in (20-a), there is a question of what else the deductive system has to be blind to when evaluating economy. For example, in (22-a), just like (20-a), the cardinality of the subject is two and therefore any LF that generates the collective reading for (22-a) will be equivalent to one of its deletion alternatives. In order to predict that the collective reading is licensed in (22-a), I therefore have to assume that the deductive system is unable to see the numeral *two* in (22-a) when proving equivalence. Given this assumption, the deductive system won't be able to prove equivalence between (22-a) and a simpler alternative, since this equivalence doesn't hold when *two* is replaced with a different numeral.<sup>6</sup>

- (22) a. The two students danced together.  
       b. **What the deductive system sees** (at most):  
           The n students danced together.

While it is beyond the scope of this dissertation to provide a general statement of what information the deductive system has access to when evaluating economy, I have shown here that this system has to minimally be blind to the identity of individual-denoting DPs and to the value of numerals.

Turning to the other readings of *together*, we can consider the relevant economy calculations given different predicates XP, corresponding to different scope positions for *together*. As discussed in section 5.2.1, the partially distributive LF will be true as long as the subject can be divided into non-atomic individuals which  $\llbracket \text{XP} \rrbracket$  is true of (23-a). We can again consider two deletion alternatives for this LF: (i) when only *with the other* is deleted, the LF will be true as long as the subject can be divided into individuals, whether atomic or not, which  $\llbracket \text{XP} \rrbracket$  is true of (23-b) and (ii) when *with the other* and both \* operators are deleted, the LF will be true if  $\llbracket \text{XP} \rrbracket$  is true directly of the plurality denoted by the subject (23-c).

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<sup>6</sup>Thanks to Viola Schmitt and Roger Schwarzschild (p.c.) for pointing out that economy needs to be blind to numerals in examples like (22-a).

- (23) a. **Partially distributive LF:** (only non-atomic parts are considered)  
 $1 \text{ iff } \llbracket * \rrbracket (\lambda x. \llbracket XP \rrbracket (x) = 1 \wedge |x| > 1) (X \oplus Y) = 1$
- b. **LF where *with the other* is deleted:** (All parts, atomic and non-atomic, are considered)  
 $1 \text{ iff } \llbracket * \rrbracket (\llbracket XP \rrbracket) (X \oplus Y) = 1$
- c. **LF where *with the other* and both *\** operators are deleted:** (Only the entire plurality is considered)  
 $1 \text{ iff } \llbracket XP \rrbracket (X \oplus Y)$

Given this general schema, it is easy to show that the result with collective *together* generalizes to simultaneous and proximate *together*. For simultaneous *together*, the relevant predicate XP is given in (24-a). Plugging in this predicate in (23-a), it is straightforward to see that the partially distributive reading will be distinct from the readings in (23-b) and (23-c). In particular, (23-a) is true iff the subject can be divided into non-atomic subparts, such that all of the members of each subpart stood up at the same time. On the other hand, (23-b) is true as long as each atomic part of the subject plurality stood up and (23-c) is only true if each atomic part of the subject plurality stood up at the same time.<sup>7</sup> Just like with collective *together*, we therefore see that if we abstract away from the identity of the subject, the deletion alternatives have possibly different truth-conditions from the partially distributive LF with simultaneous *together*. We can therefore conclude that simultaneous *together* is licensed by the generalized economy constraint.<sup>8</sup> It is straightforward to see that the same logic applies to proximate *together*, where the relevant predicate XP is given in (24-b).

- (24) a. **Simultaneous *together*:** Jane and Mary stood up together.  
 $\llbracket XP \rrbracket = \lambda x. \exists t [\forall y \leq_{AT} x : \llbracket \text{stood up} \rrbracket (t)(y) = 1]$
- b. **Proximate *together*:** Corpse 1 and Corpse 2 lay together.  
 $\llbracket XP \rrbracket = \lambda x. \exists l \exists t [\forall y \leq_{AT} x : \llbracket \text{lay} \rrbracket (l)(t)(y) = 1]$

Finally, we have to show that LFs which generate the unattested distributive reading are ruled out by the modular generalized economy constraint. As discussed in chapter 4, this unattested distributive reading corresponds to an LF where there is a lower pluralization operator below *together*. Consider the LF in (25-a), which generates an unattested reading that is true in

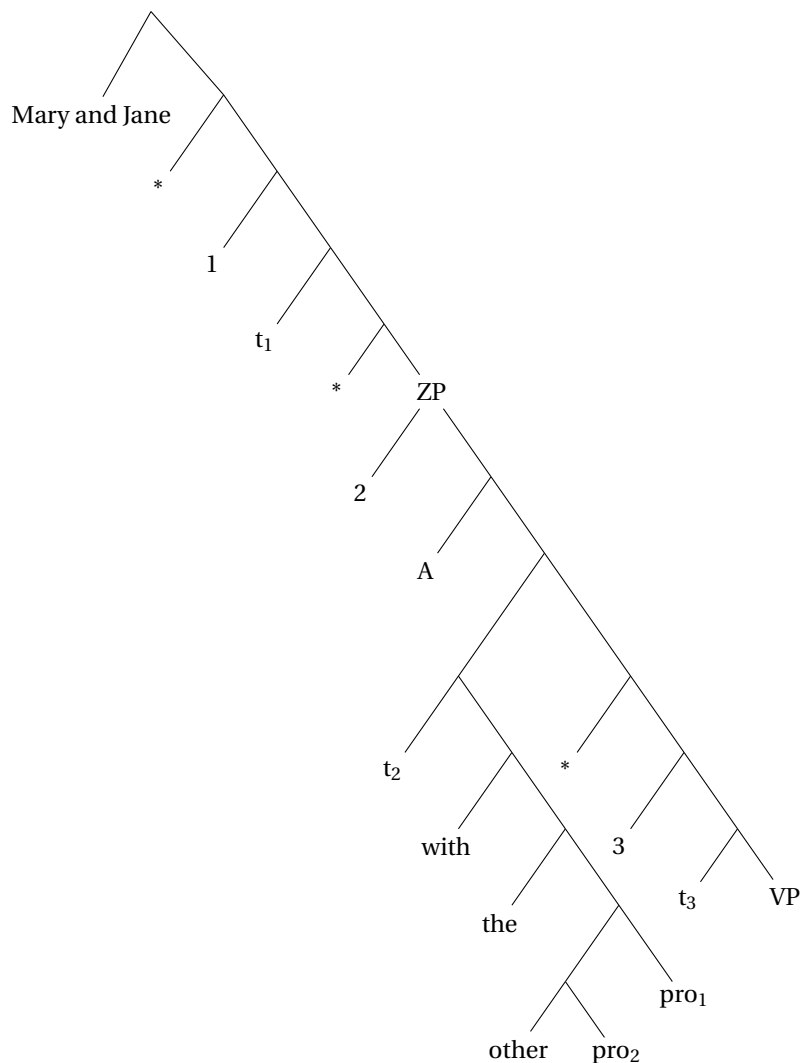
<sup>7</sup>To see that these truth-conditions are in fact different, we can consider two types of scenarios. First, in a scenario where each member of the subject plurality stood up at a different time, only (23-b) is true. Second, in a scenario where the students are divided into pairs which stood up at the same time, (23-a) and (23-b) are true but (23-c) isn't. We therefore see a three-way distinction between the truth-conditions generated by the three LFs.

<sup>8</sup>I provide more detailed derivations in Appendix D showing that generalized economy licenses the simultaneous reading.

distributive scenarios where the VP is true of Jane and of Mary separately.<sup>9</sup>

(25) **Unattested LF:**

a.



- b.  $\llbracket (25-a) \rrbracket = 1$  iff  $\llbracket * \rrbracket (\llbracket VP \rrbracket) (j \oplus m) = 1$   
iff  $\llbracket VP \rrbracket (j \oplus m) = 1 \vee \llbracket VP \rrbracket (j) = 1 \wedge \llbracket VP \rrbracket (m) = 1$

Now, regardless of the identity of the subject, we can see that the LF in (25-a) is equivalent to the LF where *with the other* is deleted: in both cases the LF is true iff the subject plurality can be divided into subparts such that  $\llbracket VP \rrbracket$  is true of each subpart.<sup>10</sup> To see this, consider the relevant

<sup>9</sup>The same reading can be generated without the higher \* operator above the binder index 1, but as we saw above this LF will always be equivalent to the result of deleting *with the other* and the \* operator above the index 2. In order to give the unattested distributive reading the best chance of satisfying generalized economy, we therefore have to consider the LF in (25-a).

<sup>10</sup>I will set aside the possibility that the entire subject denotes an individual of cardinality 1 for now and return to this in Appendix E. In particular, given the assumptions we have so far, when the subject denotes an individual of cardinality 1, as in *John danced together*, all of the LFs with *together*, including the unattested distributive LF, will

$\llbracket \text{XP} \rrbracket$  in (26), corresponding to deleting *with the other* from ZP in (25-a). Following the schema presented in section 5.2.2, the truth-conditions for the LF where *with the other* is deleted and the LF containing *with the other* are given in (26-a) and (26-b) respectively.

$$(26) \quad \llbracket \text{XP} \rrbracket = \lambda x. \llbracket * \rrbracket (\llbracket \text{VP} \rrbracket)(x) = 1$$

a. **LF where *with the other* is deleted:**

$$1 \text{ iff } \llbracket * \rrbracket (\llbracket \text{XP} \rrbracket)(X \oplus Y) = 1$$

$$\text{iff } \llbracket \text{XP} \rrbracket (X \oplus Y) = 1 \vee \dots \vee \forall z \leq_{AT} X \oplus Y : \llbracket \text{XP} \rrbracket (z) = 1$$

b. **LF containing *with the other*:**

$$1 \text{ iff } \llbracket * \rrbracket (\lambda x. \llbracket \text{XP} \rrbracket (x) = 1 \wedge |x| > 1)(X \oplus Y) = 1$$

$$\text{iff } \llbracket \text{XP} \rrbracket (X \oplus Y) = 1 \vee \dots$$

Now, the only difference between (26-a) and (26-b) is that (26-a) includes additional disjuncts in which some of the subparts that  $\llbracket \text{XP} \rrbracket$  applies to are atomic. (26-b), on the other hand, allows for partial distributivity but does not include disjuncts where the predicate applies to atomic subparts. As shown in (27), given the cumulative  $\llbracket \text{XP} \rrbracket$  in (26), all of the other disjuncts in (26-a) entail the first collective disjunct, thus rendering (26-a) and (26-b) equivalent.

(27) **Proof that (26-a) and (26-b) are equivalent for the  $\llbracket \text{XP} \rrbracket$  in (26):**

a. Since  $*$  makes predicates cumulative,

$$\forall x, y : \llbracket * \rrbracket (\llbracket \text{VP} \rrbracket)(x) = 1 \wedge \llbracket * \rrbracket (\llbracket \text{VP} \rrbracket)(y) = 1 \rightarrow \llbracket * \rrbracket (\llbracket \text{VP} \rrbracket)(x \oplus y) = 1$$

b. Therefore, all of the other disjuncts in both (26-a) and (26-b) entail the first disjunct.

For the final disjunct in (26-a) for example:

$$[\forall z \leq_{AT} X \oplus Y : \llbracket * \rrbracket (\llbracket \text{VP} \rrbracket)(z) = 1] \rightarrow \llbracket * \rrbracket \llbracket \text{VP} \rrbracket (X \oplus Y)$$

c. Both (26-a) and (26-b) are therefore equivalent to the first disjunct: they're true iff

$$\llbracket * \rrbracket (\llbracket \text{VP} \rrbracket)(X \oplus Y) = 1$$

I have therefore shown that the deductive system can prove equivalence between the LF in (25) and the LF resulting from deleting *with the other*. The LF in (25) therefore violates generalized economy. Thus, the modularity assumptions that are required to license the collective, simultaneous and proximate readings do not overgenerate: LFs where there is a lower  $*$  operator giving rise to a distributive reading continue to be ruled out by the modular economy constraint proposed here.

More generally, any cumulative XP where the distributive disjuncts entail the disjunct where

---

give rise to a contradiction, since the presupposition of *the other* can't be met. On the other hand, the LF without *together* will simply be true iff John danced. Given that the distributive LF no longer violates economy in this case, there is a question of whether the deductive system can prove equivalence for a sentence like *the students danced*, given the possibility that *the students* denotes an individual of cardinality 1.

XP applies to the plurality as a whole will result in a violation of generalized economy (28). This general result will be useful when evaluating the predictions of generalized economy for cumulative *together* in section 5.2.3.

- (28) **General result:** The deductive system can prove that (26-a) and (26-b) are equivalent if  $\forall x, y: \llbracket \text{XP} \rrbracket(x) = 1 \wedge \llbracket \text{XP} \rrbracket(y) = 1 \rightarrow \llbracket \text{XP} \rrbracket(x \oplus y) = 1$

### 5.2.3 Generalized Economy and cumulative *together*

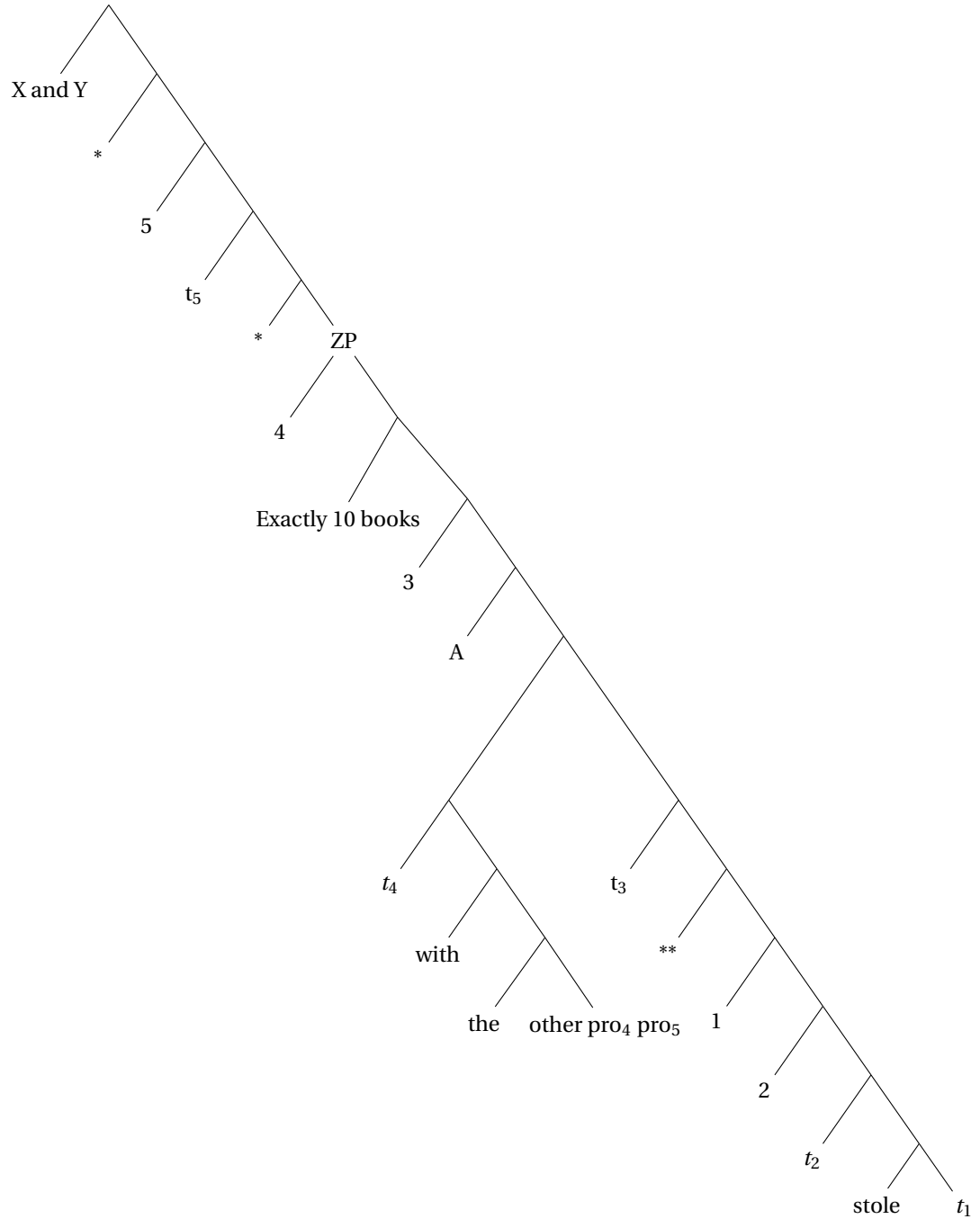
I will now turn to the predictions of generalized economy for the licensing of cumulative *together*. As discussed in chapter 4, *together* gives rise to a cumulative reading with a variety of objects, exemplified in (29).<sup>11</sup> In this section, I will show that without any additional mechanisms, generalized economy incorrectly predicts that cumulative *together* is only licensed in (29-a), when the object is a non-monotonic quantifier. I return to the objects in (29-b)-(29-e) in section 5.3, where I show that economy can be satisfied if we consider LFs where an independently motivated local implicature is computed below \*.

- (29) **Objects that license cumulative *together*:**
- a. Jane and Mary together stole **exactly 10 books**.
  - b. Jane and Mary together stole **10 books**.
  - c. Jane and Mary together stole **at least 10 books**.
  - d. Jane and Mary together stole **all of the books**.
  - e. Jane and Mary together stole **more than 10 books**.

Starting with the non-monotonic objects, we can see that generalized economy licenses the cumulative reading in (29-a) in the same way as the collective, simultaneous and proximate readings discussed in section 5.2.2. In particular, consider the LF in (30-a), where *together* takes scope above a lower \*\* operator, thus deriving the desired cumulative reading.

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<sup>11</sup>I set aside cumulative readings with objects headed by *every* for the purposes of this dissertation, which don't follow straightforwardly from the framework assumed here, where cumulative readings are due to \*\*. I leave it to future work to determine whether the account in terms of economy proposed here will be compatible with different proposals for how cumulative readings with *every* are generated (e.g. Schein, 1993; Kratzer, 2003; Champollion, 2010; Haslinger & Schmitt, 2018; Chatain, 2021).



- (30) a.
- b.  $\llbracket XP \rrbracket = \lambda x. \llbracket \text{exactly 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1$ <sup>12</sup>
- c.  $\llbracket (30\text{-a}) \rrbracket = 1$  iff
- $\llbracket * \rrbracket (\lambda x. (\llbracket \text{exactly 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1) \wedge |x| > 1)(X \oplus Y) = 1$

Abstracting away from the identity of the two conjuncts, (30-a) is predicted to be true as long as  $X \oplus Y$  can be divided into non-atomic sub-parts such that each sub-part cumulatively stole exactly 10 books (30-c). When the subject is *Jane and Mary*, since there are no non-atomic proper

<sup>12</sup>Again, XP here is the result of deleting *with the other* from ZP



sub-parts of the subject, (30-a) will simply be true iff Jane and Mary cumulatively stole exactly 10 books.

Now, in order to verify that (30-a) is licensed by generalized economy, we again have to ensure that (30-a) can't be proven to be equivalent to any of its deletion alternatives. Given the predicate  $\llbracket XP \rrbracket$  in (30-b), we can follow the general schema in (23) to see the resulting truth-conditions when *with the other* is deleted (31-a) and when *with the other* and both \* operators are deleted (31-b). (31-a) is true iff  $X \oplus Y$  can be divided into sub-parts, including atomic subparts, which each cumulatively stole exactly 10 books.<sup>13</sup> On the other hand, (31-b) is only true if  $X \oplus Y$  cumulatively stole exactly 10 books.

(31) a. **LF where *with the other* is deleted:**

$$1 \text{ iff } \llbracket * \rrbracket (\lambda x. \llbracket \text{exactly 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1)(X \oplus Y) = 1$$

b. **LF where *with the other* and both \* operators are deleted:**

$$1 \text{ iff } (\lambda x. \llbracket \text{exactly 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1)(X \oplus Y) = 1$$

To see that there are choices of X and Y which make the truth-conditions in (31-a) and (31-b) distinct from the partially distributive truth-conditions in (30-c), consider the two scenarios in (32). Consider also a choice of X and Y such that  $X \oplus Y$  is the plural individual  $j \oplus m \oplus b \oplus s$ . Now, in scenario 1, (31-a) is true, while both (30-a) and (31-b) are false. This is because only (31-a) has a purely distributive disjunct which is true if each atomic part of  $X \oplus Y$  stole exactly 10 books. Turning to Scenario 2, both (30-a) and (31-a) are true here, since the 4 individuals can be divided into pairs such that each pair cumulatively stole exactly 10 books. On the other hand, (31-b) remains false since the 4 individuals cumulatively stole 20 books, not 10 books. We therefore see that (30-a) gives rise to distinct truth-conditions from both deletion alternatives in (31-a) and (31-b).

(32)  $\llbracket X \text{ and } Y \rrbracket = j \oplus m \oplus b \oplus s$

a. **Scenario 1:** Each of j,m,b and s stole exactly 10 distinct books.

b. **Scenario 2:** Each of j,m,b and s stole exactly 5 distinct books.

We can therefore conclude that the LF in (30-a) satisfies generalized economy, thus correctly predicting that the cumulative reading in (29-a) is licensed.<sup>14</sup>

Turning to the other types of objects in (29), the partially distributive LF analogous to (30-a) turns out to violate generalized economy. To see this, consider the relevant predicate XP with these different objects in (33). All of the predicates in (33) are cumulative, such that if they are

<sup>13</sup>For an atomic sub-part x,  $\llbracket \text{exactly 10 books} \rrbracket (\lambda x. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(x)(a)) = 1$  iff x alone stole exactly 10 books.

<sup>14</sup>Again, it is easy to see that all of the other deletion alternatives give rise to distinct truth-conditions. For example, deleting *with the other* and one \* operator will result in the same truth-conditions as (31-a) and deleting the A operator will result in a presupposition that (30-a) lacks.

true of a pair of two individuals, they are also true of their sum (34). For example, if Jane and Mary cumulatively stole at least 10 books and Bill and Sue cumulatively stole at least 10 books, then it is necessarily true that Jane and Mary and Bill and Sue stole at least 10 books. The cumulative reading with *exactly 10*, on the other hand, clearly does not have this property: if Jane and Mary cumulatively stole exactly 10 books and Bill and Sue cumulatively stole exactly 10 books, then it clearly doesn't follow that Jane and Mary and Bill and Sue stole exactly 10 books.

(33) a. **At least 10:**

$$\llbracket \text{XP} \rrbracket = \lambda x. \llbracket \text{at least 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1$$

b. **More than 10:**

$$\llbracket \text{XP} \rrbracket = \lambda x. \llbracket \text{more than 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1$$

c. **All:**

$$\llbracket \text{XP} \rrbracket = \lambda x. \llbracket \text{All of the books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1$$

(34) **All of the XPs in (33) are cumulative, for example:** for any x and y,

$$(\llbracket \text{at least 10 books} \rrbracket (\lambda z. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(z)(x)) = 1 \wedge \llbracket \text{at least 10 books} \rrbracket (\lambda z. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(z)(y)) = 1) \rightarrow \llbracket \text{at least 10 books} \rrbracket (\lambda z. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(z)(x \oplus y)) = 1$$

Given the general result in (28), we can conclude that the LF analogous to (30-a) with these objects violates generalized economy (following the same logic as given in section 5.2.2 for the unattested distributive reading). To see this more concretely with an example, consider the truth-conditions for (30-a) with *at least 10* in (35-a). Since the predicate that \* applies to is already cumulative, all of the other disjuncts will entail the disjunct where the predicate applies directly to the entire plurality  $X \oplus Y$ . The same is true of the LF where *with the other* is deleted, as shown in (35-b). Both (35-a) and (35-b) are therefore true iff  $X \oplus Y$  cumulatively stole at least 10 books. The deductive system can therefore prove that the LF in (30-a) with *at least 10* is equivalent to one of its deletion alternatives, thus leading to a violation of generalized economy.

(35) **At least 10 derivations:**

a. **(30-a) with at least 10:**

1 iff  $\llbracket (30-a) \rrbracket = 1$  iff

$$\llbracket * \rrbracket (\lambda x. (\llbracket \text{at least 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1) \wedge |x| > 1)(X \oplus Y) = 1$$

$$\text{iff } (\lambda x. \llbracket \text{at least 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1)(X \oplus Y) = 1 \vee \dots$$

$$\text{iff } (\lambda x. \llbracket \text{at least 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1)(X \oplus Y) = 1$$

b. **LF where with the other is deleted:**

$$1 \text{ iff } \llbracket * \rrbracket (\lambda x. \llbracket \text{at least 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1)(X \oplus Y) = 1$$

$$\text{iff } (\lambda x. \llbracket \text{at least 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1)(X \oplus Y) = 1$$

To summarize, generalized economy seems to make the following incorrect prediction at this point: cumulative *together* is only licensed with non-monotonic quantifiers. The problem is that when the object is upward-entailing, such as *at least 10* or *all*, the relevant predicate XP is cumulative. As a result, the LF that generates the cumulative reading with *together* turns out to be equivalent to the result of deleting *together*, leading to a violation of generalized economy. To make the right prediction for cumulative *together* with these objects, we therefore need a way to make XP non-cumulative. In section 5.3, I argue that this can be achieved by inserting an exhaustification operator below \*, yielding a strengthened, non-cumulative predicate that allows for the cumulative reading to be generated without violating economy.

### 5.3 Exhaustification and cumulative *together*

In order to show how a local implicature can rescue the cumulative readings from violating economy, I will start with a simple case, where the object is a bare numeral such as *10 books*. In this case, the relevant implicature can be derived by the standard strengthening of *10 books* to *exactly 10 books*. I then return to the other objects discussed in section 5.2.3, such as *more than 10*, where it is more difficult to see what the implicature would be. I argue that we can generate an implicature based on alternatives where the subject is replaced with its subparts, thus allowing cumulative *together* to be licensed with the upward-entailing objects discussed in section 5.2.3.

Assuming an *at least* semantics for numerals (e.g. Horn, 1972), the same problem that we saw with *at least 10* and *more than 10* in section 5.2.3 arises for the cumulative reading with *10 books*. Namely, the predicate XP will be cumulative here (36): if x cumulatively stole at least 10 books and y cumulatively stole at least 10 books then  $x \oplus y$  cumulatively stole at least 10 books. As a result, the LF with *together* will be equivalent to the one without *together*, leading to a violation of generalized economy.

- (36) **Cumulative XP with 10 books:** for any x and y,  

$$(\llbracket 10 \text{ books} \rrbracket (\lambda z. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(z)(x)) = 1 \wedge \llbracket 10 \text{ books} \rrbracket (\lambda z. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(z)(y)) = 1) \rightarrow$$

$$\llbracket 10 \text{ books} \rrbracket (\lambda z. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(z)(x \oplus y)) = 1$$

Now, unlike with *at least 10*, we know that the semantics for bare numerals has to be supplemented with a mechanism that generates an *exactly* reading. In particular, a sentence like (37) gives rise to an inference that Jane stole exactly 10 books. This reading is generally assumed to arise via an implicature that Jane didn't steal 11 books (37) (Horn, 1972; Spector, 2013a, a.o.). Now, if we are able to generate this implicature locally in the scope of \*, the predicate XP will no longer be cumulative, just as we saw with *exactly 10* above. This will allow us to predict that

the cumulative reading of *together* is licensed with *10 books*, while maintaining that the basic meaning of numerals is an *at least* meaning.

(37) Jane stole 10 books.

**Implicature:** Jane didn't steal 11 books.

More specifically, I assume that implicatures are generated by a grammatical exhaustification operator *exh* (Chierchia, Fox, & Spector, 2012). I assume for simplicity here that *exh* asserts all of the alternatives that are not entailed by the prejacent are false, as shown in (38). Now, assuming that the alternatives for (39) consist of replacing *10* with other numerals (39-a), the *exactly 10* reading is predicted after applying *exh*. In particular, the prejacent asserts that Jane stole at least 10 books and negating the stronger numeral alternatives asserts that Jane didn't steal at least 11 books, giving rise to the desired *exactly* reading.

(38)  $\text{Exh}_{ALT}(\lambda w. \llbracket \phi \rrbracket^w) = 1$  iff  $\llbracket \phi \rrbracket = 1 \wedge \forall \psi \in ALT [\llbracket \phi \rrbracket \not\Rightarrow \llbracket \psi \rrbracket] \rightarrow \llbracket \psi \rrbracket = 0$

(39)  $\text{Exh}_{ALT}(\text{Jane stole 10 books})$

a.  $Alt = \{ \text{Jane stole 9 books, Jane stole 11 books...} \}$

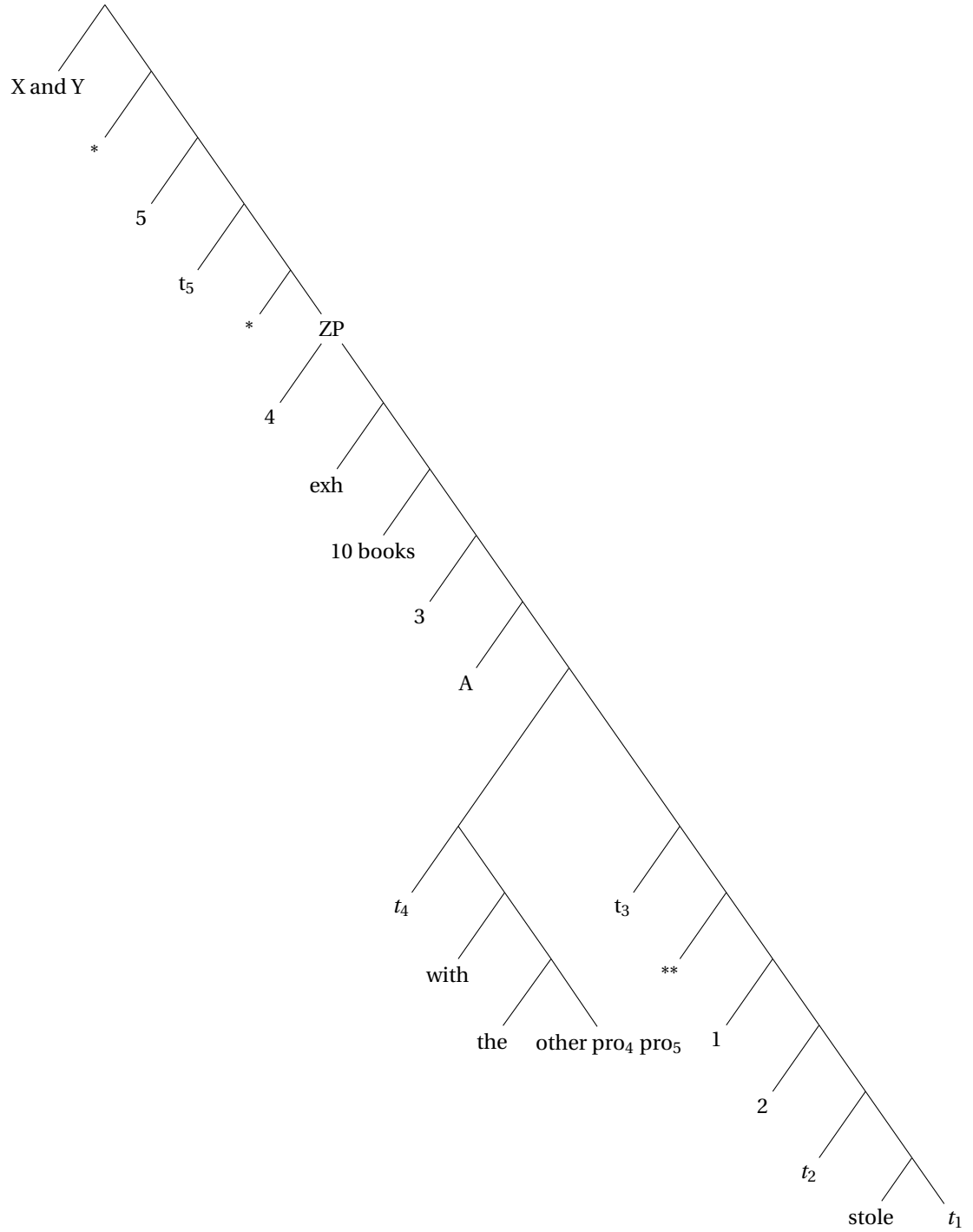
b.  $\text{Exh}_{ALT}(\llbracket \phi \rrbracket) = 1$  iff

$\llbracket \text{Jane stole 10 books} \rrbracket = 1 \wedge \llbracket \text{Jane stole 11 books} \rrbracket = 0$

iff Jane stole exactly 10 books.

We can now see how inserting *exh* locally below \* predicts that cumulative *together* is licensed in (40-a). To see this, consider the LF for (40-a) in (40-b), where we again abstract away from the identity of the two conjuncts. Here, *exh* takes scope above *10 books* but below the higher \* operator. (40-b) is predicted to be true iff X and Y can be divided into non-atomic subparts, each of which stole 10 but not 11 books (i.e. exactly 10 books). The predicate XP in (40-c) is no longer cumulative: for example, if a stole 10 books but not 11 and b stole 10 books but not 11, it clearly doesn't follow that a ⊕ b cumulatively stole 10 books but not 11.

(40) a. Jane and Mary together stole 10 books.



- b.
- c.  $\llbracket \text{XP (the result of deleting } \textit{with the other} \text{ from ZP)} \rrbracket =$   
 $\lambda x. \llbracket 10 \text{ books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1 \wedge \llbracket 11 \text{ books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 0$
- d.  $\llbracket (40\text{-b}) \rrbracket = 1$  iff  
 $\llbracket * \rrbracket (\lambda x. (\llbracket 10 \text{ books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1) \wedge (\llbracket 11 \text{ books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 0) \wedge |x| > 1) (X \oplus Y) = 1$

Just like with *exactly 10*, we therefore see that the LF in (40-b) satisfies generalized economy. In particular, given the XP in (40-c), (41-a) gives the truth-conditions where *with the other* is deleted and (41-b) the truth-conditions where *with the other* and both \* operators are deleted. Just like with *exactly 10*, it is straightforward to see that (41-a) and (41-b) are truth-conditionally distinct from (40-b) when the cardinality of the subject is greater than 2. We can therefore conclude that (40-b) satisfies generalized economy.<sup>15</sup>

- (41) a. **LF where *with the other* is deleted:**  

$$1 \text{ iff } \llbracket * \rrbracket (\lambda x. \llbracket 10 \text{ books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1 \wedge \llbracket 11 \text{ books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 0)(X \oplus Y) = 1$$
- b. **LF where *with the other* and both \* operators are deleted:**  

$$1 \text{ iff } (\lambda x. \llbracket 10 \text{ books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1 \wedge \llbracket 11 \text{ books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 0)(X \oplus Y) = 1$$

I have therefore shown that with bare numerals, the LF with *exh* satisfies the generalized economy constraint, allowing for the cumulative reading to be generated without violating economy. We can now turn to the remaining objects discussed in section 5.2.3. Unlike the bare numerals, the relevant strengthening here can't be derived by simply negating alternatives with different numerals. For example, (42-a) does not give rise to an implicature that Jane didn't steal more than 11 books and (42-b) does not give rise to an implicature that Jane didn't steal at least 11 books. Similarly, it is not clear that the sentence with *all* in (42-c) gives rise to any implicature.

- (42) a. Jane stole more than 10 books.  
 $\nrightarrow$  Jane didn't steal more than 11 books.
- b. Jane stole at least 10 books.  
 $\nrightarrow$  Jane didn't steal at least 11 books.
- c. Jane stole all of the books.

I will argue that when the subject is plural, the cumulative reading with these objects can give rise to an implicature that the cumulative predicate is not true of any proper subpart of the subject plurality (43). For example, the cumulative reading of (44) will give rise to an implicature that neither Jane nor Mary alone stole more than 10 books.

- (43) X and Y stole more than 10 books.  
**Implicature:** No proper part of  $X \oplus Y$  cumulatively stole more than 10 books.

<sup>15</sup>As for the other deletion alternatives, it is easy to see that adding *exh* does not interfere with the economy calculations. Namely, deleting *exh* clearly changes the truth-conditions, since the LF without *exh* will have an at least interpretation for the numeral.

(44) Jane and Mary stole more than 10 books.

**Implicature:** Jane didn't steal more than 10 books and Mary didn't steal more than 10 books.

Since sentences like (44) also allow for a distributive reading, it is difficult to test directly whether the purely cumulative reading gives rise to the implicature. I will therefore show that the interaction of scalar implicatures with Hurford's constraint allows us to detect the implicature here. In particular, Chierchia et al. (2012) argue that local scalar implicature can rescue apparent violations of Hurford's constraint, which rules out disjunctions where one disjunct entails the other. To see this, consider first a standard case that exemplifies Hurford's constraint in (45-a): here, the first disjunct entails the second one and the sentence is infelicitous. Now, consider (45-b). Here, the second disjunct (*Mary read all of the books*) entails the basic meaning of first disjunct, without an implicature (*Mary read at least some of the books*), yet the sentence is felicitous. Chierchia et al. argue that this is because exhaustification applies in the first disjunct and gives rise to a local implicature that breaks the entailment between the two disjuncts. In particular, when *some* is strengthened to *some but not all* in the first disjunct, it is no longer entailed by the second disjunct, thus avoiding a violation of Hurford's constraint (46).

(45) a. #Jane either lives in Paris or in France.

b. Jane either read some or all of the books.

(46) a. First disjunct:  $\text{exh}(\text{Jane read some of the books}) = \text{Jane read some but not all of the books}$ .

b. Second disjunct: Jane read all of the books.

We can therefore use Hurford's constraint to test whether the cumulative parse for a sentence like *Jane and Mary stole more than 10 books* in fact has the posited implicature in (44). Consider the disjunction in (47): here the second disjunct entails the cumulative reading of the first disjunct without the implicature: if each of Jane and Mary stole more than 10 books then they cumulatively stole more than 10 books. Yet, the sentence seems to be felicitous. This can be explained if the implicature in (44) is in fact computed in the first disjunct: the second disjunct in (47) does not entail the strengthened meaning with the implicature (*that Jane and Mary cumulatively stole more than 10 books but neither of them did so alone*). We therefore have independent evidence that this implicature does in fact arise with *more than 10*.

(47) Either Jane and Mary stole more than 10 books or Jane stole more than 10 books and Mary stole more than 10 books.

We can see parallel evidence for the same implicature with the different types of upward-entailing

objects that license cumulative readings. For example, in (48-a), I will argue that the first disjunct is strengthened with the implicature that *neither Jane nor Mary alone read all of the books*, thus breaking the entailment between the two disjuncts.

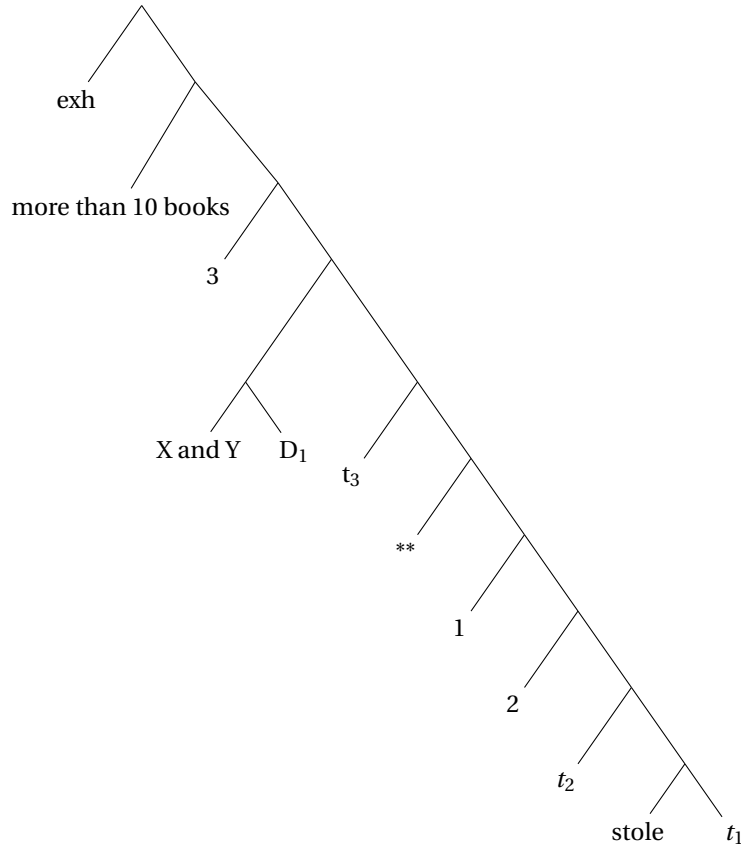
- (48) a. Either Jane and Mary read all of the books or Jane read all of the books and Mary read all of the books.  
 b. Either Jane and Mary read at least 10 books or Jane read at least 10 books and Mary read at least 10 books.

Now that I have established that these implicatures are available, there is a question of how to generate them via a principled set of alternatives. While there might be several ways of generating this implicature, I propose for concreteness that there is a domain restriction variable  $D$  that merges with the subject and always returns a subpart of the plural individual that the subject denotes. I then argue that the implicature described above can be derived by considering alternatives where the index on the domain variable is different. In order to ensure that this variable only returns subparts of the individual that it applies to, I propose that  $D$  applies to an individual and returns the plurality formed by intersecting the atomic parts of the subject with some predicate determined by the assignment function. If this intersection is empty, I assume that  $D$  simply returns the individual it applies to. The corresponding lexical entry is given in (49).

$$(49) \quad \llbracket D_n \rrbracket^g = \lambda x. \begin{cases} \bigoplus \{y : y \leq_{AT} x \wedge y \in g(n)\} & \text{if } \{y : y \leq_{AT} x \wedge y \in g(n)\} \neq \emptyset \\ x & \text{otherwise} \end{cases}$$

We can now see how this allows us to derive the desired implicature. I assume that by default, the subject merges with a domain variable  $D_1$  whose contribution is trivial (i.e. where  $g(1)$  has all atomic parts of the subject in its extension). The only role of this variable is therefore to generate the alternatives where the index 1 is replaced with other indices. In particular, for each proper subpart of the subject plurality, there will be an index  $n$  where  $g(n)$  includes only that subpart. We will therefore have alternatives which are true if these proper subparts cumulatively stole more than 10 books. Since these alternatives are not entailed by the prejacent,  $\text{exh}$  will negate them, giving rise to the desired implicature. In particular, the truth-conditions in (50-d) are predicted for (50-a): (50-a) is predicted to be true if  $X$  and  $Y$  cumulatively stole more than 10 books, but no proper subpart of  $X \oplus Y$  cumulatively stole more than 10 books.





- (50) a.   
 b.  $\llbracket D_1 \rrbracket^g(X \oplus Y) = X \oplus Y$    
 c.  $\text{Alt}_{exh} = \{ [X \text{ and } Y D_2] ** \text{ stole more than 10 books}, [X \text{ and } Y D_3] ** \text{ stole more than 10 books} \dots \}$    
 d.  $\llbracket (50-a) \rrbracket^g = 1$  iff   
 $\llbracket \text{more than 10 books} \rrbracket (\lambda x. ** \llbracket \text{stole} \rrbracket (x) (X \oplus Y)) = 1 \wedge$    
 $\forall z < X \oplus Y : \llbracket \text{more than 10 books} \rrbracket (\lambda x. ** \llbracket \text{stole} \rrbracket (x) (z)) = 0$

As in the bare numeral case, when *exh* is inserted below \*, the resulting predicate XP becomes non-cumulative. The LF with *exh* therefore generates the cumulative reading without violating generalized economy. To see this, consider the relevant XP in (51-a).<sup>16</sup> Now, it is clear that  $\llbracket XP \rrbracket$  is not cumulative here. For example, in a scenario where a stole 11 books and b stole 11 other books,  $\llbracket XP \rrbracket (a)$  is true and  $\llbracket XP \rrbracket (b)$  is true but  $\llbracket XP \rrbracket (a \oplus b)$  is not true.

- (51) a.  $\llbracket XP \rrbracket =$    
 $\lambda x. \llbracket \text{more than 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket (y) (x))) = 1 \wedge$    
 $\forall z < x : \llbracket \text{more than 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket (y) (z))) = 0$    
 b. **LF in (40-b) with *more than 10*:**

<sup>16</sup>I won't repeat the LF here with *more than 10*. The LF will simply be identical to the one in (40-b) with *10 books* replaced with *more than 10 books*.

- 1 iff  $\llbracket * \rrbracket (\lambda x. \llbracket \text{more than 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1 \wedge$   
 $\forall z < x : \llbracket \text{more than 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(z)) = 0 \wedge |x| > 1)(X \oplus Y) = 1$
- c. **LF where *with the other* is deleted:**  
1 iff  $\llbracket * \rrbracket (\lambda x. \llbracket \text{more than 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1 \wedge$   
 $\forall z < x : \llbracket \text{more than 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(z)) = 0)(X \oplus Y) = 1$
- d. **LF where *with the other* and both \* operators are deleted:**  
1 iff  $(\lambda x. \llbracket \text{more than 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1 \wedge$   
 $\forall z < x : \llbracket \text{more than 10 books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(z)) = 0)(X \oplus Y) = 1$

To see that the LF parallel to (51-b) with *more than 10* satisfies generalized economy, we can again take  $j \oplus m \oplus b \oplus s$  to be the subject plurality and consider two types of scenarios in (52). In scenario 1, (51-b) is false: for any non-atomic individual, there is a proper subpart of it that stole more than 10 books. On the other hand, (51-c) is true since the distributive disjunct is true iff each atomic part of the plurality stole more than 10 books. In scenario 2, (51-b) is true but (51-d) is false, since there are proper parts of the plurality  $j \oplus m \oplus b \oplus s$  that stole more than 10 books. We therefore see that (51-b) is not truth-conditionally equivalent to any of its deletion alternatives, thus satisfying generalized economy.

- (52) a. **Scenario 1:** Each of  $j, m, b$ , and  $s$  stole 11 (distinct) books.  
b. **Scenario 2:** Each of  $j, m, b$ , and  $s$  stole 6 (distinct) books.

This result with *more than 10* generalizes to the other upward-entailing quantifiers discussed in section 5.2. For example, with *all*, the relevant  $\llbracket XP \rrbracket$  with *exh* is given in (53). Again, it is clear that (53) is not cumulative: if  $a$  stole all of the books and  $b$  stole all of the books,  $\llbracket XP \rrbracket(a)$  is true and  $\llbracket XP \rrbracket(b)$  is true but  $\llbracket XP \rrbracket(a \oplus b)$  is false.<sup>17</sup> Following the same logic above for *more than 10*, we can therefore conclude that the LF in (40-b) with *all* satisfies generalized economy.

- (53)  $\llbracket XP \rrbracket =$   
 $\lambda x. \llbracket \text{all of the books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(x)) = 1 \wedge$   
 $\forall z < x : \llbracket \text{all of the books} \rrbracket (\lambda y. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket)(y)(z)) = 0$

To summarize, I showed in this section that with certain objects, like bare numerals, *more than n*, *at least n* and *all*, it is necessary to generate a local implicature below  $*$  in order to license cumulative *together* without violating economy. This makes the prediction that with objects where the necessary implicature can't be generated, the cumulative reading should be blocked

<sup>17</sup>We can imagine that books can be stolen and returned and then stolen again, such that it can both be true that  $a$  stole all of the books and  $b$  stole all of the books, but this point can be made more generally with other predicates like *read*, where it is more natural to imagine contexts where  $a$  read all of the books and  $b$  read all of the books. I stick with *steal* here for consistency and because this predicate makes it easier to isolate the cumulative reading.

with *together*.<sup>18</sup> In section 5.4, I argue that the implicature based on subpart alternatives is blocked when the object gives rise to a homogeneity gap. I show that these objects in fact lack a cumulative reading with *together*, thus providing evidence for the crucial interaction between the local implicature and generalized economy.

## 5.4 Homogeneity and cumulative *together*

### 5.4.1 The homogeneity-cumulativity generalization

As we saw in sections 5.2 and 5.3, cumulative *together* is licensed with a variety of objects, exemplified in (54). Interestingly, definite plural objects and definite singular collective nouns disallow the analogous cumulative reading (55). For example, (55-a) doesn't have a reading which is true if Jane stole half of the books and Mary independently stole half of the books. Similarly, (55-b) and (55-c) are not true if Jane stole half of the books in the syntax collection and Mary independently stole the other half. Crucially, this blocking of the cumulative reading occurs only in the presence of *together*; the counterparts without *together* in (55-d) and (55-e) are true in the scenarios described above. We therefore need an account of how *together* blocks the cumulative readings with these objects.

(54) **Objects that license cumulative *together*:**

**Context:** Jane and Mary don't know each other, they each walked into the bookstore, which contains collections of books, and stole some of the books.

- a. Jane and Mary together stole **exactly 10 books**.
- b. Jane and Mary together stole **10 books**.
- c. Jane and Mary together stole **more than 10 books**.
- d. Jane and Mary together stole **all of the books**.

(55) **Objects that don't license cumulative *together*:**

**Context:** Same as (54)

- a. #Jane and Mary together stole **the books**.
- b. # Jane and Mary together stole **the syntax books**.
- c. # Jane and Mary together stole **the syntax collection**.
- d. Jane and Mary stole the books.

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<sup>18</sup>One might expect that cumulative *together* can be blocked even with the objects that in principle license the implicature, by creating contexts where the implicature doesn't arise. I discuss this in section 5.5, where I show that this expectation is not borne out. In particular, cumulative *together* appears to be licensed with these objects even when the implicature is not in fact computed. I argue that this is expected under a modular view of generalized economy, where it is blind to contextual parameters (see Fox, 2000; Fox & Hackl, 2006).

- e. Jane and Mary stole the syntax collection.

I argue in section 5.4.2 that the reason that cumulative *together* is not licensed in (55-a) and (55-b) is because the definite plural object gives rise to a homogeneity effect that blocks the local implicature that is needed for cumulative *together* to satisfy generalized economy. In particular, when the object is a universal quantifier, we were able to get strengthening below *together*, by negating the subpart alternatives, as shown in (56-a). On the other hand, the analogous strengthening seems to give rise to a contradiction when the object is a definite plural, as shown in (56-b): it is infelicitous to assert that Jane and Mary stole the books and to also assert that no part of Jane and Mary stole the books.

- (56) a. Jane and Mary stole all of the books, but Jane didn't steal all of the books and Mary didn't steal all of the books.  
b. # Jane and Mary stole the books, but Jane didn't steal the books and Mary didn't steal the books.

To show how homogeneity is responsible for the contrast in (56), I will first briefly introduce homogeneity with definite plurals and homogeneity removal with *all*. Consider the example in (57): here we can observe that neither (57-a) nor its negated counterpart in (57-b) is true in a scenario where Mary read only some of the books (J. D. Fodor, 1970). On the other hand, the counterpart with *all* does not give rise to an analogous truth-value gap: the negated counterpart of (58-b) is simply true iff (58-a) isn't. This explains the contrast between (56-a) and (56-b): in (56-a) it is consistent with Jane and Mary cumulatively stealing all of the books that neither Jane nor Mary individually stole all of the books. On the other hand, (56-b) is expected to give rise to a contradiction: the cumulative reading in the first sentence entails that each of Jane and Mary stole at least some books, but the second sentence says that neither Jane nor Mary read *any* books (due to homogeneity), therefore contradicting the first sentence. Therefore, the negation of the subpart alternatives in the definite plural case gives rise to a contradiction.

- (57) **Homogeneity with definite plurals:**  
a. *Mary read the books* **is true iff** Mary read all of the books.  
b. *Mary didn't read the books* **is true iff** Mary read none of the books.  
c. **Neither is true iff** Mary read only some of the books.
- (58) **Homogeneity removal with *all*:**  
a. Mary read all of the books.  
b. Mary didn't read all of the books.  
c. **No Gap:** Whenever (58-a) is not true, (58-b) is true.

I flesh out this argument in section 5.4.2, where I show that the fact that the subpart alternatives here can't be negated consistently with the preadjacent predicts that *together* is not licensed in (55-a) and (55-b), thus predicting a connection between whether or not the object gives rise to homogeneity and whether or not cumulative readings are licensed. This generalization is given in (59) and successfully predicts that, in addition to cumulative *together* being licensed with *all*, it is also licensed with numeral indefinite objects, such as *more than 10* and *at least 10*, which don't give rise to homogeneity.<sup>19</sup> In particular, as shown in (60-a), when the object is *more than 10*, the subpart alternatives can be negated without giving rise to a contradiction. This is because *Jane read more than 10 of the books* and its negated counterpart give rise to complementary truth-conditions, just like with the homogeneity-remover *all* in (58). Similarly, in (60-b), we see that *at least 10* also doesn't license homogeneity and therefore allows for the subpart alternatives to be negated without contradiction. We therefore see that all of the objects we discussed where cumulative *together* is licensed lack a homogeneity gap, as predicted by (59).

- (59) **Homogeneity-Cumulativity Generalization:** Cumulative *together* is licensed only if the object does not give rise to a homogeneity gap.
- (60) a. **No homogeneity with *more than 10*:**  
Jane and Mary stole more than 10 of the books, but Jane didn't steal more than 10 of the books and Mary didn't steal more than 10 books.
- b. **No homogeneity with *at least 10*:**  
Jane and Mary stole at least 10 books, but Jane didn't steal at least 10 books and Mary didn't steal at least 10 books.

In addition to the cases discussed above, the homogeneity-cumulativity generalization predicts further contrasts when we look at subatomic cumulativity. For example, in (61-a) we see that *together* allows for a cumulative reading with respect to parts of the collection, where Jane stole some parts of the collection and Mary independently stole the other parts. This sub-atomic

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<sup>19</sup>As pointed out by Viola Schmitt (p.c.), cumulative *together* seems to be more easily available when the object is a conjoined DP, with more than two conjuncts, than when the object is a definite plural. For example, (i) seems to have a reading where it is true even if Jane and Mary have no connection to each other and Jane stole book 1 and book 2 and Mary independently stole book 3 and book 4. The account I will propose will predict that the cumulative reading in (i) is only licensed if the object doesn't give rise to a homogeneity gap. One possible explanation for the contrast between (i) and (55-a) is that homogeneity is easier to locally accommodate when the object is a conjoined DP, removing the homogeneity gap. In fact, it has been observed that when *and* is stressed, conjunctions in English don't give rise to a homogeneity gap under negation (Szabolcsi & Haddican, 2004). More generally, my account will predict that speakers who accept (i) will not get a homogeneity gap with a sentence like *Jane didn't steal book 1, book 2, book 3 and book 4*. It remains to be seen whether this prediction is in fact borne out.

- (i) Jane and Mary together stole book 1, book 2, book 3 and book 4.

cumulative reading, on the other hand, is not licensed in (61-b), where *whole* is removed. Interestingly, just like *all* removes homogeneity with respect to atomic parts, *whole* has been argued to remove homogeneity with respect to subatomic parts (Löbner, 2000). We can see this in the contrast between (62) and (63): while in (62) we see a homogeneity gap where neither (62-a) nor (62-b) is true if Mary stole only some of the books from the syntax collection, the counterpart with *whole* in (63), just like *all*, lacks homogeneity.

- (61) **Context:** Jane and Mary don't know each other, they each walked into the bookstore, which contains collections of books, and stole some of the books.
- a. Jane and Mary together stole the whole syntax collection.
  - b. #Jane and Mary together stole the syntax collection.
- (62) **Sub-atomic homogeneity with definite singulars:**
- a. *Mary stole the syntax collection is true iff* Mary stole all of the books in the syntax collection.
  - b. *Mary didn't steal the syntax collection is true iff* Mary didn't steal any of the books from the syntax collection.
  - c. **Neither is true iff** Mary stole only some of the books from the syntax collection.
- (63) **Homogeneity removal with *whole*:**
- a. *Mary stole the whole syntax collection is true iff* Mary stole all of the books in the syntax collection.
  - b. *Mary didn't steal the whole syntax collection is true iff* Mary didn't steal all of the books in the syntax collection.
  - c. **No Gap:** Whenever (63-a) is not true, (63-b) is true.

Just as we saw above with the contrast between *all* and definite plurals, we can see here that while negating the subpart alternatives for definite singulars results in a contradiction (64-b), inserting the homogeneity remover *whole* allows for negating the subpart alternatives without contradiction (64-a). We can therefore conclude that the contrast between (61-a) and (61-b) provides additional support for the homogeneity-cumulativity generalization.

- (64) a. Jane and Mary stole the whole syntax collection, but Mary didn't steal the whole syntax collection and Jane didn't steal the whole syntax collection.
- b. #Jane and Mary stole the syntax collection, but Mary didn't steal the syntax collection and Jane didn't steal the syntax collection.

Now that I have established that there is a connection between whether or not the object gives

rise to homogeneity and whether or not cumulative *together* is licensed, we can turn to deriving the homogeneity-cumulativity generalization.

### 5.4.2 Deriving the generalization

As mentioned in section 5.4.1, any LF for cumulative *together* without *exh* violates generalized economy when the object is a definite plural. This is because the relevant  $\llbracket \text{XP} \rrbracket$  corresponding to the cumulative reading, given in (65-b), is cumulative (65-c). Following the general result in (28), we can therefore conclude that even the LF where there are two  $*$  operators above XP, as in (25-a), will violate generalized economy, just as we saw with *all* in sections 5.2 and 5.3.

- (65) a. Jane and Mary together stole the books  
 b.  $\llbracket \text{XP} \rrbracket = \lambda x. \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket) (\llbracket \text{the books} \rrbracket) (x) = 1$   
 c.  **$\llbracket \text{XP} \rrbracket$  is cumulative:**  
 $\forall x, y : \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket) (\llbracket \text{the books} \rrbracket) (x) = 1 \wedge \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket) (\llbracket \text{the books} \rrbracket) (y) = 1$   
 $\rightarrow \llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket) (\llbracket \text{the books} \rrbracket) (x \oplus y) = 1$

In order to have an LF which generates the cumulative reading without violating generalized economy, we therefore need to find a way to strengthen the predicate in (65-b) into a non-cumulative predicate. In what follows, I will show that homogeneity blocks any such strengthening from occurring. Generalized economy will therefore correctly predict that cumulative readings are not licensed when the object gives rise to homogeneity.

For concreteness, I assume, following the discussion in chapter 2, that homogeneity is due to a truth-value gap in the lexical entry of pluralization operators, such as  $*$  and  $**$  (Schwarzschild, 1994; Bar-Lev, 2025). More specifically, I assume following Bar-Lev (2025) that  $*$  and  $**$  have the trivalent lexical entries in (66-a) and (66-b) respectively.<sup>20</sup>

- (66) a. **Homogeneity with 1-place predicates:**  

$$\llbracket (* \rrbracket (P)) (x) = \begin{cases} 1 & \text{iff } \exists Q_{\langle e, t \rangle} [\forall y [Q(y) = 1 \rightarrow P(y) = 1] \wedge x = \oplus Q] \\ 0 & \text{iff } \neg \exists x' : x' \leq x \wedge P(x') = 1 \\ \# & \text{otherwise} \end{cases}$$
  
 b. **Homogeneity with 2-place predicates:**

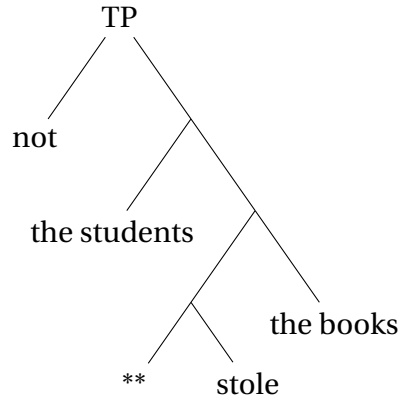
<sup>20</sup>The falsity conditions that Bar-Lev in fact assumes require that there is no part of  $x$  that is part of an individual that satisfies the predicate (i.e. no overlap). For simplicity, I assume here that *steal* is a distributive predicate and is true only of atoms, thus making the no overlap requirement equivalent to the simpler falsity conditions in (66) and (66-b). I provide the general result with non-distributive predicates in Appendix F, showing that homogeneity blocks the local implicature in the same way once we provide the full falsity conditions of  $*$  and  $**$  for non-distributive predicates.

$$(\llbracket ** \rrbracket (P))(x)(y) = \begin{cases} 1 & \text{iff } \exists Q[\llbracket \forall x', y' [Q(x')(y') = 1 \rightarrow P(x')(y') = 1] \wedge x = \\ & \oplus \lambda x. \exists y [Q(x)(y) = 1] \wedge y = \oplus \lambda y. \exists x [Q(x)(y) = 1]] \\ 0 & \text{iff } \neg \exists x', y' : x' \leq x \wedge y' \leq y \wedge P(x')(y') = 1 \\ \# & \text{otherwise} \end{cases}$$

Looking at an example with a 2-place predicate in (67), the falsity condition of **\*\*** correctly predicts that (67-a) is true iff no student stole any book, as shown in (67-b). The positive counterpart of (67), on the other hand, is as before predicted to be true iff every student stole at least one book and every book was stolen by at least one student.<sup>21</sup> We therefore predict a gap where neither (67) nor its positive counterpart is true. For example, in a scenario where the students stole half of the books, neither (67) nor the counterpart without negation is predicted to be true (they're both predicted to be undefined). We therefore see that the truth-value gap in the lexical entry of **\*\*** in (66-b) successfully predicts the non-complementary truth-conditions of the positive and negative sentences with 2-place predicates.

(67) The students didn't steal the books.

a. LF:



b.  $\llbracket (67-a) \rrbracket = 1$  iff  $(\llbracket ** \rrbracket (\llbracket \text{stole} \rrbracket))(\llbracket \text{the books} \rrbracket)(\llbracket \text{the students} \rrbracket) = 0$   
 iff  $\neg \exists x, y : y \leq \llbracket \text{the books} \rrbracket \wedge x \leq \llbracket \text{the students} \rrbracket \wedge \llbracket \text{stole} \rrbracket (y)(x) = 1$

Before seeing how homogeneity blocks the local implicature with cumulative *together*, we first need to ask how exhaustification behaves in general when the alternatives denote trivalent propositions. In particular, there is a question of whether exhaustification demands that the alternatives are false or simply that they're not true when the alternatives have a homogeneity gap.

To answer this question, we can consider a basic example where there is a definite plural in the scope of *some* and ask how the *some but not all* implicature interacts with homogeneity. If exhaustification only requires excluded alternatives to be not true, we would get an implicature

<sup>21</sup> Assuming again that *steal* is a distributive predicate.



that *not all of the students read all of the books* (68-a). This option therefore does not predict any interaction between the homogeneity gap and scalar implicatures. On the other hand, if exhaustification requires that excluded alternatives are false, the stronger implicature that *not all of the students read any books* is predicted (68-b).<sup>22</sup>

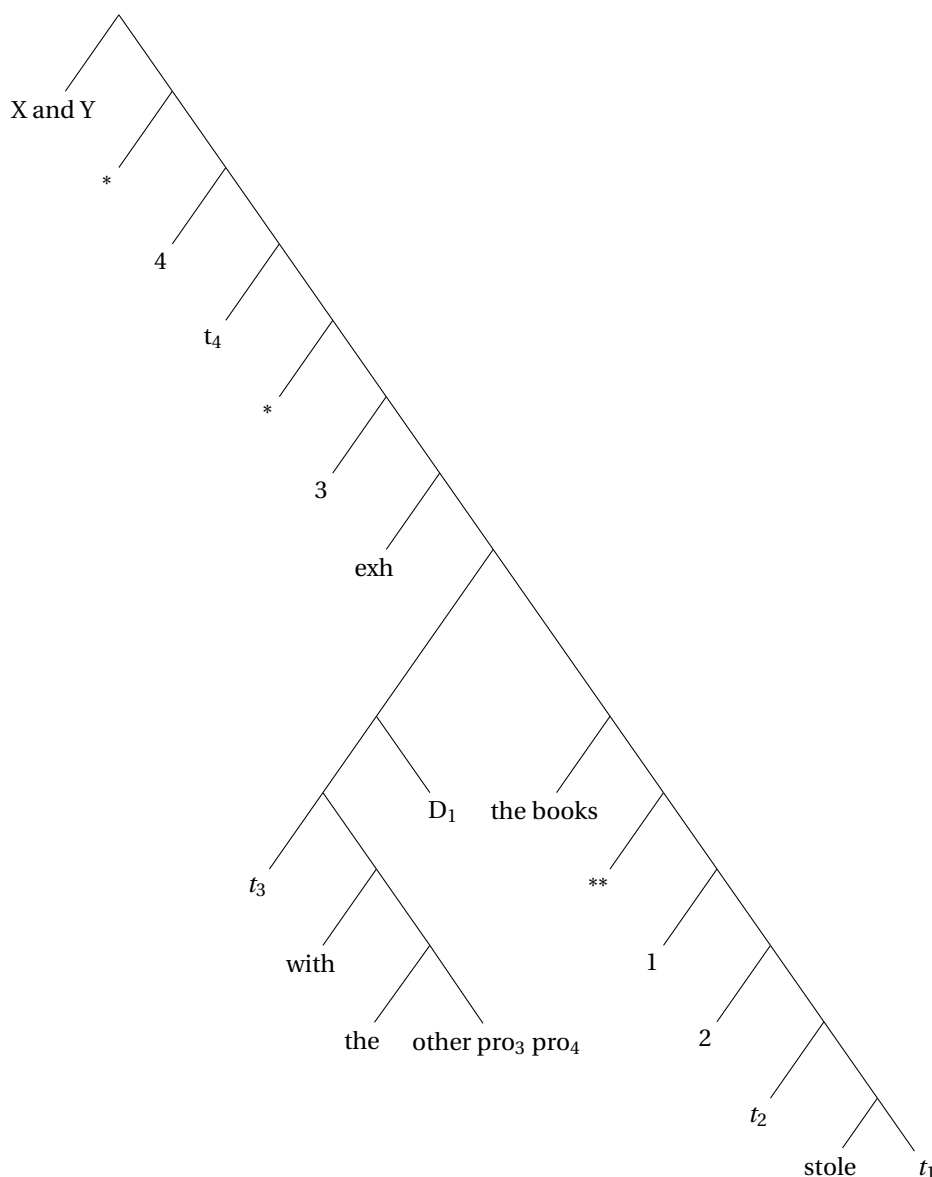
- (68) a. **Option 1: Innocent exclusion only demands that alternatives are not true**  
 $Exh(\llbracket \text{Some of the students read the books} \rrbracket) = 1$  iff  
 $\llbracket \text{Some of the students read the books} \rrbracket = 1 \wedge \llbracket \text{All of the students read the books} \rrbracket \neq 1$   
iff Some of the students read all of the books and not all of the students read all of the books.
- b. **Option 2: Innocent exclusion demands that alternatives are false**  
 $Exh(\llbracket \text{Some of the students read the books} \rrbracket) = 1$  iff  
 $\llbracket \text{Some of the students read the books} \rrbracket = 1 \wedge \llbracket \text{All of the students read the books} \rrbracket = 0$   
iff Some of the students read all of the books and not all of the students read any books.

The prediction of Option 2 seems to be borne out, as shown in (69). In the scenario in (69), where all of the students read at least one book, *some of the students read all of the books* is acceptable (69-b), while the counterpart with the definite plural object is infelicitous (69-a). This contrast is only predicted by Option 2, where the definite plural gives rise to the stronger implicature in (70-a), compared to the implicature of the counterpart with *all* in (70-b). We therefore have evidence that, when homogeneity gives rise to a trivalent proposition in the scope of *exh*, excluding the alternatives with *exh* requires that the alternatives be false rather than simply not true. This will allow us to predict that definite plural objects and *all* objects behave differently with respect to how *exh* interacts with the economy constraint, since they have different falsity conditions.

- (69) **Scenario:** The TA is looking at the list of books that each student read in the class. They see that all of the students read at least some of the books, and that some of them read all of the books. They report to the professor:
- a. # Some of the students read the books.  
b. Some of the students read all of the books.
- (70) a. Some of the students read the books.  
**Implicature:** At least one student read none of the books.  
b. Some of the students read all of the books.  
**Implicature:** Not all of the students read all of the books.

<sup>22</sup>I'm assuming here that homogeneity projects with a Strong Kleene logic from the scope of *not all*.

We are now finally in a position to evaluate whether exhaustification can rescue cumulative *together* from violating generalized economy when the object is a definite plural. Consider the relevant LF with *exh* in (71-a).



- (71) a. b.  $Alt_{exh} = \{ [t_3 \text{ with the other } D_2] ** \text{ stole the books, } [t_3 \text{ with the other } D_3] ** \text{ stole the books ...} \}$

Crucially, given that \*\* has a homogeneity gap when it applies to the definite plural object *the books*, *exh* won't be able to exclude any alternatives here. In particular, the cumulative truth-conditions corresponding to the prejacent of *exh* entail that each atomic part of *x with the other* stole at least one book. Excluding the subpart alternatives corresponding to restricting the subject to proper subparts of *x with the other* demands that no atomic part of *x with the*

*other* stole any book. The local implicature therefore results in a contradiction and can't exclude any of the alternatives. To see this with an example, consider (72), where the *x with the other* DP denotes a plural individual Z and suppose that the domain variable D<sub>2</sub> restricts the DP to a proper subpart R. Now, the prejacent of *exh* in (72-a) entails that each individual which is a part of Z stole a book (and therefore that every individual which is a part of R stole a book). Given homogeneity, the alternative with D<sub>2</sub> is false iff no subpart of R stole a book. Excluding (72-b) will therefore contradict the prejacent in (72-a) and *exh* is therefore unable to exclude this alternative.

- (72) **Suppose**  $\llbracket [t_3 \text{ with the other } \text{pro}_3 \text{ pro}_4] D_1 \rrbracket^g = Z$  **and take an arbitrary** D<sub>2</sub> **such that**  
 $\llbracket [t_3 \text{ with the other } \text{pro}_3 \text{ pro}_4] D_2 \rrbracket^g = R$ , **where**  $R < Z$ :
- a.  $\llbracket [t_3 \text{ with the other } \text{pro}_3 \text{ pro}_4] D_1 \rrbracket^{**} \text{ stole the books} \rrbracket^g = 1 \rightarrow$   
 $\forall x \leq_{AT} Z : \exists y \leq_{AT} \llbracket \text{the books} \rrbracket : \llbracket \text{stole} \rrbracket(y)(x) = 1$
  - b.  $\llbracket [t_3 \text{ with the other } D_2] \rrbracket^{**} \text{ stole the books} \rrbracket^g = 0$  **iff**  
 $\neg \exists x \leq_{AT} R : \exists y \leq_{AT} \llbracket \text{the books} \rrbracket : \llbracket \text{stole} \rrbracket(y)(x) = 1$

Since *exh* can't exclude the subpart alternatives in (71-a), the predicate XP remains cumulative when *exh* is inserted, therefore leading to a violation of generalized economy.

To summarize, I argued in this section that *exh* is unable to give rise to the necessary strengthening to license cumulative *together* when there is a definite plural in object position. In particular, the presence of a homogeneity gap with respect to the object blocks strengthening, since strengthening would give rise to a contradiction, as shown again in the paraphrase in (73). More generally, this predicts that for any object which gives rise to homogeneity, the subpart alternatives won't be negatable without contradiction, thus predicting that *exh* can't rescue cumulative *together* from violating economy with these objects. We have therefore derived the homogeneity-cumulativity generalization outlined in section 5.4.1.

- (73) # Jane and Mary cumulatively stole the books, but Jane didn't steal the books and Mary didn't steal the books.

### 5.4.3 Interim summary

I have therefore shown so far in this chapter that generalized economy predicts the full distribution of readings of *together*. In particular, I showed that the unattested distributive reading is successfully ruled out, while collective, simultaneous and proximate *together* are always licensed. For cumulative *together*, I argued that whether the economy constraint is satisfied depends on certain properties of the object DP. First, if the object DP is a non-monotonic quantifier, cumulative *together* is straightforwardly licensed. On the other hand, when the object DP is an

upward-entailing quantifier, cumulative *together* requires a local *exh* to strengthen the predicate that \* applies to in order for it to satisfy the economy constraint. Finally, we saw in this section that when the object gives rise to homogeneity, this necessary strengthening is blocked and cumulative *together* is correctly predicted to be blocked by the economy constraint. The economy constraint therefore makes correct and intricate predictions regarding the licensing of different readings with *together*.

## 5.5 Economy and modularity

This chapter so far has relied on a modular generalized economy constraint, which assumes that the deductive system doesn't access all of the information that goes into evaluating truth-conditions in a given context. In particular, in order to give a general economy constraint that applies beyond the distribution of *together*, I had to assume that the deductive system is blind to the identity of individual-denoting DPs. The goal of this section is to show that this modularity is independently needed, even if we assume the narrower version of the economy constraint from chapter 4, which applies only to the deletion of *together*. Section 5.5.1 argues that the relevant economy calculation is blind to the common ground, and section 5.5.2 argues that it is also blind to contextual parameters that determine which alternatives are relevant for calculating an implicature.

These arguments parallel existing arguments for an encapsulated deductive system (DS) with respect to which certain constraints are evaluated (Fox, 2000; Gajewski, 2002; Fox & Hackl, 2006; Chierchia, 2013). In section 5.5.3, I show that the modularity properties of economy with *together* are parallel to what Fox (2000) argues for scope economy. I then argue that scope economy can be derived as a special case of the modular generalized economy constraint proposed in this chapter. The resulting picture is one in which economy constraints, including both scope economy and economy with *together*, are evaluated relative to a deductive system which only has access to a subset of the information involved in computing truth-conditions.

### 5.5.1 Blindness to the common ground

To see how the economy constraint interacts with the context, consider the preliminary example in (74). In (74), A's utterance makes it clear that it is common ground that neither Jane nor Mary lifted the piano alone. Given this common ground, the collective reading of *together* in (74-b) is contextually equivalent to the counterpart without *together* in (74-c). In particular, while (74-b) is true iff Jane and Mary lifted the piano collectively (75-a), the LF without *together* (and where \* remains) will be true iff either Jane and Mary lifted the piano together or separately (75-b).

Since all the worlds in the context are such that Jane and Mary didn't lift the piano separately, the intersection of the context in (75-c) with (75-a) and the intersection with the counterpart without *together* in (75-b) will be equivalent: as shown in (75-d), if we restrict to worlds in the common ground, both (74-b) and (74-c) are true iff Jane and Mary lifted the piano together.<sup>23</sup>

- (74) a. A: All I know is that Jane didn't lift the piano alone and Mary didn't lift the piano alone. Do you have any more information about the competition?  
 b. B: Jane and Mary lifted the piano together.  
 c. B': Jane and Mary lifted the piano.
- (75) a.  $\llbracket \text{Collective LF for (74-b)} \rrbracket = 1$  iff  $\llbracket \text{lift the piano} \rrbracket (j \oplus m) = 1$   
 b.  $\llbracket \text{LF without } together \rrbracket = 1$  iff  
 $(\llbracket \text{lift the piano} \rrbracket (j) = 1 \wedge \llbracket \text{lift the piano} \rrbracket (m) = 1) \vee \llbracket \text{lift the piano} \rrbracket (j \oplus m) = 1$   
 c.  $C = \lambda w. \llbracket \text{lift the piano} \rrbracket^w (j) = 0 \wedge \llbracket \text{lift the piano} \rrbracket^w (m) = 0$   
 d.  $C \cap \lambda w. \llbracket \text{Collective LF for (74-b)} \rrbracket^w = C \cap \lambda w. \llbracket \text{LF without } together \rrbracket^w = \lambda w. \llbracket \text{lift the piano} \rrbracket (j \oplus m) = 1$

We can therefore conclude that the notion of equivalence that is relevant for evaluating the economy constraint is logical equivalence rather than contextual equivalence. In particular, despite the collective LF for (74-b) being contextually equivalent to the counterpart without *together*, the economy constraint is not violated in (74-b), as evidenced by the fact that B's response is perfectly felicitous. In other words, the evaluation of the economy constraint has to be blind to contextual information, such that in order to violate economy, the LF with *together* has to be equivalent to the LF without *together* **in any context** and not just in the actual context of the utterance. This is formulated in (76).

(76) **Modular economy constraint:**

If  $\forall C : C \cap \lambda w. \llbracket \text{LF with } together \rrbracket^w = C \cap \lambda w. \llbracket \text{LF without } together \rrbracket^w$ , then the LF with *together* is infelicitous.

We can see the blindness to context for the other readings of *together*, beyond the collective one above. To see this for the cumulative reading, consider the example in (77). Here, it is entailed

<sup>23</sup>One might ask how we can be sure that the sentence in B is evaluated with respect to the context where it is common ground that neither Jane nor Mary lifted the piano alone. In particular, one could ignore the information provided by A when evaluating B's response. We can ensure that this information can't be ignored by having B utter both sentences: I knew that neither Jane nor Mary lifted the piano alone, but today I learned that they lifted it together. Here, B is presupposing the information that makes the LF with *together* contextually equivalent to the one without *together* in the same sentence in which collective *together* is asserted. Yet, the sentence with *together* remains felicitous. We can therefore definitively conclude that collective *together* is licensed even when the LF with *together* is contextually equivalent to the one without *together*

by the context that each of Jane and Mary shook hands with less than 5 politicians. This makes the cumulative reading of (77-b) and the counterpart without *together* in (77-c) contextually equivalent: out of the worlds compatible with the context, both (77-b) and (77-c) are true iff Jane and Mary cumulatively shook hands with more than 5 politicians.

- (77) a. A: All I know is that Jane shook hands with less than 5 politicians and Mary shook hands with less than 5 politicians. Do you have any more information about what happened at the meeting?
- b. B: Jane and Mary together shook hands with more than 5 politicians.
- c. B': Jane and Mary shook hands with more than 5 politicians.

This conclusion that the economy constraint is blind to contextual information is surprising if we think of economy in pragmatic terms. In particular, one might have thought that our economy constraint follows in some way from the Gricean maxim of manner, which requires, all else being equal, that speakers be as brief as possible. In fact, Meyer (2013) formulates a principle, which she calls Efficiency, based on this Gricean idea. This principle is given in (78) and rules out LFs which are equivalent to simpler LFs that result from deleting a number of nodes. This constraint can be thought of as follows: a speaker can't utter  $\phi$  when  $\phi$  provides the same information as a more brief utterance  $\psi$ , violating the Maxim of Manner. This way of conceptualizing (78) only makes sense if the notion of equivalence in (78) is contextual rather than logical equivalence. In particular, the Maxim of Manner is presumably violated in the same way if  $\phi$  and a simpler alternative  $\psi$  are contextually equivalent but not logically equivalent. This is because in the given context, there is a simpler alternative that provides the same information as  $\phi$ .

- (78) a. **Efficiency:**  
An LF  $\phi$  is ruled out if there is a distinct competitor  $\psi$  such that (i)  $\psi < \phi$  and (ii)  $\llbracket \psi \rrbracket = \llbracket \phi \rrbracket$
- b.  $\psi < \phi$  iff  $\psi$  derived from  $\phi$  by deleting a number of nodes.

Interestingly, the behavior of the economy constraint on *together* is different from other cases of redundancy that follow from (78). For example, conjunctions where one of the conjuncts can be deleted without changing the truth-conditions have been observed to be infelicitous (e.g. Schlenker, 2009). In (79-a), the second conjunct entails the first one and the sentence is infelicitous. The infelicity of (79-a) follows from Efficiency: (79-a) has a simpler alternative, where the first conjunct is deleted, which has the same truth-conditions as (79-a). The notion of equivalence that is relevant for ruling out redundant conjunctions here has to be contextual rather than logical equivalence. For example, in (79-b), the second conjunct contextually entails the first one but doesn't logically entail it. In particular, the conjunction in (79-b) is equivalent to

the second conjunct, *Jane lives in Paris*, only in a context which entails that Paris is in France, yet the conjunction in (79-b) is infelicitous. This follows only if Efficiency rules out LFs which are contextually equivalent to simpler LFs.

- (79) a. # Jane read more than 5 of the books and she read 7 of them.  
 b. #Jane lives in France and in Paris.

At this point, it appears that in order to maintain our results for the distribution *together*, we have to reject the pragmatic principle of efficiency in (78), or at the very least assume that Efficiency can be violated in certain environments. In particular, if we apply the context-sensitive Efficiency principle to the deletion of *together*, we would incorrectly predict that *together* is infelicitous in (74-b) for example, where the LF with and without *together* are contextually equivalent. My account for *together* therefore requires an alternative explanation for patterns which have been argued to follow from Efficiency, such as the infelicity of (79-a) and (79-b). I leave it to future work to provide such an account.

### 5.5.2 Blindness to alternative pruning

In addition to blindness to the common ground, I will argue that the economy constraint has to be blind to certain parameters that determine which alternatives are considered when computing implicatures. In particular, as we saw in section 5.3, in order to make the LF with cumulative *together* have distinct truth-conditions from the LF without *together* with certain objects, such as *all* and *more than 10*, *exh* has to give rise to a local implicature that strengthens the meaning of the predicate that \* applies to. At the same time, we know that the generation of scalar implicatures depends on contextual factors, raising a question of whether contexts or environments which block scalar implicatures no longer license cumulative *together*. I will show in this section that cumulative *together* is licensed even when the relevant implicature is blocked in the context, thus providing additional evidence for modularity.

To see this, consider the examples in (80) and (81). Starting with (80-a), the first sentence asserts that the speaker doesn't know whether individually Jane and Mary shook hands with all of the politicians. Now, as we saw in section 5.3, in order to avoid violating economy, the cumulative reading has to be strengthened to assert that Jane didn't shake hands with all of the politicians and that Mary didn't shake hands with all of the politicians. The problem is that the desired strengthened reading contradicts the first sentence, as shown in (80-b). Similarly in (81-a), the required strengthening in (81-b) results in a contradiction. We therefore see that in both (80-a) and (81-a), strengthening is blocked, yet the sentences remain felicitous.

- (80) a. I don't know if Jane shook hands with all of the politicians or if Mary did, but Jane

and Mary together shook hands with all of the politicians.

b. **Paraphrase with the required strengthening:**

#I don't know if Jane shook hands with all of the politicians or if Mary did, but Jane and Mary together shook hands with all of the politicians and neither of them alone shook hands with all of the politicians.

(81) a. I don't know if Jane shook hands with more than 10 politicians or if Mary did, but Jane and Mary together shook hands with more than 10 politicians.

b. **Paraphrase with the required strengthening:**

#I don't know if Jane shook hands with more than 10 politicians or if Mary did, but Jane and Mary together shook hands with more than 10 politicians and neither of them alone shook hands with more than 10 politicians.

I will argue that despite the fact that the absence of an implicature makes the LF without *together* equivalent to the one with *together* in (80-a) and (81-a), the economy constraint is not violated. In particular, it has been argued that the calculation of an implicature is sensitive to a relevance parameter which allows for exhaustification to be computed relative to only a subset of the possible alternatives of the prejacent (e.g. Fox & Katzir, 2011). I argue, following similar modularity considerations in Fox (2000), that the acceptability of (80-a) and (81-a) follows if the economy constraint is modular, such that it does not have access to the value of this parameter when evaluating equivalence between the LF with and without *together*. Therefore, despite the parameter having to be set in such a way that no implicature is generated, the economy constraint will not be able to see this. Instead, economy will only rule out cumulative *together* if no matter what value this parameter is set to, the LF with *together* will be equivalent to the one without *together*.

To see this in a bit more detail, consider the example in (82), which gives rise to an implicature that John didn't read more than 5 books, resulting in an exactly 5 interpretation. In the context in (82), we see that (82-a) gives rise to a weaker implicature that John didn't read 10 or more books. In particular, in the context in (82), it is only relevant whether John read at least 5 or at least 10 books and (82-a) can have a reading where it simply means that John read between 5 and 9 books. This is evidenced by the oddness of the response in (82-b).

(82) **Scenario:** For this class, students can either read 5 books throughout the semester and take a final exam or they can read 10 books and they won't have to take a final exam. The TA, who sees that John is on the list of students who have to take the final exam, says to the professor:

a. John read 5 books, so he has to take the exam.



- John didn't read 10 books.
- b. Professor: # No you're wrong, he actually read 6 books.
- c. **Relevant alternatives:** { John read 10 books }

We therefore see that generation of scalar implicatures in general depends on context. We can formalize this by having *exh* take a relevance parameter *R* as an argument, which denotes a set of propositions that are relevant in the given context. *exh* will then compute an implicature based only on the subset of its formal alternatives that denote propositions in *R*, as shown in (83).

$$(83) \quad \text{Exh}_R(\phi) \\ \text{ALT}_{\text{Exh}_R} = \{\psi : \psi \in \text{Alt}(\phi) \wedge \llbracket \psi \rrbracket \in R\}$$

Now, the data in (80-a) and (81-a) follows if we assume that the economy constraint does not have access to the contextually-determined value for *R* when attempting to prove that the LF with *together* is equivalent to the LF without *together*. More specifically, in (80-a), in order to avoid a contradiction, *R* can't include the propositions corresponding to *Jane shook hands with all of the politicians* and to *Mary shook hands with all of the politicians*. This will allow us to have an LF with *exh* which does not generate the problematic implicature in (80-a). At the same time, in order for the economy constraint to be violated, the deductive system has to be able to prove that the LF for the smallest clause dominating *together* is equivalent to the result of deleting *together*, regardless of the value of *R*. This can't be proven in (80-a). In particular, given an alternative value of *R* where it does contain the propositions corresponding to the subpart alternatives, *exh* will strengthen the predicate that \* applies to and break the equivalence between the LFs with and without *together*, as we saw in section 5.3. The modular version of the economy constraint therefore correctly predicts that cumulative *together* is licensed in (80-a) and (81-a).

The prediction is therefore that economy will rule out cumulative *together* if the LF with and without *together* are equivalent regardless of what the value of *R* is. We saw that this predicts that cumulative *together* is always licensed when the object is headed by *more than 10* or *all*, regardless of whether an implicature is in fact computed. On the other hand, when the object is a definite plural, as in (84), the LF with cumulative *together* will always be equivalent to the counterpart without *together*, regardless of the value of *R*. In particular, since the alternatives corresponding to *Jane shook hands with the politicians* and *Mary shook hands with the politicians* can't be negated without contradicting the prejacent, no implicature can be computed in (84) and therefore, regardless of the value of *R*, both (84) and the counterpart without *together* are predicted to be true if Jane and Mary cumulatively shook hands with all of the politicians. We therefore see that in (84), unlike with *all* and *more than 10*, the economy constraint can prove

equivalence without having access to the value of R, thus correctly predicting that cumulative *together* in (84) violates economy.

(84) Jane and Mary together shook hands with the politicians.

I have therefore provided independent evidence that the economy constraint has to be modular, such that it is blind to the common ground and to the value of the relevance parameter R. As we saw throughout this chapter, the modularity of the economy constraint was also crucial in formulating a general economy constraint, repeated in (85), that applies beyond the distribution of *together*. At this point, we can ask whether we see other phenomena which are governed by this modular generalized economy constraint. I turn to one such case in section 5.5.3, where I argue that Fox's (2000) scope economy can be seen as a special case of the generalized economy constraint in (85).

(85) a. **Generalized Economy Constraint:**

An LF  $\phi$  is ruled out if the deductive system can prove that there is a competitor  $\psi$  such that (i)  $\psi < \phi$  and (ii)  $\llbracket \psi \rrbracket = \llbracket \phi \rrbracket$

b. An LF  $\psi < \phi$  if  $\psi$  can be derived from  $\phi$  by deleting a number of nodes.

### 5.5.3 Scope economy as a special case of generalized economy

Fox (2000) argues for an economy constraint that determines whether covert movement of quantifiers is licensed in different environments. In particular, Fox argues that covert movement that is not needed independently for syntactic reasons or to avoid a type mismatch is only possible if the LF with movement gives rise to different truth-conditions than the LF without movement (86). Interestingly, Fox shows that his scope economy constraint is modular in the same way that I argued for economy with *together* above. The goal of this section is to briefly present Fox's arguments for scope economy and its modular properties and then show that scope economy follows from the generalized economy constraint proposed in this chapter.

(86) **Scope Economy:** Covert scope-shifting operations can apply only if they affect semantic interpretation (i.e., only if inverse-scope and surface-scope are semantically distinct).

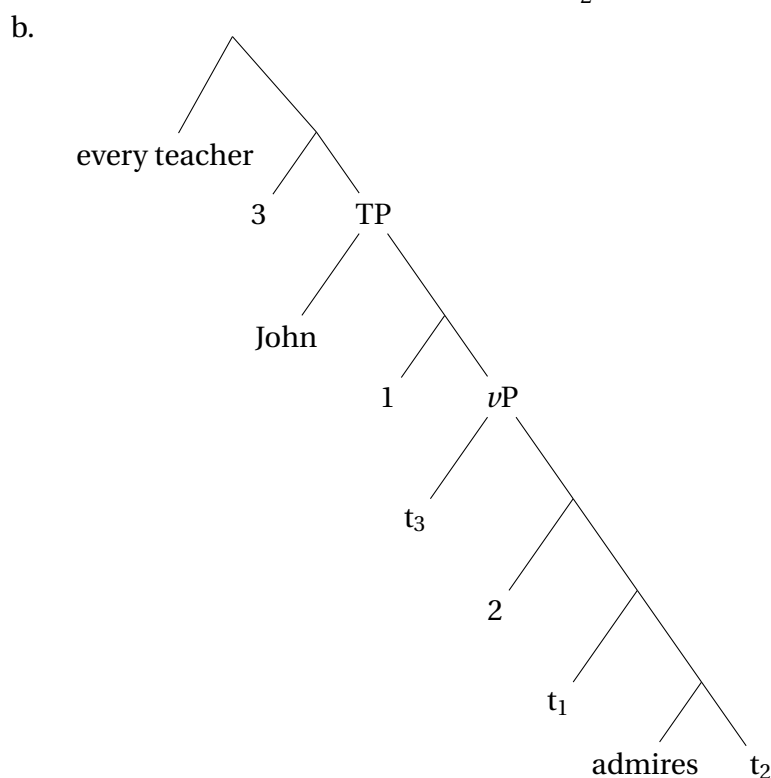
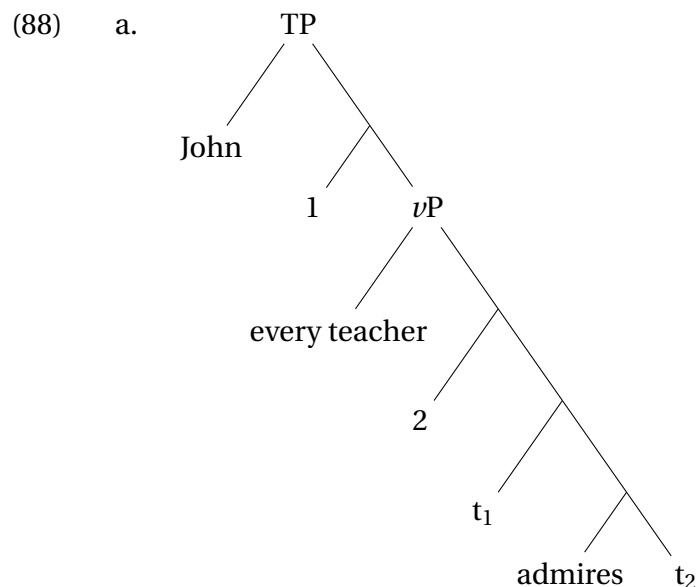
While I won't be able to present the full evidence for scope economy in detail here, I will briefly illustrate Fox's argument from parallelism in ellipsis in (87). In particular, since covert movement that violates scope economy by definition does not result in any truth-conditional effect, it is not possible to directly detect whether this movement is ruled out or not. Fox shows that this can be probed indirectly by using ellipsis to detect the scope of a given DP. In particular, consider the contrast between (87-a) and (87-b): while in (87-a), the object can take inverse scope over the

subject in the first sentence, this is not possible in (87-b).

- (87) a. A boy admires every teacher. A girl does too.  
A boy > Every teacher      Every teacher > a girl  
b. A boy admires every teacher. John does too.  
A boy > Every teacher      ✗Every teacher > a boy

Fox argues that the contrast in (87) follows from scope economy, coupled with an independently motivated assumption that when the object takes scope above the subject in the first sentence, the object necessarily also takes scope above the subject in the second sentence with ellipsis. In particular, Fox argues that in (87-b), scope economy prevents *every teacher* from taking scope above *John* thus forcing the object in the first sentence to also take scope below the subject. On the other hand, scope economy allows *every teacher* to take scope above *a girl* in (87-a), thus predicting that inverse scope is also licensed in the first sentence.

To see how scope economy prevents *every teacher* from taking scope above *John* in *John admires every teacher*, we must first make appropriate assumptions about what movement is independently needed for syntactic reasons and type reasons. Fox assumes that the subject always moves from spec,  $\nu$ P to spec, TP and that a quantifier object has to QR to a higher specifier of  $\nu$ P in order to be interpreted. This results in the surface scope LF in (88-a), where John takes scope above *every teacher*. In order to generate the inverse scope LF, *every teacher* therefore has to covertly move to a position higher than the subject in spec, TP, as shown in (88-b). Now, the LF in (88-b) is ruled out by the scope economy principle in (86). In particular, both (88-a) and (88-b) are true iff John admires every teacher. The covert movement of *every teacher* to a position higher than *John* in (88-b) therefore does not affect semantic interpretation and is correctly ruled out by scope economy.



Now, Fox (2000) argues that scope economy has to be evaluated relative to a formal deductive system that does not have access to all of the information involved in computing truth-conditions. I present one of these examples in (89). In (89), the inverse scope reading, where *more than half* takes scope above negation, and the surface scope reading turn out to be equivalent: they are both true iff Rob speaks 4 or less of the languages spoken in Sydney. Interestingly, this equivalence depends on the fact that the numeral 9 is odd and that it is not possible to speak half a language. To see this, consider the alternative example in (90), where 9 is replaced with

the even numeral 8. Here, the inverse scope and the surface scope are no longer equivalent. In particular, if Rob speaks exactly 4 of the 8 languages, the inverse scope LF is false, while the surface scope LF is true.

- (89) Rob doesn't speak more than half of the 9 languages spoken in Sydney.
- a. **Inverse scope:** True iff there are more than 4.5 languages spoken in Sydney that Rob doesn't speak.  
= Rob speaks 4 or less of the languages spoken in Sydney.
  - b. **Surface scope:** True iff it is not the case that Rob speaks more than 4.5 languages.  
= Rob speaks 4 or less of the languages spoken in Sydney.
- (90) Rob doesn't speak more than half of the 8 languages spoken in Sydney.
- a. **Inverse scope:** False if Rob speaks 4 languages.
  - b. **Surface scope:** True if Rob speaks 4 languages.

Fox shows that (89) does in fact license an inverse scope LF where *more than half* takes scope above negation, despite the inverse scope reading being equivalent to the surface scope one. Fox argues that this can be explained if we assume that scope economy is blind to the identity of the numeral 9 and crucially to the fact that 9 is odd. We therefore see that scope economy has to be evaluated relative to a system that only has access to a subset of the information that goes into computing the truth-conditions of (89) in a given context.

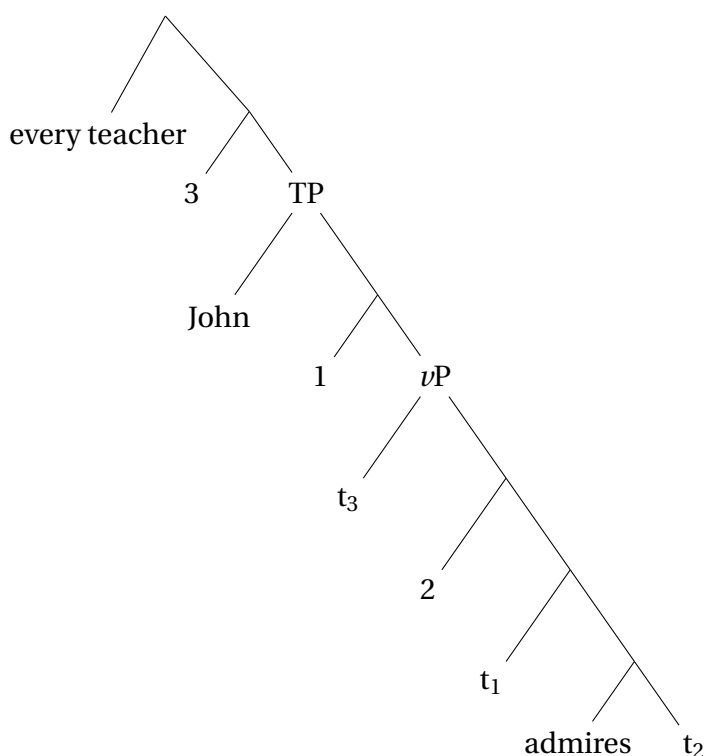
Fox (2000) provides several other examples to motivate the conclusion that scope economy is modular, which I won't be able to get into here. I will briefly note though that scope economy has to be blind to contextual information and to preceding information in the sentence in the same way that I argued for economy with *together* in section 5.5.1. For example, in (91), since there is only one student who has access to the books in the library, the inverse scope reading of the second sentence *for every book there is a student who drew on it* is contextually equivalent to the surface scope *there is a specific student that drew on every book*. Yet, inverse scope is licensed in the first sentence in (91-a): (91-a) is true in a scenario where a different professor drew on every book. Similarly, in (92), given the first sentence, the inverse scope and surface scope reading in the ellipsis site are equivalent, yet (92) allows for *every book* to take scope above *a professor*.

- (91) **Context:** We know that only one student could have access to the books in the library, so if something happened to them and a student did it, it would have to be them. Someone says:
- a. A professor drew on every book and a student did too.
- (92) I knew that only one student could have had access to the books in the classroom, but

today I learned that a professor drew on every book and a student did too.

We can therefore conclude that scope economy is modular in the same way as the economy constraint with *together* above. This raises the question of whether scope economy can be made to follow from the generalized modular economy principle given in (85). Consider again the LF in (93) which is ruled out by scope economy. If we assume the copy theory of movement, where the trace  $t_3$  is in fact a lower copy of *every teacher*, then the surface scope LF in (88-a) can be derived from (93) by deleting the higher copy of *every teacher* and the index 3. We therefore see that (93) is ruled out by generalized economy: the deductive system can prove that (93) is equivalent to a simpler alternative where both *every teacher* and the index 3 are deleted.

(93)



More generally, whenever covert movement which is not independently needed does not result in any truth-conditional effect, there will always be a simpler alternative (where covert movement did not occur) which is truth-conditionally equivalent to the LF with movement. Thus, we can tentatively conclude that the scope economy principle in (86) can be thought of as a special case of generalized economy. It remains to be seen whether all of the properties of scope economy, such as its locality properties, are parallel with those I have argued for with *together*, but the parallels discussed in this section seem promising and suggest that there is in fact a general grammatical economy constraint that rules out LFs which are equivalent to simpler LFs.

## 5.6 Conclusion

This chapter provided a detailed account of how the disambiguating role of *together* follows from a generalized economy constraint in the grammar, coupled with a uniform compositional meaning for *together*. More specifically, I proposed that the distribution of readings with *together* is governed by a meaning-sensitive economy constraint that rules out LFs when the deductive system can prove that they are equivalent to simpler LFs.

One central finding of this chapter is that this economy constraint, which has consequences beyond the distribution of *together*, must be modular in nature. In particular, I argued that when proving equivalence for the purposes of economy, the deductive system must be blind to the identity and cardinality of individual-denoting DPs, to the common ground, and to contextual parameters involved in alternative pruning. These modularity assumptions are crucially not specific to *together*, but are shared with other grammatical economy constraints like scope economy. We thus begin to see evidence for a unified general economy principle that has consequences for a variety of different phenomena.

This chapter also showed that generalized economy makes fine-grained predictions regarding the licensing of cumulative *together*. Namely, I showed that with a certain class of object DPs, a local implicature is necessary in order to ensure that the cumulative reading with *together* satisfies economy. This predicts that in environments where this local implicature is not available, cumulative *together* will be blocked. I argued that this prediction is borne out with objects that give rise to a homogeneity gap. More specifically, I showed that homogeneity blocks the necessary local implicature, thereby correctly predicting that objects that give rise to homogeneity do not license cumulative *together*.

More broadly, the analysis of *together* developed in this chapter illustrates the role of meaning-sensitive grammatical constraints in determining which scope configurations are licensed in a particular construction. Namely, the independently motivated economy constraint allowed us to develop an account of *together* where its different readings are not specified in the lexical entry of *together* itself. Rather, they emerge from the interaction between a simple compositional semantics, independently motivated pluralization operators, and a modular economy constraint.

## Appendix A: Word order with comitatives and *together*

The analysis I've developed for *together* in this chapter and chapter 4 assumes that *together* is decomposed into a coordinator *with* and a reciprocal anaphor. For the collective reading of *together*, I proposed that the *with the other* conjunction is generated in spec,  $\nu P$  and only the first conjunct moves to a higher position. On the other hand, I argued that the other readings of *together*, including the cumulative, proximate and simultaneous ones, require movement of the entire *x with the other* conjunction from spec,  $\nu P$  to different higher scope positions. As mentioned in chapter 4, there is a question here of how this analysis of *together* and comitatives being conjunctions in spec,  $\nu P$  derives the correct word order for sentences with *together* and with comitative *with*.

Focusing on the collective reading, both collective *together* and *with*-phrases appear to be pronounced after the VP, as shown in (94) and (95). This is surprising given both Ionin and Matushansky's proposal for comitatives and my proposal here for *together*. In particular, it is generally assumed that specifiers are pronounced to the left of the head, thus predicting that *with Mary* in (94) and *together* in (95), which are both assumed to be in spec,  $\nu P$ , should be pronounced to the left of the verb.

- (94) a. Jane lifted the piano with Mary.  
b. \*Jane with Mary lifted the piano.  
c. \*Jane lifted with Mary the piano.
- (95) a. Jane and Mary lifted the piano together.  
b. ?Jane and Mary together lifted the piano.  
c. \*Jane and Mary lifted together the piano.

While I don't have a decisive solution to this issue, I want to discuss some possible solutions here. First, it could be that the VP moves overtly to a position higher than spec,  $\nu P$ , thus resulting in the specifier in spec,  $\nu P$  being pronounced to the right of the VP. A major problem with this analysis is that there is no evidence that this type of movement occurs in English. The second possibility is that the specifier corresponding to the base position of the comitative conjunction is in fact linearized to the right of its head. Although this goes against standard assumptions that specifiers are linearized on the left, Fox and Pesetsky (2009) argue that certain specifiers, namely inner specifiers, are sometimes linearized on the right. One possibility is to therefore assume that the *x with the other* is always an inner specifier, where the first conjunct moves to the outer specifier position before moving to spec, TP. I leave it for future work to work out the details of such an analysis and to consider the word order of with *together* cross-linguistically.

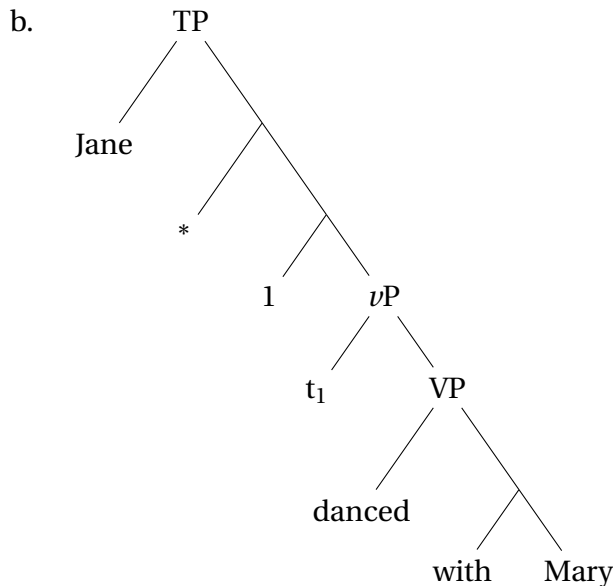


## Appendix B: Modifier analysis of *together*

The analyses of both comitatives and *together* offered here can be recast as modifiers rather than DP coordination, while maintaining the key insights of the interactions between economy and the distribution of *together* argued for here. This will allow the proposal here to be more in line with existing analyses of *together* (e.g. Lasersohn, 1990; Schwarzschild, 1994; Moltmann, 2004). In addition, a modifier analysis will alleviate the concerns regarding word order above, which stem from assuming that *together* is in a lower subject position. Nevertheless, I will argue below that there are conceptual reasons for preferring the analysis offered here where, following Ionin and Matushansky, *with* simply denotes a coordinator that has the same meaning as *and*.

To see how the proposal can be recast with a modifier meaning, consider the lexical entry for *with* in (96-a), which replicates the effect of having the predicate apply to a lower conjunction without committing to there being a syntactically represented comitative conjunction. Again, if we assume that the DIST operator can't apply to the predicate before the *with Mary* modifier applies to it, then we predict the desired collective truth-conditions for *Jane danced with Mary*. The LF and corresponding truth-conditions are given in (96-b) and (96-d) respectively, and we see that just like in the case of the conjunctive analysis, this allows the predicate to apply to the sum of *j* and *m* directly, predicting the desired truth-conditions.

(96) a.  $\llbracket \text{with} \rrbracket = \lambda x. \lambda P. \lambda y. P(x \oplus y) = 1$



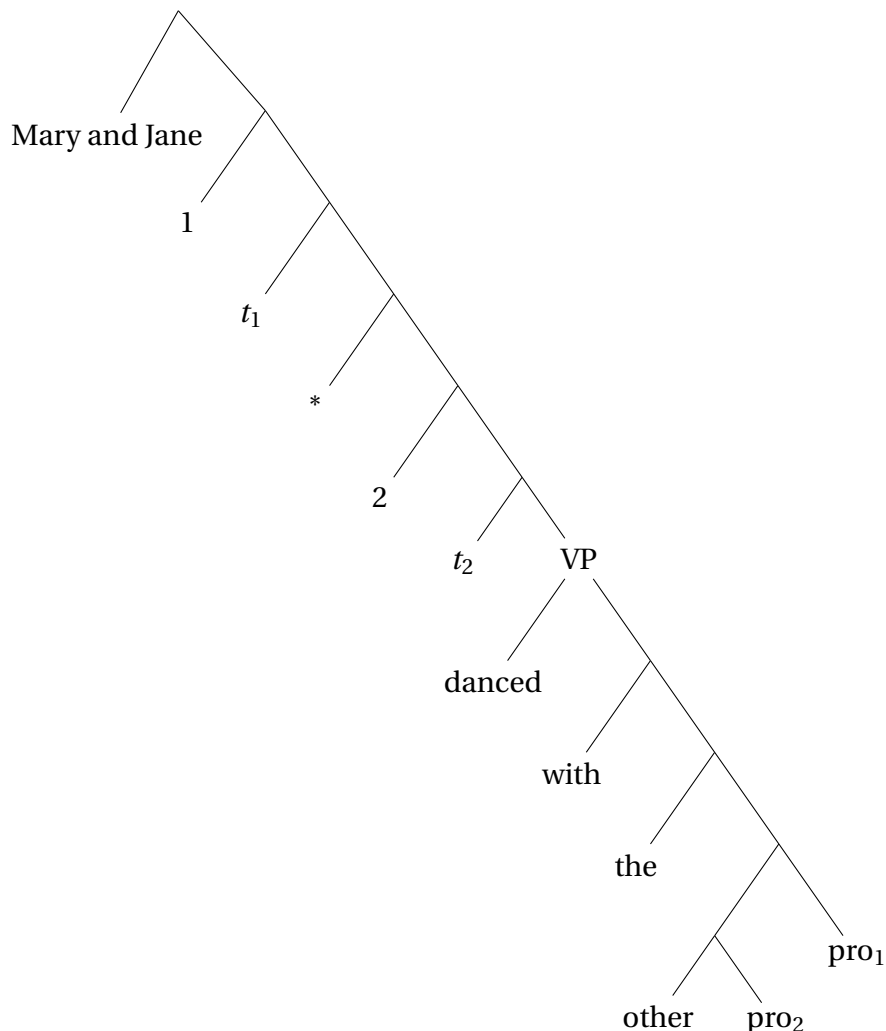
c.  $\llbracket \text{VP} \rrbracket = \lambda x. \llbracket \text{dance} \rrbracket (x \oplus m) = 1$

d.  $\llbracket (96\text{-b}) \rrbracket = 1$  iff  $\llbracket \text{dance} \rrbracket (j \oplus m) = 1$  iff Jane and Mary danced together.

Similarly, we can get the collective reading with *together* by assuming that it is decomposed

into the modifier *with* in (96-a) and a reciprocal anaphor and that it takes scope below all distributivity/pluralization operators. The relevant LF is given in (97-a) and the corresponding collective truth-conditions are given in (97-b).

(97) a.



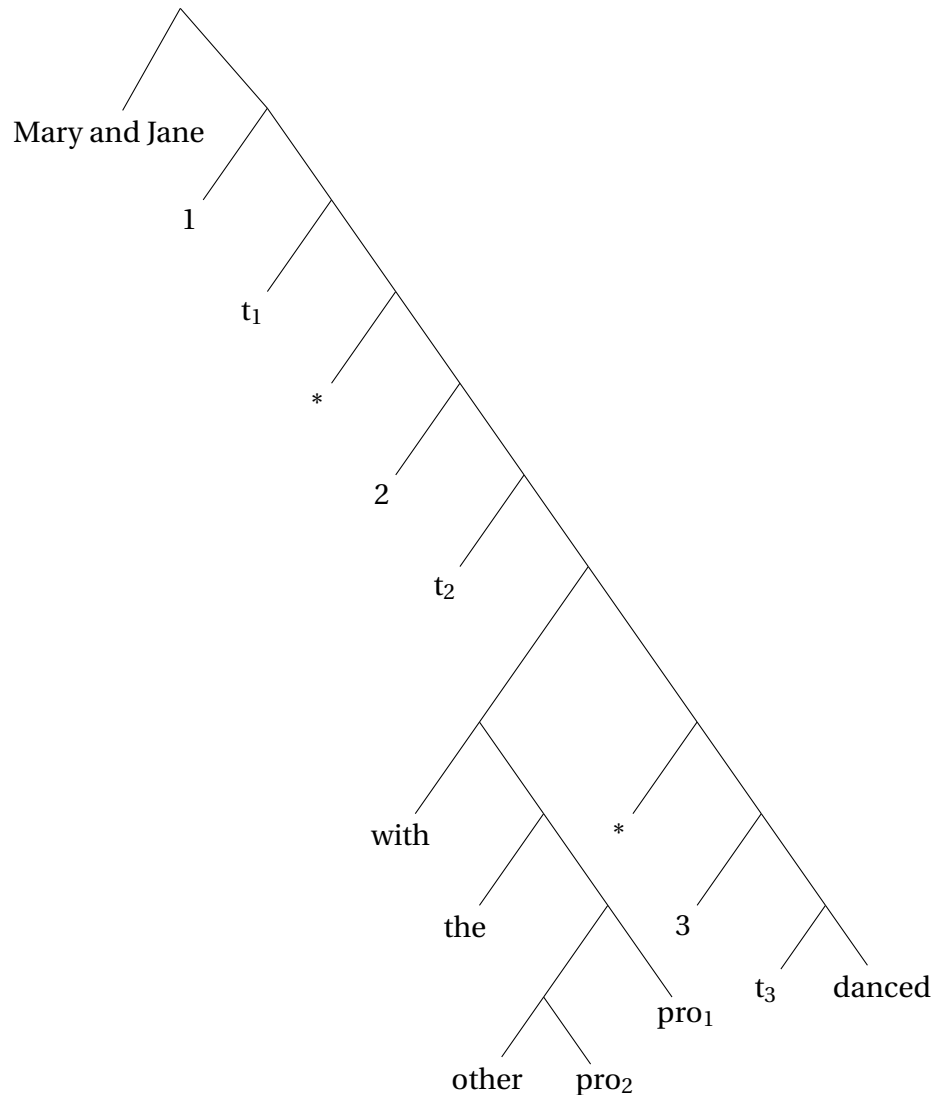
b.  $\llbracket (97-a) \rrbracket = 1$  iff  $\llbracket \text{dance} \rrbracket (m \oplus j) = 1 \wedge \llbracket \text{dance} \rrbracket (j \oplus m) = 1$   
 iff  $\llbracket \text{dance} \rrbracket (j \oplus m) = 1$  iff Jane and Mary danced together.

As for the economy results, while I won't go into this in detail here, it is easy to see that an economy constraint on *with the other* will correctly rule out the unattested distributive reading with *together*. In particular, consider the LF in (98-a): here *with the other* takes scope above a lower *\** and modifies a starred predicate. Since the modifier takes scope above *\**, the truth-conditions in (98-b) are predicted: (98-a) is predicted to be true iff each of Jane and Mary danced. Now, if the modifier *with the other* is deleted, it is straightforward to see that the truth-conditions won't change: the LF without *together* will also be true iff each of Jane and Mary danced. The economy constraint will therefore rule out the unattested LF in (98-a) in the same way that we

ruled out the LF above where the *x with the other* conjunction moved to a position above a lower \*.

(98) **Unattested LF:**

a.



b.  $\llbracket (98-a) \rrbracket = 1$  iff  $\llbracket \text{dance} \rrbracket(j) = 1 \wedge \llbracket \text{dance} \rrbracket(m) = 1$

For the purposes of this chapter, I will continue adopting Ionin and Matushansky's analysis of comitatives as involving DP coordination for a couple of reasons. First, I believe that this analysis offers a more elegant account of the parallels between comitatives and *together* on one hand and the Lebanese Arabic subject construction on the other. In particular, the Lebanese Arabic double subject construction shows plainly that collective readings arise when a conjoined DP is stranded below all pluralization operators. The analysis of *together* as an *x with the other* conjunction predicts the same result, while maintaining that *together* underlyingly also involves

individual conjunction. While building sum formation into a meaning of *together* as a modifier can predict the same result, it renders the parallel between conjunction and *together* accidental.

Another reason to prefer the analysis of *with* as DP conjunction is type economy considerations. In particular, Heim (2017) suggests an economy constraint on possible lexical meanings that rules out higher type lexical items which can be defined in terms of lexical items of lower types. More specifically, in order to violate type economy the higher type meaning has to be definable in terms of the lower type one using only lambdas, variables and function application. This principle is given in (99). To see this principle in action consider the example in (100) with negation. Heim shows that type economy rules out predicate negation in (100-b), since this meaning is definable in terms of sentential negation in (100-a), as shown in (100-c) (see Hirsch, 2017 for additional arguments that the meanings of connectives should be restricted in this way).

- (99) **Type Economy Principle:** If  $F$  is definable in terms of an  $F'$  with a simpler type than  $F$ , then  $F$  is not a possible denotation for a morpheme in natural language.
- a. A function  $F$  has a simpler type than a function  $F'$  iff the type-label of  $F$  is shorter than the type-label of  $F'$ .
  - b. A function  $F$  is definable in terms of a function  $F'$  iff  $F$  can be described by an expression that contains only variables, lambdas, parentheses (for functional application) and expressions that name  $F'$ .
- (100) **Type economy and negation:**
- a. **Type**  $\langle t, t \rangle$  **lexical entry:**  $\llbracket \text{not}_1 \rrbracket = \lambda t.1$  iff  $t = 0$
  - b. **Type**  $\langle \langle e, t \rangle, \langle e, t \rangle \rangle$  **lexical entry:**  $\llbracket \text{not}_2 \rrbracket = \lambda P.\lambda x.P(x) = 0$
  - c.  **$\llbracket \text{not}_2 \rrbracket$  is definable in terms of  $\llbracket \text{not}_1 \rrbracket$ :**  $\llbracket \text{not}_2 \rrbracket = \lambda P.\lambda x.\llbracket \text{not}_1 \rrbracket(P(x)) = 1$

Now, Type Economy also rules out our higher type lexical entry for *with*, repeated in (101-b). In particular, this type  $\langle e, \langle \langle e, t \rangle, \langle e, t \rangle \rangle \rangle$  meaning for *with* is definable in terms of the simpler type  $\langle e, \langle e, e \rangle \rangle$  meaning I offered earlier in the section, where *with* simply takes two individuals and returns their sum (101-a). This is illustrated in (101-c). If we assume Type Economy as a general principle of grammar, we therefore are committed to the analysis of *with* as a DP coordinator and consequently to the analysis of *together* described earlier in this section.

- (101) **Type economy and *with***
- a. **Type**  $\langle e, \langle e, e \rangle \rangle$  **lexical entry:**  $\llbracket \text{with}_1 \rrbracket = \lambda x.\lambda y.x \oplus y$
  - b. **Type**  $\langle e, \langle \langle e, t \rangle, \langle e, t \rangle \rangle \rangle$  **lexical entry:**  $\llbracket \text{with}_2 \rrbracket = \lambda x.\lambda P.\lambda y.P(x \oplus y) = 1$
  - c.  **$\llbracket \text{with}_2 \rrbracket$  is definable in terms of  $\llbracket \text{with}_1 \rrbracket$ :**  $\llbracket \text{with}_2 \rrbracket = \lambda x.\lambda P.\lambda y.P(\llbracket \text{with}_1 \rrbracket(x)(y)) = 1$

## Appendix C: Economy and Distributivity

In this chapter, I showed that the economy constraint on *together* predicts the correct distribution of readings if we assume that pluralization is due to a \* operator which delivers weak disjunctive readings. Of course, it has been argued in the literature that in certain contexts and environments, we can get readings that are stronger than what \* predicts, such as purely distributive readings (Gillon, 1987; Schwarzschild, 1994, 1996, a.o.). To see this, consider the example in (102): here (102-a) is judged true (on at least one reading) in the scenario. Crucially, in order for (102-a) to be true, the antecedent has to be interpreted distributively. In particular, in this scenario it is true that Bill will be happy if each of Jane and Mary lift the piano separately (102-b), but it is not true that Bill will be happy as long as Jane and Mary either lift the piano together or separately (102-c). The truth-conditions predicted by \* are therefore too weak to capture the correct truth-conditions in (102-a). In particular, \* predicts that (102-a), just like (102-c), is not true in the given scenario.

- (102)   **Scenario:** Bill wants to see if Jane and Mary are strong. He gives them the option to lift the piano together or to each lift it separately, but he will only be happy if they each lift it separately.
- a.   If Jane and Mary lift the piano, Bill will be happy.
  - b.   If each of Jane and Mary lift the piano, Bill will be happy.
  - c.   # If Jane and Mary lift the piano either together or separately, Bill will be happy.

In what follows, I will discuss how these readings can be generated while maintaining the economy results from this chapter. In particular, I show that if we allow for an operator that can generate these stronger readings, such as the distributivity operator DIST, the specific economy constraint on the deletion of *with the other* given in chapter 4 is not able to rule out all of the unattested LFs with *together*.<sup>24</sup> Nevertheless, I argue that once we consider the generalized economy constraint proposed in this chapter, the unattested readings are successfully ruled out even in the presence of DIST.

I will start with presenting the issue that arises if we assume that there is a distinct DIST operator.<sup>25</sup> Now, at first sight, it seems that if we allow for both \* and DIST in the lexicon, we

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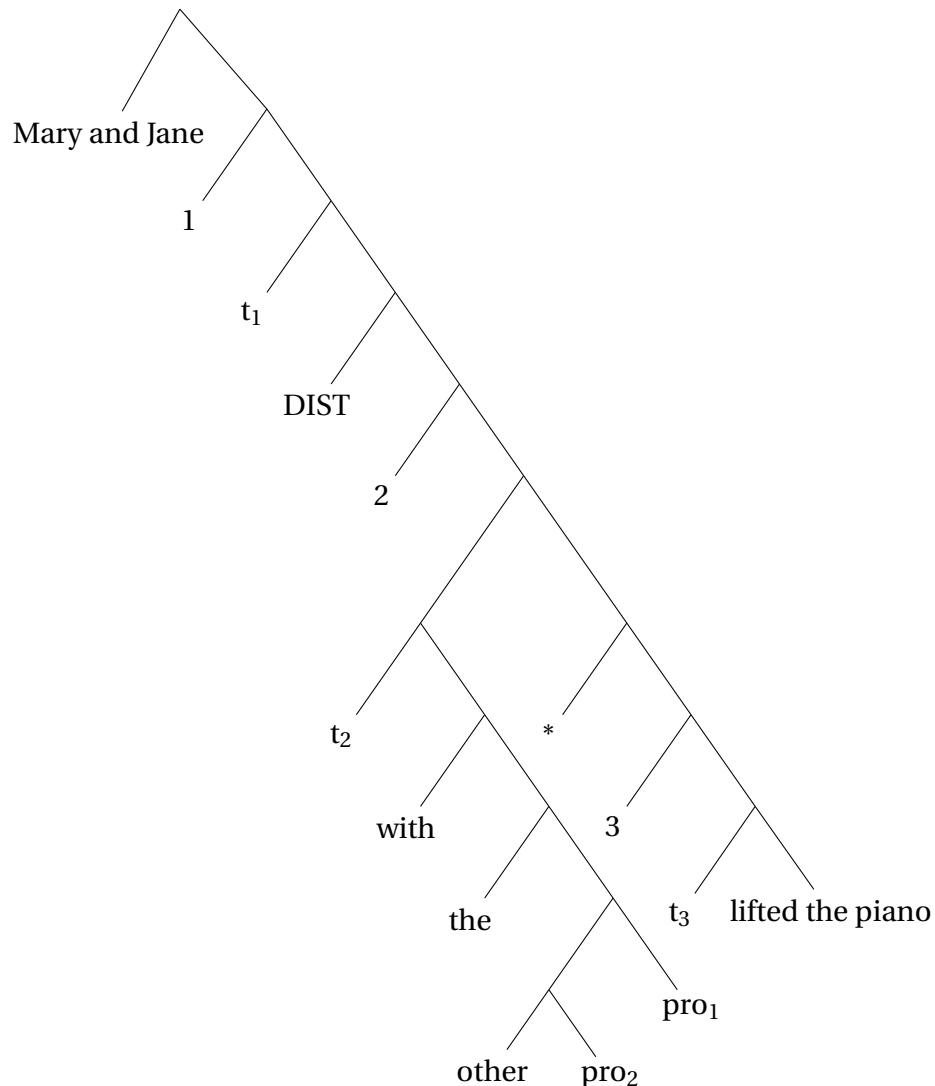
<sup>24</sup>Note that there are other ways to generate these strong readings, without positing a separate distributivity operator. For example, we could allow for a cover variable to restrict \*, following Schwarzschild (1994) among others. While I won't discuss covers here, the solution I propose below for DIST will extend to other ways of generating purely distributive readings, such as with covers.

<sup>25</sup>Note that we can similarly show that in certain contexts, partially distributive readings are available which are also stronger than the weak truth-conditions that \* predicts. This account will therefore have to assume that there are separate operators responsible for partial distributivity (see Champollion, 2016, for one example of such a theory).

incorrectly predict that the economy constraint licenses an undesirable reading with *together* which is true in a scenario where Jane and Mary lifted the piano separately. To see this, consider the possible LF in (103-b) for (103-a): here DIST is inserted above *with the other* but \* is inserted below. Now, (103-b) is true iff either Jane and Mary lifted the piano together or if they each did so separately. Unlike the cases where we only have \*, the counterpart where *with the other* is deleted in (104-a) gives rise to different truth-conditions here. In particular, since DIST takes scope above \*, deleting *together* results in the distributive truth-conditions in (104-b): (104-a) is true iff each of Jane and Mary lifted the piano. Now, (103-b) and (104-a) clearly have different truth-conditions: only (103-b) is true in a scenario where Jane and Mary lifted the piano together. The economy constraint is therefore unable to rule out the LF in (103-b).

(103) a. Jane and Mary lifted the piano together.

b.

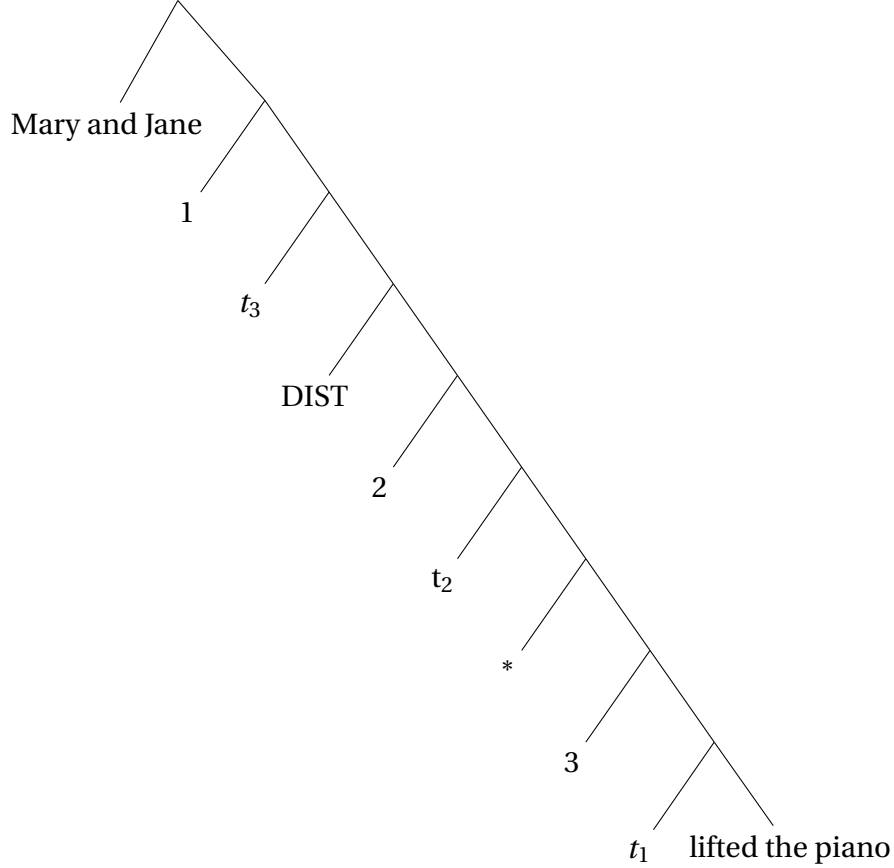


c.  $\llbracket (103-b) \rrbracket = 1$  iff

$$(\llbracket \text{lifted the piano} \rrbracket(j)=1 \wedge \llbracket \text{lifted the piano} \rrbracket(m)=1) \vee \llbracket \text{lifted the piano} \rrbracket(j \oplus m)=1$$

(104) **Result of deleting *with the other*:**

a.



b.  $\llbracket (104-a) \rrbracket = 1$  iff

$$\llbracket \text{lifted the piano} \rrbracket(j)=1 \wedge \llbracket \text{lifted the piano} \rrbracket(m)=1$$

I have therefore shown that if we allow for \* and DIST to both be freely inserted, the economy constraint on the deletion of *together* incorrectly predicts that *Jane and Mary lifted the piano together* has a reading which is true if each of Jane and Mary lifted the piano separately.

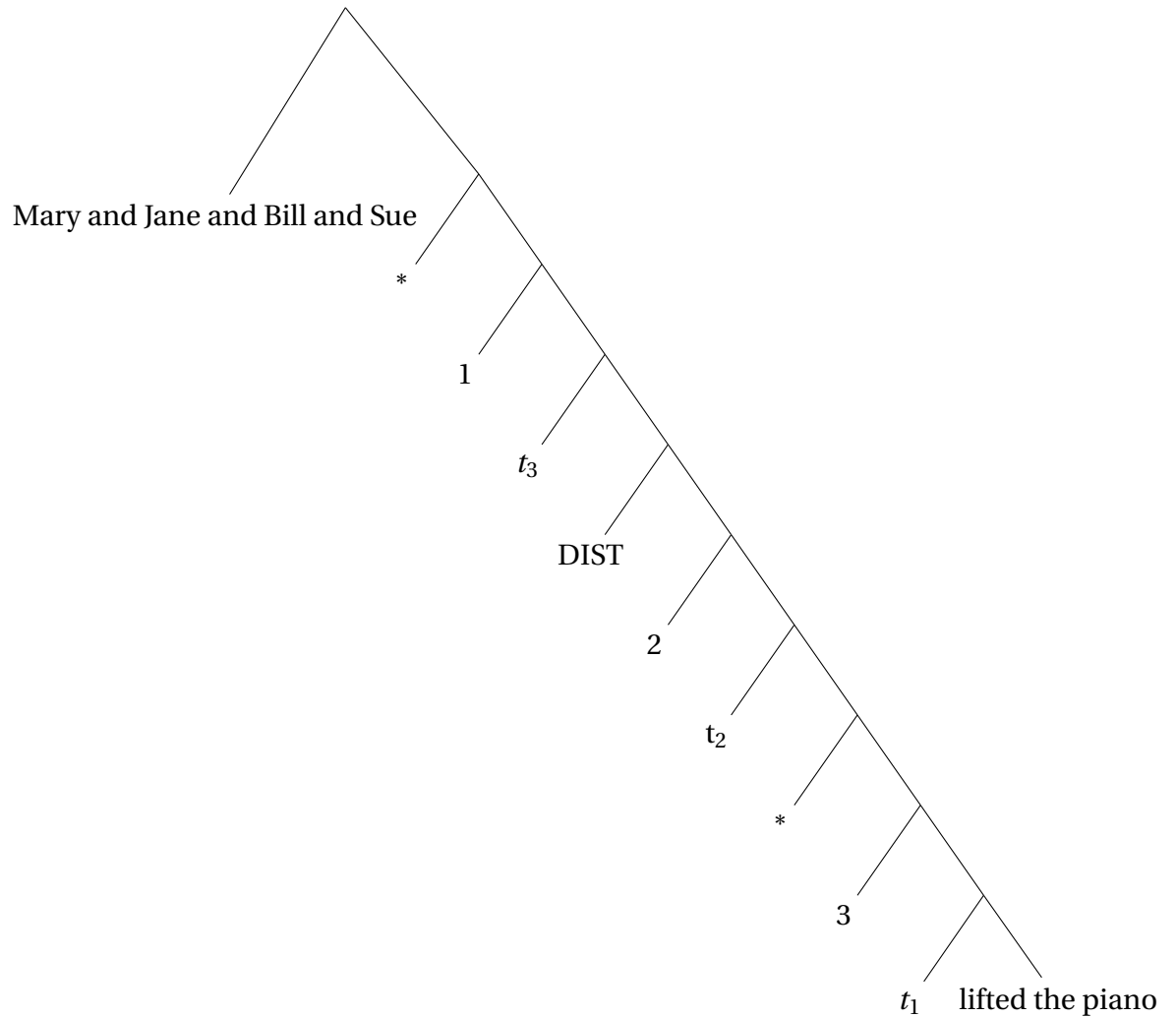
Interestingly, the generalized economy constraint proposed in this chapter is able to rule out the unattested distributive reading, even in the presence of a distributivity operator. In particular, the LF in (103-b) is equivalent to the LF where both DIST and *with the other* are deleted. As shown in (105), when DIST and *together* are deleted, \* applies to the predicate *lift the piano* and the resulting predicate applies to the subject, giving rise to the same underspecified truth-conditions as (103-b).

(105)  $\llbracket (103-b) \text{ without DIST and } with\ the\ other \rrbracket = 1$  iff  $*\llbracket \text{lift the piano} \rrbracket(j \oplus m)=1$

Unlike in the case we saw in section 5.2 with the collective reading, blindness to cardinality is

unable to rescue the distributive reading from violating economy here. In particular, consider the LF in (106-a), which allows for partial distributivity over the higher subject. Since there is a lower \* operator below *together* here, (106-a) is again predicted to be true iff the starred predicate is true of the plurality denoted by the subject, as shown in (106-b). These are the same truth-conditions we would get from deleting both DIST and *with the other*. Therefore, the generalized economy constraint is violated here, correctly predicting that *Jane and Mary lifted the piano together* does not have a reading which is true in a distributive scenario where Jane and Mary lifted the piano separately.

(106) a.



b.  $\llbracket (106-a) \rrbracket = 1$  iff  $\llbracket \text{lift the piano} \rrbracket (j \oplus m \oplus b \oplus s) = 1$

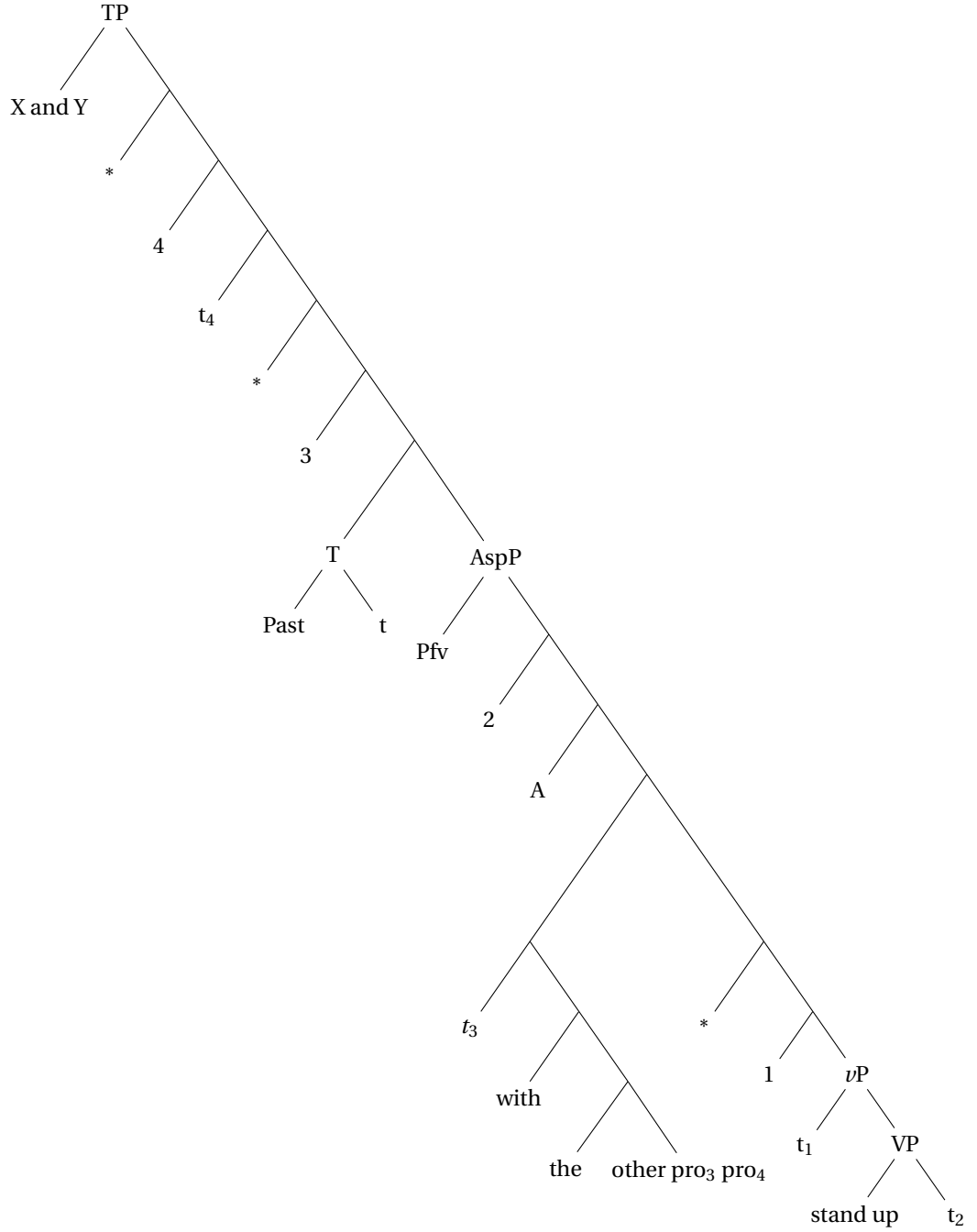


## Appendix D: Generalized economy with simultaneous *together*

The goal of this appendix is to simply show that generalized economy predicts that the simultaneous and proximate readings with *together* are licensed in the same way as the collective reading. Starting with the simultaneous reading, consider the LF in (107-b) and the associated truth-conditions in (107-c). Just like we saw with the collective reading, (107-b) allows for partial distributivity to non-atomic subparts but still demands that members of each subpart stood up at the same time. Now, deleting *with the other* results in an LF which is true as long as each atomic part of  $X \oplus Y$  stood up (108-a), and deleting both *with the other* and the two higher \* operators results in an LF which is true only if all of the atomic parts of  $X \oplus Y$  stood up at the same time (108-b). We therefore again see a three-way distinction in truth-conditions between the LF in (107-b) and the two deletion alternatives in (108-a) and (108-b) when  $X \oplus Y$  has cardinality greater than 2.

- (107)    a.    Jane and Mary stood up together.

b.



- c.  $\llbracket (107-b) \rrbracket = 1$  iff  $\llbracket * \rrbracket (\lambda x. \exists t' \subseteq t : \llbracket * \rrbracket (\llbracket \text{stood up} \rrbracket (t'))(x) \wedge |x| > 1)(X \oplus Y) = 1$   
iff  $X \oplus Y$  can be divided into non-atomic sub-parts such that the members of each sub-part stood up at the same time.

(108)

a. **LF where *with the other* is deleted:**

1 iff  $\llbracket * \rrbracket (\lambda x. \exists t' \subseteq t : \llbracket * \rrbracket (\llbracket \text{stood up} \rrbracket (t'))(x))(X \oplus Y) = 1$

b. **LF where *with the other* and the two higher \* operators are deleted:**

1 iff  $(\lambda x. \exists t' \subseteq t : \llbracket * \rrbracket (\llbracket \text{stood up} \rrbracket (t'))(x))(X \oplus Y) = 1$

Just as we saw with the collective reading, no other deletion alternative is able to derive the same partially distributive truth-conditions as (107-b). We can therefore conclude that the LF in (107-b) satisfies generalized economy. Generalized economy therefore predicts that simultaneous *together* is always licensed. While I won't give the parallel derivation with proximate *together*, it is easy to see how the same logic applies with the existential quantifier over location in that case.

## Appendix E: When the subject denotes an atomic individual

So far, I've been looking mainly at examples where the subject is *Jane and Mary*, where we know that the subject denotes an individual of cardinality 2. In this section, I will show that if we allow the subject to denote an individual of cardinality 1, this raises a problem for the generalized economy constraint proposed in this chapter. This issue is particularly serious given that I argued in this chapter that the generalized economy constraint must be blind to the cardinality of the subject. While I don't have a full solution to this issue at the moment, I present the problem in this section and discuss a tentative way of addressing it by relating it to a larger puzzle with how the presuppositions of definites like *the other* behave in conjunctions.

Sauerland (2003) argues that plural marking on nouns within definites does not carry any requirement that the definite refers to a plural individual. For example (109-a) can be true in the context in (109) even if some dads have one child, as long as they brought that child to the field trip. Similarly (109-b) is true if there are exactly 2 dads who brought their child or children to the field trip. Sauerland therefore concludes that definite plurals can refer to atomic individuals and attributes the infelicity of plural marking in contexts like (110) to competition with the singular form. Crucially, in a different context where it is not common ground whether John has one or multiple children the unembedded definite plural becomes felicitous, as shown in (111). Here the definite singular is infelicitous and therefore the definite plural does not compete with the singular and is able to refer to an atomic individual in worlds where John has only one child.

(109) **Context:** A teaching assistant is in charge of calling the dads from three families to ask if their children are coming to the field trip. Dad 1 has 3 kids, Dad 2 has 2 kids and Dad 3 only has one.

- a. Every dad said that his children are coming to the field trip.
- b. Exactly 2 of the 3 dads said that their children are coming to the field trip.

(110) **Context:** A and B both know that John has one child. A says to B:

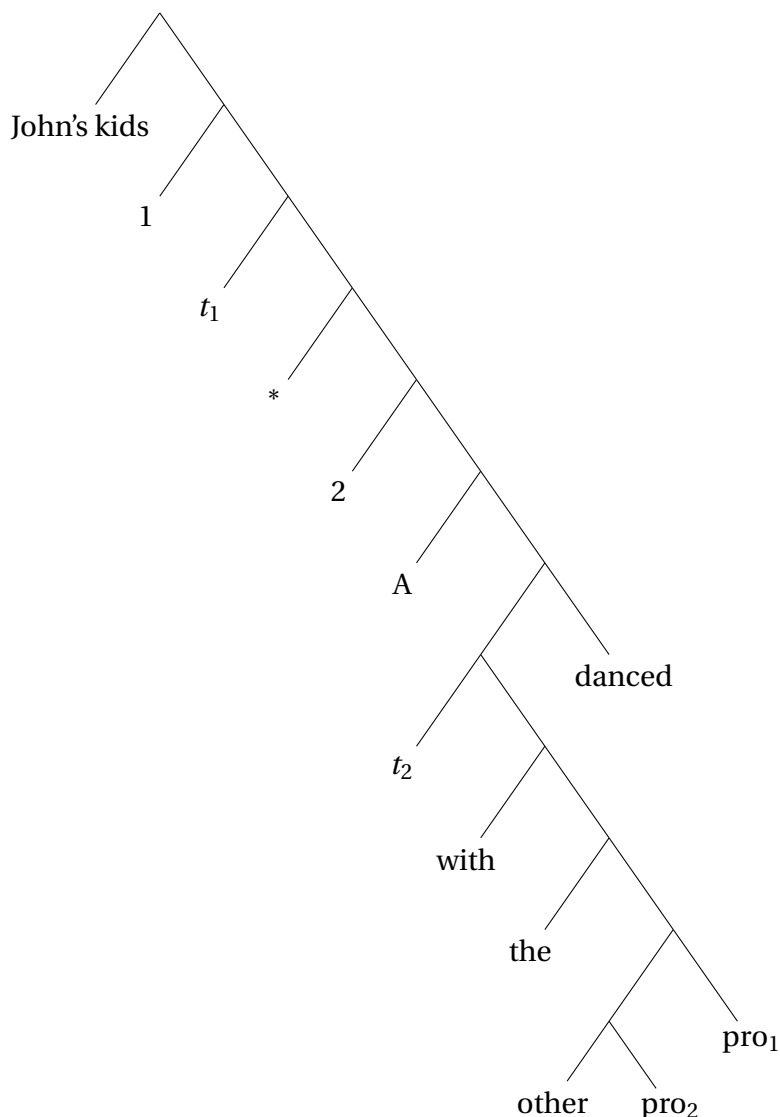
- a. #John's children are coming to the field trip.
- b. John's child is coming to the field trip.

- (111) **Context:** A and B don't know how many children John has. Each parent has to fill out one form for their family saying whether they allow their children to go to the field trip. The form doesn't indicate how many children the parent has. A sees that John filled yes on the form. A says to B:
- a. John's children are coming to the field trip.
  - b. # John's child is coming to the field trip.

This raises the question of what our analysis for *together* predicts if we allow for the subject to denote an atomic individual. Crucially, the presupposition of *the other* will not be met in a world where the subject denotes an atomic individual. Given that we assumed that the presupposition of *the other* is locally accommodated, we therefore predict that the collective LF for (112) will be false in a world where there is only one student. To see this, consider the LF in (112-a). As argued above, *the other* presupposes that the first argument of *other* is a proper subpart of the second. This requirement can't be met when the subject denotes an atomic individual, since both arguments will have to denote the same atomic individual. The resulting truth-conditions for (112-a) are therefore given in (112-b): (112-a) is predicted to be true iff John has more than one kid and they danced together.

- (112) John's kids danced together.

a.



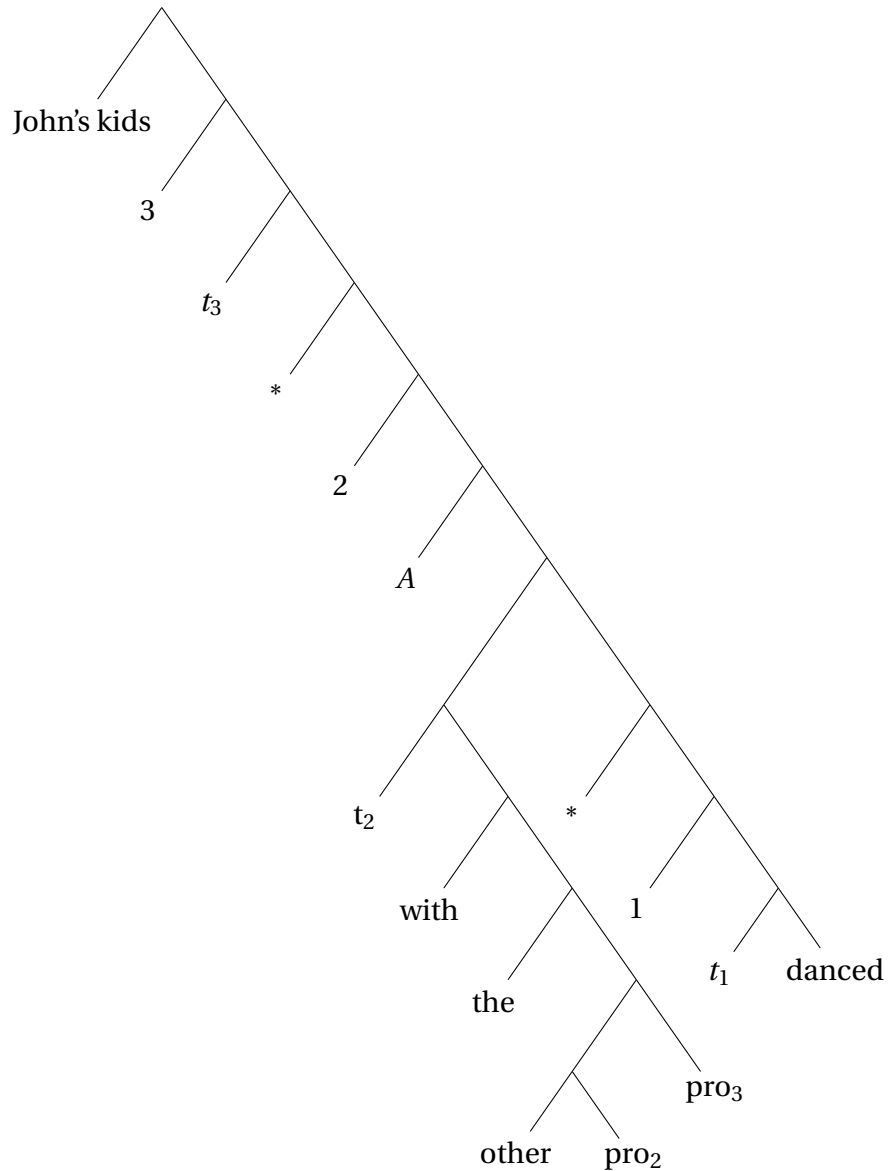
- b.  $\llbracket (112-a) \rrbracket = 1$  iff  $\llbracket \text{John's kids} \rrbracket > 1 \wedge \llbracket \text{dance} \rrbracket (\llbracket \text{John's kids} \rrbracket) = 1$   
iff John has more than one kid and they danced together.

We therefore see that the LF with *together* in addition to forcing a collective reading, is only true if the subject denotes an individual of cardinality greater than 1. This additional requirement will pose a problem for our economy constraint. In particular, since *together* adds this non-atomicity requirement, even the unattested LFs where *together* takes scope above a lower \* operator will be licensed by the economy constraint. To see this, consider again the unattested LF in (113-a): here *with the other* takes scope above a \* operator and (113-a) is predicted to be true iff John has multiple kids and each of them danced. Now, as shown in (114), the counterpart without *together* will also be true only if all of John's kids danced but will not have the additional requirement that John has multiple kids. We therefore see that the unattested LF with *together* in (113-a) is no longer equivalent to the result of deleting *together*. The economy constraint therefore incorrectly

predicts that (113-a) is a licensed LF with *together*.

(113) **Unattested LF:**

a.



b.  $\llbracket (113-a) \rrbracket = 1$  iff  $\llbracket \text{John's kids} \rrbracket > 1 \wedge \llbracket * \rrbracket (\llbracket \text{dance} \rrbracket) (\llbracket \text{John's kids} \rrbracket) = 1$   
iff John has multiple kids and they all danced.

(114) **Result of deleting *together*:**  $\llbracket (113-a) \text{ without } together \rrbracket = 1$  iff  
 $\llbracket * \rrbracket (\llbracket \text{dance} \rrbracket) (\llbracket \text{John's kids} \rrbracket) = 1$  iff all of John's kids danced.

While I don't have a definitive solution to this problem at this point, I want to point out that there is a more general puzzle with how *together* interacts with atomic subjects. In particular, in certain environments it appears that the sentence with *together* is in fact true when the subject

denotes an atomic individual, unlike what is predicted by the account outlined above. To see this consider the example in (115). Here, the scope in both (115-a) and (115-b) is true of dads who have one child as long as that child picked less than 20 apples. For example, (115-a) is true if every dad who has multiple children made sure that together they picked less than 20 apples and every dad who has one child made sure that child picked less than 20 apples. Similarly in (116), we see the same pattern in an unembedded environment where we don't know whether the subject denotes an atomic individual. Clearly, (116-a) is true even if John has only one child as long as that child picked less than 20 apples.

(115) **Context:** Children are going on an apple picking field trip with their dads. Some dads have one child and some have multiple.

- a. Every dad has to make sure that his children together pick less than 20 apples.
- b. Exactly 3 of the dads made sure that their children together picked less than 20 apples.

(116) **Context:** Every parent on the field trip has to fill out a form indicating if their family in total picked more than or less than 20 apples. We're not sure if John has multiple children or one child. A sees that John marked that his family picked less than 20 apples so they say:

- a. John's children together picked less than 20 apples.

At this point, one might wonder whether this pattern provides evidence against the compositional analysis of *together* as *with each other* that I pursued in this chapter. In particular, what we can conclude from (115) and (116) is that sentences with *together* are judged true in worlds where the presupposition of *the other* is not met. Clearly, reciprocals in general don't behave this way. In particular, if we look at reciprocals in object position, they are clearly judged to be not true when the subject denotes an atomic individual. For example, (117-a) clearly can't be true in a world where John has one child, as evidenced by the fact that (117-b) is judged to be a contradiction.

- (117)
- a. John's children hugged each other.
  - b. #I don't know if John has one or multiple children, but I know that John's children hugged each other.

In what follows, I will argue that the puzzle we see in (115) and (116) can be resolved while maintaining the compositional semantics for *together* that I argued for in this chapter and chapter 4. In particular, I will show that this puzzle can be seen as part of a larger puzzle relating to how definites behave in conjunctions when their presuppositions are not met. To see this, it

will be useful to first characterize the puzzle in (115) and (116) a bit more abstractly. Focusing on (116), the LF that is predicted by my compositional analysis is given in (118). Now, when John has only one child, the extension of *other* will necessarily be empty and *the other* will be undefined. Yet, the sentence is judged to be true. The result that we want to get is that *x and the other* is simply equivalent to *x* when *the other* is undefined. This will allow the sentence to be true under the conditions paraphrased in (118-a) when the subject denotes an atomic individual, thus predicting that (116) is true if John has one child and that child picked less than 20 apples.

(118) **LF paraphrase:**

John's children are each such that *x* and the other cumulatively picked less than 20 apples.

a. **When John has one child:**

John's children are each such that *x* cumulatively picked less than 20 apples.

b. **Desired result:** When the extension of *other* is empty *x and the other* simply denotes the individual *x*.

Interestingly, we see a pattern with conjunction in general that can be characterized in a parallel way. To see this, consider first the example in (119): here we see that the scope in (119-a) and (119-b) is taken to be true of dads who only have children but no siblings as long as the children picked less than 20 apples. Similarly in the context in (120), (120-a) is true even if John has no siblings as long as his children picked less than 20 apples. This issue is parallel to what we saw above with *together*: we want *John's children and his siblings* to simply be equivalent to *John's children* in worlds where John has no siblings, rather than simply being undefined due to the extension of *siblings* being empty. This is exactly what we saw with *together*, where we wanted *x with the other* to simply denote *x* when the extension of *other* is empty.

(119) **Context:** A group of dads are going on an apple picking field trip with their children. Some dads have one child and some have multiple. Several of the dads also brought one or more of their siblings on the trip. Each dad has to make sure that the rest of their family together picks less than 20 apples.

a. Every dad made sure that his children and his siblings picked less than 20 apples.

b. Exactly 3 of the dads made sure that their children and their siblings picked less than 20 apples.

(120) **Context:** Every parent on the field trip has to fill out a form indicating if their family in total picked more than or less than 20 apples. We know that John brought his children to the field trip, but we're not sure if he has any siblings. A sees that John marked that his family picked less than 20 apples so they say:



- a. John's children and his siblings picked less than 20 apples.

Resolving this general puzzle with conjunctions of definite plurals where one of the conjuncts is undefined is beyond the scope of this dissertation. Nevertheless, we can see how a solution to this problem can also address the issue of how the economy constraint interacts with the possibility that the subject denotes an atomic individual. In particular, this problem for the economy constraint arose because the analysis I proposed for *together* predicts that the sentence with *together* will always be false (or undefined) when the subject denotes an atomic individual. The counterpart without *together*, on the other hand, can be true when the subject denotes an atomic individual. This made the undesirable prediction that even the unattested distributive reading of *together*, paraphrased in (121-a) does not violate the economy constraint: if there is only one child and they danced, the counterpart without *together* in (121-b) is true but (121-a) isn't. Now, if we assume that conjunctions where one of the conjuncts is undefined can simply be reduced to the other conjunct, this problem no longer arises. In particular, this assumption predicts that (121-a), just like (121-b), is true if there is one child and they danced, thus giving the desired result that (121-a) and (121-b) are truth-conditionally equivalent.

(121) a. **Unattested LF Paraphrase:**

The children are each such that them and the other each danced.

b. **Counterpart without *together*:**

The children each danced.

I have therefore suggested that the puzzle with subjects that denote atomic individuals with *together* can be reduced to a more general puzzle relating to how conjunctions of definite plurals behave when the presupposition of one of the conjuncts is not met. Of course, at this point this discussion is very tentative, as we don't yet have an understanding of how the more general puzzle with conjunctions can be resolved. Nevertheless, I think this parallel between conjunctions and *together* is promising given the analysis of *together* as coordination that I've been arguing for.

## Appendix F: Homogeneity and non-distributive predication

In section 5.4.2, I showed that with distributive predicates, homogeneity blocks the subpart implicature, thus predicting that cumulative *together* is not licensed when the object gives rise to homogeneity. The goal of this section is to show that this result generalizes to predicates which allow for collective readings.

To see this, I will start by providing the full trivalent lexical entries for \* and \*\* argued for by Bar-Lev (2025) in (122-a) and (122-b). Starting with (122-a), the falsity conditions require that

there is no overlap between any part of  $x$  and an individual which  $P$  is true of. Similarly, the falsity conditions in (122-b) require that there is no pair of individuals  $x''$  and  $y''$  which overlap with  $x$  and  $y$  and which  $P$  is true of. This no overlap requirement is stronger than the preliminary falsity conditions given in section 5.4.2.

(122) a. **Homogeneity with 1-place predicates:**

$$(\llbracket * \rrbracket (P))(x) = \begin{cases} 1 \text{ iff } \exists Q_{\langle e, t \rangle} [\forall y [Q(y) = 1 \rightarrow P(y) = 1] \wedge x = \oplus Q] \\ 0 \text{ iff } \neg \exists x' \leq x [\exists x'' : x' \leq x'' \wedge P(x'') = 1] \\ \# \text{ otherwise} \end{cases}$$

b. **Homogeneity with 2-place predicates:**

$$(\llbracket ** \rrbracket (P))(x)(y) = \begin{cases} 1 \text{ iff } \exists Q [\forall x', y' [Q(x')(y') = 1 \rightarrow P(x')(y') = 1] \wedge x = \oplus \lambda x. \exists y [Q(x)(y) = 1] \wedge y = \oplus \lambda y. \exists x [Q(x)(y) = 1]] \\ 0 \text{ iff } \neg \exists x' \leq x, y' \leq y [\exists x'', y'' : x' \leq x'' \wedge y' \leq y'' \wedge P(x'')(y'') = 1] \\ \# \text{ otherwise} \end{cases}$$

Motivation for these stronger falsity conditions comes from examples like (123). In a scenario where Jane and Mary lifted the piano collectively, *Jane didn't lift the piano* is not true. This is captured by the trivalent lexical entry in (122-b), if  $**$  applies to the predicate before it applies to its arguments. In particular, the LF with  $**$  will be false iff Jane is not part of any plurality that lifted the piano (124-b). These falsity conditions are not met in the scenario, since  $\llbracket \text{lift} \rrbracket (\llbracket \text{the piano} \rrbracket)(j \oplus m)$  is true in (123-a).

(123) **Scenario:** Jane and Mary together lifted the piano.

a. # Jane didn't lift the piano.

(124) a. **In the scenario in (123):**  $\llbracket \text{lift} \rrbracket (\llbracket \text{the piano} \rrbracket)(j \oplus m) = 1$

b.  $\llbracket \text{Jane} ** \text{ lifted the piano} \rrbracket = 0$  iff  $\neg \exists x' \leq \llbracket \text{the piano} \rrbracket, y' \leq j [\exists x'', y'' : x' \leq x'' \wedge y' \leq y'' \wedge \llbracket \text{lift} \rrbracket (x'')(y'') = 1]$   
iff  $\neg \exists y'' : j \leq y'' \wedge \llbracket \text{lift} \rrbracket (\llbracket \text{the piano} \rrbracket)(y'')$

Now, we can see that the result in (72) regarding the exclusion of subpart alternatives generalizes to collective predicates with the trivalent entry for  $**$  in (122-b). Consider (125), which is parallel to the original case with a distributive predicate in (72). The cumulative LF is predicted to entail that each atomic subpart of  $Z$  is part of an individual that lifted some subpart of  $\llbracket \text{the pianos} \rrbracket$  (collectively) (125-a). On the other hand, the subpart alternative in (125-b) is only false if there is no subpart of  $R$  which is part of an individual that lifted a subpart of the pianos. This entails that there is no part of  $R$  that overlaps with any individual that lifted a part of the pianos, thus contradicting (125-a).

- (125) **Suppose  $\llbracket [t_3 \text{ with the other } \text{pro}_3 \text{ pro}_4] D_1 \rrbracket^g = Z$  and take an arbitrary  $D_2$  such that  $\llbracket [t_3 \text{ with the other } \text{pro}_3 \text{ pro}_4] D_2 \rrbracket^g = R$ , where  $R < Z$ :**
- a.  $\llbracket [t_3 \text{ with the other } \text{pro}_3 \text{ pro}_4] D_1 \rrbracket^{**} \text{ lifted the pianos} \rrbracket^g = 1 \rightarrow$   
 $\forall x \leq_{AT} Z [\exists y \leq \llbracket \text{the pianos} \rrbracket [\exists x' : x \leq x' \wedge \llbracket \text{lifted} \rrbracket (y)(x') = 1]$
  - b.  $\llbracket [t_3 \text{ with the other } D_2] \rrbracket^{**} \text{ lifted the pianos} \rrbracket^g = 0$  iff  
 $\neg \exists x \leq R, y \leq \llbracket \text{the pianos} \rrbracket [\exists x', y' : x \leq x' \wedge y \leq y' \wedge \llbracket \text{lift} \rrbracket (y')(x') = 1]$   
 $\rightarrow \neg \exists x \leq_{AT} R, y \leq \llbracket \text{the pianos} \rrbracket [\exists x' : x \leq x' \wedge \llbracket \text{lifted} \rrbracket (y)(x') = 1]$

Excluding the subpart alternative in (125-b) therefore contradicts the prejacent of *exh* in (125-a), just as we saw with distributive predicates in section 5.4.2. We can therefore conclude that cumulative *together* is always ruled out with objects that give rise to homogeneity, regardless of whether the predicate is distributive or not.



# Chapter 6

## Economy and the double subject construction

### 6.1 Introduction

This chapter explores the possibility that the distribution of readings in the Lebanese Arabic double subject construction is governed by the same kind of economy constraint that I proposed for *together* in chapter 5. Namely, chapters 4 and 5 developed an analysis of *together* as denoting a *with the other* coordination, whose different readings are derived by allowing the coordinated DP to take scope in different positions. Chapter 2 argued that the double subject construction similarly involves a conjoined DP (the lower subject), which gives rise to parallel interpretations depending on where it takes scope. In both constructions, the relevant conditions on where the coordinated DP can take scope can't be stated as purely syntactic constraints; rather, they have to depend on certain semantic properties of the associated LFs. In this chapter, I show how the generalized economy constraint proposed in chapter 5 might be extended to account for the distribution of readings in the double subject construction, explaining some of the parallels between the two constructions.

More specifically, I show that the distribution of readings in the double subject construction closely parallels what we saw with *together* in chapters 4 and 5. First, just as with *together*, LFs where the lower subject takes scope above a lower pluralization operator and gives rise to a distributive reading have to be ruled out. Second, the same homogeneity-cumulativity generalization we saw in chapter 5 manifests itself in the double subject construction: the double subject construction only allows for a cumulative reading when the object doesn't give rise to homogeneity. These parallels therefore call for a unified mechanism for restricting the distribution of readings in the two constructions.

I argue that the distribution of readings in the DSC can be captured by similar economy considerations. Nevertheless, there are some challenges to providing a unified analysis of the DSC and the distribution of *together* in terms of generalized economy. First, the notion of competition that is needed to predict the distribution of readings in the DSC has to be more general than what I proposed in chapter 5 for *together*, where only strict deletion alternatives are considered. Second, unlike *together*, the DSC does not license simultaneous readings, a difference which does not follow from economy. I leave a full account of these differences as an open question for future work.

The rest of the chapter is structured as follows. Section 6.2 presents in more detail the distribution of readings in the double subject construction, as described above. In section 6.3, I show how the economy constraint might be extended to predict the distribution of readings in the double subject construction. I argue that in order to successfully predict the licensing conditions on cumulative readings in the DSC, it is crucial that the economy constraint be blind to the identity/cardinality of the higher subject. Section 6.4 discusses some outstanding issues, including the lack of a simultaneous reading in the double subject construction and some differences in how the two constructions interact with collective predicates. Finally, section 6.5 concludes.

## 6.2 The distribution of readings in the DSC

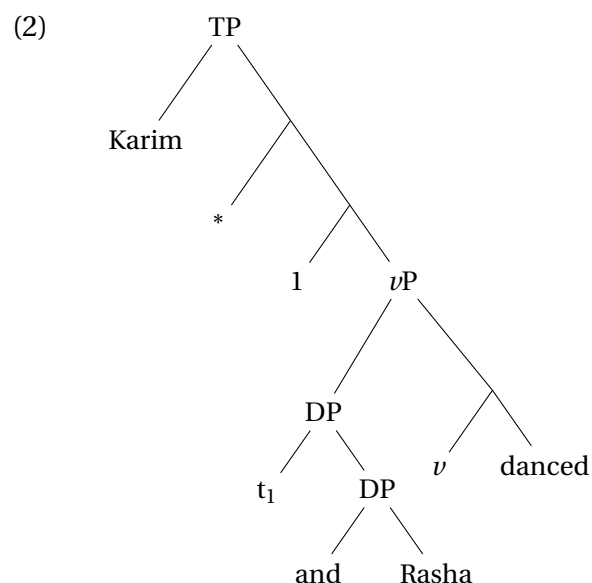
### 6.2.1 Ban on distributive readings

As shown in chapter 2, with simple 1-place predicates like *dance*, the double subject construction allows only for the collective reading given in (1), which is true iff Karim and Rasha danced together. Crucially, the sentence in (1), unlike the counterpart with standard conjunction, is not true in a scenario where Karim and Rasha danced separately. In chapter 2, I argued that this restriction is predicted if (i) pluralization operators, which are the only sources of distributivity, can apply only to predicates derived by movement and (ii) the lower subject can't move covertly from its base position in this environment.

- (1) Karim raʔas howwe w Rasha.  
       Karim danced him and Rasha.  
       Karim and Rasha danced together.

To see again how the proposal in chapter 2 predicts the desired reading in (1), consider the LF for (1) in (2). Here, since the conjoined DP remains in its base position, \* can't apply to the predicate below it and therefore there can be no distributive reading over the conjoined DP. Assuming as

I've been doing throughout this dissertation that *dance* is only true of a plurality if they danced together, (2) predicts the desired reading in (1).



Of course, it is crucial to prevent the lower subject in (2) from moving covertly, otherwise it can take scope in an intermediate position where \* can apply below it, predicting an unattested distributive reading for (1). In chapter 2, I argued that this requirement that the lower subject remains in its base position can't be a general one that applies in all environments. In particular, as we will see in section 6.2.2, when we look at 2-place predicates like *lift*, covert movement of the lower subject will have to be licensed with certain objects in order to predict the availability of cumulative readings. The question, then, is what controls the distribution of covert movement, such that it is banned in (2) but licensed in other environments.

## 6.2.2 The homogeneity-cumulativity generalization

Turning to the distribution of readings in the double subject construction when we have a 2-place predicate like *lift*, we saw in chapter 2 that when the object is a numeral quantifier like *exactly 20*, the double subject construction allows for a cumulative reading, in addition to the familiar collective reading. This is illustrated in (3): (3-a) is false in a distributive scenario where each of Karim and Hadi lifted 20 pianos, but it is true in the cumulative scenario in (3-b) where each of Karim and Hadi independently lifted 10 pianos, resulting in them cumulatively lifting exactly 20 pianos.

- (3) Karim ḥimil howwe w Hadi ṣishreen piano ṣalʔad.  
 Karim lifted him and Hadi 20 piano exactly.

- a. **Distributive Scenario:** Karim and Hadi have never met each other. They each independently lift 20 pianos. **False**
- b. **Cumulative Scenario:** Karim and Hadi don't know each other. Karim lifted 10 pianos and Hadi independently lifted 10 other pianos. **True**
- c. **Collective Scenario:** Karim and Hadi are working together to lift pianos. They go to each piano and lift it together. They do this for exactly 20 pianos. **True**

Interestingly, the availability of the cumulative reading here depends on the object DP in the same way that I argued in chapter 5 for cumulative *together*. As shown in (4-a)-(4-c), the cumulative reading is licensed with a variety of objects, such as *all*, *more than n* and bare numerals, just like we saw with *together*. Crucially, definite plural objects don't seem to allow for a cumulative reading: (4-d) can't be used to describe a scenario in which cumulatively Karim and Rasha stole all of the books.

- (4) **Context:** Karim and Rasha don't know each other, they each walked into a room with collections of books and stole some of the books.
- a. Karim saraʔ howwe w Rasha ʕashr ketob.  
Karim stole him and Rasha ten books.  
Karim and Rasha together stole 10 books.
  - b. Karim saraʔ howwe w Rasha aktar men ʕashr ketob.  
Karim stole him and Rasha more than 10 books.  
Karim and Rasha together stole more than 10 books.
  - c. Karim saraʔ howwe w Rasha kel l ketob.  
Karim stole him and Rasha all the books.  
Karim and Rasha together stole all of the books.
  - d. #Karim saraʔ howwe w Rasha l ketob.  
Karim stole him and Rasha the books.  
Intended: Karim and Rasha together stole all of the books.

As shown in chapter 5, the main difference between the universal object in (4-c) and the definite plural object in (4-d) is that the latter gives rise to a homogeneity gap, while in the former, *all* removes homogeneity. We can therefore hypothesize that like *together*, the Lebanese Arabic double subject construction only allows for a cumulative reading when the object doesn't give rise to homogeneity.

Additional evidence for this connection between homogeneity and cumulativity comes from considering sub-atomic cumulativity. When the object is a definite singular, as in (5-b), the double subject construction is not true in a scenario where each of Karim and Rasha independently stole half of the collection. On the other hand, this reading is available in (5-a), when the definite singular is modified by *kel*.



- (5) **Context:** Karim and Rasha don't know each other, they each walked into a room with collections of books and stole some of the books.
- a. Karim sara? howwe w Rasha kel l collection.  
Karim stole him and Rasha all the collection.  
Karim and Rasha together stole the whole collection.
  - b. #Karim sara? howwe w Rasha l collection.  
Karim stole him and Rasha the collection.  
Intended: Karim and Rasha together stole the whole collection.

Just like *whole* in English, *kel* removes sub-atomic homogeneity when it applies to a singular definite. To see this, consider first the reading we get from negating the definite singular object in (6-a): (6-a) is only true if Karim stole no part of the collection, while its positive counterpart is only true if Karim stole every book in the collection. We therefore see that the definite singular object gives rise to a homogeneity gap with respect to parts of the collection: if Karim stole half of the books in the collection, neither (6-a) nor its positive counterpart is true. Now, the counterpart in (6-b) where *kel* is inserted has no homogeneity gap. In particular, (6-b) is true as long as Karim didn't steal all of the books from the collection, thus removing the homogeneity gap in (6-a). We therefore see that the pattern in (5) is in line with the homogeneity-cumulativity generalization.

- (6) a. **Sub-atomic Homogeneity with *the collection*:**  
Karim ma sara? l collection.  
Karim not steal the collection.  
Karim stole **no part** of the collection.
- b. **Homogeneity removal with *kel*:**  
Karim ma sara? kel l collection.  
Karim not steal all the collection.  
Karim didn't steal the whole collection.

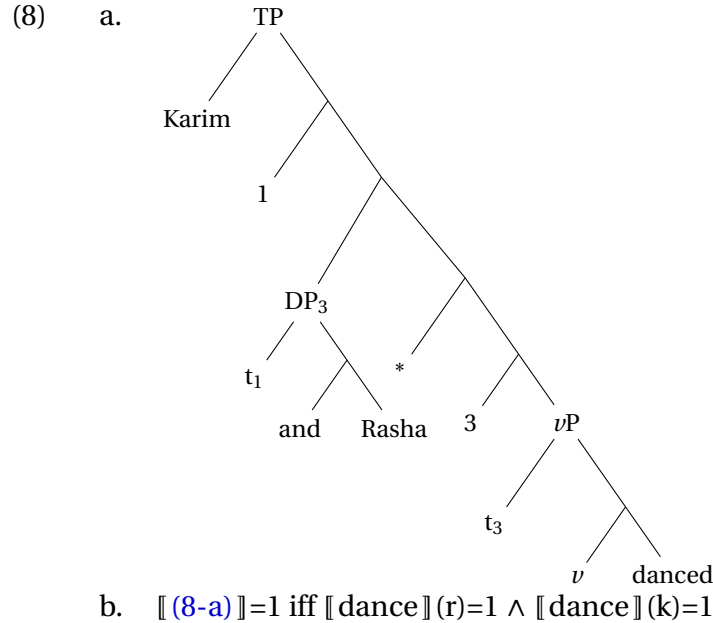
I will therefore take the connection between homogeneity and the availability of cumulative readings, repeated in (7), to be a desideratum for any successful theory of the Lebanese Arabic double subject construction. In section 6.3, I show how the economy constraint proposed for *together* might be extended to account for the homogeneity-cumulativity generalization, in addition to the ban on distributive readings described in section 6.2.1.

- (7) **Homogeneity-Cumulativity Generalization:** A cumulative reading in the Lebanese Arabic double subject construction is licensed only if the object does not give rise to a homogeneity gap.

## 6.3 The role of economy

### 6.3.1 Blocking distributive readings

To see how economy can account for the distribution of readings in the double subject construction, consider first the LF in (8-a). Here, the lower subject moves covertly to a position above \*, predicting an unattested reading which is true iff each of Karim and Rasha danced (8-b).



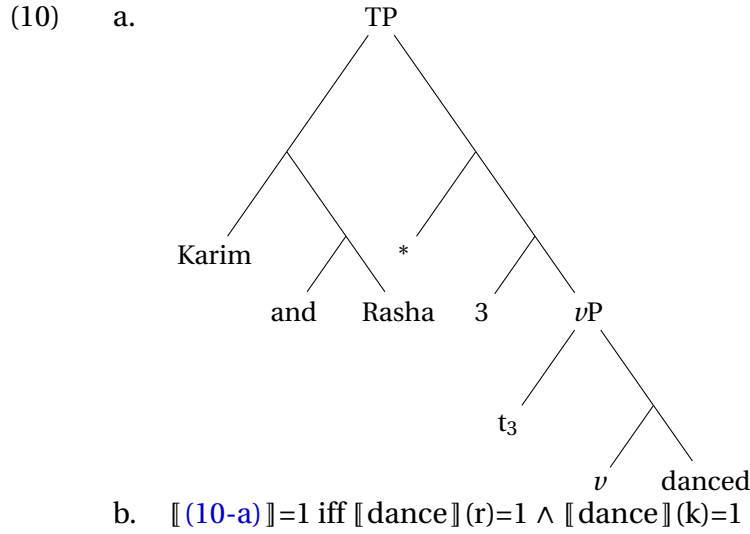
As proposed in chapter 5, the generalized economy constraint, repeated in (9), will rule out (8-a) if the deductive system can prove that its truth-conditions are equivalent to those of a simpler LF. I want to suggest that (8-a) competes with the simpler LF in (10-a) with standard conjunction, where the conjoined DP moves directly to spec,TP, generating the same distributive reading. The constraint, as it is stated in (9), does not predict that (10-a) will count as a simpler competitor to (8-a) for the purposes of evaluating economy. Namely, since the conjoined DP is in spec,TP in (10-a) and in an intermediate position in (8-a), we can't straightforwardly derive (10-a) from (8-a) by deleting the higher *Karim* and the index 1.<sup>1</sup>

(9) a. **Generalized Economy Constraint:**

An LF  $\phi$  is ruled out if the deductive system can prove that there is a competitor  $\psi$  such that (i)  $\psi < \phi$  and (ii)  $\llbracket \psi \rrbracket = \llbracket \phi \rrbracket$

<sup>1</sup>If we delete *Karim* and the index 1 from (8-a), we would get an LF where there is no DP in spec,TP. If we assume that T has an EPP feature, then this LF will be ungrammatical. One possible solution is to assume that there is no obligatory EPP in LA, but this will fail to predict the fact, discussed in section 6.3.4, that the DSC does allow for distributive readings over the lower subject when it moves overtly to an intermediate EPP position.

- b. An LF  $\psi < \phi$  if  $\psi$  can be derived from  $\phi$  by deleting a number of nodes.



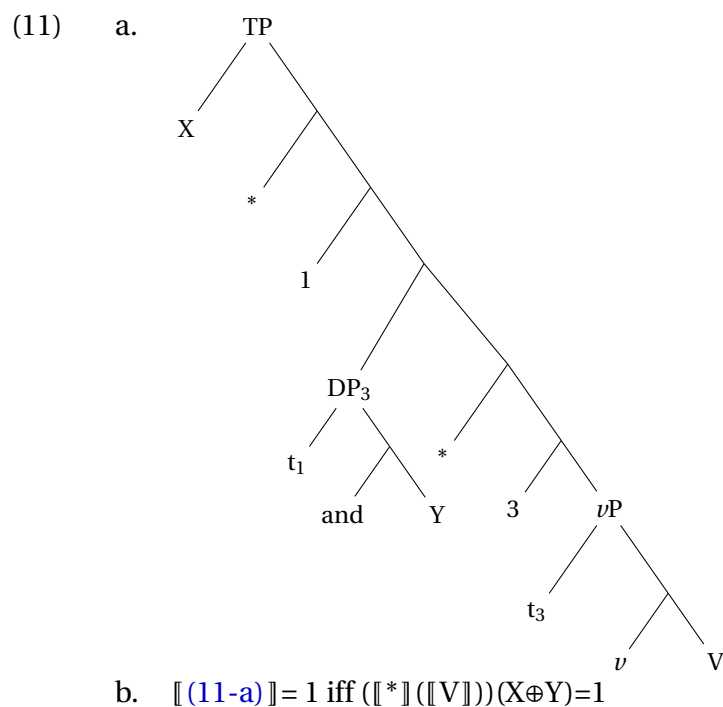
In order to predict that (8-a) in fact competes with (10-a) for the purposes of economy, we therefore have to formulate a notion of simpler alternative which is not sensitive to the fact that the conjoined DP is in spec,TP in (10-a) and in a lower intermediate position in (8-a). I'd like to tentatively suggest an alternative derivational notion which considers an LF  $\psi$  to be simpler than  $\phi$  iff a proper subset of the merge steps that generate  $\phi$  are needed to generate  $\psi$ . Assuming a copy theory of movement, this will predict that (10-a) counts as a simpler alternative to (8-a). In (8-a), *Karim and Rasha* is merged twice and *Karim* is merged an additional time, while in (10-a), *Karim and Rasha* is merged twice, while there is no additional merge of *Karim*. These LFs otherwise involve the merging of the same elements. We can therefore conclude that (8-a) is strictly more complex than (10-a). This formulation would continue to consider deletion alternatives as simpler LFs but also subsumes cases like (8-a)-(10-a), where certain elements are merged in different positions.

I will leave a detailed exploration of this alternative formulation of economy and its consequences to future work. What's important here is that an LF where the conjoined DP moves to an intermediate position before the first conjunct moves to spec,TP always competes with a simpler alternative where the conjoined DP moves directly to spec,TP.

As discussed in chapter 5, in order to definitively show that the distributive reading is ruled out by generalized economy, we have to (i) prove that the LF in (8-a) is equivalent to (10-a) regardless of the cardinality of the two conjuncts and (ii) show that there is no alternative LF that can generate the distributive reading in the DSC without violating economy. Starting with (i), it is straightforward to see that even if the two conjuncts *Karim* and *Rasha* are replaced with plural individuals, (8-a) will still be equivalent to (10-a). In particular, both LFs are predicted to be true

if each member of the plurality denoted by the conjoined DP danced, regardless of whether they did so individually, in subgroups or all together. Therefore, generalized economy will rule out the LF in (8-a).

As for (ii), we can consider alternative LFs that generate the unattested reading. In particular, the same underspecified reading is predicted if there is another \* operator below the higher subject in (8-a), as shown in (11-a). Again, (11-a) is equivalent to the simpler LF illustrated in (10-a), where the conjoined DP moves directly to spec,TP. In particular, both (11-a) and (10-a) are true iff the plural individual denoted by the conjoined DP can be divided into parts which the unpluralized predicate is true of.



I have therefore suggested in this section that the generalized economy constraint can be extended to rule out the unattested distributive reading with the double subject construction, thus predicting that with 1-place predicates like *dance*, the double subject construction is true only in collective scenarios.

### 6.3.2 Licensing cumulative readings with non-monotonic objects

To see what the economy constraint predicts for cumulative readings in the double subject construction, I will start by looking at the case where the object is a non-monotonic quantifier *exactly n*. I will show that the modular generalized economy constraint successfully predicts that (12) allows for a cumulative reading. Just like in the case of *together* in chapter 5, licensing the

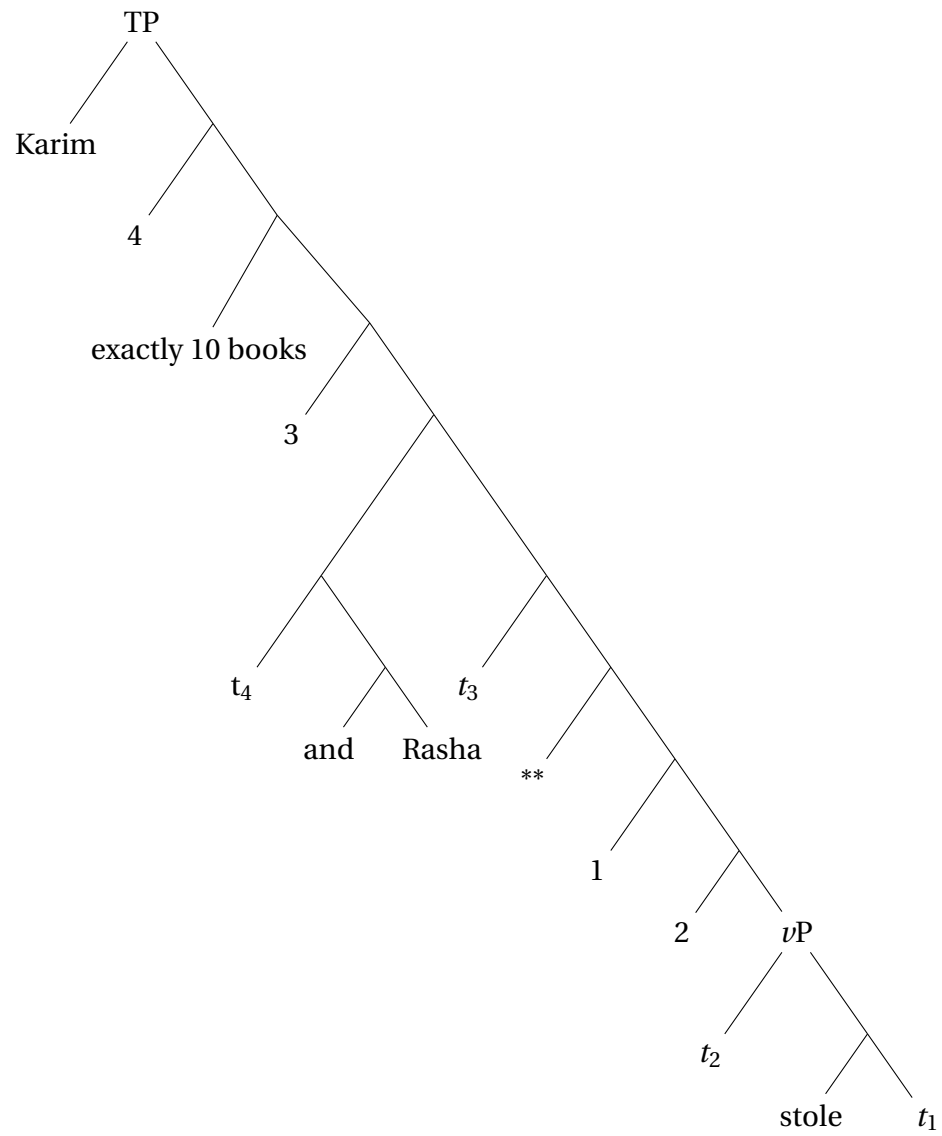
cumulative reading will be less straightforward with upward-entailing quantifiers like *all* or *more than n*. I therefore return to these other objects after showing the results with *exactly 10*.

- (12) Karim saraʔ howwe w Rasha ʕashr ketob ʕal-ʔadd.  
 Karim stole him and Rasha ten books exactly.  
 Karim and Rasha together stole exactly 10 books.

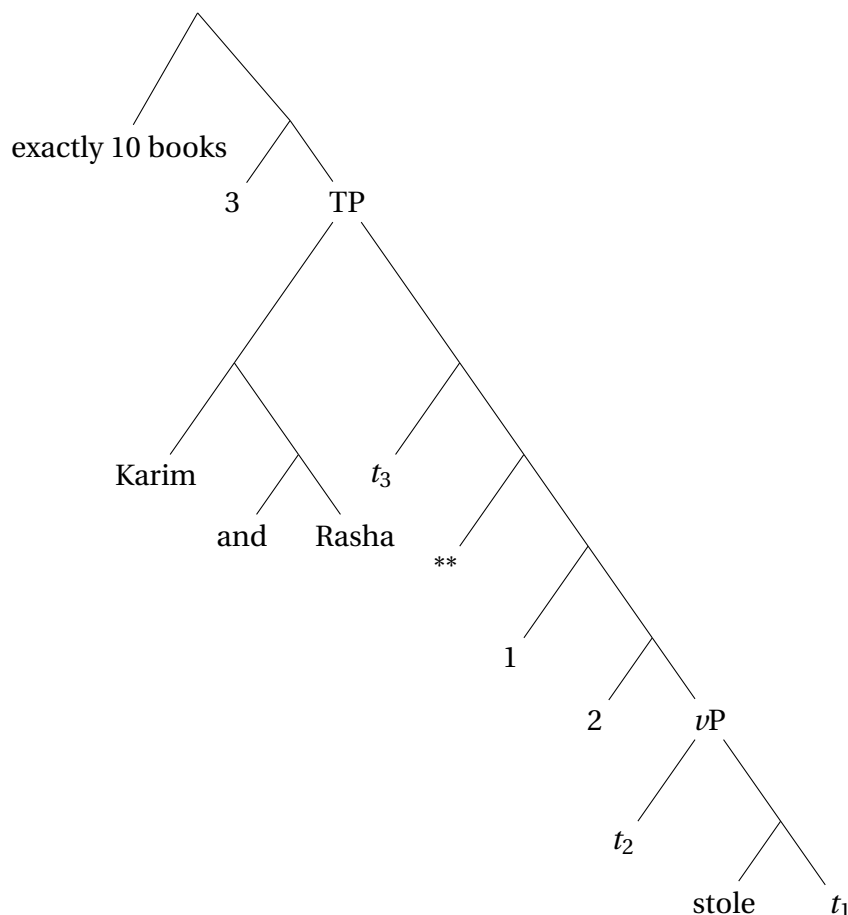
Consider first the LF in (13-a) that generates the cumulative reading in (12). This LF appears to be ruled out by the economy constraint in the same way as the distributive reading in section 6.3.1. In particular, there is a simpler LF in (13-b), where the entire conjoined DP moves directly to spec,TP rather than to an intermediate position and where \*\* can take scope below this conjoined DP in spec,TP. Just like we saw with the distributive reading in section 6.3.1, the LF in (13-a) involves an additional step of merging *Karim* and the binder index 4 and otherwise merges the same elements as (13-b). We therefore see that (13-a) has a simpler competitor in (13-b) which is truth-conditionally equivalent to it: both (13-a) and (13-b) are true iff Karim and Rasha cumulatively stole exactly 10 books. As it stands, the economy constraint therefore seems to predict that the cumulative reading of (12) is ruled out.

(13)

a.



b.



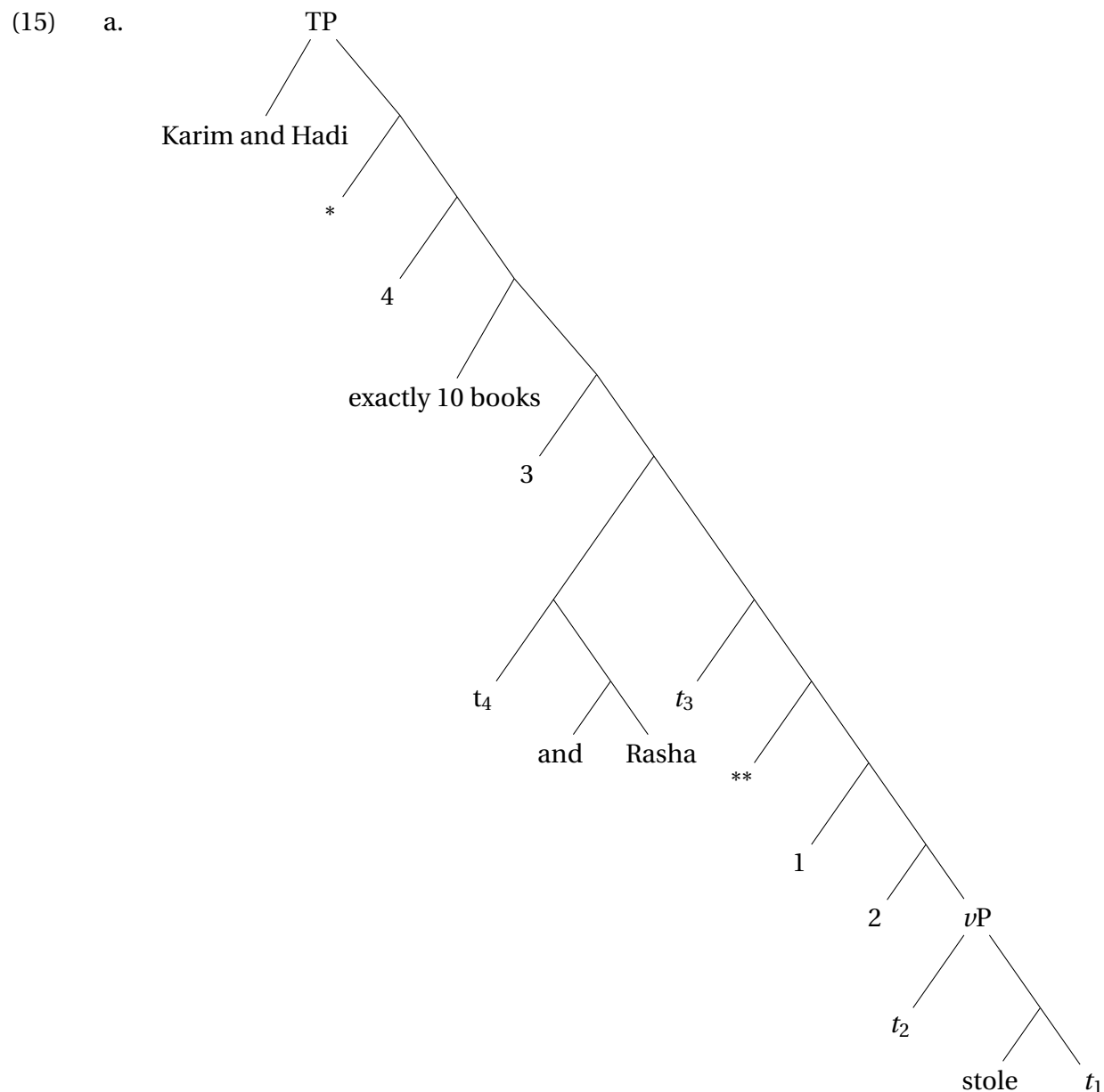
Again, in order for the generalized economy constraint to block a given reading, we have to consider all of the LFs that can generate that reading and ensure that for each of them generalized economy can prove equivalence with a simpler LF, regardless of the cardinality of the two conjuncts. In what follows, I will show that when the higher subject denotes an individual of cardinality greater than 1, we can generate an LF that gives rise to the desired cumulative reading without being equivalent to any simplification alternative. I therefore conclude that generalized economy licenses the cumulative reading in (12).

To see this, consider the example in (14), where the higher subject is *Karim and Hadi*. Now, (14), in addition to being true in the purely cumulative scenario in (14-a), is also true in the scenario in (14-b) where we get distributivity over the higher subject and a lower cumulative reading.

(14) Karim w Hadi sara?o henne w Rasha fashr ketob fal-?add  
Karim and Hadi stole them and Rasha ten books exactly.

- a. **Scenario 1:** Karim and Hadi and Rasha together stole exactly 10 books. **True**
- b. **Scenario 2:** Hadi and Rasha together stole exactly 10 books and Karim and Rasha together stole exactly 10 books. **True**

The reading of (14) which is true in Scenario 2 can be generated by having a \* operator take scope below the higher subject in (13-a). The resulting LF is given in (15-a) and the associated truth-conditions in (15-b): (15-a) is predicted to be true iff either Karim and Rasha and Hadi together stole exactly 10 books or if each of Karim and Hadi is such that him and Rasha together stole exactly 10 books. Now, it is clear that (15-a) does not violate the economy constraint. In particular, (15-a) will compete with the LF in (16), where the entire conjoined DP *Karim and Rasha and Hadi* moves directly to spec,TP. (16) will generate a purely cumulative reading which is true iff Karim and Rasha and Hadi together stole exactly 10 books. It is therefore clear that (15-a) and (16) are not truth-conditionally equivalent: while (15-a) is true in a scenario where each of Karim and Hadi and Rasha stole exactly 5 (different) books, (16) is false in such a scenario.







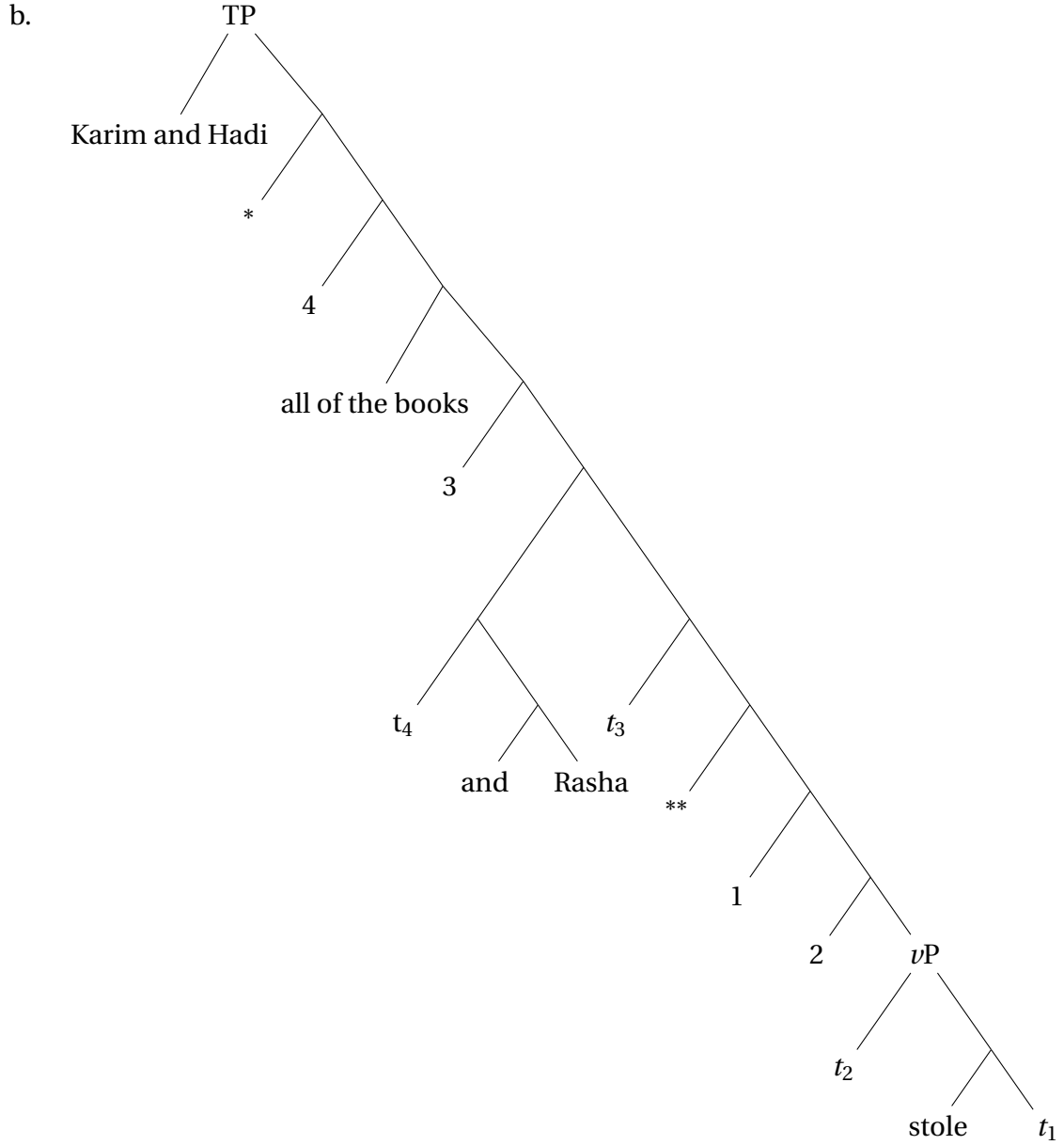
of the lower subject seems to depend on certain semantic properties of the sentence, a result which a purely syntactic constraint, such as the ban on improper sub-extraction discussed in chapter 2, can't deliver.

### 6.3.3 Deriving the homogeneity-cumulativity generalization

I will now turn to the predictions of the economy constraint for other object DPs, such as *all*, *more than n* and definite plurals. For upward-entailing quantifiers, such as *all* and *more than n*, I will argue that just like with *together* in chapter 5, a local implicature due to *exh* will be necessary in order to avoid violating economy. This will allow me to predict that objects that give rise to homogeneity, which block this necessary strengthening, don't license a cumulative reading in the double subject construction. I therefore argue in this section that the homogeneity-cumulativity generalization can be derived in a parallel way to the analogous generalization with English *together*.

For concreteness, I will use *all* as a case-study here and compare it to the definite plural object. The result with *all* will generalize to other quantifiers following the same recipe we saw in chapter 5. Unlike with *exactly n*, without strengthening, the cumulative reading with *all* violates economy even when the subject has cardinality greater than 1. To see this, consider the LF in (17-b) for (17-a): (17-b) is predicted to be true iff either Rasha and Karim and Hadi together stole all of the books or if Rasha and Hadi together stole all of the books and Karim and Rasha together stole all of the books (17-c). Notice that the second disjunct here entails the first one. The disjunction in (17-c) is therefore equivalent to the first disjunct: (17-b) is therefore true iff Karim and Hadi and Rasha together stole all of the books.

- (17) a. Karim and Hadi stole them and Rasha all of the books.



- c.  $\llbracket (17-b) \rrbracket = 1$  iff
- $$\llbracket \text{All of the books} \rrbracket (\lambda x. ** \llbracket \text{stole} \rrbracket (x)(k \oplus r \oplus h)) = 1 \vee [\llbracket \text{All of the books} \rrbracket (\lambda x. ** \llbracket \text{stole} \rrbracket (x)(k \oplus r)) = 1$$
- $$\wedge \llbracket \text{All of the books} \rrbracket (\lambda x. ** \llbracket \text{stole} \rrbracket (x)(h \oplus r)) = 1]$$
- iff  $\llbracket \text{All of the books} \rrbracket (\lambda x. ** \llbracket \text{stole} \rrbracket (x)(k \oplus r \oplus h)) = 1$

Now, as in the case of exactly  $n$  in (16), (17-b) will compete with an LF where the entire conjunction *Karim and Hadi and Rasha* moves to spec,TP directly, licensing a  $**$  operator below it and deriving a purely cumulative reading. The corresponding truth-conditions are given in (18): the simpler LF parallel to (16) will be true iff Karim and Hadi and Rasha together stole all of the books. Now, given that the second disjunct in (17-c) entails the first one, (17-c) turns out to be equivalent to (18): they both give rise to a purely cumulative reading.

(18) **Competitor with \*\* below spec,TP:**

$\llbracket (16) \text{ with } all \rrbracket = 1$  iff  $\llbracket \text{All of the books} \rrbracket (\lambda x. ** \llbracket \text{stole} \rrbracket (x)(k \oplus r \oplus h)) = 1$

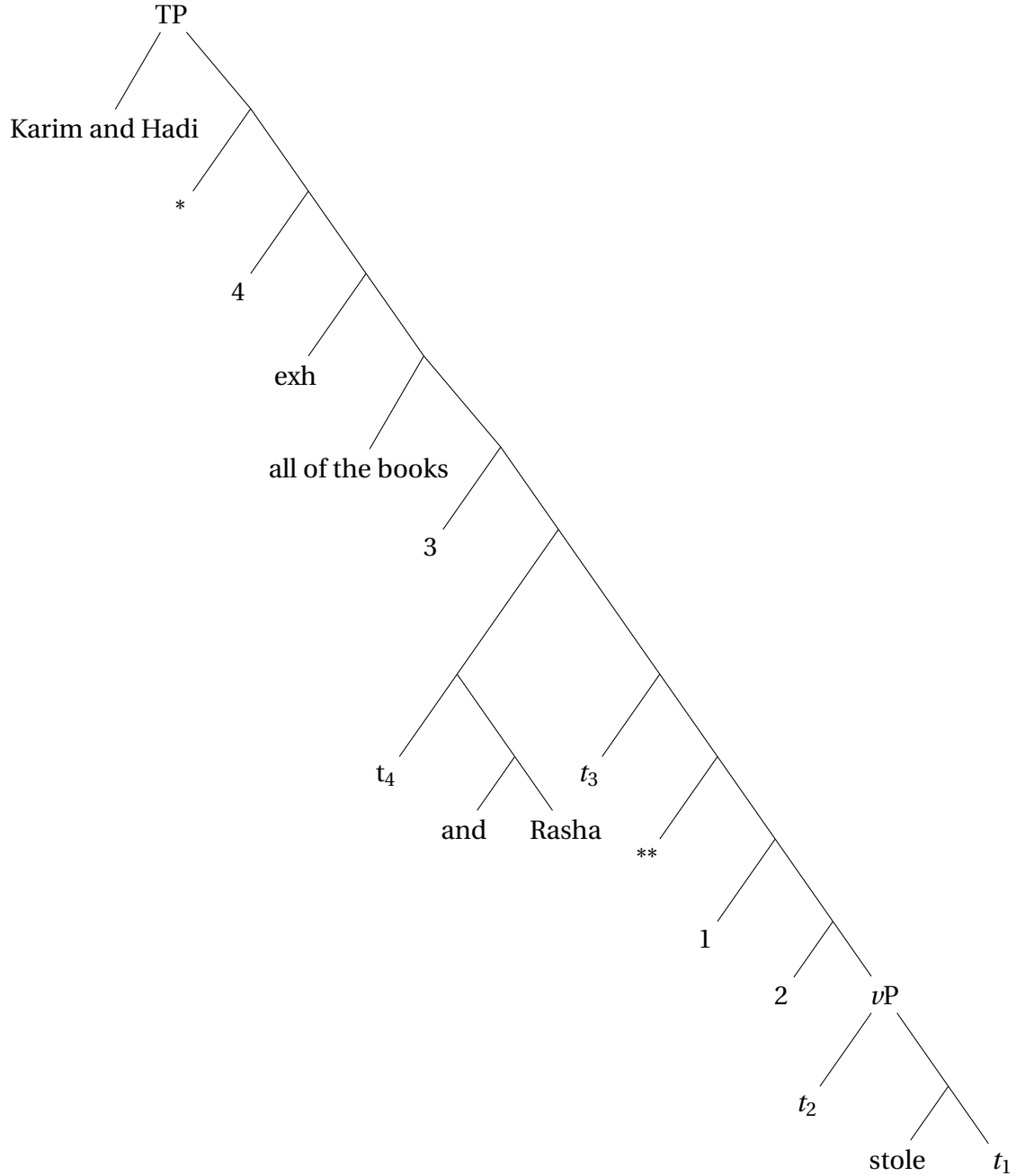
In order to avoid violating the economy constraint, we therefore have to find a way to break the entailment pattern between the two disjuncts in (17-c). As argued in chapter 5, this can be done by inserting *exh* locally below the higher \*. More specifically, the relevant implicature will strengthen the predicate by asserting that the predicate is not true for any proper part of the plurality that it applies to (19).

(19) **Strengthened Predicate below \*:**

$\lambda y. \llbracket \text{All of the books} \rrbracket (\lambda x. ** \llbracket \text{stole} \rrbracket (x)(y)) = 1 \wedge \forall y' < y : [\llbracket \text{All of the books} \rrbracket (\lambda x. ** \llbracket \text{stole} \rrbracket (x)(y')) = 0]$

Now, the LF with *exh* given in (20-a) will give rise to the truth-conditions in (20-b): (20-a) is true iff either Karim and Hadi and Rasha together stole all of the books and no single individual or pair of them stole all of the books or if Karim and Rasha together stole all of the books and Hadi and Rasha together stole all of the books but no single person out of the three alone stole all of the books. Here, the second disjunct in (20-b) is in fact incompatible with the first one, breaking the problematic entailment pattern between the two disjuncts. We can see that the economy constraint is therefore no longer violated here. The competitor which gives rise to a purely cumulative reading will have distinct truth-conditions from (20-b): While (20-b) is true if Karim and Rasha together stole all of the books and Hadi and Rasha together stole all of the books, (20-c) is false in such a scenario.

(20) a.



b.  $\llbracket (20-a) \rrbracket = 1$  iff

$\llbracket \text{All of the books} \rrbracket (\lambda x. \llbracket \text{stole} \rrbracket (x)(k \oplus h \oplus r)) = 1 \wedge \forall y < k \oplus h \oplus r: [\llbracket \text{All of the books} \rrbracket (\lambda x. \llbracket \text{stole} \rrbracket (x)(y)) = 0]$   
 $\vee [\llbracket \text{All of the books} \rrbracket (\lambda x. \llbracket \text{stole} \rrbracket (x)(k \oplus r)) = 1 \wedge \forall y < k \oplus r: [\llbracket \text{All of the books} \rrbracket (\lambda x. \llbracket \text{stole} \rrbracket (x)(y)) = 0]$   
 $\wedge \llbracket \text{All of the books} \rrbracket (\lambda x. \llbracket \text{stole} \rrbracket (x)(h \oplus r)) = 1 \wedge \forall y < h \oplus r: [\llbracket \text{All of the books} \rrbracket (\lambda x. \llbracket \text{stole} \rrbracket (x)(y)) = 0]$

c. **Competitor with \*\* below spec,TP:**

$\llbracket \text{Competitor where the entire conjunction moves directly to spec,TP} \rrbracket = 1$  iff

$\llbracket \text{All of the books} \rrbracket (\lambda x. \llbracket \text{stole} \rrbracket (x)(k \oplus h \oplus r)) = 1$   
 $\wedge \forall y < k \oplus h \oplus r: [\llbracket \text{All of the books} \rrbracket (\lambda x. \llbracket \text{stole} \rrbracket (x)(y)) = 0]$

I have therefore shown that just like with cumulative *together*, the local implicature below \* is necessary in order to license a cumulative reading in the double subject construction. Now, as shown in chapter 5, this implicature based on subpart alternatives can't be generated when the object gives rise to homogeneity. This is because negating the subpart alternatives in this case will give rise to a contradiction. While I won't repeat the full discussion from chapter 5 for how homogeneity interacts with the required strengthening, we can see that with the cumulative reading of *all* in (21-a), we can negate the subpart alternatives without contradiction, while the counterpart with the definite plural in (21-b) gives rise to a contradiction.

- (21) a. Jane and Mary and Sue stole all of the books, but Jane and Mary didn't steal all of the books.  
 b. #Jane and Mary and Sue stole the books, but Jane and Mary didn't steal the books.

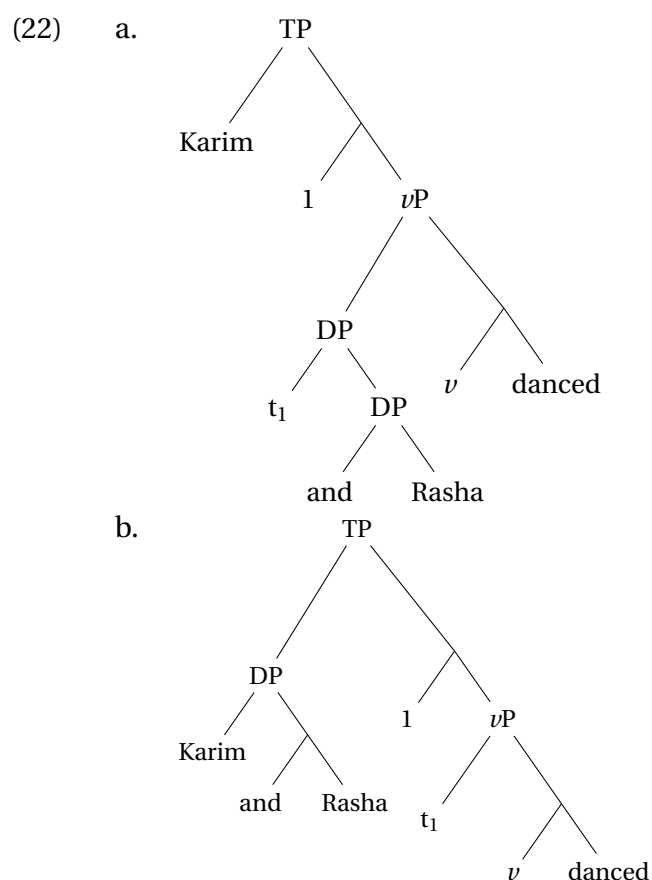
Since the local implicature can't arise with definite plural objects due to homogeneity, the cumulative reading with these objects can't be rescued from violating economy, predicting that the cumulative reading is not licensed when the object gives rise to homogeneity. I have therefore accounted for the homogeneity-cumulativity generalization in the double subject construction in a parallel way as the analogous generalization with cumulative *together*.

### 6.3.4 The relationship between movement and distributivity

So far in this section, I showed that if we adopt an appropriate notion of a simpler competitor, the economy constraint successfully predicts that the double subject construction blocks distributive readings and allows for a cumulative reading only under certain conditions. I've been assuming throughout the section that pluralization operators apply only to predicates derived by movement, as argued in chapter 2. The goal of this section is to show that this constraint on \* is still crucial to predicting the desired distribution of readings, given the economy constraint. To do this, I will start by showing that if we allow \* to apply directly to lexical predicates, the generalized economy constraint will incorrectly predict that the DSC allows for distributive readings over the lower subject. I then turn to the more intricate cases discussed in chapter 2, where the double subject construction allows for distributive readings when there is an intermediate A position that the lower subject can move to overtly. I argue that the economy constraint, coupled with the restriction on the distribution of \*, predicts the relationship between overt movement and the availability of a distributive reading when auxiliaries are present, thus accounting for the full pattern presented in chapter 2.

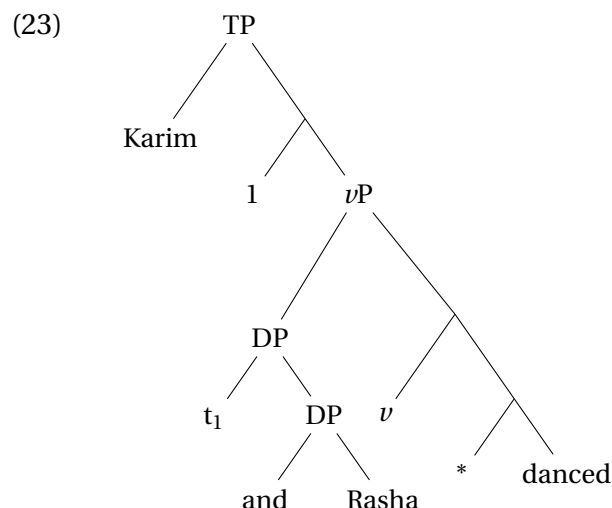
To see why the restriction on \* is crucial, consider first the collective LF in (22-a) for the double subject construction. Here, there is no simpler grammatical LF that can generate the collective

reading. In particular, I assume as discussed in chapter 2 that there is an agreement probe on T that requires either the first conjunct or the entire conjunction to move to spec,TP. Therefore, the only alternative LF that generates the same collective reading is the one with standard conjunction given in (22-b). Note that, unlike in the case of the unattested distributive reading in section 6.3.1, (22-b) clearly does not count as a simpler alternative to (22-a). In particular, in (22-b), the conjoined DP is merged twice and in (22-a), the conjoined DP is merged once and Karim is merged an additional time in spec,TP. Therefore neither of the two LFs involves a subset of the merge steps that are involved in the other. We can therefore conclude that (22-a) doesn't compete with (22-b) for the purposes of the economy constraint, thus correctly predicting that the collective LF in (22-a) is licensed by economy.



Now, suppose that the \* operator could apply directly to the verb *dance* in (22-a). The result would be the LF in (23) which is true iff each of Karim and Rasha danced. Again, this LF will not compete with an LF like (22-b) where the entire conjunction moves to spec,TP. In fact, there will be no simpler alternative for (23) which gives rise to the same distributive truth-conditions. In particular, the higher subject *Karim* can't be deleted, since movement to spec,TP is obligatory. Deleting the \* operator is possible but it will give rise to collective truth-conditions which are distinct from the distributive ones corresponding to (23). Therefore, the economy constraint

alone will incorrectly predict that the LF in (23) is licensed.



It is therefore crucial to supplement the economy constraint with the restriction on the distribution of pluralization operators argued for in chapters 2 and 3 and repeated in (24). The constraint in (24) ensures that the LF in (23) can't be generated, thus allowing us to maintain that the economy constraint blocks distributive readings in the double subject construction.

- (24) **Restriction on pluralization operators:** Pluralization operators, like \* and \*\*, can only apply to predicates derived by movement.

We can now turn to the cases discussed in chapter 2 where a distributive reading is in fact available in the double subject construction. This is exemplified in (25-a): here the lower subject intervenes between the auxiliary and the main verb and the auxiliary agrees with the higher subject while the main verb agrees with the lower subject. We can contrast this with (25-b), where the lower subject follows the main verb and both the auxiliary and the verb agree with the higher subject. While (25-a) is true in a distributive scenario where Karim and Rasha danced separately, (25-b) isn't.

- (25) a. Karim *ke:n* howwe w Rasha *ʔam yereʔso*.  
 Karim was.3SG him and Rasha PROG dance3PL.  
 Karim and Rasha were dancing.
- b. Karim *ke:n* *ʔam yerʔos* howwe w Hadi.  
 Karim was.3SG.M PROG dance.3SG.M him and Hadi.  
 Karim was dancing with Hadi.

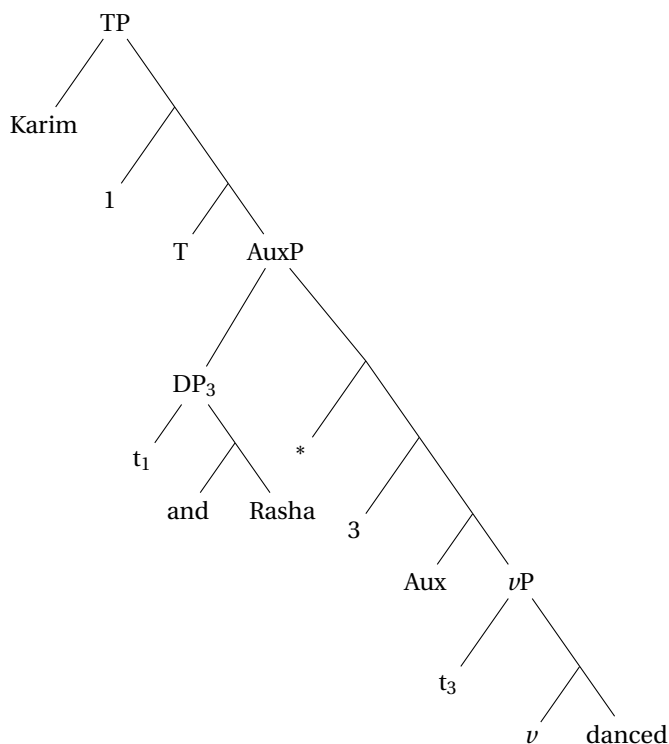
I proposed in chapter 2 that when the auxiliary *ke:n* is present, there are two agreement probes, one corresponding to agreement on the main verb and one to agreement on the auxiliary. I argued that both of these probes obligatorily trigger movement of the DP that they agree with



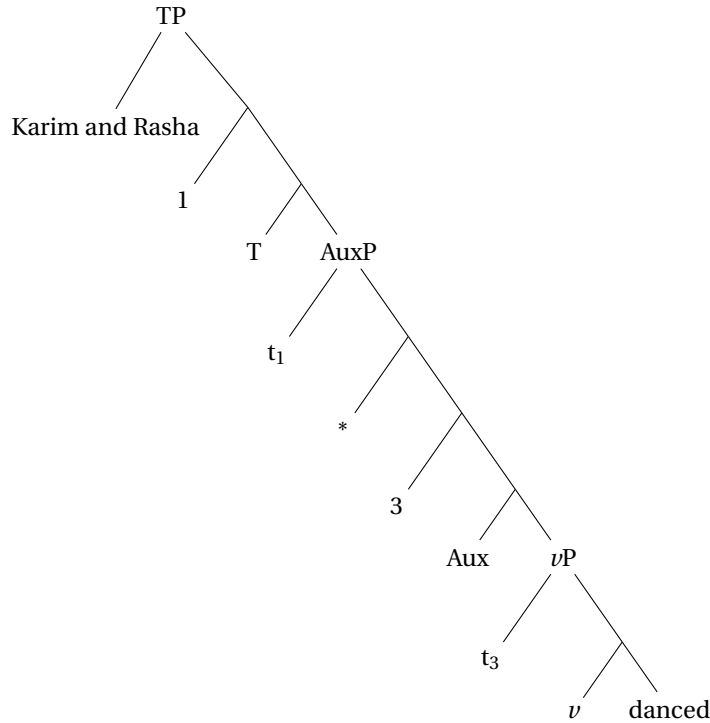
to the specifier of the head hosting the probe. Therefore in (25-a), *him and Rasha* moves to an intermediate specifier position corresponding to agreement with the main verb and then *Karim* moves to spec,TP, corresponding to agreement with the auxiliary.

We can now see why the distributive reading in (25-a) is not ruled out by the economy constraint. The LF that derives the distributive reading for (25-a) is given in (26-a): here the conjoined DP moves to an intermediate position allowing \* to be inserted below it. Now, this LF is very similar to the one that I argued was ruled out by the economy constraint in section 6.3.1, where the conjoined DP moved covertly rather than overtly to an intermediate position. In section 6.3.1, intermediate covert movement was ruled out because there was an alternative simpler LF with standard coordination where the entire conjoined DP moved directly to spec,TP. This alternative is not available here: as mentioned above, movement to the intermediate position is obligatorily triggered by the lower agreement probe when the auxiliary is present. The LF for standard coordination is therefore given in (26-b). Notice that (26-b), like (26-a), involves two steps of movement. Unlike what we saw with (8-a) in section 6.3.1, (26-a) therefore does not involve an additional step of merge, when compared to ((26-b)). We can therefore conclude that there is no simpler LF that generates the distributive reading in (26-a), thus successfully predicting that overt movement to an intermediate A position licenses a distributive reading in the DSC.

(26) a. **Intermediate overt movement:**



b. **Distributive reading with standard coordination:**



I have therefore shown in this section that the economy constraint, coupled with the restriction that pluralization operators apply only to predicates derived by movement, derives the full distribution of readings that we've seen so far in the double subject construction. I have also shown several parallels between the double subject construction and *together* which are captured in the same way by this unified analysis with the economy constraint. In particular, both constructions ban distributive readings (at least in the simple cases) and only license cumulative readings when the object doesn't give rise to homogeneity. In the next section, I will discuss a couple of differences between the two constructions which raise challenges for this unified analysis.

## 6.4 Challenges for a unified analysis

The goal of this section is to discuss two differences between the double subject construction and *together* that are not captured by the unified analysis explored so far. Section 6.4.1 shows that the DSC seems to lack the simultaneous reading that *together* has. Section 6.4.2 discusses some differences with respect to how the double subject construction and *together* interact with collective predicates like *argue*.

### 6.4.1 The simultaneous reading

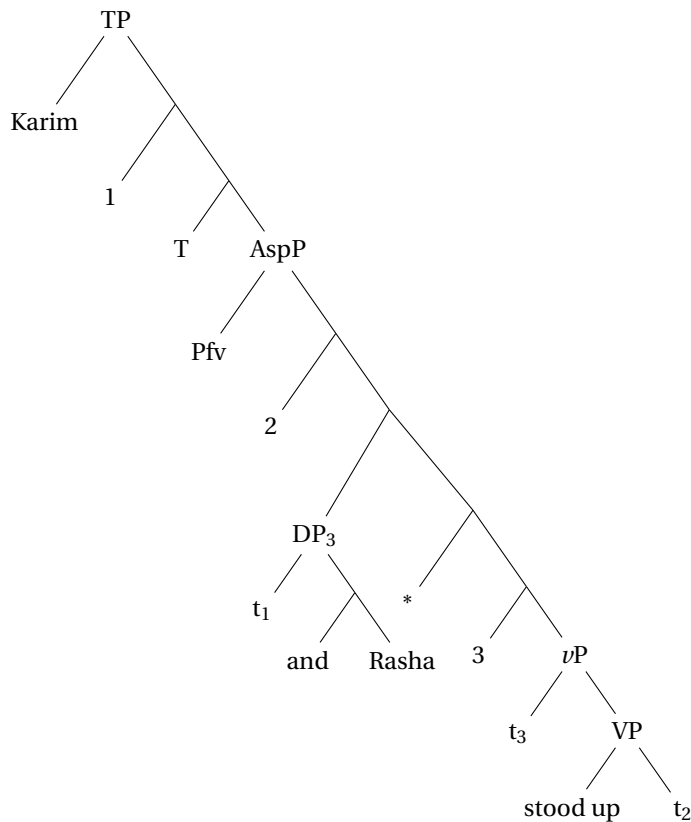
While we saw that the double subject construction and *together* give rise to the same cumulative and collective readings, they appear to differ when we look at the simultaneous reading. In particular, while *together* licenses a simultaneous reading (Lasersohn, 1990), the double subject construction seems to lack this reading. This is shown in (27). In this context, where Karim and Hadi independently stood up at the same time, both English *together* and the Lebanese Arabic counterpart *maʕ baʕed* are felicitous and true (27-b). On the other hand, the double subject construction is not true in this context (27-a).

(27) **Context:** Karim and Hadi were both watching a play. Karim was sitting in the front and Hadi in the back. They both stood up at the same time to leave the play in the middle.

- a. #Karim wiʔif                      howwe w    Hadi.  
    Karim stood-up.3SG him    and Hadi.  
    Intended: Karim and Hadi stood up at the same time.
- b. Karim w    Hadi weʔfo                      maʕ baʕed.  
    Karim and Hadi stood-up.3PL with each-other.  
    Karim and Hadi stood up together.

This difference between *together* and the double subject construction is very puzzling for the account pursued so far in this dissertation. In particular, just like we saw with *together* in chapter 4, the simultaneous reading can simply be generated in the double subject construction by moving the lower subject to a position above the lowest \* operator but below a higher \* operator above perfective Aspect. This LF is given in (28).

(28)



It is not clear how this LF can be ruled out by the economy constraint. In particular, suppose that (28) competes with an alternative LF with standard coordination where the conjoined DP moves directly to spec,TP. This alternative LF can't give rise to the same simultaneous truth-conditions. If there is a \* operator below the conjoined DP in spec,TP, we would get a distinct reading that distributes over times. On the other hand, if there is no \* operator, we would again get a distinct collective reading. Therefore, in order to generate the simultaneous reading in (28) that conjoined DP has to move to the intermediate position below Aspect. We can therefore conclude that (28) appears to be licensed by the generalized economy constraint.

In order to block the simultaneous reading in the DSC, we therefore have to assume that movement of the conjoined DP to the intermediate position in (28) is ruled out for independent reasons. At the same time, given that *maḡ baḡed* in (27-b) does license a simultaneous reading, we have to assume that the *x with the other* coordinated DP can in fact move to the same intermediate position. At this point, the source of this difference remains unclear. In particular, the licensing of the relevant covert movement must somehow depend on the internal composition of the coordinated DP, an issue which I will have to leave open.

### 6.4.2 Collective predicates

Another difference between the double subject construction and English *together* concerns how they interact with a certain class of collective predicates. To see this, let's first observe the pattern in (29) with the collective predicate *argue* in Lebanese Arabic. As evidenced by (29-a) and (29-b), this predicate seems to require a plural subject. This can be accounted for if we assume that *argue* is lexically collective; it can only be true of plural arguments. Both the double subject construction and the Lebanese Arabic counterpart of *together*, *maʕ baʕed*, are licensed with this predicate, giving rise to the same collective reading we see in (29-a).

- (29) a. Karim w Rasha tshe:raʕo.  
Karim and Rasha argued.3PL.  
Karim and Rasha argued.
- b. #Karim tshe:raʕ.  
Karim argued.3SG.
- c. Karim tshe:raʕ howwe w Rasha.  
Karim argued.3SG.M him and Rasha.  
Karim argued with Rasha.
- d. Karim w Rasha tshe:raʕo maʕ baʕed.  
Karim and Rasha argued.3PL with each-other.  
Karim and Rasha argued with each other.

While the collective reading in (29-c) is predicted to be licensed by the economy constraint, there is a question of what the economy constraint predicts for (29-d). In particular, given that *argue* is a lexically collective predicate, the same truth-conditions that we get in (29-d) can be obtained by deleting *with the other*. Note that this is true regardless of the cardinality of the subject: as shown in chapters 4 and 5 the contribution of *together* was to ensure that we don't get a purely distributive reading where the predicate applies to the atoms of the plurality denoted by the subject. The contribution of *together* is therefore trivial when the predicate itself is never true of atomic individuals, which is the case with *argue*, as evidenced by the infelicity of (29-b). If the deductive system has access to the information that *argue* is a collective predicate, the economy constraint will therefore rule out the collective reading in (29-d). In order to predict that *together* is in fact licensed in (29-d), we therefore have to assume that the deductive system is blind to the fact that the predicate in (29-d) is a collective predicate.

Note that predicates like *argue* in (29) behave differently from the predicates we considered in chapter 2, like *kiss*, which alternate between a two-place variant that takes a DP argument and a seemingly 1-place predicate. As shown in chapter 2 these predicates are infelicitous in the double subject construction (30-b). These predicates are also infelicitous with *together* and comitatives (30-d). I therefore conclude, as argued in chapter 2, that while *kiss* in (30-a) appears

at first sight to be a 1-place collective predicate, it is in fact a 2-place predicate with a covert reciprocal in object position (Hackl, 2002). I set these predicates aside for now and return to them in chapter 7, where I provide a novel scope-based analysis of these reciprocal alternations.

- (30) a. Karim w Rasha be:so.  
 Karim and Rasha kissed.3PL.  
 Karim and Rasha kissed.
- b. #Karim be:s howwe w Rasha.  
 Karim kissed.3SG.M him and Rasha.
- c. Karim w Rasha be:so baʕed.  
 Karim and Rasha kissed.3PL each-other.  
 Karim and Rasha kissed each other.
- d. Karim w Rasha be:so maʕ baʕed.  
 Karim and Rasha kissed.3PL with each-other.

Given the analyses of *together* as a *with the other* coordination proposed in chapters 4 and 5, we might expect English *together* to pattern the same as the Lebanese Arabic double subject construction with respect to genuinely collective predicates like *argue*. In particular, *argue* in English gives rise to a collective reading with comitatives (31-a) and with *with each other* (31-b). Given that I argued that *together* is decomposed into a comitative *with* and a reciprocal anaphor, the analysis I propose predicts that *together* should give rise to the same collective reading. This prediction is not borne out, as shown by the infelicity of (31-c). We therefore see a clear difference between *together* and the Lebanese Arabic double subject construction with lexically collective predicates like *argue*: while the Lebanese Arabic double subject construction is licensed with these predicates (29-c), English *together* isn't (31-c). This difference is not expected if we treat both constructions as involving the predicate applying to a coordinated DP in spec, *vP*.

- (31) a. Jane argued with Mary.  
 b. Jane and Mary argued with each other.  
 c. #Jane and Mary argued together.

One might argue that the *with* in (31-a) and (31-b) is not in fact a comitative *with* and rather introduces the internal argument of a 2-place predicate *argue* here. The seemingly 1-place variant in *Jane and Mary argued* can then be assumed to have a covert reciprocal in object position, just like *kiss* in (30-a). While this analysis might be argued for if we look only at English, the fact that the Lebanese Arabic counterpart *tshe:raʕ* is licensed with the double subject construction raises doubts about the analysis. In particular, we saw that the double subject construction is infelicitous with predicates that participate in reciprocal alternation, but it is in fact felicitous with predicates like *argue* which allow for a comitative *with*. This suggests that at least in

Lebanese Arabic, the analysis of *argue* as a collective 1-place predicate is the correct one. Of course, it is possible that the Lebanese Arabic *tshē:raʿ* is a lexically collective predicate while the English counterpart *argue* is underlyingly a 2-place predicate, but at this point there is no independent motivation for making this distinction. I therefore have to leave it to future work to investigate the full range of differences between *together* and the double subject construction.

## 6.5 Conclusion

This chapter showed that the same type of economy considerations that are needed to predict the distribution of readings with *together* also explain why covert movement of the lower subject in the DSC is blocked in some environments but licensed in others. More specifically, I presented some parallels between the distribution of readings with *together* and with the DSC that suggest that a unified analysis of these two constructions is desirable. I explored a possible way to extend the generalized economy constraint proposed in chapter 5 to account for these parallels, thus paving the way to understanding the restrictions on which readings are available in each construction within a unified framework. These parallels therefore provide additional motivation for my analysis of *together*, which allows for both the double subject construction and *together* to be uniformly characterized as involving a conjunction where the first conjunct moves to a higher position. At the same time, there are important differences between the two constructions, discussed in section 6.4, which pose a challenge for this unified picture. It therefore remains to be seen whether we can give a unified analysis of these two constructions in terms of economy.





# Chapter 7

## Covert reciprocals: a scope-based analysis of reciprocal alternations

This chapter is based on [Wehbe \(2025\)](#), published in *Natural Language Semantics*. I include the paper here with only very minor edits.

The chapter discusses reciprocal alternations, another case-study where the question of the division of labor between lexical semantics and compositional mechanisms is very relevant. As we saw briefly in chapter 2, the Lebanese Arabic double subject construction provides evidence against the lexical analysis of reciprocal alternations, where *hug* in a sentence like *Jane and Mary hugged* is a 1-place collective predicate. In particular, I showed that the 1-place variants of predicates like *hug* are infelicitous in the DSC, while collective predicates in general are licensed in the same construction. I argued that this provides evidence for a syntactic analysis of reciprocal alternations where the 1-place variants have a covert reciprocal in object position. The goal of this chapter is to therefore propose a scope-based syntactic analysis of these alternations and provide additional evidence for such an analysis.

### 7.1 Introduction

This chapter is concerned with a class of verb alternations, termed reciprocal alternations ([Levin, 1993](#); [Winter, 2018](#)). This alternation is exemplified in (1), where the seemingly 1-place predicate *date* in (1-a) is equivalent to the corresponding 2-place predicate with a reciprocal object (1-b) (and to the corresponding conjunction in (1-c)). One characterizing property of this alternation, which will be useful as we look at different types of predicates, is that the 1-place variant generally requires a plural subject, as shown by the infelicity of (2).<sup>1</sup>

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<sup>1</sup>Of course there are predicates like *talk* which seem to participate both in reciprocal alternations and have a *talk to someone* reading in their 1-place variant. Thus, when they have a singular subject, the *talk to someone* reading is

- (1)    a. Jane and Mary dated.  
       b. Jane and Mary dated each other.  
       c. Jane dated Mary and Mary dated Jane.
- (2)    \*Jane dated.

Given cases like (1), we can formulate two possible preliminary analyses of the logical relationship between the 1-place and 2-place variants of predicates which participate in reciprocal alternations. On the lexical ambiguity view, we can posit that there are two lexical items with the phonological exponent /date/, a 1-place and a 2-place predicate, which are related by a certain logical relation (e.g. Winter, 2018). We can then give a general statement of the logical relationship between the two variants for all predicates which participate in the reciprocal alternation, as shown in (3).

- (3)    Given two predicates  $P_{1\text{-place}}$  and  $P_{2\text{-place}}$  with the same phonological realization, where  $P_{1\text{-place}}$  and  $P_{2\text{-place}}$  participate in a reciprocal alternation, and given any two individuals, a and b:

$$\llbracket P_{1\text{-place}} \rrbracket (a \oplus b) \Leftrightarrow \llbracket P_{2\text{-place}} \rrbracket (a)(b) \wedge \llbracket P_{2\text{-place}} \rrbracket (b)(a)$$

On this preliminary version of the lexical perspective, we would expect that the 1-place variant, exemplified in (1-a), is always truth-conditionally equivalent to the paraphrase with conjunction, exemplified in (1-c), where each conjunct has the two place predicates applying to the individuals (here Jane and Mary) in a different order.

Alternatively, one can posit a syntactic analysis of this alternation, following Hackl (2002), where what looks like a 1-place variant in (1-a) is in fact the same 2-place predicate in (1-b) and (1-c) but with a covert reciprocal in object position. On this preliminary version of this perspective, the LFs for the covert and the overt reciprocal are identical, with the only difference being whether or not the reciprocal object is pronounced. The truth-conditional equivalence between (1-a) and (1-b) is therefore predicted.

The simple versions of the syntactic and the lexical ambiguity approaches laid out above both face a challenge from predicates which participate in the same type of alternation, yet do not exhibit truth-conditional equivalence between the 1-place variant on one hand and the overt reciprocal and conjunctive paraphrases on the other (see Gleitman et al., 1996; Winter, 2018, among many others). For example, the predicate *hug* in (4) seems to show the same pattern as *date* in (1), in that it has a 1-place and 2-place variant where the 1-place variant requires a plural subject. Furthermore, it is tempting to paraphrase (4-a) with the overt reciprocal in (4-c). Yet, (4-a) doesn't in fact have the same truth-conditions as (4-c) and (4-d).

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still available.

- (4) a. Jane and Mary hugged.
- b. \*Jane hugged.
- c. Jane and Mary hugged each other.
- d. Jane hugged Mary and Mary hugged Jane.

To see the relevant truth-conditional difference, consider the context in (5) from Winter (2018): in this context, the variant with no overt reciprocal (5-a) is not true, while the one with the overt reciprocal in (5-b) and the conjunction paraphrase in (5-c) are both true. A simple syntactic analysis where (5-a) involves a covert *each other* therefore incorrectly predicts that (5-a) and (5-b) are truth-conditionally equivalent. Similarly, as pointed out by Winter, a simple lexical ambiguity analysis along the lines of what is laid out in (3) also fails to predict this difference.

- (5) Context: Jane hugged Mary while Mary was sleeping and then Jane fell asleep and Mary woke up and hugged her.
- a. #Jane and Mary hugged.
- b. Jane and Mary hugged each other.
- c. Jane hugged Mary and Mary hugged Jane.

Winter (2018) takes the data in (5) as evidence for a more involved ambiguity view where different types of predicates give rise to different relationships between the 1-place and 2-place variants. In this chapter, I argue for an alternative view on these differences in terms of scope and argue that these differences can therefore be captured within a syntactic account that treats the 1-place variants as covert reciprocals.

More specifically, the main goals of this chapter are to (i) propose that differences between the 1-place variant and the overt reciprocal in contexts like (5) can be reduced to differences in scope-taking possibilities and (ii) argue for an analysis of (5-a) as a covert reciprocal which, unlike the overt counterpart, has to be bound at the lowest possible scope position. Covert reciprocals are therefore predicted to allow for a proper subset of the readings that overt reciprocals, which can be bound from any scope position, allow for. I show in section 7.3 that, given certain assumptions about clausal architecture, a range of cases like (5), where the overt reciprocal is true and the covert reciprocal isn't, correspond to contexts where only the reading where the reciprocal is bound above another scope-taking operator is true. Thus, the low-scope restriction on covert reciprocals explains why only the overt reciprocal is true in such contexts.

At this point, it is worth pointing out that the goals of this chapter are modest, in that it will be beyond my scope to go through *all* differences between covert and overt reciprocals and analyze them in terms of scope. Rather, I will focus on a few case-studies like (5), propose a detailed scope-based analysis for them, and argue that the resulting perspective provides a

more explanatory account for the variety of inferences they give rise to than prominent lexical alternatives. The hope is that this syntactic analysis can then be extended in future work to cases where the relevant scopal configurations might be less well-understood. Moreover, the main focus of the present chapter will be only on reciprocal alternations in English, where the covert reciprocal lacks any overt marking, but I will briefly discuss in section 7.5 how the analysis proposed here might fit into the larger typology of reciprocal constructions cross-linguistically.

Note that Heim et al. (1991) also characterize certain differences in truth-conditions between overt reciprocals and the seemingly 1-place counterparts in terms of differences in scope, in particular cases where the reciprocal is embedded under an attitude predicate like *say* (see Carlson (1998) for a more explicit discussion of this difference and the consequences for an analysis of the 1-place variants). Heim et al. observe that when reciprocals are embedded under a scope-taking predicate, they give rise to a certain type of ambiguity, depending on where the reciprocal takes scope. Consider the example in (6): the long-distance reciprocal reading, paraphrased in (6-a), can be informally characterized as having the reciprocity target the *saying*. For example, (6) is true in a context like (7), where neither of Mary and Jane said that the other person hugged them but they each said that they hugged the other person. On the other hand, in the low-scope reciprocal reading, paraphrased in (6-b), reciprocity targets the embedded verb *hug*.

- (6) Jane and Mary said that they hugged each other.
  - a. **Long-distance reciprocal:** Jane said that she hugged Mary and Mary said that she hugged Jane.
  - b. **Low-scope reciprocal:** Jane and Mary said that Jane hugged Mary and Mary hugged Jane.
- (7) **Context:** Jane said that she hugged Mary but that Mary didn't hug her back. Mary said that she hugged Jane but that Jane didn't hug her back.

Now, consider the 1-place variant in (8). Unlike the overt reciprocal counterpart in (6), (8) is not true in the context in (7). It therefore seems that the 1-place variant can only have what we're referring to as the low-scope reading in (6-b). Heim et al. (1991) argue that this is because *each* in the overt reciprocal can take scope above *say*, allowing us to get the long-distance reciprocal reading. On the other hand, the 1-place variant in (8) only has the reading in (6-b), corresponding to the case where the overt reciprocal is bound below *say*.

- (8) Jane and Mary said that they hugged.

While Heim et al. assume that the variants without a covert reciprocal lack the long-distance

readings due to them simply not being reciprocals but instead 1-place predicates, I will argue for a principle that accounts for the scope differences while maintaining that (8) involves a reciprocal object. I then extend a similar strategy to cases where we see truth-conditional differences but where there is no overt scope-taking element, as in (5). For example, I argue that in (5), the overt reciprocal allows for a similar non-local reading, where the reciprocal is bound above aspect. On the other hand, the 1-place counterpart in (5) only has the reading corresponding to binding below aspect, thus explaining the truth-conditional difference.

The rest of the chapter is structured as follows. In section 7.2, I discuss in more detail Winter (2018) lexical account and provide a preliminary version of the covert reciprocal approach that builds on Heim et al.'s analysis of reciprocals. In section 7.3, I lay out some truth-conditional differences between overt and covert reciprocals and propose a modified syntactic analysis that derives these differences from an assumption that only covert reciprocals have to be bound at the lowest possible position. In section 7.4, I provide evidence for the syntactic approach and against the lexical one by showing that just like overt reciprocals, covert reciprocals give rise to homogeneity and non-maximality. These inferences are systematically derived under an analysis where overt and covert reciprocals have the same LFs (modulo the low-scope requirement) but require additional stipulations under various lexical approaches. Finally, section 7.5 discusses some further issues and section 7.6 concludes.

## 7.2 Syntactic and lexical accounts of reciprocal alternations

### 7.2.1 The basic syntactic account

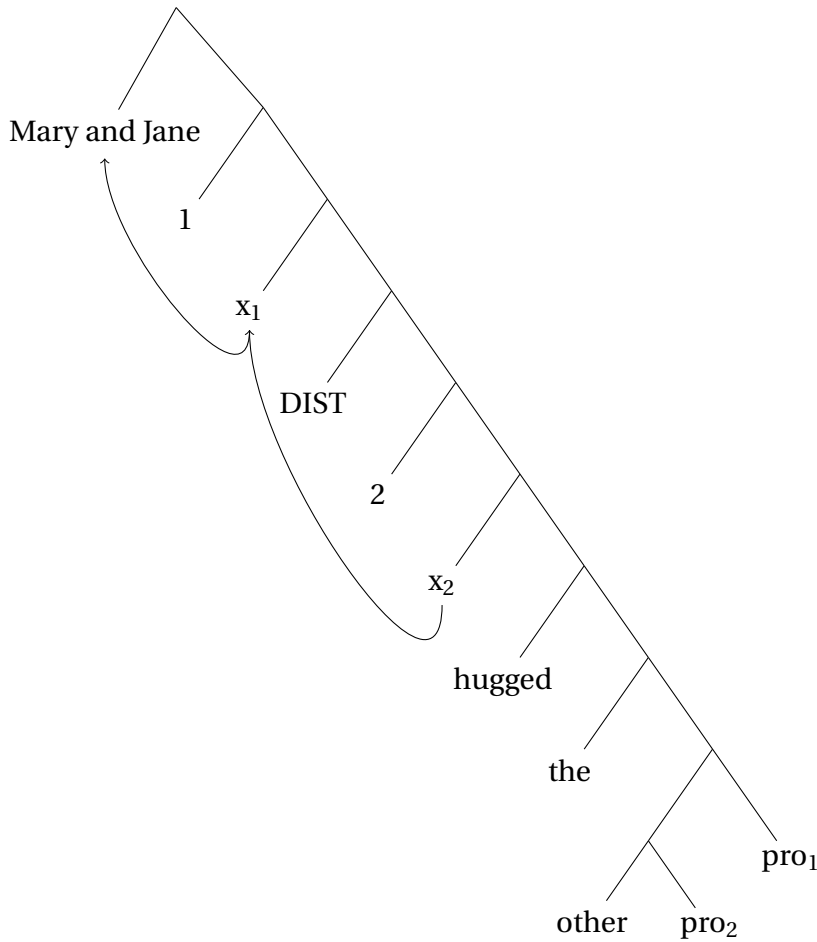
For the purposes of the chapter, I will be assuming a decompositional account for overt reciprocals (Heim et al., 1991; Roberts, 1991).<sup>2</sup> More specifically, I adopt (a slightly simplified version of) the implementation of this general approach from Sauerland (1998). Under this account, reciprocals involve covert operators which are interpreted above the predicate and responsible for distributivity and a *the other* DP in object position. This approach in particular, where *each other* is treated as a definite DP, relies on certain parallels between the interpretations of reciprocals and predication involving a plural definite more generally (Beck, 2001).

- (9) a. Mary and Jane hugged each other.  
b. Mary and Jane (each) hugged the other.

<sup>2</sup>I only look at overt reciprocals with *each other* in this chapter and set aside other reciprocal anaphors like *one another*. As far as I know, the judgements with *each other* generalize to *one another*, but I haven't investigated this systematically.

The LF for (9-a) under this approach is given in (10). The LF in (10) is generated as follows: (i) the subject DP *Mary and Jane* moves (as indicated by the lower arrow) and binds the first argument of *the other* ( $\text{pro}_2$ ); (ii) the distributivity operator is inserted counter-cyclically between the initial landing site of *Mary and Jane* and the binder 2; (iii) the DP *Mary and Jane* moves again (as indicated by the higher arrow) and binds the second argument of *other*. This results in the first argument of *other* being interpreted distributively due to DIST, while the second argument is interpreted collectively.<sup>3</sup> The lexical entry for *other* is given in (12), where  $\llbracket \text{the other } x \ y \rrbracket$  denotes the individual that results from removing  $x$  from  $y$ . The resulting truth-conditions for (10) are given in (13): (10) is predicted to be true if Mary saw Jane and Jane saw Mary.

(10)



(11)  $\llbracket \text{DIST} \rrbracket = \lambda f_{\langle e, t \rangle}. \lambda x. \forall y [y \leq_{AT} x \rightarrow f(y) = 1]$

<sup>3</sup>The LF in (10) uses a distributivity operator and not the pluralization operator \*. For the simple cases discussed in this chapter, this distinction will not matter, but see Sauerland (1998) for arguments that the pluralization operator is needed here. Sauerland assumes, following Schwarzschild (1994), that the pluralization operator \* is in fact obligatorily present as the sister of every node of type  $\langle e, t \rangle$ . The LF in (10) therefore omits another \* operator above the index 1. Note that the contribution of this operator would be trivial here: for the presupposition of  $\llbracket \text{the others} \rrbracket$  to be met,  $\text{pro}_1$  has to be interpreted collectively.

- (12) a.  $\llbracket \text{other} \rrbracket = \lambda x. \lambda y. \lambda z. z \leq y \wedge \neg \exists x' : x' \leq x \wedge x' \leq z$   
 b.  $\llbracket \text{the} \rrbracket (\llbracket \text{other} \rrbracket (x)(y)) = y \oplus x$ , defined iff  $x < y$ ,  
 (where  $y \oplus x$  is the maximal individual  $z$  s.t.  $z \leq y \wedge \neg \exists x' : x' \leq x \wedge x' \leq z$ )  
 c.  $\llbracket \text{the others} \rrbracket (j)(j \oplus m) = m$
- (13)  $\llbracket \text{Mary and Jane hugged each other} \rrbracket = 1$  iff  
 $\lambda y. ((\text{DIST}(\lambda x. \llbracket \text{hugged} \rrbracket (x)(y \oplus x))) (y)) (m \oplus j) = 1$  iff  
 $(\llbracket \text{hugged} \rrbracket (m)(j) \wedge \llbracket \text{hugged} \rrbracket (j)(m))$

Now, the most straightforward version of the syntactic account is to simply assume that (14) has the same LF as the overt reciprocal in (13), with the only difference being that the reciprocal object is not pronounced. This would of course predict, incorrectly as we saw in the introduction, that (14) and the overt counterpart in (9-a) have the same truth-conditions. The goal of section 7.3 will be to propose a version of this syntactic account which can predict the relevant differences in truth-conditions between (14) and (9-a) as due to differences in the positions from which the two arguments of *the other* can be bound.<sup>4</sup>

- (14) Jane and Mary hugged.

## 7.2.2 Winter's lexical account

For the purposes of comparing the syntactic account to a concrete example of a lexical account, I will outline Winter's (2018) account here.<sup>5</sup> Winter (2018) argues that there are two types of predicates that participate in reciprocal alternations: (i) symmetric predicates, which exhibit "plain reciprocity" and (ii) non-symmetric predicates which exhibit "preferential reciprocity". Symmetric predicates are predicates where switching the order of their arguments does not affect the overall truth-conditions. This is illustrated for the symmetric predicate *date* in (15-a). These predicates, Winter argues, also exhibit the plain 'reciprocal' logical relationship between the 1-place and the two-place variants in (15-b). On the other hand, there are predicates like *hug*, which participate in this alternation, yet are not symmetric (16-a). Winter shows that these predicates in general do not exhibit the simple reciprocal logical relationship between the 2-place and the 1-place variant (16-b), as I illustrated for *hug* in the introduction. This reciprocity-symmetry generalization is formulated in (17).

<sup>4</sup>One thing to note at this point is that not all predicates which can take an overt reciprocal object also allow for a variant where the reciprocal object is not pronounced. For the purposes of this chapter, I will simply assume that the licensing of a covert reciprocal object is an idiosyncratic property of certain predicates and set aside the question why only certain predicates license this alternation.

<sup>5</sup>See section 7.4.3 for a discussion of alternative lexical accounts, in particular Dimitriadis (2008).

- (15) a. Jane dated Mary  $\Leftrightarrow$  Mary dated Jane.  
 b. Jane and Mary dated  $\Leftrightarrow$  Jane dated Mary and Mary dated Jane
- (16) a. Jane hugged Mary  $\nrightarrow$  Mary hugged Jane  
 b. Jane and Mary hugged  $\nrightarrow$  Jane hugged Mary and Mary hugged Jane
- (17) **Reciprocity-Symmetry Generalization:** A reciprocal alternation between a unary-collective predicate P and a binary predicate R is plain if and only if R is truth-conditionally symmetric. (Winter, 2018)

Winter takes this generalization as evidence that the source of the reciprocal alternation is different for symmetric and non-symmetric predicates. He proposes that with symmetric predicates like *date*, the 2-place predicate is logically derived from the collective 1-place predicate as illustrated in (18). This predicts that these predicates are necessarily symmetric, since the sum operation  $\oplus$  is commutative and the order of the two arguments in  $\llbracket \text{date}_{2\text{-place}} \rrbracket$  therefore does not affect the truth-conditions.

$$(18) \quad \llbracket \text{date}_{2\text{-place}} \rrbracket = \lambda x. \lambda y. \llbracket \text{date}_{1\text{-place}} \rrbracket (x \oplus y)$$

On the other hand, for non-symmetric predicates, Winter argues that they do not exhibit generalizable logical relationships between the 1-place collective predicate and the 2-place variant. Instead, he argues different types of predicates exhibit a variety of lexical relationships between the two variants. For example, as we saw in the introduction, predicates like *hug* seem to have a simultaneity requirement in their 1-place variant, where if Jane hugged Mary at one time and Mary hugged Jane at a later time, *Jane and Mary hugged* is not true. The inference in (19) therefore seems to hold for *hug*. On the other hand, with predicates like *divorce* and *break up*, the 1-place variant is true as long as the 2-place variant is true in one of the directions, as illustrated in (20-a). The analogous inference clearly doesn't hold for *hug* (20-b).

(19) Jane hugged Mary and Mary hugged Jane at the same time  $\Rightarrow$  Jane and Mary hugged.

- (20) a. Jane broke up with Mary  $\Rightarrow$  Jane and Mary broke up.  
 b. Jane hugged Mary  $\nrightarrow$  Jane and Mary hugged.

This lexical analysis therefore built on the assumption that the relationship between the 1-place and 2-place predicates for non-symmetric predicates is not systematic and therefore has to be at some level encoded as part of the lexical meaning. If this is true, then it would constitute evidence against a syntactic account where there is no reason to expect that the relationship between the 1-place and 2-place variant depends simply on the lexical content of the predicate. In the next section, I argue that these differences between the covert and the overt reciprocals



can in fact be characterized systematically. In particular, I show that the differences can be characterized in terms of differences in the scope of reciprocal binding. I argue that this allows us to maintain a modified syntactic analysis, where covert reciprocals are simply reciprocals which require local binding of the two arguments of *the other*.

For the purposes of evaluating whether the syntactic account can systematically predict the logical relationships between the 1-place and the 2-place variants, it is useful to categorize Winter's case-studies into two separate categories: (i) environments where the truth-conditions of the overt reciprocal differ from those of the 1-place variant and (ii) environments where the truth-conditions for the overt reciprocal and the 1-place variant are identical but differ from the paraphrase with conjunction. For example, the contexts with *hug* in (5) and with *kiss* in (21) are instances of the first type, where the 1-place variant is false (21-a) but both the overt reciprocal (21-b) and the conjunction (21-c) are both true. On the other hand, the cases where *break up* is true even though only one person broke up with the other are of the second type, where both the 1-place variant (22-a) and the covert reciprocal (22-b) pattern together, in contrast with the conjunctive paraphrase (22-c).

(21) **Context:** Mary kissed Jane on the arm and Jane kissed Mary on the arm.

- a. #Jane and Mary kissed.
- b. Jane and Mary kissed each other.
- c. Jane kissed Mary and Mary kissed Jane.

(22) **Context:** Jane and Mary broke up and Jane initiated the break up.

- a. Jane and Mary broke up.
- b. Jane and Mary broke up with each other.
- c. # Jane broke up with Mary and Mary broke up with Jane.

While cases of the second type will come up throughout the chapter, the main focus of this chapter will be differences of the first type. This is because under the syntactic analysis which is outlined in section 7.2.1, the 1-place variants are taken to be reciprocals and not conjunctions. Therefore, the major challenge for this type of account are cases like (21), where what is claimed to be a covert reciprocal differs in truth-conditions from the overt counterpart. Cases like (22) on the other hand, while interesting in their own right, provide a puzzle that a general theory of overt reciprocals has to address and not a challenge for the covert reciprocal analysis per se.

### 7.3 A scope-based proposal for covert reciprocals

In this section, I will propose a modified version of the covert reciprocal analysis outlined in section 7.2.1. In particular, I propose that unlike overt reciprocals, where binding of the arguments of *the other* is free to occur from any position, covert reciprocals require this binding to occur as low as possible (23). I argue, using several case-studies, that the differences in meaning between overt and covert reciprocals, which at first sight seem to be lexical differences, can in fact be characterized systematically in terms of scope.

- (23) **Restriction on Covert Reciprocity:** The *each other* in a reciprocal can be elided only if the two arguments of *other* are bound at the lowest possible positions which avoid a contradiction.

As a starting point, in order to appreciate the impact of (23), let's observe that this principle straightforwardly captures the differences between overt and covert reciprocals for the cases that Heim et al. discuss, namely cases where the reciprocal is embedded under an attitude predicate. This difference is illustrated with the context in (24), where only the overt reciprocal is true.

- (24) **Context:** Jane said that she hugged Mary but that Mary didn't hug her back. Mary said that she hugged Jane but that Jane didn't hug her back.
- a. Jane and Mary said that they hugged each other.
  - b. # Jane and Mary said that they hugged.

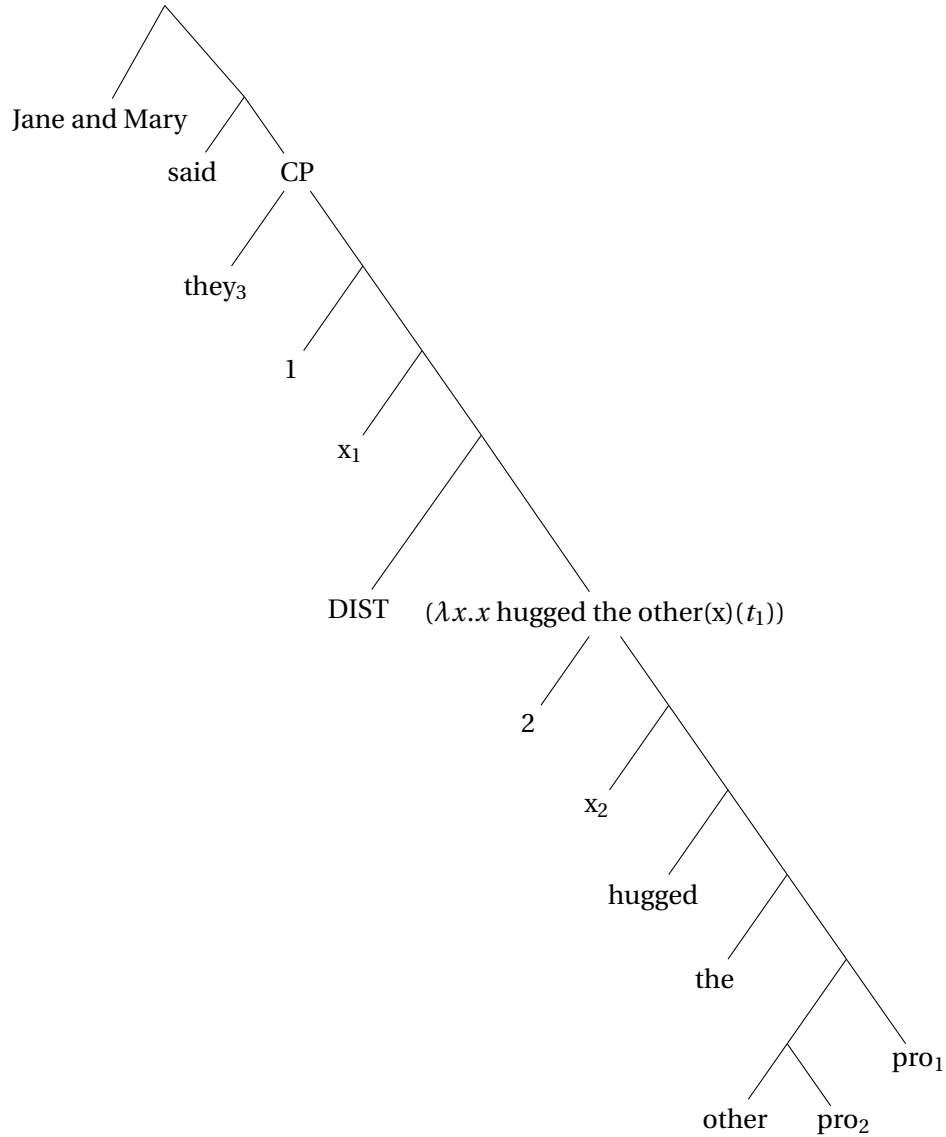
Heim et al. (1991) argue that overt reciprocals like (25) are ambiguous between a long-distance reciprocal reading in (25-a) and a low-scope reading in (25-b). The data in (24) can therefore be taken as evidence that covert reciprocals only have the low-scope reading in (25-b) (Carlson, 1998 a.o.). In what follows, I show that within the Sauerland (1998) implementation of the decomposition analysis, the LF where the arguments of *the other* are bound below *say* corresponds to the low-scope reading in (25-b), while the long-distance reading in (25-a) can be derived by binding the arguments of *the other* above *say*. The restriction in (23) then correctly predicts that the covert reciprocal only have the reading in (25-b) and therefore is not true in the context in (24).

- (25) Jane and Mary said that they hugged each other.
- a. **Long-distance reciprocal:** Jane said that she hugged Mary and Mary said that she hugged Jane.
  - b. **Low-scope reciprocal:** Jane and Mary said that Jane hugged Mary and Mary hugged Jane.

In the low-scope reciprocal reading, the relevant predicate that needs to be reciprocated is *hug*, while for the long-distance reciprocal reading, reciprocity needs to target the *saying* predicate. In case of the low-scope reciprocal the relevant distributivity operator needs to apply to the predicate in (26-a) to ensure that the proposition embedded under *say* corresponds to *each of Jane and Mary hugged the other*. On the other hand, in the long-distance reciprocal, the relevant distributivity operator has to apply to the predicate in (26-b), which has *say* in its scope.

- (26)    a. Low-scope predicate:  $\lambda x.x \text{ hugged the other}(x)(x_1)$   
           b. Long-distance predicate:  $\lambda x.x \text{ said that } x \text{ hugged the other } (x)(x_1)$

I will begin by showing that when both arguments of *the other* are bound below *say*, the operator that distributes to the atoms of *Jane and Mary* necessarily applies to the predicate in (26-a). To see this let's consider the LF in (27-a). Here, since both arguments of *the other* are bound below *say*, DIST applies to the predicate in (26-a) and the proposition that  $\llbracket \text{say} \rrbracket$  applies to is  $\lambda w.$  *they*<sub>3</sub> hugged each other in *w*, as required to derive the low-scope reciprocal reading in (25-b). For concreteness, I assume here that *say* is a distributive predicate and therefore gives rise to distributive entailments. Furthermore, I assume that the subject of the embedded clause (*they*) co-refers with *Jane and Mary*. The truth-conditions in (27-c), which correspond to the low-scope reading in (25-b) are derived.



- (27) a.
- b.  $g(3) = j \oplus m$
- c.  $\llbracket (27-a) \rrbracket^{g,w} = 1$  iff  $\llbracket \text{said} \rrbracket^{g,w}(\lambda w'. \llbracket \text{CP} \rrbracket^{w'})(j \oplus m) = 1$  iff
- $\llbracket \text{said} \rrbracket^{g,w}(\lambda w'. \llbracket \text{CP} \rrbracket^{w'})(j) = 1 \wedge \llbracket \text{said} \rrbracket^{g,w}(\lambda w'. \llbracket \text{CP} \rrbracket^{w'})(m) = 1$  iff
- $\llbracket \text{said} \rrbracket^w(\lambda w'. \llbracket \text{hugged} \rrbracket^{w'}(j)(m) = 1 \wedge \llbracket \text{hugged} \rrbracket^{w'}(m)(j) = 1)(j) = 1 \wedge$
- $\llbracket \text{said} \rrbracket^w(\lambda w'. \llbracket \text{hugged} \rrbracket^{w'}(j)(m) = 1 \wedge \llbracket \text{hugged} \rrbracket^{w'}(m)(j) = 1)(m) = 1$  iff
- (informal paraphrase) Mary said that Mary hugged Jane and that Jane hugged Mary and Jane said that Mary hugged Jane and that Jane hugged Mary

The principle in (23) predicts that the LF in (27-a) is the only available LF for the covert reciprocal. This is because  $\text{pro}_2$  is bound at the lowest possible position it could be bound (above the lowest node of type t). The second argument can't be bound from the same position, otherwise a contradiction would arise (since  $\llbracket [\text{the other } [x] [y]] \rrbracket$  presupposes that  $x < y$ ). In order

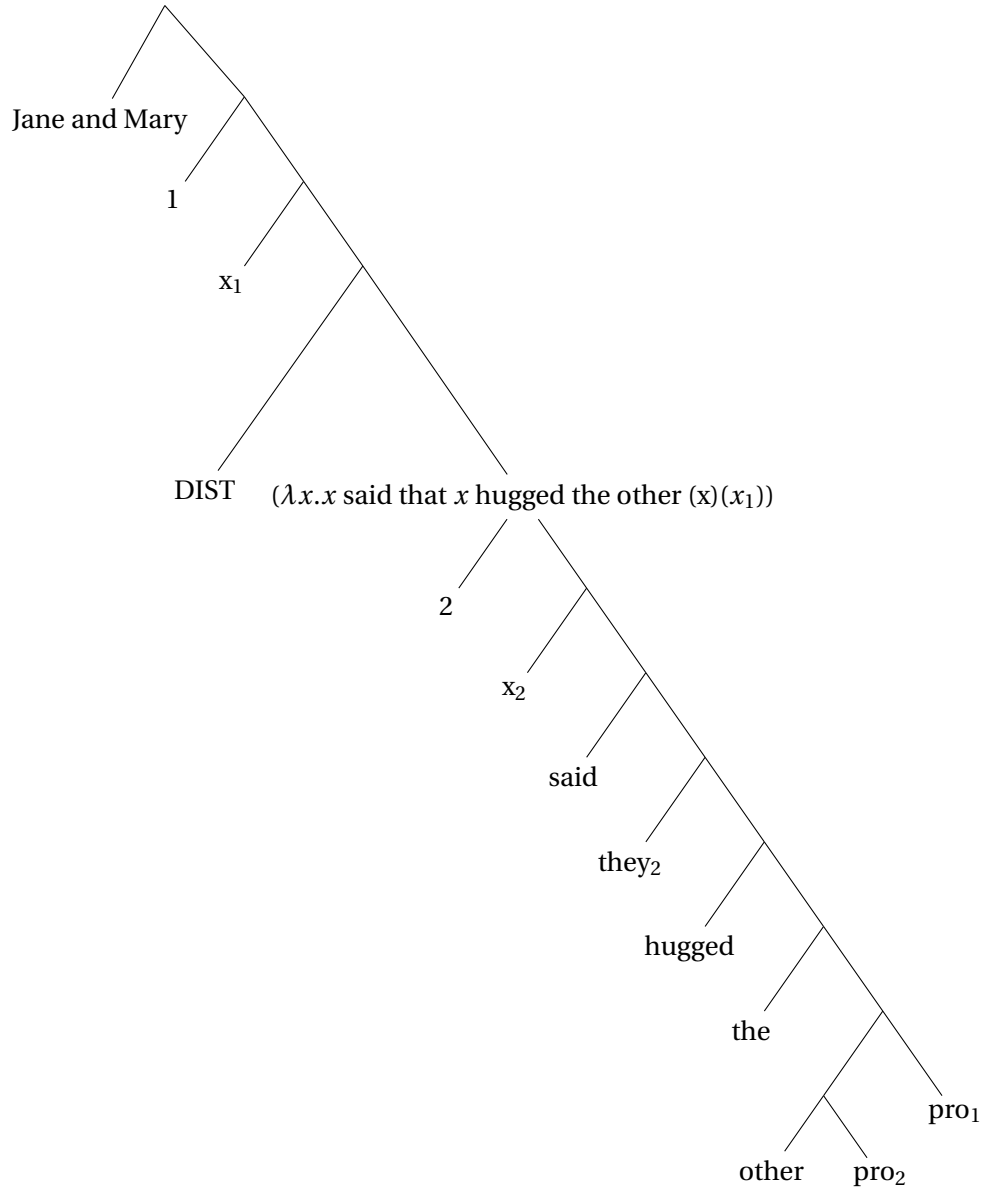
to avoid a contradiction, a distributivity operator has to be inserted between the positions where the two arguments are bound. Therefore,  $\text{pro}_1$  in (27-a) is bound at the lowest possible position it could be bound while avoiding a contradiction, satisfying the principle in (23).

The long-distance reading, on the other hand, can be derived if we allow the arguments of *the other* to be bound above *say*. This is illustrated in the LF in (28-a). Notice here that DIST applies to the predicate in (26-b) and therefore we get that (28-a) is true iff Mary said that she hugged Jane and Jane hugged that she hugged Mary (28-b). The long-distance reading can therefore be derived when we allow the arguments of *the other* to be bound above *say*, but not when they are bound below *say*.<sup>6</sup> Since in covert reciprocals, according to the restriction in (23), require the arguments of *the other* to be bound below *say*, (23) correctly accounts for the fact that covert reciprocals only have the low-scope reading in (25-b).<sup>7</sup>

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<sup>6</sup>Note that the long-distance reading can be derived, under certain assumptions, if only the first argument of *the other* is bound below *say*, assuming that the lower distributivity operator is trivial and that there can be a higher distributivity operator that applies to the predicate in (26-b). Having this as an LF for covert reciprocals would still violate the condition in (23) and would give rise to the same truth-conditions as (28-a). For simplicity, I will therefore only consider LFs where both arguments of *the other* are bound together, relative to other scope-taking elements.

<sup>7</sup>An anonymous reviewer raised the question of whether we might expect a principle like *Be Brief* to ensure that the covert reciprocal will be preferred when expressing the low-scope LF. I assume that the notion of complexity that is relevant for evaluating competition between different utterances is the complexity of the corresponding LFs. Since the LFs for the covert and overt reciprocals are equally complex here, we therefore do not expect that there would be a preference for one over the other.



- (28) a.
- b.  $\llbracket (28\text{-a}) \rrbracket^w = 1$  iff
- $[\lambda y. ((\text{DIST}(\lambda x. \llbracket \text{said} \rrbracket^w(\lambda w'. \llbracket \text{hugged} \rrbracket^{w'}(\llbracket \text{the other} \rrbracket(x)(y))(x)(x))(y)))(j \oplus m) = 1$  iff
- $\llbracket \text{said} \rrbracket^w(\lambda w'. \llbracket \text{hugged} \rrbracket^{w'}(j)(m))(m) = 1 \wedge \llbracket \text{said} \rrbracket^w(\lambda w'. \llbracket \text{hugged} \rrbracket^{w'}(m)(j))(j) = 1$
- iff
- (informally) Mary said that Mary hugged Jane and Jane said that Jane hugged Mary.

We therefore see that the principle in (23) accounts for the differences in truth-conditions between overt and covert reciprocals under embedding predicates like *say*. Of course, there is a question here about the mechanism through which the restriction in (23) arises, which I will set aside for the purposes of the paper. I will simply assume that some predicates, like *hug*, license ellipsis of their reciprocal internal argument under the condition that the reciprocal is bound at

the appropriate position. This raises the issue of how ellipsis of an argument can be conditioned on a property of the larger LF in which it occurs (i.e. where the binding occurs), a non-trivial issue which I leave for future work.

In the rest of this section, I will extend the strategy above to other cases where there is no overt scope-taking element like *say*. I argue that nevertheless, these cases do involve a (covert) scope-taking element and just like with *say* above, give rise to a scope ambiguity in the overt reciprocal. On the other hand, in the covert reciprocal, only the low-scope reading is licensed, explaining the apparent differences in truth-conditions.

To see the general logic of how these differences will be derived, consider the schema in (29). Suppose that we have a two-place predicate  $P$  with a reciprocal object and a subject that denotes the plural individual  $a \oplus b$ . Furthermore, suppose that we have a quantifier  $Q$  that quantifies over some argument of  $P$  and that the reciprocal binding can take place either above or below  $Q$ . When reciprocal binding takes place below  $Q$ , the quantifier outscopes the relevant distributivity operator and the quantifier therefore applies to the conjunction of  $P(a)(b)(\alpha)$  and  $P(b)(a)(\alpha)$ , as shown in (29-a). On the other hand, when reciprocal binding takes place above  $Q$ , the relevant distributivity operator outscopes this quantifier and therefore the quantifier applies to each of the conjuncts  $P(a)(b)(\alpha)$  and  $P(b)(a)(\alpha)$  separately, as shown in (29-b). As a result of the restriction in (23), the covert reciprocal will only have the reading corresponding to (29-a) and therefore, given a quantifier which is not scopally commutative with conjunction, the difference in truth-conditions between (29-a) and (29-b) will result in a difference in meaning between the covert reciprocal and the overt counterpart.

(29) **General Schema:**

- a. Low-scope distributivity:  $Q(\lambda\alpha. P(a)(b)(\alpha) \wedge P(b)(a)(\alpha))$
- b. High-scope distributivity:  $Q(\lambda\alpha. P(a)(b)(\alpha)) \wedge Q(\lambda\alpha. P(b)(a)(\alpha))$

I will focus in the rest of this section on two such quantifiers that I argue are responsible for certain differences in truth-conditions between overt and covert reciprocals. The first is an existential quantifier over times, which I propose is contributed by perfective aspect. I show that scopal differences with respect to this quantifier derive the simultaneity requirement of covert reciprocals that was discussed above. In particular, the relevant schema as applied to this quantifier is given in (30). The covert reciprocal will only have the low-scope reading corresponding to (30-a), thus requiring that the two directions of the reciprocal to hold at the same time  $t$ . On the other hand, the overt reciprocal will also have a high-scope reading corresponding to (30-b), thus allowing for the two directions of the reciprocal to occur at different times.

(30) **Schema for simultaneity:**

- a. Low-scope distributivity :  $\exists t : P(a)(b)(t) \wedge P(b)(a)(t)$
- b. High-scope distributivity:  $\exists t : P(a)(b)(t) \wedge \exists t' : P(b)(a)(t)$

The second quantifier I propose is an analogous existential quantifier that quantifies over locations rather than times. I argue that this quantifier is responsible for a spatial overlap requirement that we see in the covert reciprocal with certain predicates. For example, consider the example in (31), discussed in Carlson (1998) and attributed to Leila Gleitmann. Here, in a context where Jane kissed Mary on one location (*Mary's arm*) and Mary kissed Jane on a different location (*Jane's arm*), the covert reciprocal is not true (31-a), while the overt counterpart is (31-b). I take this difference to be captured by the relative scope of reciprocal binding with respect to spatial quantifier I propose, as illustrated in (32). In particular, analogously to the temporal counterpart in (30), the covert reciprocal will only have the low-scope reading in (32-a), thus requiring that the two directions of the reciprocals to occur at the same location.

(31) **Context:** Mary kissed Jane on the arm and Jane kissed Mary on the arm (at the same time).

- a. #Jane and Mary kissed.
- b. Jane and Mary kissed each other.

(32) **Schema for spatial overlap:**

- a. Low-scope distributivity :  $\exists l : P(a)(b)(l) \wedge P(b)(a)(l)$
- b. High-scope distributivity:  $\exists t : P(a)(b)(l) \wedge \exists t' : P(b)(a)(l)$

We can also go beyond existential quantifiers and consider cases where it is less straightforward to see how scope differences with respect to distributivity give rise to the differences in truth-conditions. For example, within the temporal domain, we see also see truth-conditional differences with the imperfective, as shown in (33). If the imperfective was simply a universal quantifier over times, we would not expect to see these differences, since a universal quantifier would be scopally commutative with the distributivity operator, which is another universal quantifier. In particular, we would expect both (33-a) and (33-b) to be false in (33), since it is not true that at all times during the day, both Jane was hugging Mary and Mary was hugging Jane. The fact that the overt reciprocal is true in (33) raises questions about what kind of quantifier the imperfective denotes. Giving an analysis of the contrast in (33) is beyond the scope of this chapter, but the hope is that once we have an understanding of how the reading in (33-b) arises, the differences between (33-a) and (33-b) can then be similarly accounted for in terms of scope.

(33) **Context:** All day today, Jane and Mary were alternating going to sleep and when one of



them fell asleep the other would hug them.

- a. # Jane and Mary were hugging all day.
- b. Jane and Mary were hugging each other all day.

The purpose of this section will simply be to investigate in detail the two relatively simple case-studies outlined above and show that what looks like lexical differences between covert and overt reciprocals in those cases can be reanalyzed in terms of scope. I leave the exploration of further cases with different quantifiers for future work.

### 7.3.1 Covert reciprocals and simultaneity

The first case-study I consider is the simultaneity requirement that we observe with covert reciprocals with simple predicates like *hug*. Consider again the example in (34): while the overt reciprocal in (34-b) is true when Jane hugged Mary at one time and Mary hugged Jane at a different time, the covert counterpart in (34-a) isn't.

- (34) Context: Jane hugged Mary while Mary was sleeping and then Jane fell asleep and Mary woke up and hugged Jane.
- a. #Jane and Mary hugged.
  - b. Jane and Mary hugged each other.

I take (34) to show that covert reciprocals have a simultaneity requirement, while overt reciprocals do not. In particular, given a predicate *R* and two individuals *A* and *B* with a covert reciprocal of the form *A and B R'ed*, the event where *A Rs B* and where *B Rs A* have to occur at the same time. On the other hand, in the overt reciprocal variant, the two events could occur either at the same time or at different times. The same kind of truth-conditional difference can be seen with other predicates beyond *hug*, for example with *kiss* (35) and *talked to/spoke to* (36). This difference therefore demands a principled explanation.

- (35) **Context:** Jane kissed Mary on the cheek while she was sleeping and then she fell asleep and Mary woke up and kissed her on the cheek.
- a. # Jane and Mary kissed.
  - b. Jane and Mary kissed each other.
- (36) **Context:** Mary was in a coma 2 weeks ago. John went to visit her and he was telling her some stories. Last week, Mary woke up from her coma, but John got into a coma. Mary went to visit him in turn and was telling him some stories.
- a. John and Mary talked to/ spoke to each other.

- b. #John and Mary talked/spoke.

I will argue that just like there were multiple scope positions from which the overt reciprocal could be bound when embedded under *say*, there are also multiple scope positions where the overt reciprocal can be bound here in the unembedded cases. In particular, all of the examples above have perfective aspect, which introduces an existential quantifier over times, and binding of the arguments of *the other* in the overt reciprocal can occur either above or below aspect.

I make the following assumptions about tense and aspect: (i) predicates like *hug* take a time interval as their first argument, in addition to the standard individual arguments; (ii) tense is referential, such that *Past* for example takes a contextually supplied time and presupposes that this time is before the utterance time (37-b) (Partee, 1973); (iii) I assume a simplified analysis of perfective aspect, where it existentially quantifies over points in time interval that tense picks out.<sup>8</sup>

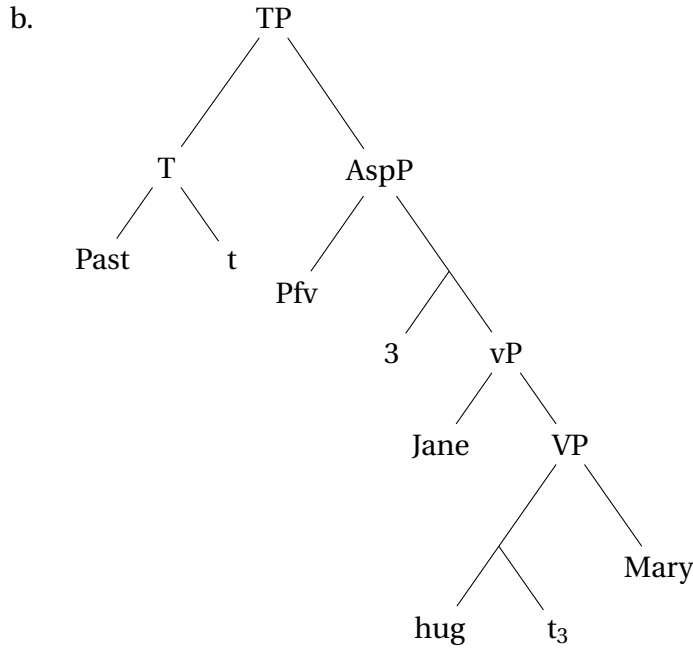
- (37) a.  $\llbracket \text{hug} \rrbracket = \lambda t. \lambda x. \lambda y. y \text{ hugged } x \text{ at } t$   
 b.  $\llbracket \text{Past} \rrbracket^c = \lambda t : t < t_c. t$   
 c.  $\llbracket \text{Pfv} \rrbracket = \lambda P. \lambda t. \exists t' \in t : P(t') = 1$

To see how this works with a concrete example, consider (38). Predicate abstraction takes place below aspect, resulting in the necessary predicate of times for *pfv* apply to. The *t* that *Past* takes as an argument is a contextually specified time interval.  $\llbracket \text{TP} \rrbracket$  is therefore predicted to be true iff John hugged Mary at some time within the contextually specified time interval.

- (38) a. Jane hugged Mary.

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<sup>8</sup>This simplified denotation for the perfective captures the main insights of the standard analysis where perfective places the event time as a subpart of the reference time (Bhatt & Pancheva, 2005).

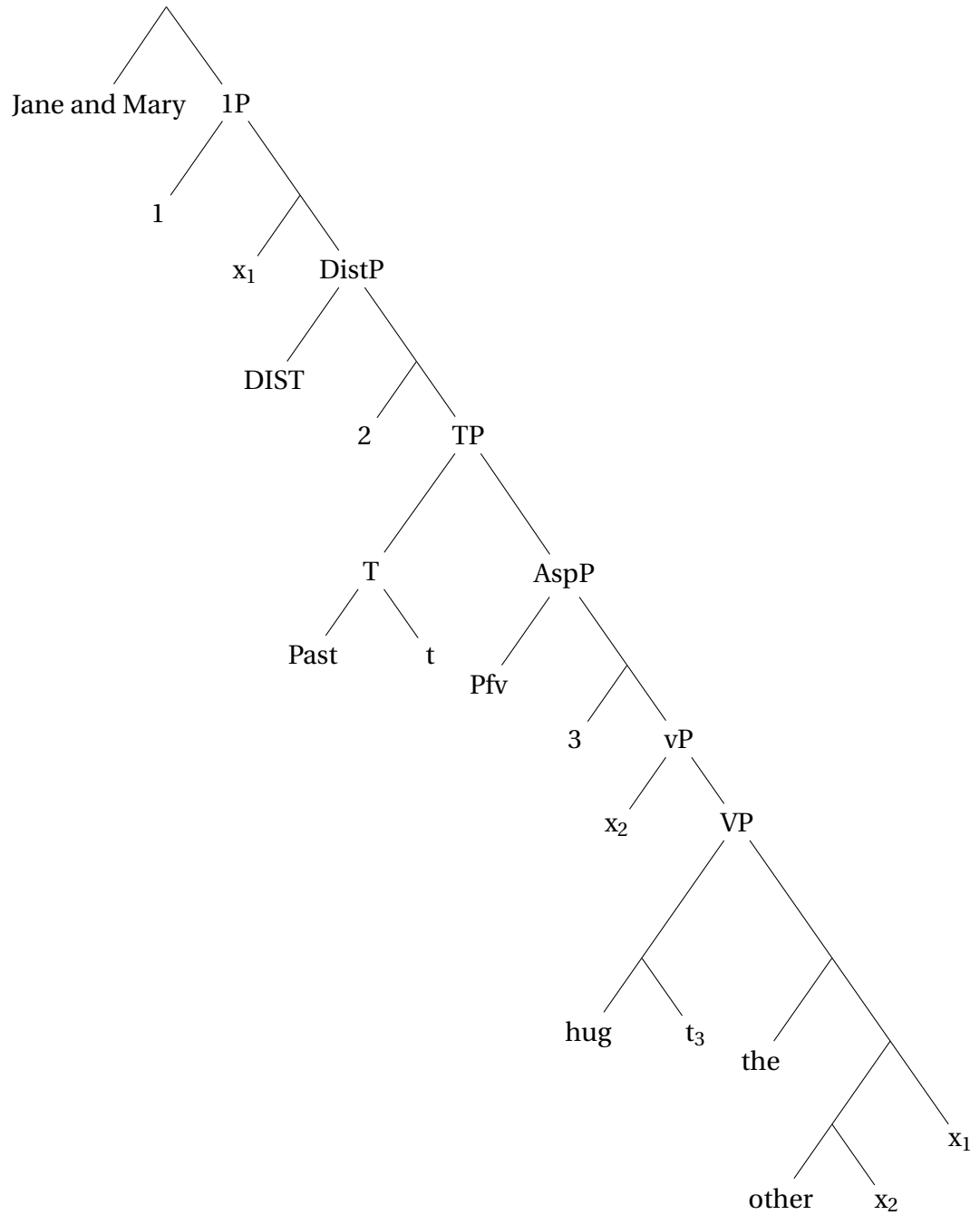


c.  $\llbracket \text{TP} \rrbracket^c = 1$  iff  $\exists t' \in t : \llbracket \text{hug} \rrbracket(t')(m)(j) = 1$ , defined iff  $t < t_c$

Now, we can derive the possible LFs for the overt reciprocal in (39). Namely, just like with *say* above, I assume that the arguments of *the other* are free to be bound from any scope position. We therefore expect that there will be LFs corresponding to (39) where the reciprocal is bound below aspect and ones where the reciprocal is bound above aspect.

(39) Jane and Mary hugged each other.

I will begin with the LF with binding above aspect. Since tense does not involve any quantification, the relative scope of distributivity with respect to tense does not affect the truth-conditions. I therefore only consider the LF in (40-a), where *Jane and Mary* moves twice above tense to bind the arguments of *the other*. Under this LF the relevant distributivity to the individuals *Jane* and *Mary* takes place above the existential quantifier contributed by aspect, and therefore the truth-conditions in (40-c) are predicted.



- (40) a.   
 b.  $\llbracket \text{TP} \rrbracket^c = 1$  iff  $\exists t' \in t : \llbracket \text{hug} \rrbracket(t')(x_1 \ominus x_2)(x_2) = 1$ , defined iff  $t < t_c$    
 c.  $\llbracket (40\text{-a}) \rrbracket^c = 1$  iff  $\exists t' \in t : \llbracket \text{hug} \rrbracket(t')(j)(m) = 1 \wedge \exists t'' \in t : \llbracket \text{hug} \rrbracket(t'')(m)(j) = 1$ , where  $t < t_c$

We predict that (40-c) is true as long as Jane hugged Mary at some time and Mary hugged Jane at some time, within the contextually salient past time. The informal paraphrase in (41) makes clear how distributivity taking scope above aspect gives rise to this non-simultaneous reading.

- (41) **Non-simultaneous reading:** For each of Mary and Jane there is some past time such that

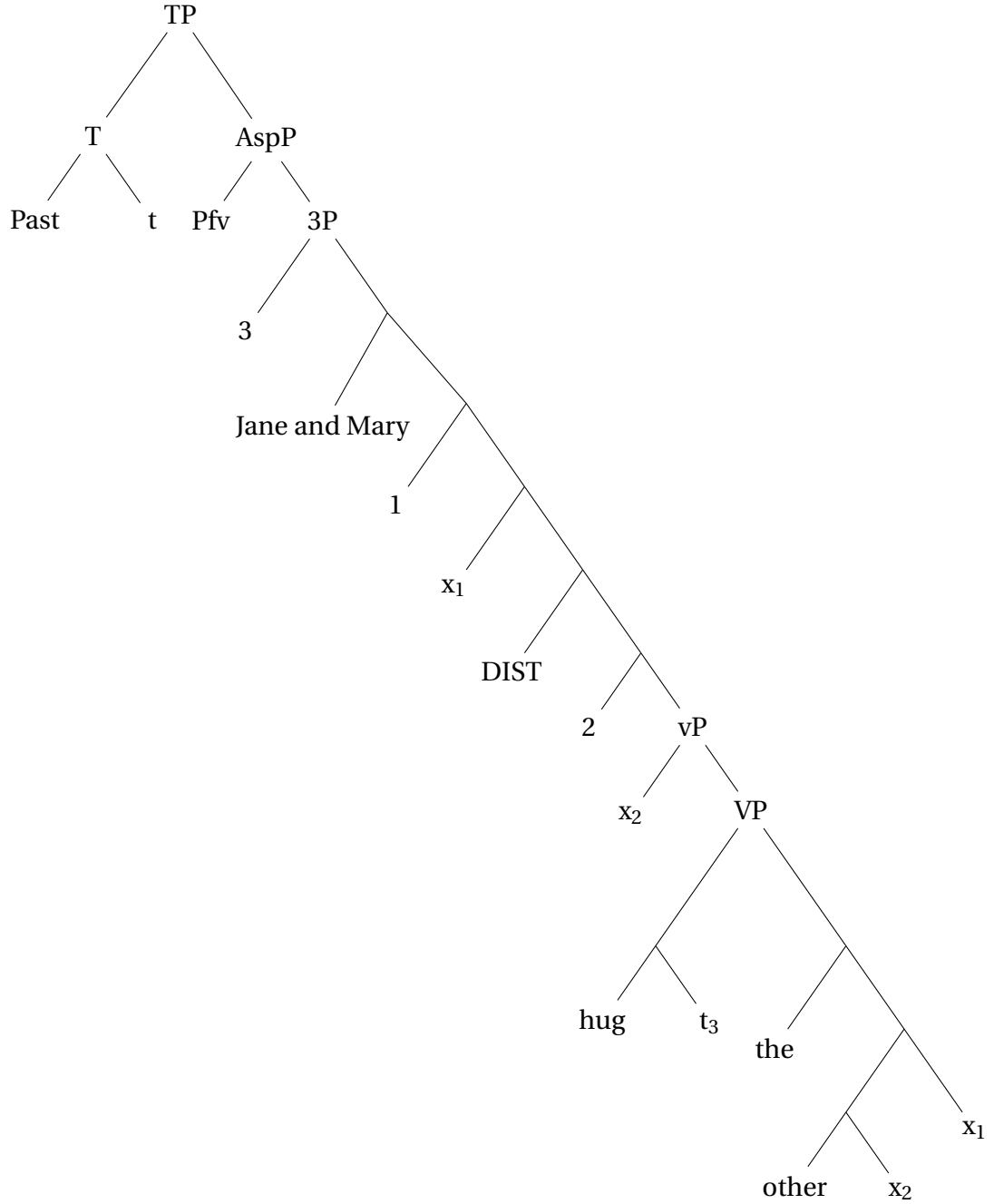
they hugged the other at that time.

The LF where binding occurs below tense and aspect is given in (42-a).<sup>9</sup> Here, distributivity to *Mary* and *John* takes scope below the existential quantifier over times and therefore the resulting truth-conditions require that there is a time when both Jane was hugging Mary and Mary was hugging Jane. The predicted truth-conditions are given in (42-c): informally, (42-c) is true if there is some time within the past time interval such that Mary was hugging Jane throughout that time and Jane was hugging Mary throughout that time. We therefore have a distinct LF that only picks out the simultaneous reading of the reciprocal.

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<sup>9</sup>Again, I only consider the LF here where both arguments of *the other* are bound below aspect. In fact, in this case if the second step of movement of *John and Mary* was to a position above aspect, we would still get the simultaneous reading.

(42) a.



b.  $\llbracket 3P \rrbracket = \lambda t. \llbracket \text{hug} \rrbracket(t)(m)(j) = 1 \wedge \llbracket \text{hug} \rrbracket(t)(j)(m) = 1$

c.  $\llbracket TP \rrbracket^c = 1$  iff  $\exists t' \in t : \llbracket \text{hug} \rrbracket(t')(m)(j) = 1 \wedge \llbracket \text{hug} \rrbracket(t')(j)(m) = 1$ , defined iff  $t < t_c$

(43) **Simultaneous reading:** There is some past time such that each of Mary and Jane hugged the other at that time.

Given these assumptions above about the possible LFs for reciprocals, we can now see that the low-scope restriction on covert reciprocity straightforwardly predicts the truth-conditional differences between covert and overt reciprocals with respect to simultaneity. In particular, the

overt reciprocals are ambiguous between the LF in (40-a) and (42-a) and therefore allows for a non-simultaneous reading. On the other hand, covert reciprocals only have the LF in (42-a), where the binding of the reciprocal arguments occurs as low as possible. Therefore, covert reciprocals are correctly predicted to only have the simultaneous reading.

There is independent evidence that overt reciprocals are in fact ambiguous between a simultaneous and possibly non-simultaneous LF, as I proposed in this section. Since the simultaneous reading is a special case of the non-simultaneous one, we have to embed the reciprocal in a downward-entailing environment in order to diagnose this ambiguity. Consider (44), where the reciprocal is in the restrictor of a universal quantifier.<sup>10</sup> Given the context in (44), the simultaneous reading, paraphrased in (44-a-i) is true. On the other hand, the possibly non-simultaneous reading, paraphrased in (44-a-ii) is not true, since Jane and Mary each hugged the other at some time, yet they didn't get a prize. Since the overt reciprocal in (44-a) is judged to be true in this context (under one reading), there has to be an LF corresponding to (44-a) that is only true when the hugging events were simultaneous.

- (44) Context: There's a reality TV show observing couples after a fight. Couples 1-3 hugged and made up after the fight. Couple 4 was Jane and Mary. Jane hugged Mary after the fight but Mary didn't want to hug her back. Later when Jane was taking a nap, Mary went and hugged her. Couples 1-3 got a prize, but Jane and Mary didn't.
- a. Every couple who hugged each other the day of the fight got a prize.
- (i) **Simultaneous reading:** Every couple who hugged the day of the fight got a prize. (True)
  - (ii) **Non-simultaneous reading:** Every couple whose members each hugged the other at some time got a prize. (Not True)

We therefore have evidence that overt reciprocals are ambiguous, as outlined above in this section. The truth-conditional differences with respect to simultaneity between covert and overt reciprocals then follow straightforwardly, given our low-scope assumption for covert reciprocals. I have therefore shown in this section that what looks like a lexical difference between covert and overt reciprocals can be systematically accounted for as a difference in scope-taking possibilities, given independently motivated assumptions about clausal architecture.

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<sup>10</sup>The reason we can't use negation as the downward-entailing environment here is that the simultaneous and non-simultaneous reading collapse under negation due to homogeneity. I therefore use the restrictor of *every*, an environment where homogeneity does not have to project, which allows us to detect the ambiguity.

### 7.3.2 Spatial overlap requirement

Looking beyond the temporal domain, we can observe other differences between covert and overt reciprocals. One such case, presented in Carlson (1998) and attributed to Leila Gleitman, is given in (45). Here, the covert reciprocal in (45-a) is not true, while the overt counterpart is. Note that this can't be reduced to the simultaneity requirement for covert reciprocals discussed in section 7.3.1. In particular, (45-a) remains not true in this context, even if the events of Mary kissing Jane and of Jane kissing Mary occur at the same time. In what follows, I will suggest that this difference is also due to a difference in scope but with respect to a quantifier over locations rather than over times.

- (45) **Context:** Mary kissed Jane on the arm and Jane kissed Mary on the arm (at the same time).
- a. #Jane and Mary kissed.
  - b. Jane and Mary kissed each other.

First, note that just like with the simultaneity requirement in section 7.3.1, there is evidence independent of (45) that the overt reciprocal is ambiguous between a reading which is true in (45) and one which, like the covert counterpart, is only true if Jane and Mary kissed each other on the mouth. To see this consider the context in (46). In this context, only the *kiss on the lips* reading is true, since Jane and Mary did in fact each kiss the other yet they didn't get a prize. Nevertheless, there seems to be a reading of (46-a) which is true in this context. This shows that the reading in (46-a-i) is in fact a possible reading for the overt reciprocal. As with the case of simultaneity above, we therefore need an account of the ambiguity in the overt reciprocal.

- (46) **Context:** There's a reality TV show observing couples after a fight. Couples 1-3 kissed after the fight. Couple 4 was Jane and Mary. After the fight, Jane kissed Mary on the arm and Mary kissed Jane on the arm at the same time. Couples 1-3 got a prize, but Jane and Mary didn't.
- a. Every couple who kissed each other the day of the fight got a prize.
    - (i) **Kiss on lips reading:** Every couple who kissed the day of the fight got a prize. (True)
    - (ii) **Kiss somewhere reading:** Every couple where each of them kissed the other got a prize. (Not True)

In section 7.3.1, I argued that there is an existential quantifier over time and that the scope of distributivity with respect to that quantifier determines whether we get a simultaneous or a non-simultaneous reading. I believe that the ambiguity evidenced by in (46), and by extension



the truth-conditional difference exemplified in (45), can be accounted for in an analogous way, by extending this strategy to the spatial domain.

In particular, we can informally characterize the truth-conditional difference in (45) as follows: while the overt reciprocal allows the two kissing events to occur at different locations, the covert counterpart requires the two events to occur at the same location.<sup>11</sup> It is reasonable to characterize the context in (45) as one where the two kissing events occur at different locations (one on Mary's arm and one on Jane's arm) and since the covert reciprocal requires spatial overlap, only the overt reciprocal is felicitous in this context.

To extend the strategy with aspect above, I propose that there is a similar existential quantifier that quantifies over locations rather than time intervals. I assume similarly that *kiss* takes a location argument in addition to the time interval argument and two individual arguments (47-a). Additionally, there is an existential quantifier over location (47-b). Now, in the covert reciprocal, distributivity will have to take scope below this existential quantifier over times and therefore we predict that the two events of Jane kissing Mary and Mary kissing Jane have to occur at the same location. This predicts that the covert reciprocal is not true in (45-a), while the overt counterpart is.

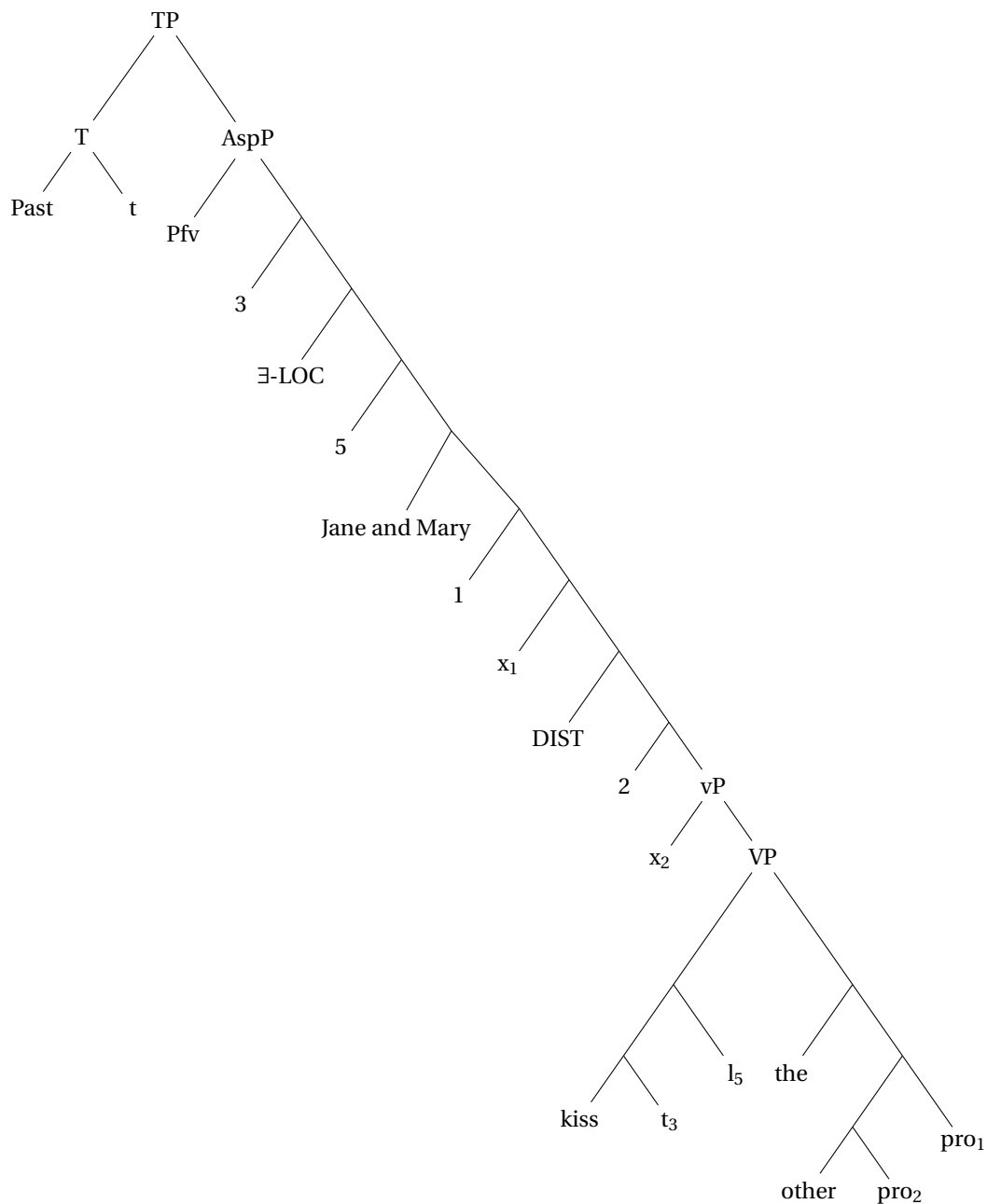
- (47) a.  $\llbracket \text{kissed} \rrbracket = \lambda t. \lambda l. \lambda x. \lambda y. y \text{ kissed } x \text{ at time } t \text{ at location } l$   
 b.  $\llbracket \exists\text{-LOC} \rrbracket = \lambda P_{\langle l, t \rangle}. \exists l : P(l) = 1$

To see this in more detail, consider the LF in (48-a) with lowest binding, corresponding to the covert reciprocal. I assume for concreteness here that  $\exists\text{-LOC}$  takes scope below Aspect, but I leave it to future work to investigate the relative scope of these two quantifiers and whether there is independent evidence for a particular configuration. Both the existential quantifier over times contributed by the perfective and the existential quantifier over locations take scope above the relevant distributivity operator, and the truth-conditions in (48-b) are predicted for (48-a). Namely (48-a) is predicted to be true as long as Mary kissed Jane and Jane kissed Mary at the same time and the same location. Assuming that the *same location* requirement is met when Jane and Mary kiss on the lips and not when they each kiss the other's arm, the fact that the covert reciprocal is not true in (45) is predicted.<sup>12</sup>

<sup>11</sup>Some people accept (31-a) in a context where Jane and Mary kissed on the cheek. While I won't be able to discuss this in detail, it is conceivable that the *same location* requirement can have different granularities, such that if they kiss on the cheek it counts as one location, but if they kiss each other on the arms, it doesn't.

<sup>12</sup>If we assume that perfective takes scope above the existential quantifier over location, as in the LF in (48-a), we would expect to have LFs where the relevant reciprocal binding occurs above  $\exists\text{-LOC}$  and below Pfv. This would give rise to a reading, paraphrased in (i), which allows for different locations but requires temporal simultaneity. Moreover, assuming that  $\exists\text{-LOC}$  can't take scope above Pfv, there will be no LF which requires spatial overlap but not temporal simultaneity (Same location, different times). It is difficult to test whether such intermediate-scope readings in fact exist as distinct readings, since the resulting reading would be weaker than the low-scope reading (Same time and location) but stronger than the high-scope reading (possibly different time and location). I will

(48) a.



- b.  $\llbracket (40-a) \rrbracket^c = 1$  iff  $\exists t' \in t \exists l : \llbracket \text{kiss} \rrbracket(t')(l)(j)(m) = 1 \wedge \llbracket \text{kiss} \rrbracket(t')(l)(m)(j) = 1$ , where  $t < t_c$

I have therefore discussed two case-studies (simultaneity and spatial overlap) and showed that the low-scope restriction on covert reciprocals can account for the differences in truth-

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therefore not provide evidence that this exists as a distinct reading here but simply note that this is a prediction of the account proposed here.

- (i) Intermediate binding: There is a time such that each of Mary kissed the other at a (possibly different) location at that time.

conditions between overt and covert reciprocal in both of them, given reasonable assumptions about clausal architecture. So far, I have been focusing on predicates and contexts where we do see a truth-conditional difference between overt and covert reciprocals, but I have not touched on the question, featured prominently in Winter's proposal, of which types of predicates give rise to this difference. In the next subsection, I will turn to this question.

### 7.3.3 On the role of the predicate

In this subsection, I will show that my analysis given above predicts certain generalizations regarding which types of predicates give rise to apparent truth-conditional differences between covert and overt reciprocals. In particular, I show that for symmetric predicates, the high-scope and low-scope readings of the overt reciprocal will always be truth-conditionally equivalent and therefore the scope-based proposal given above correctly predicts that there is no truth-conditional difference with the covert counterpart which only has the low-scope LF. I discuss the relationship between what the proposal here predicts and Winter's Reciprocity-Symmetry generalization.

I will begin by discussing how symmetric predicates neutralize the apparent truth-conditional differences between covert and overt reciprocals. I argue that the account given in this section predicts the generalization in (49), which says that overt and covert reciprocals are always truth-conditionally equivalent when the predicate is symmetric.

- (49) **Symmetry and Reciprocity:** Given a predicate  $R$  which takes two individual arguments (possibly in addition to a number of non-individual arguments) and allows for a covert reciprocal object: the covert and overt reciprocal are truth-conditionally equivalent if  $\forall x, y, \alpha_1 \dots, \alpha_n : R(\alpha_1)(\dots)(\alpha_n)(x)(y) \Leftrightarrow R(\alpha_1)(\dots)(\alpha_n)(y)(x)$ .

To derive this, consider again the general schema in (50) for the scope ambiguity that overt reciprocals give rise to in the presence of a quantifier  $Q$  that binds one of the arguments of the predicate  $P$ . Now, suppose that the predicate  $P$  is symmetric and therefore has the property defined in (51-a). We can then show that the low-scope and high-scope readings corresponding to (50-a) and (50-b) are equivalent for any such predicate. The derivation of this is given in (51), but more informally, since the two directions of the reciprocal are equivalent, the conjunction of the two directions is equivalent to one of the conjuncts regardless of whether it takes scope above or below the quantifier. We can therefore delete the second conjunct in both (50-a) and (50-b) without changing the truth-conditions. Therefore, (50-a) and (50-b) are equivalent for a symmetric predicate.

- (50) **General Schema:**

- a. Low-scope distributivity:  $Q(\lambda\alpha_1.P(\alpha_1)(..)(\alpha_n)(a)(b) \wedge P(\alpha_1)(..)(\alpha_n)(b)(a))$
  - b. High-scope distributivity:  $Q(\lambda\alpha_1.P(\alpha_1)(..)(\alpha_n)(a)(b)) \wedge Q(\lambda\alpha_1.P(\alpha_1)(..)(\alpha_n)(b)(a))$
- (51)
- a. A predicate is symmetric iff  $\forall x, y, \alpha_1, \dots, \alpha_n : P(\alpha_1)(..)(\alpha_n)(x)(y) \Leftrightarrow P(\alpha_1)(..)(\alpha_n)(y)(x)$
  - b. Low-scope distributivity reading with a symmetric predicate:  
 $Q(\lambda\alpha_1.P(\alpha_1)(..)(\alpha_n)(a)(b) \wedge P(\alpha_1)(..)(\alpha_n)(b)(a)) = Q(\lambda\alpha_1.P(\alpha_1)(..)(\alpha_n)(a)(b))$   
(since  $P(\alpha_1)(..)(\alpha_n)(a)(b) = P(\alpha_1)(..)(\alpha_n)(b)(a)$  for any value of  $\alpha_1$ )
  - c. High-scope distributivity reading with a symmetric predicate:  
 $Q(\lambda\alpha_1.P(\alpha_1)(..)(\alpha_n)(a)(b)) \wedge Q(\lambda\alpha_1.P(\alpha_1)(..)(\alpha_n)(b)(a)) = Q(\lambda\alpha_1.P(\alpha_1)(..)(\alpha_n)(a)(b))$   
(Justification:  $P(\alpha_1)(..)(\alpha_n)(a)(b) = P(\alpha_1)(..)(\alpha_n)(b)(a)$  for any value of  $\alpha_1$  and therefore  $Q(\lambda\alpha_1.P(\alpha_1)(..)(\alpha_n)(a)(b)) = Q(\lambda\alpha_1.P(\alpha_1)(..)(\alpha_n)(b)(a))$ )
  - d. The high-scope distributivity reading and the low-scope distributivity reading are therefore equivalent for an arbitrary choice of quantifier and arbitrary individuals  $a$  and  $b$ .

Now, we can conclude that with a symmetric predicate, all LFs for the overt reciprocal will give equivalent truth-conditions, regardless of where the reciprocal arguments are bound. The overt reciprocal will therefore not have a reading that is distinct from that of the covert reciprocal, which only has the lowest-scope LF. We have therefore derived the generalization in (49). This generalization seems to hold in general, and is closely related to Winter's reciprocity-symmetry generalization, which I will discuss later in this section. For example, with *date*, a symmetric predicate, we don't see any truth-conditional differences between the overt and the covert reciprocal, both in the unembedded case (52) and under an attitude predicate like *say*.

- (52)
- a. Jane and Mary dated.
  - b. Jane and Mary dated each other.
- (53)
- a. Jane and Mary said that they dated.
  - b. Jane and Mary said that they dated each other.

Another case where we expect to see no truth-conditional differences between overt and covert reciprocals in environments is when there are no quantifiers that bind arguments of the predicate with respect to which a scope ambiguity can arise in the overt reciprocal. For the purpose of illustrating this, let's assume that the only possible covert quantifiers are quantifiers over times and over locations, as outlined in section 7.3.1 and 7.3.2 respectively.

One potential candidate to see if the effects of these quantifiers can be neutralized is to look at stative predicates which do not depend on location, in the present tense. In what follows, I consider the predicate *is a confidante* which, just like *hug* and *kiss*, is non-symmetric and has a

covert reciprocal variant. I show that in the present tense, there is no truth-conditional difference between the overt reciprocal and the covert counterpart, which is expected because *confidante* will simply be a property that holds of an individual at the utterance time and will therefore not have any arguments that are bound by a quantifier over times or locations.

The relational noun *confidante* is not symmetric: if Jane confides in Mary but Mary doesn't confide in Jane, (54-a) is true while (54-b) is not. In this environment, the truth-conditions for (54-a) for example simply depend on whether the property of being a confidante of Mary holds of Jane at the utterance time. It is therefore plausible to conclude that existential quantifiers over time or location do not play a role in deriving these truth-conditions. If this is true then we expect to see no truth-conditional differences between the overt and the covert reciprocal variants, and this is in fact what we observe. Consider the covert and overt reciprocal counterparts in (55-a) and (55-b). Here, there doesn't seem to be any truth-conditional difference between (55-a) and (55-b).

- (54) a. Jane is a confidante of Mary.
- b. Mary is a confidante of Jane.
  
- (55) Jane and Mary are more than friends..
- a. They are confidantes.
- b. They are confidantes of each other.

Note that since this predicate is not symmetric, when the scope ambiguity is due to an embedding predicate like *say*, we expect that the difference between overt and covert reciprocals will resurface. This is exactly what we observe. Consider the scenario in (56): here, only the long-distance reading of the reciprocal is true since each of Mary and Jane only said that they are a confidant of the other person. And indeed, the covert reciprocal in (56-a) is not true in this scenario, while the overt counterpart in (56-b) is.

- (56) **Context:** Jane said that she is a confidante of Mary, but that she doesn't consider Mary a confidante of hers. Mary similarly said that she is a confidante of Jane but that she doesn't consider Jane to be a confidante of hers.
- a. #Jane and Mary said that they are confidantes.
- b. Jane and Mary said that they are confidantes of each other.

There is a question here about how the generalizations outlined in this section, in particular the one regarding symmetry, relate to Winter's Reciprocity-Symmetry generalization, restated in (57).<sup>13</sup> Note that this generalization is formulated as equivalence between the covert reciprocal

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<sup>13</sup>Again, I'm focusing only here on pluralities of two individuals and assuming in (57) that x and y are atomic

(1-place variant) and a logical conjunction consisting of the binary predicate with the two arguments in both orders. As I discussed above, under the theory given here, what is relevant is the relationship between overt reciprocals and covert reciprocals, rather than the relationship between covert reciprocals and the conjunctive paraphrase. The generalization in (49) can therefore be seen as a variant of Winter's generalization that applies to the difference between covert and overt reciprocals.

(57) **Reciprocity-Symmetry Generalization:** Given a reciprocal alternation between a unary collective predicate  $P$  and a binary predicate  $R$ :

$$\forall x, y : P(x \oplus y) \Leftrightarrow P(x)(y) \wedge P(y)(x) \quad \text{iff} \quad \forall x, y : R(x)(y) \Leftrightarrow R(y)(x)$$

While Winter's generalization is bi-directional, the generalization predicted in (49) regarding symmetry goes only in one direction. Namely, the account I propose does not predict that only symmetric predicates give rise to truth-conditional equivalence between overt and covert reciprocals. If we accept the conclusion above that we do in fact get truth-conditional equivalence in the present with *confidante* without symmetry, then the one-directional generalization in (49) is in fact the correct one for the relationship between overt and covert reciprocals.

### 7.3.4 Interim conclusion

In this section, I argued for a version of the syntactic account for covert reciprocals, where covert reciprocals differ from their overt counterparts only in having to be bound as low as possible. Under this account, overt reciprocals are always ambiguous, depending on where the reciprocal is bound, while the covert counterparts simply pick out one of the readings of the overt reciprocals, namely the one with narrow-scope binding.

Before delving into further predictions of this approach, I would like to point out a generalization regarding all of the cases discussed in this section. Abstracting away from the details of the particular case-studies, an immediate prediction is that there will never be a context where the overt reciprocal is not true (on any reading), while the covert reciprocal is, since the covert reciprocal corresponds to one of the readings of the overt counterpart. A lexical account which treats covert reciprocals as 1-place collective predicates does not have a principled explanation for this generalization, since the 1-place and 2-place variants of the predicate are simply two different lexical items. If this generalization is borne out, it would therefore provide strong evidence for the scope-based account.

While this prediction seems to be borne out in the limited set of cases discussed in this section, there seem to be some counterexamples, at least at the surface level.<sup>14</sup> Consider the

individuals.

<sup>14</sup>I would like to thank an anonymous reviewer for pointing this out.

example in (58): this appears to violate our generalization, since the covert reciprocal is true in this context, while the overt reciprocal isn't. Nevertheless, it is possible that this only breaks the generalization at the surface level. For example, suppose that (58-a) was in fact ambiguous between a covert reciprocal variant of (58-b) and a *got divorced* variant, where the argument corresponding to the causer of the divorce is passivized. Then, it could be that the variant which is true in (58-b) is not in fact the covert reciprocal, maintaining that the generalization above holds for covert reciprocals. While this is not a satisfying solution at this point and much remains to be investigated, the example serves to illustrate that the predicted generalization may not always be straightforward to test. It therefore remains to be seen whether this is a correct prediction for all covert reciprocals.

- (58) Context: Jane decided to divorce Mary, but Mary was very unhappy with the divorce and wanted to stay married to Jane.
- a. Jane and Mary divorced.
  - b. #Jane and Mary divorced each other.

Note at this point that one might accept that the differences discussed in this section are due to scope but nevertheless formulate a modified lexical account that accepts that some of the differences between the 1-place variants on one hand and overt reciprocals and conjunctions on the other might be related to scope. In particular, the lexical account can maintain that the 1-place variants are collective predicates and assume that they relate to the 2-place variants as outlined in (59). Now, since this relation between the 1-place variant and the reciprocal conjunction is a lexical one, it will correspond to an LF where conjunction takes scope below all other operators. On the other hand, we would expect that in syntactic conjunction, conjunction can apply to higher nodes rather than to the predicate itself, giving rise to differences in truth-conditions between syntactic conjunction and the logical conjunction in (59).

- (59) Given two predicates  $P_{1\text{-place}}$  and  $P_{2\text{-place}}$  with the same phonological realization, where  $P_{1\text{-place}}$  and  $P_{2\text{-place}}$  participate in a reciprocal alternation, and given two individuals,  $a$  and  $b$ :

$$\forall l, t : \llbracket P_{1\text{-place}} \rrbracket(t)(l)(a \oplus b) \Leftrightarrow \llbracket P_{2\text{-place}} \rrbracket(t)(l)(a)(b) \wedge \llbracket P_{2\text{-place}} \rrbracket(t)(l)(b)(a)$$

For example, assuming that the only relevant quantifier is perfective aspect, an existential quantifier over times as above, this would predict that *Jane and Mary hugged* is true as long as there is a time when Jane hugged Mary and Mary hugged Jane at that time, as shown in (60-a). On the hand, in the case of syntactic conjunction, we can assume that conjunction targets AspP and therefore that the existential quantifier over times takes scope below conjunction. As a result the truth-conditions in (60-b) are predicted for syntactic conjunction. Under this lexical account,

the truth-conditional difference between the 1-place variants and the conjunctive paraphrase with respect to simultaneity can therefore be predicted simply from the lexical generalization in (59).

- (60) a.  $\llbracket \text{Jane and Mary hugged}_{1\text{-place}} \rrbracket = 1$  iff  $\exists t' \in t : \llbracket \text{hugged}_{1\text{-place}} \rrbracket(t')(j \oplus m)$   
 $= 1$  iff  $\exists t' \in t : \llbracket \text{hugged}_{2\text{-place}} \rrbracket(t')(j)(m) \wedge \llbracket \text{hugged}_{2\text{-place}} \rrbracket(t')(m)(j)$   
 b.  $\llbracket \text{Jane hugged}_{2\text{-place}} \text{ Mary and Mary hugged}_{2\text{-place}} \text{ Jane} \rrbracket =$   
 $1$  iff  $\exists t' \in t : \llbracket \text{hugged}_{2\text{-place}} \rrbracket(t')(j)(m) \wedge \exists t'' \in t : \llbracket \text{hugged}_{2\text{-place}} \rrbracket(t'')(m)(j)$

This type of account would be a lexical account that, unlike Winter's (2018), tries to maintain a systematic logical relationship between the 1-place and the 2-place variants for all predicates. Of course, since this account equates the 1-place variant with logical conjunction, this account will face problems in the cases discussed by Winter (2018), for example, where the covert reciprocal patterns with the overt reciprocals but not with a conjunctive paraphrase.

In the next section, I present some additional data and argue that covert reciprocals, modulo the scope requirement, systematically pattern like overt reciprocals and unlike logical conjunction. In particular, I show that covert reciprocals exhibit homogeneity and non-maximality. I argue that these patterns are systematically predicted by the version of the compositional account of reciprocals outlined in section 7.2 (Sauerland, 1998; Beck, 2001) and therefore take them to provide strong evidence that covert reciprocals are in fact reciprocals. While the lexical account can potentially postulate additional lexical rules to account for these inferences, I take the fact that covert reciprocals systematically give rise to the full range of readings that overt reciprocals give rise to as evidence that they are best treated as low-scope reciprocals.

## 7.4 Evidence for the syntactic account

On the account proposed in section 7.3, covert reciprocals only differ from overt reciprocals in requiring that the arguments of *the other* are bound at the lowest possible scope position. This analysis therefore predicts that, setting this requirement aside, covert reciprocals should give rise to the full range of inferences that overt reciprocals give rise to. In particular, Sauerland (1998) and Beck (2001) argue that reciprocals involve covert pluralization/distributivity operators and therefore exhibit phenomena associated with these operators in plural predication more generally. In this section, I discuss two such phenomena - homogeneity and non-maximality - and show how they follow from Sauerland's (1998) analysis which I outlined in section 7.2. I show that in both cases, covert reciprocals pattern with the overt counterparts, as predicted by the account proposed in section 7.3. I argue that this provides evidence against a lexical account which requires additional stipulations to account for the relevant inferences.



### 7.4.1 Homogeneity

The first case where we can see the contribution of pluralization operators in the LFs for covert reciprocals concerns the interpretation of reciprocals in downward-entailing environments. In particular, both overt and covert reciprocals give rise to homogeneity effects, just like plural definites more generally.

Plural predication gives rise to an apparent truth-value gap known as homogeneity (J. D. Fodor, 1970). This is illustrated in (61): neither (61-a) nor its negated counterpart in (61-b) is true in the context in (61), where only one of the two students smiled. Note that the counterpart with an overt *each* in (61-c) lacks homogeneity.

- (61)    **Context:** There are two students, Jane and Mary. Jane smiled but Mary didn't.
- a.    # The two students smiled.
  - b.    # The two students didn't smile.
  - c.    The two students didn't each smile.

An analogous truth-value gap is observed with overt reciprocals (Dotlačil, 2010; Kobayashi, 2021). This is illustrated in (62): neither (62-a) nor its negated counterpart in (62-b) seems to be true in the context in (62). I show below that this pattern is predicted if we assume following Sauerland (1998) that distributivity in (62-b) is due to a covert operator and not the *each* in *each other*. Note again that the counterpart here where distributivity is due to an overt *each* lacks homogeneity (62-c).

- (62)    **Context:** Jane hugged Mary while Mary was sleeping.
- a.    # Jane and Mary hugged each other.
  - b.    # Jane and Mary didn't hug each other.
  - c.    Jane and Mary didn't each hug the other.

Both homogeneity with plural definites more generally and with reciprocals can be accounted for by assuming a truth-value gap in the denotation of DIST (Schwarzschild, 1994; Križ, 2015). I start by illustrating how homogeneity works in the simple case in (61). I assume that DIST has the truth and falsity conditions given in (63) (Schwarzschild, 1994), where after DIST applies to a predicate, the resulting predicate is only false of a plurality if it is false of all atomic subparts of it. Now, assuming that DIST applies to  $\llbracket \text{smile} \rrbracket$  in (61), this correctly predicts that *Jane and Mary smiled* is only false if neither Jane nor Mary smiled (64).

- (63)    For a distributive predicate P,  $\llbracket \text{DIST} \rrbracket (P)(x) =$
- a.    1 iff  $\forall y \leq_{AT} x : P(y) = 1$

- b.  $0 \text{ iff } \neg \exists y \leq_{AT} x : P(y) = 1$
- c.  $\# \text{ otherwise}$

$$(64) \quad \llbracket \text{DIST} \rrbracket (\llbracket \text{smiled} \rrbracket) (m \oplus j) = 0 \text{ iff } \llbracket \text{smiled} \rrbracket (m) = 0 \wedge \llbracket \text{smiled} \rrbracket (j) = 0$$

Abstracting away from the contribution of Aspect for the moment, we can now derive the truth-conditions for the negated reciprocal in (62-b). Again, assuming that negation simply switches truth and falsity, this will correspond to the falsity conditions for the original LF given in (10). The predicted truth-conditions are given in (65): (62-b) is predicted to be true if neither Jane hugged Mary nor Mary hugged Jane. We have therefore derived the homogeneity effect with reciprocals by using independently motivated assumptions about the distributivity operator more generally.

$$(65) \quad \llbracket \text{Mary and Jane hugged each other} \rrbracket = 0 \text{ iff} \\ \llbracket \text{Mary and Jane } 1 \ x_1 \text{ DIST } 2 \ x_2 \text{ hug [the other } x_2 \ x_1 \text{]} \rrbracket = 0 \text{ iff} \\ \lambda y. ((\text{DIST}(\lambda x. \llbracket \text{hug} \rrbracket (x)(y \ominus x))) (y)) (m \oplus j) = 0 \text{ iff} \\ (\llbracket \text{hug} \rrbracket (m)(j) = 0 \wedge \llbracket \text{hug} \rrbracket (j)(m) = 0)$$

In order to see the predictions of the account given in section 7.3 with respect to homogeneity, we have to derive the truth-conditions for the two LFs where distributivity takes scope above and below aspect. It turns out that the same falsity-conditions are predicted under both the simultaneous and non-simultaneous LF: they are both false iff there is no time when Jane hugged Mary and no time when Mary hugged Jane. To see this let's first start with the falsity-conditions for the non-simultaneous LF in (66). Here, distributivity takes scope above aspect, and the homogeneity gap in (63) therefore ensures that (66) is false iff for neither Jane nor Mary is there a time when they hugged the other person.

$$(66) \quad \textbf{Non-simultaneous LF: } \llbracket \text{Mary and Jane } 1 \ x_1 \text{ DIST } 2 \text{ Pfv } 3 \ x_2 \llbracket \text{hug } t_3 \rrbracket \llbracket \text{the other } x_2 \ x_1 \rrbracket \rrbracket = 0 \\ \text{iff } \neg \exists t' \subseteq t : \llbracket \text{hug} \rrbracket (t')(m)(j) = 1 \wedge \neg \exists t' \subseteq t : \llbracket \text{hug} \rrbracket (t')(j)(m) = 1$$

Deriving the resulting falsity-conditions for the simultaneous LF is less straightforward. Here, since the distributivity operator takes scope below the existential quantifier over times, we must ask how homogeneity projects from the scope of this quantifier. I will show below that the final truth-conditions in (67) are predicted if we assume that homogeneity projects with a strong Kleene logic. In particular, this predicts that the simultaneous LF is false iff for neither Jane nor Mary is there a time when they hugged the other person, just like the non-simultaneous counterpart in (66).

$$(67) \quad \textbf{Simultaneous LF: } \llbracket \text{Pfv } 3 \llbracket \text{Jane and Mary} \rrbracket 1 \ x_1 \text{ DIST } 2 \ x_2 \llbracket \text{hug } t_3 \rrbracket \llbracket \text{the other } x_2 \ x_1 \rrbracket \rrbracket = 0$$

$$\text{iff } \neg \exists t' \subseteq t : \llbracket \text{hug} \rrbracket (t')(m)(j) = 1 \wedge \neg \exists t' \subseteq t : \llbracket \text{hug} \rrbracket (t')(j)(m) = 1$$

Strong Kleene gives us the projection pattern in (68) when PFV applies to a trivalent predicate P. Crucially, the result is false only if P is false at all times. This projection pattern derives the right prediction when there is an overt negated existential quantifier over times, like in (69): (69) is correctly predicted to be true iff *for all times t, Mary read the books is false* (i.e. if there is no time when Mary read any of the books).

(68) Given an arbitrary trivalent predicate P of type  $\langle i, t \rangle$ ,

$$\llbracket \text{Pfv} \rrbracket (P) = \begin{cases} 1 & \text{iff } \exists t : P(t) = 1 \\ 0 & \text{iff } \forall t : P(t) = 0 \\ \# & \text{otherwise} \end{cases}$$

(69) Mary has never read the books.

$\approx$  There is no time when Mary read any of the books.

Given this rule for projection out of negated existential, we can now derive the final truth-conditions for (67). The simultaneous LF is predicted to be false iff for all times within the contextually specified time interval both *Jane hugged Mary* and *Mary hugged Jane* are false or equivalently that there is no time when Jane hugged Mary and no time when Mary hugged Jane, as represented in the truth-conditions in (67). We therefore predict that, under both the simultaneous and the non-simultaneous LFs, the negated overt reciprocal in (62-b) is only true if neither Jane hugged Mary nor Mary hugged Jane at any time within the contextually specified time interval. This therefore correctly derives that (62-b) is not true in (62) on both LFs.

Turning to covert reciprocals, since they only have the simultaneous LF, they are predicted to be false according to the falsity-conditions in (67). Assuming again that negation simply switches truth and falsity, we predict the truth-conditions paraphrased in (70) for covert reciprocals and their negated counterpart, with the truth-value gap given in (70-c).

- (70) a.  $\llbracket \text{Jane and Mary hugged} \rrbracket$  is true iff there is some past time such that each of Jane and Mary hugged the other.  
 b.  $\llbracket \text{Jane and Mary didn't hug} \rrbracket$  is true iff there is no past time such that either of Jane and Mary hugged the other.  
 c. Neither is true iff (only one of Jane and Mary hugged the other) or (Jane hugged Mary at one time and Mary hugged Jane at a different time).

We can now test whether covert reciprocals do in fact have the predicted truth-value gap. Consider first the context where only one of Mary and Jane hugged the other in (71). As illustrated in

(70-c), we expect that neither the covert reciprocal nor its negated counterpart is true in such a context. This prediction is borne out: both (71-a) and (71-b) are judged not to be true in (71). Another scenario where we predict a truth-value gap is given in (72), where the two hugging events occurred at different times. Again, neither the covert reciprocal in (72-a) nor its negated counterpart in (72-b) is judged to be true here.

(71) **Context:** Jane hugged Mary while she was sleeping.

- a. # Jane and Mary hugged.
- b. # Jane and Mary didn't hug.

(72) **Context:** Jane hugged Mary while she was sleeping. Later, when Mary woke up and Jane was sleeping, Mary hugged Jane.

- a. # Jane and Mary hugged.
- b. # Jane and Mary didn't hug.

For some speakers, the truth-value gaps in (61), (71) and (72) are not as sharp as what is reported above. Namely, these speakers somewhat accept (62-b), (71-b) and (72-b) in the given contexts. One possibility is that this is due to the availability of a non-maximal reading which would make (71-b) and (72-b) true here. Since it is expected that homogeneity comes hand-in-hand with non-maximality, the possibility of non-maximal readings must be controlled for when testing for homogeneity. I return to this issue at the end of section 7.4.2.

We see that covert reciprocals, like overt reciprocals, give rise to a homogeneity truth-value gap, as predicted by the syntactic account proposed in section 7.3. Note that the lexical account is unable to account for this truth-value gap without positing additional lexical stipulations. In particular, under the lexical account, the 1-place variant of *hug*,  $\llbracket \text{hug}_{1-\text{place}} \rrbracket$ , is a collective predicate that is only defined when it applies to plural individuals. First, we can observe that such collective predicates in general do not give rise to any truth-value gap when they apply to an individual of cardinality 2. For example, consider the example in (73) with the predicate *jointly own this house*. Here, we see that (73-a) and (73-b) have complementary truth-conditions.

- (73)
- a. Jane and Mary jointly own this house.
  - b. Jane and Mary don't jointly own this house.

Furthermore, the relevant homogeneity gap relates the 1-place *hug* to the two-place counterpart applied to the atoms of the plurality  $j \oplus m$ . Access to the two-place counterpart is thus needed to formulate that there is a gap in scenarios where only Mary hugged Jane and where only Jane hugged Mary. If *hug* is in fact a collective 1-place predicate, this truth-value gap therefore requires an additional lexical that relates it to the two-place counterpart. The required stipulation

is given in (74): the collective predicate is true of  $a \oplus b$  if the 2-place variant is true of the atoms of  $a \oplus b$  in both directions and false if the 2-place variant is false in both directions.

(74) **Homogeneity stipulation:**

- a.  $\forall l, t : \llbracket P_{1\text{-place}} \rrbracket(t)(l)(a \oplus b) = 1$  iff  $\llbracket P_{2\text{-place}} \rrbracket(t)(l)(a)(b) = 1 \wedge \llbracket P_{2\text{-place}} \rrbracket(t)(l)(b)(a) = 1$
- b.  $\forall l, t : \llbracket P_{1\text{-place}} \rrbracket(t)(l)(a \oplus b) = 0$  iff  $\llbracket P_{2\text{-place}} \rrbracket(t)(l)(a)(b) = 0 \wedge \llbracket P_{2\text{-place}} \rrbracket(t)(l)(b)(a) = 0$

The lexical account therefore requires additional stipulations to account for the truth-value gap that covert reciprocals exhibit. The syntactic account, on the other hand, straightforwardly accounts for this gap as the result of homogeneity. The presence of a homogeneity gap therefore provides additional evidence for the syntactic approach.

### 7.4.2 Non-maximality

Another phenomenon that has been argued to come hand-in-hand with homogeneity is non-maximality or exception tolerance (Brissou, 1998). In particular, given an appropriate context and QUD, a predicate can be true of a definite plural without being true of all subparts of it. Consider the example in (75): in this context, the predicate *laughed* is judged to be true of the plurality denoted by *the kids*, even though only 15 out of the 20 kids laughed. Note that the availability of non-maximality is dependent on the QUD (Malamud, 2012; Križ, 2015). In particular, a non-maximal reading is only available when worlds where the sentence is true on the maximal readings (i.e. where all of the kids laughed) are in the same cell of the QUD as the actual world. In other words, the maximal reading has to answer the QUD correctly. Given the context in (75), an answer of *all of the kids laughed* would answer the question with *yes*, and in the actual situation, where only 15 of the kids laughed and the other 5 were neutral, the answer to the QUD is also presumably *yes*. Therefore, this context allows for the desired non-maximal reading of (75).

(75) **Context 1:** There are 20 kids at the party. A clown was performing, and 15 of the kids were laughing and having fun, while the other 5 seemed neutral.

- a. A: Was the clown show successful?
- b. B: The kids laughed. (Adapted from Bar-Lev, 2021)

Note that given an alternative context in (76), where again 15 kids laughed, but the other 5 were crying, the definite plural is judged to not be true (76-b). This is expected given the QUD-sensitivity of non-maximal readings. Given the context in (76), the answer to the question in (76-a) is arguably *no*, since the 5 kids that were crying make it so that the show is considered unsuccessful. The desired non-maximal interpretation here therefore provides a different answer

from the maximal one (that all of the kids laughed), which answers the question in the affirmative. Non-maximality therefore is unable to make (76-b) felicitous in this context, even though the same proportion of the kids laughed.

(76) **Context 2:** There are 20 kids at the party. A clown was performing, and 15 of the kids were laughing and having fun, while the other 5 got scared and started crying.

a. A: Was the clown show successful?

b. B: # The kids laughed.

(Adapted from Bar-Lev, 2021)

Now, the decompositional account of reciprocals predicts that, given an appropriate QUD and context, we should also see non-maximality with reciprocals. In particular, as we saw in section 7.2, the decompositional analysis creates the derived predicate in (77), where  $pro_2$  is later bound by the subject. Now, supposing that the subject is *Jane and Mary*, the analogous non-maximal reading to (75-b) will allow the reciprocal to be true when the predicate in (77) is true of only one of Jane and Mary. The resulting non-maximal reading will make the reciprocal true when only one of Jane and Mary hugged the other (78).

(77) **Relevant Predicate:**  $\lambda x. \llbracket \text{hugged} \rrbracket (x)(\llbracket \text{the other} \rrbracket (x)(pro_2))$

(78) **Non-maximal reading of reciprocals:** Jane and Mary R'ed each other is true even though only one of Jane and Mary R'ed the other.

Beginning with overt reciprocals, we see that this prediction is borne out. Consider the context in (79), adapted from the experimental stimuli in Kruitwagen, Poortman, and Winter (2017). Here, one possible QUD that one can infer is whether Jane and Mary had a positive interaction today. Even though it is false that Jane hugged Mary in this context, the overt reciprocal in (79-a) is felicitous here. We therefore see that reciprocals seem to have the non-maximal reading in (78) here. Note that given this QUD, we expect the non-maximal reading to be licensed here. The answer to the QUD here is arguably *yes* (that Jane and Mary did have a positive interaction), since Jane let Mary hug her. The maximal reading of (79-a) (that each of Mary and Jane hugged the other) also answers the question with *yes*. The QUD given here therefore licenses a non-maximal reading that makes (79-a) true in this context.

(79) **Context 2:** Jane has been crying all day. Mary, her partner went and hugged her. Jane didn't hug Mary back but she let Mary hug her and was comforted by the hug.

Possible QUD: Did Jane and Mary have a positive interaction today?

a. B: Mary and Jane hugged each other this morning.

b. B': # Jane hugged Mary this morning.

As with the simple case with a definite plural in (76), altering the context can make the reciprocal not true, even if it remains true that Mary hugged Jane. Consider (80), for example. Here, instead of Jane accepting Mary's hug, she asked Mary to go away. Presumably here, the correct answer to the QUD is *no* (that Jane and Mary did not have a positive interaction). Therefore, the non-maximal reading can't be true in this context, since it would answer the question differently from the maximal one. I have therefore shown that the availability of a weaker reading for the overt reciprocal is sensitive to the QUD, as expected if these weaker readings are in fact due to non-maximality.

- (80)    **Context 1:** Jane has been crying all day. Mary, her partner, went and hugged her. Jane was still very upset and she asked Mary to go away.  
          Possible QUD: Did Jane and Mary have a positive interaction today?
- a.   # Mary and Jane hugged each other this morning.
  - b.   #Jane hugged Mary this morning.

The fact that we get non-maximality at all with reciprocals where the relevant DP is a conjunction of two individuals is surprising, given that this in general does not seem to be possible. For example, in analogous cases with simple distributive plural predication like (81-a), it has been claimed that a non-maximal reading is not available, regardless of the context or the QUD (Križ, 2015; Bar-Lev, 2021, a.o.). Nevertheless, since as it stands there is no conclusive account for the lack of non-maximality in (81-a), it is unclear whether we would in fact expect this restriction to generalize to reciprocals. In particular, Haslinger (2024b, 2025) proposes that the non-maximal reading has to preserve entailment relationships between the maximal reading and the deletion alternatives of the sentence. In (81), this constraint blocks a non-maximal reading, since the maximal universal reading entails both of the deletion alternatives in and any weaker-than-universal reading would no longer entail both of these alternatives. On the other hand, this constraint would not rule out non-maximality with reciprocals. Here, due to the presupposition of *the other*, deleting one of the conjuncts yields an infelicitous sentence (82-b). Therefore, the sentence does not have any deletion alternatives and Haslinger's constraint is trivially satisfied. We therefore see that this account explains the fact that the reciprocal in (82-a) allows for non-maximality while the counterpart with a simple distributive predicate in (81-a) doesn't.

- (81)    a.   Jane and Mary smiled.  
             b.   Deletion alternatives: { Jane smiled, Mary smiled }
- (82)    a.   Jane and Mary hugged each other.  
             b.   \*Jane hugged each other



Turning to covert reciprocals, if the analysis given in section 7.3 is correct, we expect that, just like overt reciprocals, they should give rise to non-maximality. Namely, a covert reciprocal like *Jane and Mary hugged* should be judged true in a scenario where only one of Mary and Jane hugged the other, given an appropriate QUD that licenses the non-maximal readings. This prediction is borne out. In particular, the data in (79) and (80) can be replicated with covert reciprocals, as shown in (83) and (84). This shows that covert reciprocals give rise to a non-maximal reading which is sensitive to the QUD in the same way as the non-maximal reading with overt reciprocals.

- (83) **Context 2:** Jane has been crying all day. Mary, her partner went and hugged her. Jane didn't hug Mary back but she let Mary hug her and was comforted by the hug.  
Possible QUD: Did Jane and Mary have a positive interaction today?  
Mary and Jane hugged this morning.
- (84) **Context 1:** Jane has been crying all day. Mary, her partner, went and hugged her. Jane was still very upset and she asked Mary to go away.  
Possible QUD: Did Jane and Mary have a positive interaction today?  
# Mary and Jane hugged this morning.

Another identifying property of non-maximality is that the exceptions can't be explicitly mentioned (Kroch, 1974; Lasersohn, 1999). This is illustrated in (85-a). Even in the context in (75), which in general licenses non-maximality, mentioning the exceptions as in (85-a) makes the sentence infelicitous. Non-maximality with both overt and covert reciprocals patterns the same way, as shown in (85-b) and (85-c). Even given the context in (83), where the covert reciprocal is felicitous, mentioning the exception (that Jane didn't hug Mary back) makes the sentence infelicitous. This provides further evidence that what we see in (83) is in fact an instance of non-maximality.

- (85) a. # The kids laughed, but 5 of them didn't.  
b. # Mary and Jane hugged, but Jane didn't hug Mary.  
c. # Mary and Jane hugged each other, but Jane didn't hug Mary.

To summarize, I have argued that covert reciprocals, just like overt reciprocals, allow for non-maximal readings, as expected if they have the same LFs as overt reciprocals.<sup>15</sup> The account

<sup>15</sup>The acceptability of reciprocals in Dutch in similar asymmetrical scenarios has been tested experimentally (Kruitwagen et al., 2017; Kruitwagen, 2024). Notably, Kruitwagen (2024) shows that the effect of the manipulating the scenario, in a roughly analogous way to the non-maximal scenarios above, improves the acceptability of covert reciprocals in asymmetric scenarios significantly more than it does for overt reciprocals. At the same time, the manipulations of the scenario did improve the overall acceptability of the overt reciprocals. Kruitwagen takes this difference as evidence for the lexical account of covert reciprocals, and at first sight the fact that there is



proposed here correctly predicts non-maximality to be available even when the subject denotes an individual of cardinality 2. This is because the distributivity operator applies to the 1-place predicate in (86-a): non-maximality then corresponds to cases where this predicate is true of one of the atomic subparts of the plurality but not the other. Since the 1-place collective predicate *hug* on the lexical approach by definition can't be true of atoms, when the subject has cardinality 2, there can be no analogous non-maximality. Under the lexical analysis, this non-maximal reading therefore has to be stipulated as part of the lexical entry for the 1-place collective predicate. The fact that non-maximality in covert reciprocals behaves analogously to non-maximality in plural predication more generally therefore provides additional evidence for the syntactic approach proposed here.

- (86) a. **Derived 1-place Predicate:**  $\lambda x. \llbracket \text{hugged} \rrbracket (x)(\llbracket \text{the other} \rrbracket (x)(pro_2))$   
 b. **Lexical 1-place predicate:**  $\llbracket \text{hug} \rrbracket = \lambda x : x \notin AT.x \text{ hugged}$

We can now see how to sharpen our original homogeneity data from the beginning of this section, by blocking non-maximality, which can obscure the homogeneity effect. To do this, we need a QUD where both directions of the reciprocal are relevant. This will ensure that there is no non-maximality, neither in the positive case nor under negation. Consider the context in (87): here it is relevant whether Mary hugged Jane and whether Jane hugged Mary, and we get a clear homogeneity effect. Both (87-b-i) and (87-b-ii) are clearly not true in this scenario. Similarly, with the overt reciprocal, both (87-c-i) and (87-c-ii) are not true in this scenario. Note that again the counterpart with an overt *each*, which lacks homogeneity, is true here.

- (87) **Context:** Jane and Mary are a couple in a show. They have been fighting, so the game show said they will each get a prize if they hug the other person. Jane went and hugged, but Mary was still upset so she didn't hug Jane back.
- a. Implicit QUD: Who is gonna get a prize?
- b. **Covert reciprocal:**
- (i) #Mary and Jane hugged.
- (ii) #Mary and Jane didn't hug.
- c. **Overt reciprocal:**
- (i) #Mary and Jane hugged each other.

---

this asymmetry between covert and overt reciprocals seems to pose a problem for the account proposed here. Nevertheless, we know that several factors influence how easy it is to access a non-maximal reading, including the choice of predicate. Comparing covert and overt reciprocals, the relevant derived predicate with respect to which a non-maximal reading is obtained is potentially different in the two cases. For the low-scope reading corresponding to the covert reciprocal, this is the predicate in (86-a), while for the high-scope reading of the overt reciprocal the relevant predicate is the one formed above aspect. It is therefore possible that these two different derived predicates result in different conditions regarding how easy it is to license non-maximality.

- (ii) # Mary and Jane didn't hug each other.
- d. **Overt each:** Mary and Jane didn't each hug the other.

In this section, I have argued that covert reciprocals, just like their overt counterparts, give rise to homogeneity and non-maximality. I showed that these readings can be derived straightforwardly by the compositional approach outlined in section 7.2. On the other hand, there is no reason to expect these systematic parallels between overt and covert reciprocals on the lexical analysis. I therefore take the data discussed in this section as providing strong evidence for the syntactic analysis of covert reciprocals given in section 7.3, where they only differ from the overt counterpart in having a low-scope restriction.

### 7.4.3 Alternative accounts

So far, the only alternative I discussed to the scope-based account proposed here is Winter's lexical account, which can be characterized as a purely lexical account in the sense that verbs like *hug* are simply taken to be ambiguous between a 1-place and a 2-place variant. Specifically, Winter's account allows for the alternation between the 1-place and 2-place variants to look different for different predicates, thus requiring us to stipulate homogeneity and non-maximality for each predicate individually. Other existing lexical accounts propose a systematic way to relate the 2-place and 1-place variants (e.g. [Siloni, 2001](#); [Dimitriadis, 2008](#)). In this section, I discuss [Dimitriadis's \(2008\)](#) account of the alternation for concreteness and argue that the arguments from explanatory adequacy given against lexical accounts in this section can be extended to any account that does not rely on common compositional ingredients (i.e. pluralization operators/DIST) to derive covert reciprocals, overt reciprocals and non-collective plural predication more generally.

[Dimitriadis \(2008\)](#), building on [Siloni \(2002\)](#), proposes that covert reciprocals denote events which take a bundled Agent-Theme theta role, as defined in (88). In short, a plurality *A* composed of two atomic individuals *a* and *b* is an Agent-Theme of an event *e* if there is a specification of the event where *a* is the Agent and *b* is the Patient and another specification where *a* is the Patient and *b* is the Agent. Applying this to a sentence like *Jane and Mary kissed*, this is roughly predicted to be true if there is an atomic kissing event which has a specification where Jane kissed Mary and a specification where Mary kissed Jane. For concreteness, we can treat this mapping from the 2 place predicate to the one place predicate as arising from an operator RECIP takes a predicate like *kiss*, which takes an event argument and two individual arguments, and returns a predicate which takes an event argument and a single plural individual argument. The resulting predicate has the truth-condition given in (89-b), which maintains the result from Dimitriadis's proposal.

$$(88) \quad [\text{Agent-Theme}](e, A) = \forall x \in A \exists y[\{x, y\} = A \wedge \exists e'(e' Se \wedge \text{Agent}(e', x) \wedge \text{Patient}(e', y))]$$

$$(89) \quad \text{a. } \llbracket \text{hug} \rrbracket = \lambda e. \lambda y. \lambda x. \text{hug}(e) \wedge \text{Ag}(e)=x \wedge \text{Th}(e)=y$$

$$\text{b. } (\llbracket \text{RECIP} \rrbracket (P))(e)(A) = 1 \text{ iff}$$

$$\forall x \leq_{AT} A \exists y[x \oplus y = A \wedge \exists e'(e' Se \wedge P(e')(y)(x))]$$

Assuming that this event-based accounts predicts the desired differences between covert and overt reciprocals in the positive cases, one can easily define homogeneity as part of the lexical entry of this operator by defining its falsity conditions as in (90): (90) ensures that *Jane and Mary didn't hug* is only true if there is neither a specification of a hugging event where Mary is the Agent and Jane is the Patient, nor one where Mary is the Patient and Jane is the agent.

$$(90) \quad (\llbracket \text{RECIP} \rrbracket (P))(e)(A) = 0 \text{ iff}$$

$$\neg \exists x \leq_{AT} A \exists y[x \oplus y = A \wedge \exists e'(e' Se \wedge P(e')(y)(x))]$$

The problem with this type of approach is that homogeneity and non-maximality simply have to be stipulated as part of the lexical entry of RECIP. Namely, nothing explains why we couldn't have had an alternative operator RECIP' that simply doesn't have the homogeneity gap. Moreover, RECIP is taken to be a new source of homogeneity which is distinct from what gives rise to homogeneity in simple cases like (91). The account proposed in this chapter, on the other hand, allows us to treat pluralization operators as the only source of homogeneity and non-maximality, following Bar-Lev (2025), deriving the desired readings for covert reciprocals from independently needed properties of these operators. I therefore take the common presence of homogeneity and non-maximality in covert reciprocals and in plural predication more generally as evidence for the syntactic decompositional account pursued here.

$$(91) \quad \text{Mary read the books.}$$

## 7.5 Further issues

### 7.5.1 Reciprocal constructions cross-linguistically

The present chapter has focused exclusively on reciprocal alternations in English, where the covert reciprocal variant is identical to the overt counterpart but simply lacks the reciprocal object. Cross-linguistically, this is a special case and a variety of reciprocal constructions that participate in similar alternations have been identified (Siloni, 2001; Dimitriadis, 2004; Siloni, 2012, a.o.). For example, in French (and Romance languages more generally), the clitic 'SE'

gives rise to both reciprocal and reflexive interpretations.<sup>16</sup> The ‘SE’ reciprocal alternates with a variant which has a dedicated reciprocal anaphor in object position, giving rise to certain interpretive differences that partly parallel those we see with covert vs overt reciprocals in English (Dimitriadis, 2004; Siloni, 2012). Other languages, like Hebrew, have a dedicated form of certain verbs that expresses a reciprocal interpretation, which Siloni argues to give rise to the same interpretation as English covert reciprocals.

- (92) Pierre et Jean se sont habillés.  
 Pierre and Jean SE are dressed.  
 a. **Reflexive reading:** Pierre and Jean dressed themselves.  
 b. **Reciprocal reading:** Pierre and Jean dressed each other. (Siloni, 2012, :ex.32)
- (93) Dan ve-dina hitnašku.  
 Dan and-Dina kissed<sub>REC</sub>  
 ‘Dan and Dina kissed.’ (Siloni, 2012, :ex.54c)

While it is worth exploring in future work whether the present scope-based account can be extended to other cases of reciprocal alternations, this faces several challenges at this point. For example, Siloni (2001) claims that ‘SE’ reciprocals in French, unlike covert reciprocals in English and the Hebrew reciprocal verbs like (93), allow for readings like (95), where there are two separate events of kissing each by one of Jean and Marie on the other’s forehead. Nevertheless, Siloni shows that SE-reciprocals still lack long-distance readings with *say* that overt reciprocals allow. We therefore potentially need an account that can make more fine-grained distinctions between the scopal possibilities that different reciprocal constructions allow. Furthermore, Siloni argues that there are syntactic differences between different reciprocal constructions, an issue which the present chapter does not engage with. For example, Siloni shows that while ‘SE’ reciprocals can be formed with ECM verbs, reciprocal verbs like the Hebrew example in (93) can’t be formed with ECM verbs. Siloni argues that such syntactic differences provide evidence that reciprocal verbs in Hebrew and English covert reciprocals fall into a class of reciprocal constructions which are formed in the lexicon, and therefore a complete syntactic account should try to account for these differences.

- (94) \*Jane and Mary kissed on the forehead.

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<sup>16</sup>A couple of interesting questions arise here regarding the relationship between the reflexive and the reciprocal readings: (i) Is there genuine ambiguity or is the sentence underspecified between a reflexive and reciprocal interpretation; (ii) Are there similar restrictions on the reflexive readings as we see with the reciprocal one? These questions are relevant if one wants to extend the present account to other alternations like those between a reflexive verb and a transitive verb with a reflexive anaphor as object.

- (95) Jean et Marie se sont embrassés sur le front.  
 Jean and Marie SE are kissed on the forehead  
 'Jean and Marie kissed each other on the forehead.' (Siloni, 2012, :ex.10)

To summarize, the scope-based account for English covert reciprocals offered here argues that the interpretive differences between covert and overt reciprocals can be reduced to different scopal possibilities for where the reciprocal can be bound. It remains to be seen whether a similar scope-based theory can be extended to account for interpretive differences between different reciprocal constructions cross-linguistically and whether the syntactic differences between different reciprocal constructions that Siloni (2012) identifies can be reconciled within such an account.

### 7.5.2 Other differences between covert and overt reciprocals

The account proposed above argues that covert reciprocals differ from their overt counterpart only in the requirement that they be bound at the lowest scope. While this allows us to predict certain parallels between overt and covert reciprocals, like the presence of homogeneity and non-maximality, there remains some differences between the two constructions that the present account does not straightforwardly account for.

One such difference is in how the two types of reciprocals interact with group nouns. In particular, while certain covert reciprocals are acceptable when the subject is a group noun (96-a), the overt counterpart is unacceptable (or at least degraded). Lexical accounts have a straightforward explanation for the contrast in (96): in (96-a), the collective predicate *met* is predicated directly of the group noun, while in (96-b), since the overt reciprocal involves distributivity, it is incompatible with the group nouns. The syntactic account, on the other hand, does not have a straightforward explanation for this contrast.

- (96) a. Team A met.  
 b. \*Team A met each other.

One thing to note here is that this contrast parallels a larger asymmetry between how covert and overt distributivity interact with group nouns. In particular, (97-a) licenses a distributive inference to the members of the team, ensuring that the derived predicate *wears one blue shoe and one red shoe* is true of each of them. Arguably, this inference is due to a covert distributivity operator, which can distribute to the individual members of the team. Interestingly, the counterpart with the overt distributive marker *each* is again ungrammatical. It is therefore possible that the pluralization operator present in the decompositional LF for (96-a) allows for distributivity

with group nouns, while the counterpart when overt *each* is present blocks such distributivity.<sup>17</sup>

- (97) a. Team A wears one blue shoe and one red shoe.  
b. \*Team A each wears one blue shoe and one red shoe.

Another possible difference between covert and overt reciprocals is how they interact with comitative modifiers/arguments. In particular, while the overt reciprocal in (98-a) does not allow for the comitative *with Mary*, the sentence becomes acceptable when the reciprocal is dropped (98-b). Of course, it is possible that (98-a) simply is not a covert reciprocal and that *fought* optionally takes a comitative argument. This seems to be supported by the fact that not all predicates which have covert reciprocal variants allow for a comitative, as evidenced by *hug* in (99-a). One problem for this view is that in languages like Hebrew where the seemingly 1-place variant in reciprocal alternations is different from the standard transitive predicate, the reciprocal verb and not the standard transitive verb has to be used with the comitative (100) (Siloni, 2002; Dimitriadis, 2004). This suggests that at least when we look at languages other than English, the proposal that (98-b) is not a reciprocal does not seem straightforwardly tenable. I therefore leave the analysis of how comitatives interact with different reciprocal constructions in this framework as a problem for future work.

- (98) a. \*John fought each other with Mary.  
b. John fought with Mary.

- (99) a. \*John hugged with Mary  
b. \*John hugged each other with Mary.

- (100) Dan hitnašek im Dina.  
Dan kissed<sub>REC</sub> with Dina.  
Dan and Dina kissed.

## 7.6 Conclusion

I have argued for a theory where LFs for sentences like (101) involve a covert reciprocal in object position. I provided evidence for a syntactic analysis along the lines of Heim et al. (1991) by showing that, just like with overt reciprocals, we can diagnose the presence of pluralization/distributivity operators in the LFs for covert reciprocals.

- (101) a. Mary and Jane dated.

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<sup>17</sup>Recall that I have been assuming that it is a pluralization operator and not *each* itself that is responsible for distributivity with overt reciprocals, so it is not clear whether the parallel between (96-b) and (97-b) is in fact expected on such an account.

- b. Mary and Jane hugged.

The one stipulated assumption that this analysis relies on is the restriction in (102) that requires covert reciprocals to be bound at the lowest possible scope position. I argued that this restriction is sufficient for explaining a range of readings we obtain with covert reciprocals, without appealing to lexical stipulations.

- (102) **Restriction on Covert Reciprocity:** The *each other* in a reciprocal can be elided only if the two arguments of *other* are bound at the lowest possible position.

While proposing a fully worked out account of the source of the restriction in (102) is beyond the scope of this chapter, there are interesting parallels between (102) and certain restrictions on implicit arguments more generally. In particular, it has been argued that existential implicit arguments must take narrow scope (J. Fodor & Fodor, 1980). For example, the predicate *eat* can surface with or without an overt object (103), and we can treat the counterpart without an overt object in (103-b) as having a covert existential object. Just like we saw with covert reciprocals, there are differences in truth-conditions between the overt and covert existential in (103-b). This is illustrated in (104). While (104-a), with the overt *something*, can have a reading where there is something specific that exactly half of the students ate, (104-b) lacks this reading. This can be explained if overt *something* can take scope above *Exactly half*, while covert *something* has to take narrow scope, as noted in Fodor and Fodor (1980).

- (103) a. John ate the cake.  
b. John ate (something).
- (104) a. Exactly half of the students ate something. half >> something; something >> half  
b. Exactly half of the students ate. half >> Something; \*something >> half

Scoping out, the present account therefore allows us to maintain a relatively simple lexical meaning for predicates which have previously been argued to be lexically collective. In particular, the different readings the predicates give rise to are predicted as a result of different scopal interactions with independently needed pluralization operators. This raises questions about the division of labor between lexical semantics and the compositional semantics in plural predication more generally. For example, we can wonder whether other inferences which have been attributed to lexical collectivity, like the collective readings of (105), can also be reduced to interactions between a relatively bleached lexical semantics and appropriate pluralization operators. Moreover, while the chapter focuses only on reciprocals where the subject denotes an individual of cardinality 2, reciprocals with larger pluralities have been argued to give rise to a wider range of readings (Dalrymple, Kanazawa, Kim, Mchombo, & Peters, 1998, a.o.). Beck

(2001) similarly attributes this variability to different choices of pluralization operators and to independently needed properties of these operators, allowing us to do without certain lexical stipulations. The current work can therefore be seen as part of a larger body of work whose goal is to reduce the role of lexical stipulations in plural predication, relying instead on a few common compositional ingredients and their scopal interactions.

(105) Jane and Mary lifted the piano.



# Chapter 8

## Conclusion

This thesis has developed a scope-based syntactic framework for plural predication, where cumulative inferences, reciprocal alternations and the different readings of *together* arise from scope interactions between plural DPs, pluralization operators and a small inventory of other scope-taking elements. I have therefore argued that none of these inferences are encoded in the lexical meanings of predicates. For cumulative inferences, I showed through various diagnostics that their availability depends on movement and that they must therefore be generated by syntactic pluralization operators that are parasitic on movement. For reciprocal alternations, I argued that covert reciprocals involve the same pluralization operators responsible for cumulative inferences and proposed an appropriate syntactic account that relies on these operators. Finally, I argued that the different readings of *together* should be derived from having plural DPs take scope in different positions, yielding a more explanatory account than lexical alternatives. Across these constructions, I proposed that the resulting scope configurations have to be constrained by independently needed principles, including a generalized economy constraint and a constraint requiring implicit arguments to take low scope.

At this point, I have therefore proceeded by exclusion, looking at a particular set of inferences and arguing that they are not encoded in the lexicon. This raises broader questions about what the lexicon does in fact encode in this domain. In this concluding chapter, I address this question by offering some conjectures about possible restrictions on what the lexicon can encode, both within plural predication and beyond. First, I provide some tentative reasons to think that even certain cases of collective predication, which I have so far assumed are lexically encoded, have to be generated in the syntax. This raises the more radical possibility that lexical predicates in natural language only have singular individuals in their domain and that all plural predication is mediated by syntactic operators. Second, I turn to the more general question of whether there are universal constraints that limit what can be encoded into the lexical meanings of predicates. I suggest a possible economy constraint on lexicalization, which optimizes the expressive power

of the lexicon, relative to its size. These considerations of economy explain the ban on lexical cumulativity and have consequences beyond plural predication, for example for the question of whether antonyms have to be syntactically decomposed.

## 8.1 What does the lexicon encode?

While this dissertation has attributed a variety of inferences in plural predication to scope interactions between pluralization operators, I have still been assuming that the lexicon plays an important role in encoding when a predicate is true primitively of a plurality. For example, given a predicate like *dance*, I've been assuming that the lexical meaning of the predicate ensures that  $\llbracket \text{dance} \rrbracket$  is true of a plurality iff they danced together. Collective readings with *together* therefore arise when a lexical predicate applies directly to a plurality. This is in contrast with other readings, such as the simultaneous one with predicates like *stood up*, which I argued involve pluralization operators applying above the predicate. In this section, I'd like to discuss some parallels between the collective reading and the other readings of *together* which suggest that even the collective reading might not be encoded in the lexical semantics of predicates.

- (1)  $\llbracket \text{dance} \rrbracket(a \oplus b) = 1$  iff  $a$  and  $b$  danced together.

One inference pattern which we see with predicates like *stood up* concerns the relationship between when the sentence with *together* is true and when the predicate is true of the individual parts of the subject plurality. In particular, given any two individuals  $x$  and  $y$ ,  *$x$  and  $y$  stood up together* entails that  *$x$  stood up* and that  *$y$  stood up* (2). These distributive entailments follow straightforwardly from my analysis of simultaneous *together*. As shown in chapters 4 and 5, the truth-conditions in (3) are predicted for the simultaneous reading where *together* takes scope above the lowest pluralization operator but below Aspect. Now it's clear that (3) entails that there is a time such that Mary stood up at that time and a time such that Jane stood up at that time, thus accounting for the inference pattern in (2).

- (2) **Distributivity inferences with simultaneous readings:**

- a. Jane and Mary stood up together.  $\rightarrow$  Jane stood up and Mary stood up.
- b. Jane and Mary yelled together.  $\rightarrow$  Jane yelled and Mary yelled.

- (3)  $\llbracket \text{Jane and Mary stood up together} \rrbracket =$   
 $1$  iff  $\exists t' \subseteq t : \llbracket \text{stood up} \rrbracket(t')(j) = 1 \wedge \llbracket \text{stood up} \rrbracket(t')(m) = 1$ , where  $t$  is a contextually salient past time interval.

Given the assumption that the collective meaning of a predicate like *dance* is encoded in

the lexicon, we might expect that unlike the simultaneous reading above, there is no general pattern with respect to the relationship between when such a predicate is true collectively of a plurality and when it is true of its atomic parts. This expectation does not seem to be borne out. In particular, we see that a variety of 1-place predicates license distributive inferences from the collective reading, in parallel to what we saw with the simultaneous reading in (2).<sup>1</sup> The framework laid out so far has to attribute these inferences to the lexical semantics of predicates like *dance*. The parallel between these inferences and the ones we get via the syntax in (2) therefore raises the question of whether at least some readings which I attribute to lexically specified collective readings should in fact be reanalyzed as arising from scope interactions between plural DPs and pluralization operators.

(4) **Distributivity inferences with collective readings:**

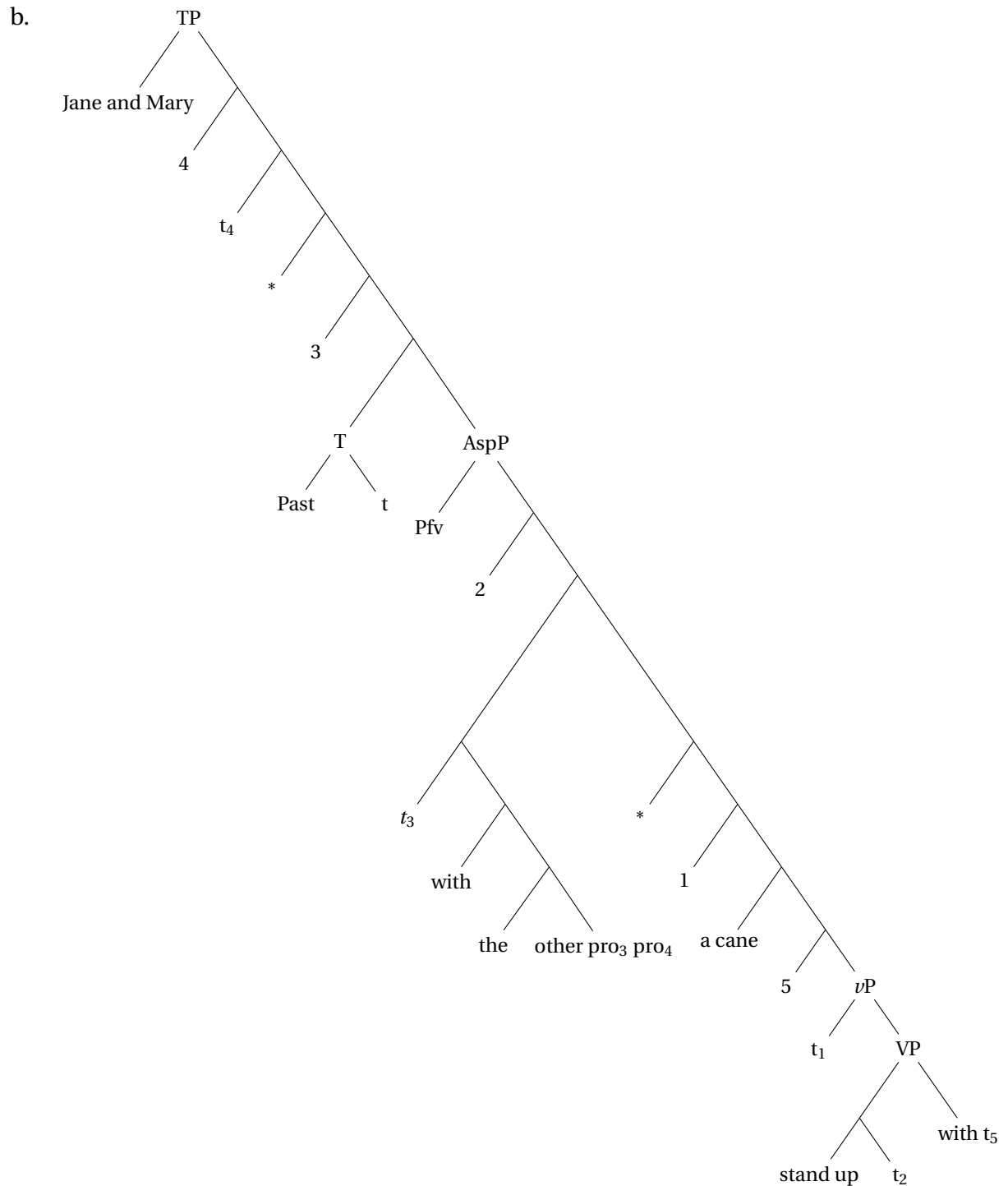
- a. Jane and Mary danced together. → Jane danced and Mary danced.
- b. Jane and Mary ate together. → Jane ate and Mary ate.
- c. Jane and Mary sang together. → Jane sang and Mary sang.
- d. Jane and Mary played together. → Jane played and Mary played.

Another parallel between the simultaneous reading and certain collective readings that suggests that the latter should be derived in the syntax concerns the interaction of *together* with the availability of a co-variation reading with respect to a lower indefinite. To see this first with the simultaneous reading, consider the example in (5-a): (5-a) has a reading where each of Jane and Mary used a different cane to stand up but they did so at the same time. To get this reading, there needs to be a distributivity/pluralization operator that takes scope below the *x with the other* conjunction and above *a cane*. At the same time, in order to ensure the desired simultaneous reading, this operator has to take scope below Aspect. The resulting LF is given in (5-b): (5-b) is true iff there is a time such that each of Jane and Mary stood up at that time using a cane.

- (5) a. Jane and Mary stood up together with a cane.

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<sup>1</sup>Note that the pattern in (4) seems to hold in general for the collective readings of underspecified 1-place predicates, but clearly does not generalize to two-place predicates like *lift the piano*, where *Jane and Mary lifted the piano together* doesn't entail that each of Jane and Mary lifted the piano. Moreover, this pattern does not hold of certain collective predicates like *be numerous* or *be three in number*, which clearly don't license distributive inferences. Nevertheless, if this pattern holds in general for underspecified 1-place predicates, then it points towards the need for an explanation that relies on the compositional mechanisms that I've argued for in this dissertation.



Interestingly, this same kind of distributive reading over a lower indefinite arises with the collective reading of predicates like *eat* and *lift the piano*. For example, (6-a) is true in a scenario where Jane and Mary ate together and they each used one hand to eat. Similarly, (7-a) is true if Jane and Mary lifted the piano together and they each used one hand to contribute to the lifting. Now, these readings in (6-a) and (7-a) are puzzling if we assume that the collective reading requires

the predicate to apply directly to the plurality with no intervening pluralization operators. In particular, in order to get distributivity over the indefinite, there needs to be a distributivity operator below the *with the other* coordination, but this distributivity operator will also ensure that the predicate applies to the atomic parts of the plurality, blocking the collective reading. On the other hand, if the collective reading, like the simultaneous one, was due to a pluralization operator that takes scope below some other scope-taking element, the readings in (6-a) and (7-a) can be straightforwardly accounted for just like in (5-b).

- (6) **Scenario:** Jane and Mary sat down at a table to eat together. They each decided to tie one hand behind their back and eat using only one hand.
  - a. Jane and Mary ate together with one hand.
- (7) **Scenario:** Jane and Mary were in a competition where you have to lift a piano using only one hand. Jane and Mary collaborated to lift one piano, each using one hand.
  - a. Jane and Mary lifted the piano together with one hand.

Of course there might be alternative ways of thinking of the clausal architecture that can derive the desired reading in (6-a) while maintaining that the collective reading is encoded in the lexical semantics of predicates like *eat*. For example, frameworks of plural predication that rely on event semantics allow for distributivity to target specific theta positions, which would allow for the subject to be interpreted distributively with respect to the modifier *with one hand* but collectively with respect to the eating event (e.g. [Champollion, 2016](#)). Nevertheless, the parallels between the collective and simultaneous readings in this section, within the framework that I have been relying on in this dissertation, suggest that the collective readings of predicates like *ate* and *danced* are in fact derived in the syntax.

At this point, we can therefore ask whether the lexicon ever specifies when a predicate is true primitively of a plurality. In particular, we saw that predicates that have been argued to be underspecified (i.e. can hold primitively of both singular individuals and pluralities) might turn out to require pluralization operators in order to apply to pluralities. Similarly, in chapter 7, I argued that a certain class of seemingly collective predicates, like *hug* and *meet*, are in fact two-place predicates with a reciprocal in object position. One radical possibility that should be explored is that this points to a general restriction: lexical predicates in natural language only have singular individuals in their domain and all plural predication arises solely via syntactic operators.

Of course, this general restriction faces a variety of challenges from the outset. First, there is the question of how to derive the collective readings with *danced* and *ate* via scope in the same way as the simultaneous and the proximate readings. Collective dancing and eating seem to

require some sort of joint action on the part of the members of the subject plurality and it is not clear how this inference can be derived syntactically. Second, this restriction predicts that all collective predicates, including those that don't have a 2-place counterpart, like *gather* and *numerous*, have at their core a predicate that holds only of atoms. It remains to be seen what kind of syntactic decomposition is needed to make such an account work. Finally, there is a question of how this restriction manifests itself beyond the verbal domain. For example, are collective nouns like *team* and *committee* also derived from predicates that apply to atoms or is there a different notion of atomicity that is necessary to account for these nouns.

I will have to leave all of these interesting questions for future work. At this point, we can simply speculate that there might be something inherently syntactic about plural predication that explains why we saw in the different case-studies that the role of the lexicon is very minimal in deriving different inferences in plural predication.

## 8.2 The division of labor between syntax and the lexicon

I will now turn to the question of whether the case-studies I considered in this dissertation give us any insights into general principles that constrain the division of labor between syntax and the lexicon beyond plural predication. More specifically, if we accept the framework given in this thesis, there appears to be a pattern, at least in plural predication, where the linguistic system favors a simpler lexicon, combined with a limited set of operators that generate a variety of different readings. It might therefore be useful at this point to speculate about why the contributions of the lexicon and the syntax have to be divided in this way. While a detailed investigation of this question is beyond the scope of this dissertation, the goal of this section is to suggest that these restrictions on the lexicon might follow from a lexicalization economy constraint that mediates the division of labor between syntax and the lexicon.

More specifically, I argued in chapter 2 and chapter 3 that lexical predicates in natural language can't be cumulative. This was crucial in order to predict that cumulative inferences always come from syntactic pluralization operators and never from the lexical meanings themselves. Looking beyond plural predication, there are other areas where this question of whether certain inferences are encoded in the lexicon or in the syntax has been explored. For example, it has been argued that antonym pairs, at least in the domain of gradable adjectives, have to be derived in the syntax (Rullmann, 1995; Büring, 2007; Heim, 2008, a.o.). Namely, given a pair of antonyms like *tall-short*, these syntactic accounts of antonyms argue that *short* should be syntactically decomposed into *not tall*.

I want to explore the possibility that both the ban on lexically cumulative predicates and the possible ban on lexical antonyms follow from a lexicalization economy constraint that optimizes

the expressive power of the lexicon, relative to its size, given a limited set of independently needed logical operators. Two such operators, which will be relevant to cumulative inferences and to antonyms respectively, are the pluralization operators ( $*$  and  $**$ ) and predicate negation. In particular, we saw in chapter 2 that regardless of whether one adopts lexical cumulativity or not, pluralization operators are independently needed to account for cumulative inferences over derived predicates. Similarly, we can assume that negation is a universally available logical operator.

Now, we can ask what it would take for a lexicon to be optimal relative to these operators. More specifically, we can consider two limiting conditions that make a lexicon  $L$  non-optimal. First, suppose we have an alternative lexicon  $L'$  which has the same number of lexical predicates as  $L$  but can generate, by applying these logical operators to the predicates in the lexicon, a superset of the derived predicates that  $L$  can generate. The lexicon  $L$  will then be non-optimal with respect to expressive power. Second, suppose we have an alternative lexicon  $L'$  which has a smaller number of lexical items than  $L$  but generates the same set of derived expressions by applying the logical operators. The lexicon  $L$  will then be non-optimal with respect to its size. While I leave a detailed formal formulation of these constraints to future work, I want to suggest here that these considerations plausibly rule out lexically cumulative predicates and lexical antonyms, given the independent availability of pluralization operators and negation in the lexicon.

Starting with the ban on lexically cumulative predicates, suppose we have a lexicon  $L$  with a lexically cumulative 1-place predicate  $P$ . We can construct an alternative lexicon  $L'$ , where  $P$  is replaced by a non-cumulative predicate  $P'$  which is not in  $L$ , such that  $[[*]]([P'])=P$ .<sup>2</sup> Assuming that the lexicalization economy constraint considers all of the predicates which are derived by applying  $*$  to lexical predicates, we can conclude that  $L'$  will have the same size as  $L$  but generate a superset of the predicates that  $L$  can generate. This is because  $L'$  will generate both  $P'$  as a lexical predicate and  $P$  by applying  $*$  to  $P'$ . On the other hand,  $L$  will have  $P$  as a cumulative lexical predicate and simply return the same predicate  $P$  when  $*$  is applied to  $P$ . We therefore see that economy considerations plausibly rule out a lexicon which contains both lexically cumulative predicates and pluralization operators like  $*$  which can derive cumulative predicates from non-cumulative ones.

Similarly, for antonym decomposition, suppose we have a lexicon  $L$  with a pair of 1-place predicates  $P$  and  $Q$  which are antonyms. We can construct an alternative lexicon  $L'$ , which does not contain  $Q$  but is otherwise identical to  $L$ .  $L'$  will have a smaller size than  $L$  but be able to

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<sup>2</sup>This will not be possible if  $L$  already contains all of the predicates  $P'$  such that  $[[*]](P')=P$ , but such a lexicon will clearly not be optimal, since we can simply remove  $P$  from the lexicon and generate it instead by applying  $*$  to any of these predicates  $P'$ .

derive the same set of derived predicates as L by applying predicate negation. Namely, L' can generate Q by applying negation to P. The lexicon L which lexicalizes both P and Q will therefore be non-optimal, since there is an alternative lexicon that is smaller but has the same expressive power.

Taken together, the ban on lexical antonyms and the ban on lexical cumulativity therefore appear to support the idea that there is a general pressure against lexicalizing meanings that can be generated by independently available syntactic mechanisms. Of course, this idea will have to be developed in more detail in order to make principled predictions about what kinds of meanings can be lexicalized in a single lexical item. Moreover, this raises a variety of open questions about how general this constraint might be and where we see its consequences beyond the two limited case-studies discussed in this section, questions which I hope to explore further in future work.



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